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Thanks you so much for your help. I scored 972 in my exam today. More than 90% were from your PDFs!

Dana

September 04, 2022



Thanks a lot for this updated AZ-900 Q&A. I just passed my exam and got 974, I followed both of your Az-900 videos and the 6 PDF, the PDFs are very much valid, all answers are correct. Could you please create a similar video/PDF for DP900, your content/PDF's is really awesome. The team did a really good job. Thank You 😊.

Ahamed Shibly

2 months ago



Customer support is really fast and helpful, I just finished my exam and this video along with the 6 PDF helped me pass! Definitely recommend getting the PDFs. Thank you!

October 22, 2022



Passed my exam today with 891 marks. Out of 52 questions, 51 were from certyiq PDFs including Contoso case study. Thank You certyiq team!

Henry Rome

2 months ago



These questions are real and 100 % valid. Thank you so much for your efforts, also your 4 PDFs are awesome, I passed the DP900 exam on 1 Sept. With 968 marks. Thanks a lot, buddy!

Esmaria

2 months ago



Simple easy to understand explanations. To anyone out there wanting to write AZ900, I highly recommend 6 PDF's. Thank you so much, appreciate all your hard work in having such great content. Passed my exam Today -3 September with 942 score.

Google

(Professional Cloud Network Engineer)

Professional Cloud Network Engineer

Total: **236 Questions**

Link: <https://certyiq.com/papers/google/professional-cloud-network-engineer>

Question: 1

You need to restrict access to your Google Cloud load-balanced application so that only specific IP addresses can connect.

What should you do?

- A. Create a secure perimeter using the Access Context Manager feature of VPC Service Controls and restrict access to the source IP range of the allowed clients and Google health check IP ranges.
- B. Create a secure perimeter using VPC Service Controls, and mark the load balancer as a service restricted to the source IP range of the allowed clients and Google health check IP ranges.
- C. Tag the backend instances "application," and create a firewall rule with target tag "application" and the source IP range of the allowed clients and Google health check IP ranges.
- D. Label the backend instances "application," and create a firewall rule with the target label "application" and the source IP range of the allowed clients and Google health check IP ranges.

Answer: C

Explanation:

The correct answer is **C. Tag the backend instances "application," and create a firewall rule with target tag "application" and the source IP range of the allowed clients and Google health check IP ranges.**

Here's why:

Firewall rules in Google Cloud Platform (GCP) are the primary mechanism for controlling network traffic to and from virtual machines (VMs). These rules can be configured based on various criteria including source IP ranges, target tags, and protocols.

In this scenario, the requirement is to restrict access to the load-balanced application based on source IP addresses. This aligns perfectly with the functionality of firewall rules. You create an ingress firewall rule to allow traffic only from specific IP address ranges.

Target tags are used to apply firewall rules to specific instances. By tagging the backend instances with "application," the firewall rule can be targeted only to these instances making the access restriction load balancer aware. This means any traffic coming from IPs not in the source IP list for this rule will be rejected by the firewall rule before it can get to the application running on the VM instances.

Crucially, the firewall rule also needs to allow health checks from Google's health check IP ranges so that the load balancer can determine the health of backend instances and route traffic correctly. Without allowing this health check traffic, your application will become unreachable through the load balancer.

Options A and B involve using VPC Service Controls, which primarily focus on protecting data within service perimeters and not designed to filter traffic based on source IP for applications in the manner described in the question. While VPC Service Controls can control access to services based on IPs, they are not as simple for managing general traffic to an application, and would require extra layers of complexity beyond what the question implies.

Option D is incorrect because firewall rules use target tags, not target labels.

Therefore, option C is the most appropriate and simplest method for restricting access to the application based on source IP addresses.

Authoritative Links:

Google Cloud Firewall Rules: <https://cloud.google.com/vpc/docs/firewalls>

Google Cloud Firewall Rule Concepts: <https://cloud.google.com/vpc/docs/firewalls#concepts>

Google Cloud Target Tags: https://cloud.google.com/vpc/docs/firewalls#target_tags

Google Health Check IP Ranges: <https://cloud.google.com/load-balancing/docs/health-checks#ip-ranges>

Question: 2

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Your end users are located in close proximity to us-east1 and europe-west1. Their workloads need to communicate with each other. You want to minimize cost and increase network efficiency. How should you design this topology?

- A. Create 2 VPCs, each with their own regions and individual subnets. Create 2 VPN gateways to establish connectivity between these regions.
- B. Create 2 VPCs, each with their own region and individual subnets. Use external IP addresses on the instances to establish connectivity between these regions.
- C. Create 1 VPC with 2 regional subnets. Create a global load balancer to establish connectivity between the regions.
- D. Create 1 VPC with 2 regional subnets. Deploy workloads in these subnets and have them communicate using private RFC1918 IP addresses.

Answer: D

Explanation:

Option D is the most efficient and cost-effective solution for inter-region communication within Google Cloud Platform (GCP) in this scenario. Creating a single VPC network with regional subnets in us-east1 and europe-west1 allows workloads to communicate using private RFC1918 IP addresses. This avoids the need for public IP addresses, which can incur additional costs and introduce security complexities. Private communication within a single VPC is also more efficient due to Google's internal networking infrastructure.

Options A and B involve establishing connectivity between two separate VPCs, which requires extra configuration and costs for inter-VPC routing solutions like VPN gateways (Option A) or external IP addresses (Option B). These approaches create unnecessary overhead compared to using subnets within a single VPC. Option C, using a global load balancer, is intended for directing traffic to multiple endpoints for load distribution and failover, not for general inter-instance communication; it introduces unnecessary complexities and costs for the use case described.

A single VPC facilitates simpler network management and easier implementation of consistent security policies across regions. By using private IP addresses, the traffic remains within Google's network, minimizing latency and improving performance. This approach leverages Google's underlying networking capabilities for optimal intra-VPC communication. It aligns with best practices for building cost-effective, highly-performant, and secure cloud environments.

Further Research Links:

VPC networks overview: <https://cloud.google.com/vpc/docs/vpc>

Subnets: <https://cloud.google.com/vpc/docs/subnets>

Private IP addresses: <https://cloud.google.com/vpc/docs/configure-private-google-access>

Inter-VPC networking: <https://cloud.google.com/vpc/docs/inter-vpc-connectivity>

Load Balancing Overview: <https://cloud.google.com/load-balancing/docs/load-balancing-overview>

Question: 3

CertyIQ

Your organization is deploying a single project for 3 separate departments. Two of these departments require network connectivity between each other, but the third department should remain in isolation. Your design should create separate network administrative domains between these departments. You want to minimize operational overhead.

How should you design the topology?

- A. Create a Shared VPC Host Project and the respective Service Projects for each of the 3 separate departments.
- B. Create 3 separate VPCs, and use Cloud VPN to establish connectivity between the two appropriate VPCs.
- C. Create 3 separate VPCs, and use VPC peering to establish connectivity between the two appropriate VPCs.
- D. Create a single project, and deploy specific firewall rules. Use network tags to isolate access between the departments.

Answer: C

Explanation:

The correct answer is C, creating three separate VPCs and using VPC peering for connectivity between two. This approach best addresses the requirements of network isolation and minimizing operational overhead for three departments with specific connectivity needs.

Option C aligns with the principle of least privilege by creating distinct VPCs, thereby establishing separate network administrative domains. This ensures that the third department remains completely isolated, preventing any unintended network interactions. VPC peering provides a direct, private connection between the two connected departments' networks without needing to traverse the public internet, improving performance and security. This method also avoids the complexities of setting up and maintaining VPN tunnels, reducing operational overhead.

While Shared VPC (option A) centralizes network administration, it does not easily achieve complete network isolation between service projects, as all projects share a single host project's network. Option B, using Cloud VPN, is more complex and incurs higher management overhead than VPC peering. Option D, using a single VPC with firewall rules and network tags, creates a single administrative domain, making it less robust for complete isolation and more complex to manage than multiple VPCs. Also, it's more prone to misconfigurations which can result in unwanted connectivity between departments.

VPC peering is a managed service, minimizing the need for manual intervention, and it also offers better performance compared to VPN, as traffic doesn't need to be encrypted and decrypted.

Authoritative Links for Further Research:

VPC Peering: <https://cloud.google.com/vpc/docs/vpc-peering>

Shared VPC: <https://cloud.google.com/vpc/docs/shared-vpc>

Cloud VPN: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/overview>

Question: 4

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You are migrating to Cloud DNS and want to import your BIND zone file.
Which command should you use?

- A. `gcloud dns record-sets import ZONE_FILE --zone MANAGED_ZONE`
- B. `gcloud dns record-sets import ZONE_FILE --replace-origin-ns --zone MANAGED_ZONE`
- C. `gcloud dns record-sets import ZONE_FILE --zone-file-format --zone MANAGED_ZONE`
- D. `gcloud dns record-sets import ZONE_FILE --delete-all-existing --zone MANAGED_ZONE`

Answer: C

Explanation:

The correct command to import a BIND zone file into Cloud DNS is **C. `gcloud dns record-sets import ZONE_FILE --zone-file-format --zone MANAGED_ZONE`**.

Here's why:

gcloud dns record-sets import: This is the core command within the Google Cloud SDK (gcloud) specifically designed to import DNS record sets. It signifies the intent to bring existing DNS records into Cloud DNS.

ZONE_FILE: This placeholder represents the actual file path of the BIND zone file you want to import. The command needs the path to locate the source DNS data.

--zone-file-format: This crucial flag explicitly tells gcloud that the provided input ZONE_FILE is in the BIND zone file format, which is a widely used standard for DNS zone data. Without this flag, gcloud would not interpret the file correctly.

--zone MANAGED_ZONE: This flag indicates the target Cloud DNS managed zone where the records will be imported. MANAGED_ZONE is a placeholder for the actual name of your Cloud DNS managed zone.

Option A is incorrect because it lacks the --zone-file-format flag, which is necessary for parsing the BIND zone file correctly. Option B incorrectly suggests using --replace-origin-ns, which is used to replace NS record during import. Option D uses the --delete-all-existing flag which is also incorrect, as this flag would delete all existing records before import instead of just importing the zone file, this is not mentioned in the question and it might be dangerous to use.

In essence, the command correctly informs gcloud about the data source (BIND zone file), its format, and the destination within Cloud DNS, enabling a smooth import process.

Authoritative Links:

gcloud dns record-sets import documentation: <https://cloud.google.com/sdk/gcloud/reference/dns/record-sets/import>

Cloud DNS Overview: <https://cloud.google.com/dns/docs/overview>

BIND zone file format: https://en.wikipedia.org/wiki/Zone_file

Question: 5

CertyIQ

You created a VPC network named Retail in auto mode. You want to create a VPC network named Distribution and peer it with the Retail VPC.

How should you configure the Distribution VPC?

- A. Create the Distribution VPC in auto mode. Peer both the VPCs via network peering.
- B. Create the Distribution VPC in custom mode. Use the CIDR range 10.0.0.0/9. Create the necessary subnets, and then peer them via network peering.
- C. Create the Distribution VPC in custom mode. Use the CIDR range 10.128.0.0/9. Create the necessary subnets, and then peer them via network peering.
- D. Rename the default VPC as "Distribution" and peer it via network peering.

Answer: B

Explanation:

Here's a detailed justification for why option B is the correct approach for peering a custom VPC with an auto-mode VPC in Google Cloud:

Option B is correct because it addresses the core requirements for successful VPC peering while avoiding conflicts with the auto-mode VPC's automatically created subnets. The key reason lies in the nature of auto-mode VPCs. Auto-mode VPCs automatically create subnets in each region using predefined IP address ranges. These ranges are not customizable and are fixed. If you attempt to create another auto-mode VPC for "Distribution", it would also try to create subnets using the same default ranges, resulting in overlapping CIDR blocks and making peering impossible.

Therefore, you must create a VPC in custom mode ("Distribution") which gives you full control over IP address ranges to avoid conflicts. Option B specifies using the CIDR range 10.0.0.0/9. This is a valid private IP range and does not overlap with the default ranges utilized by the 'Retail' VPC which makes it suitable for this use case. The step to "Create necessary subnets" allows you to define subnets within this non-conflicting range, essential for deploying resources within the 'Distribution' VPC. Finally, establishing peering between the "Retail" and "Distribution" networks enables network communication between resources in these separate VPCs.

Option A is incorrect because creating the Distribution VPC in auto mode will lead to IP address collisions with the Retail auto-mode VPC, rendering VPC peering impossible. Option C uses 10.128.0.0/9 which, while a valid non-overlapping CIDR, is not ideal because the subnet ranges need to be different, not just the major CIDR block. Option D is incorrect because you cannot rename the default VPC and changing its functionality won't solve the issue of overlapping IP ranges.

In essence, the answer hinges on understanding that you cannot peer two auto-mode VPCs due to overlapping subnet ranges. Custom mode VPCs provide the necessary control to define non-conflicting subnet address spaces and enable successful peering.

Supporting Links:

VPC Network Overview: <https://cloud.google.com/vpc/docs/vpc>

Auto mode and custom mode VPC networks: <https://cloud.google.com/vpc/docs/vpc#auto-vs-custom>

VPC Network Peering: <https://cloud.google.com/vpc/docs/vpc-peering>

IP address ranges in GCP: <https://cloud.google.com/vpc/docs/vpc#ip-ranges>

Question: 6

CertyIQ

You are using a third-party next-generation firewall to inspect traffic. You created a custom route of 0.0.0.0/0 to route egress traffic to the firewall. You want to allow your VPC instances without public IP addresses to access the BigQuery and Cloud Pub/Sub APIs, without sending the traffic through the firewall. Which two actions should you take? (Choose two.)

- A. Turn on Private Google Access at the subnet level.
- B. Turn on Private Google Access at the VPC level.
- C. Turn on Private Services Access at the VPC level.
- D. Create a set of custom static routes to send traffic to the external IP addresses of Google APIs and services via the default internet gateway.
- E. Create a set of custom static routes to send traffic to the internal IP addresses of Google APIs and services via the default internet gateway.

Answer: AD

Explanation:

Here's a detailed justification for why options A and D are the correct choices, and why the others are not:

Why A is Correct: Turn on Private Google Access at the subnet level.

Private Google Access allows instances within a VPC subnet that lack external IP addresses to reach Google APIs and services using Google's internal network. This avoids the need to route such traffic via a third-party firewall. Enabling Private Google Access at the subnet level ensures that instances within that subnet can access Google APIs, effectively bypassing the custom 0.0.0.0/0 route meant for general egress. This is crucial for granular control and allows specific resources to bypass custom egress pathways.

<https://cloud.google.com/vpc/docs/private-google-access>

Why D is Correct: Create a set of custom static routes to send traffic to the external IP addresses of Google APIs and services via the default internet gateway.

While Private Google Access handles the API access through an internal network, it might not cover all Google service external IP ranges, or it might not be immediately available for certain external IPs of Google APIs. To specifically direct traffic to those external IPs of APIs and services (like Cloud Pub/Sub and BigQuery) via the default internet gateway, you need custom static routes. These routes override the default 0.0.0.0/0 route. By specifying these custom routes, you send the required API traffic directly to Google's services through the internet gateway while other egress traffic uses the firewall. This enables selective traffic management. <https://cloud.google.com/vpc/docs/routes>

Why B is Incorrect: Turn on Private Google Access at the VPC level.

Private Google Access is configured at the subnet level, not the VPC level. Attempting to turn it on at the VPC level would not be a valid configuration.

Why C is Incorrect: Turn on Private Services Access at the VPC level.

Private Services Access is used for private access to Google managed services, not to general Google APIs such as BigQuery or Cloud Pub/Sub. Moreover, Private Service Access relies on shared VPC networks and does not apply in this situation.

Why E is Incorrect: Create a set of custom static routes to send traffic to the internal IP addresses of Google APIs and services via the default internet gateway.

Google APIs are generally exposed via external IP addresses (that may sometimes be reachable through the internet) that must be reached through the internet. Internal IPs of Google APIs are not exposed to VPC clients. Therefore, using internal IP addresses for static routes won't route traffic to the API endpoints and these IPs will not be reachable through the internet gateway.

In summary, enabling Private Google Access at the subnet level (A) and creating specific static routes for Google API's external IPs (D) are necessary to allow instances without public IPs to access BigQuery and Cloud Pub/Sub APIs without sending the traffic through the firewall, as described in the prompt.

Question: 7

CertyIQ

All the instances in your project are configured with the custom metadata enable-oslogin value set to FALSE and to block project-wide SSH keys. None of the instances are set with any SSH key, and no project-wide SSH keys have been configured. Firewall rules are set up to allow SSH sessions from any IP address range. You want to SSH into one instance.

What should you do?

- A. Open the Cloud Shell SSH into the instance using `gcloud compute ssh`.
- B. Set the custom metadata enable-oslogin to TRUE, and SSH into the instance using a third-party tool like putty or ssh.
- C. Generate a new SSH key pair. Verify the format of the private key and add it to the instance. SSH into the instance using a third-party tool like putty or ssh.
- D. Generate a new SSH key pair. Verify the format of the public key and add it to the project. SSH into the instance using a third-party tool like putty or ssh.

Answer: A

Explanation:

The correct answer is **A. Open the Cloud Shell SSH into the instance using `gcloud compute ssh`**. Here's why:

The scenario describes a situation where traditional SSH methods are intentionally disabled. Specifically, the custom metadata enable-oslogin is set to FALSE, which blocks OS Login features, and project-wide SSH keys are also blocked. Since no instance-specific SSH keys are configured either, standard SSH connections directly from tools like putty or ssh won't work, regardless of the firewall rules.

Cloud Shell, however, offers a unique way to establish SSH connections. When you use `gcloud compute ssh`, it leverages a different authentication mechanism that bypasses the blocked OS Login and project-wide keys. It utilizes the user's Google Cloud identity and associated permissions to generate temporary, short-lived keys and handles the SSH connection on your behalf, effectively acting as a proxy. This mechanism doesn't rely on the presence of standard SSH keys on the instance itself, making it a viable option in the described configuration.

Option B is incorrect because enabling OS Login would still not automatically create keys. It requires a configured user that is mapped to a Google identity and a login profile. Options C and D are incorrect since the project-wide keys and instance SSH metadata is blocked explicitly. These would also not solve the issue at hand. Cloud Shell's `gcloud compute ssh` command is designed to handle authentication in such situations, adhering to the principle of least privilege by not requiring any preconfigured SSH keys on instances or project.

Relevant resource for further research:

Connecting to instances using SSH: <https://cloud.google.com/compute/docs/connect/ssh>

Using OS Login: <https://cloud.google.com/compute/docs/oslogin/>

gcloud compute ssh: <https://cloud.google.com/sdk/gcloud/reference/compute/ssh>

Question: 8

CertyIQ

You work for a university that is migrating to GCP.

These are the cloud requirements:

"Ç On-premises connectivity with 10 Gbps

"Ç Lowest latency access to the cloud

"Ç Centralized Networking Administration Team

New departments are asking for on-premises connectivity to their projects. You want to deploy the most cost-efficient interconnect solution for connecting the campus to Google Cloud.

What should you do?

- A. Use Shared VPC, and deploy the VLAN attachments and Interconnect in the host project.
- B. Use Shared VPC, and deploy the VLAN attachments in the service projects. Connect the VLAN attachment to the Shared VPC's host project.
- C. Use standalone projects, and deploy the VLAN attachments in the individual projects. Connect the VLAN attachment to the standalone projects' Interconnects.
- D. Use standalone projects and deploy the VLAN attachments and Interconnects in each of the individual projects.

Answer: A

Explanation:

The correct answer is **A. Use Shared VPC, and deploy the VLAN attachments and Interconnect in the host project.**

Here's a detailed justification:

Centralized Networking: Shared VPC allows for centralized networking administration, where a host project manages the network resources and service projects utilize them. This directly addresses the "Centralized Networking Administration Team" requirement.

Cost Efficiency: Deploying a single Interconnect in the host project and sharing it across service projects through Shared VPC is more cost-effective than setting up individual Interconnects for each project. This is because you pay for the port capacity of a single interconnect rather than multiple.

10 Gbps Connectivity: Google Cloud Interconnect, when configured properly, can provide the required 10 Gbps connection between the university's on-premises network and Google Cloud.

Lowest Latency: The solution doesn't specify any latency concerns tied to different configurations, so this point isn't a deciding factor in selecting option A.

VLAN Attachment Placement: By deploying the VLAN attachments in the host project alongside the Interconnect, you simplify management and ensure that all traffic traverses through the central network. Each project (department) will have a VLAN attachment mapped to its project.

Shared VPC Mechanism: Service projects connect to the central VPC network through Shared VPC. This setup allows departments to provision and manage their own resources, while still leveraging the centralized interconnect connection.

Why other options are incorrect:

B. While using Shared VPC is correct, attaching VLAN attachments to service projects is not the best practice. The VLAN attachments should be in the Host project and the host project routes them.

C. and D. Using standalone projects would require multiple Interconnects and duplicate resources, going against the "most cost-efficient" and "centralized" requirements. This would lead to higher costs and more complex management.

In summary, using Shared VPC with the Interconnect and VLAN attachments in the host project provides a cost-effective, centralized, and scalable approach that meets all the listed requirements.

Authoritative Links:

Shared VPC Overview: <https://cloud.google.com/vpc/docs/shared-vpc>

Cloud Interconnect Overview: <https://cloud.google.com/network-connectivity/docs/interconnect/>

Cloud Interconnect Pricing: <https://cloud.google.com/network-connectivity/interconnect/pricing>

Question: 9

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You have deployed a new internal application that provides HTTP and TFTP services to on-premises hosts. You want to be able to distribute traffic across multiple Compute Engine instances, but need to ensure that clients are sticky to a particular instance across both services. Which session affinity should you choose?

- A. None
- B. Client IP
- C. Client IP and protocol
- D. Client IP, port and protocol

Answer: B

Explanation:

The correct answer is **B. Client IP**. Here's why:

The requirement is to maintain session stickiness, meaning a client is consistently directed to the same backend instance for both HTTP and TFTP services. Both services originate from the same client, so we must uniquely identify a client, not the protocols. While HTTP uses TCP and TFTP uses UDP, these are separate protocols that the application utilizes, but the underlying user or device making the request has a consistent IP address.

Option A (None): No stickiness would be enforced, distributing traffic arbitrarily and breaking the

requirement.

Option C (Client IP and protocol): This would create separate sessions based on the protocol. A client using HTTP and TFTP would likely be routed to different instances, violating the stickiness requirement.

Option D (Client IP, port, and protocol): This is overly granular. Different port numbers from the same client are expected for different services. The same application will not likely be using the same port numbers. We just need to stick users, and ports are not relevant to the users, just processes. It is unnecessary to use ports for stickiness since the same source will always use the same IP.

By choosing **Client IP**, all traffic originating from the same IP address, regardless of the protocol or port, will be directed to the same backend instance. This ensures that clients have a consistent experience across both HTTP and TFTP, fulfilling the requirements of sticky sessions. Using Client IP will keep the user connected to the same session even between protocols.

Authoritative Links:

Google Cloud Load Balancing - Session affinity: <https://cloud.google.com/load-balancing/docs/session-affinity> - This page details session affinity options and their use cases in Google Cloud Load Balancing.

Understanding HTTP and TFTP protocols: Research on HTTP and TFTP protocol to understand their differences, and how they operate.

In summary, Client IP is the only option that provides the required stickiness across different protocols coming from the same user because the request originates from the same user IP address.

Question: 10

CertyIQ

You created a new VPC network named Dev with a single subnet. You added a firewall rule for the network Dev to allow HTTP traffic only and enabled logging.

When you try to log in to an instance in the subnet via Remote Desktop Protocol, the login fails. You look for the Firewall rules logs in Stackdriver Logging, but you do not see any entries for blocked traffic. You want to see the logs for blocked traffic.

What should you do?

- A. Check the VPC flow logs for the instance.
- B. Try connecting to the instance via SSH, and check the logs.
- C. Create a new firewall rule to allow traffic from port 22, and enable logs.
- D. Create a new firewall rule with priority 65500 to deny all traffic, and enable logs.

Answer: D

Explanation:

The correct answer is **D. Create a new firewall rule with priority 65500 to deny all traffic, and enable logs.** Here's why:

The problem is that the existing firewall rule only allows HTTP (port 80) traffic. RDP, which uses port 3389, is being implicitly denied by the default deny rule. The issue isn't that logging isn't working; it's that the denied traffic isn't explicitly logged because it's handled by the implicit deny. Firewall rule logging only captures actions taken by rules you define, not the default deny action.

To capture this blocked traffic in logs, you need an explicit deny rule with logging enabled. By creating a new firewall rule with priority 65500 (highest priority, as lower numbers mean higher priority) to deny all traffic, you make the implicit deny rule ineffective. Because this rule is explicit, logging will now capture all packets that match this rule and are therefore denied, including the attempted RDP connection.

Option A (VPC flow logs) is incorrect because flow logs capture network traffic flows, not whether a firewall

rule blocked traffic. Option B (SSH) is irrelevant because the problem is with RDP, not SSH; creating a rule for SSH would only allow successful SSH connections, not log blocked traffic. Option C (allowing port 22) is also incorrect because it solves for an entirely different problem (SSH), not the RDP access issue, and it would still not log the blocked traffic. Option D is correct as it creates a specific rule with logging enabled to capture denied traffic.

Authoritative links for further research:

Google Cloud Firewall Rules Logging: <https://cloud.google.com/vpc/docs/firewall-logging>

Google Cloud Firewall Rules: <https://cloud.google.com/vpc/docs/firewalls>

VPC Flow Logs: <https://cloud.google.com/vpc/docs/flow-logs>

Question: 11

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You are trying to update firewall rules in a shared VPC for which you have been assigned only Network Admin permissions. You cannot modify the firewall rules.
Your organization requires using the least privilege necessary.
Which level of permissions should you request?

- A. Security Admin privileges from the Shared VPC Admin.
- B. Service Project Admin privileges from the Shared VPC Admin.
- C. Shared VPC Admin privileges from the Organization Admin.
- D. Organization Admin privileges from the Organization Admin.

Answer: A

Explanation:

The correct answer is **A. Security Admin privileges from the Shared VPC Admin.**

Here's why:

The scenario highlights the principle of least privilege, where users should only have the necessary permissions to perform their tasks. You currently have Network Admin permissions, which allow you to manage network resources but not firewall rules. Firewall rules are security-related, not solely network configuration. To modify firewall rules, you need specific permissions related to security management.

Security Admin Role: This role grants permissions to manage firewalls, including creating, updating, and deleting them within a project. This precisely aligns with the task you need to accomplish.

Why other options are incorrect:

B. Service Project Admin privileges: Service Project Admin permissions focus on managing resources within service projects attached to a Shared VPC. These permissions do not extend to directly modifying firewall rules of the host project.

C. Shared VPC Admin privileges: Shared VPC Admin permissions encompass more than just firewall rules and involve broad management of the Shared VPC. It's an overly permissive request for the required task.

D. Organization Admin privileges: Organization Admin is the highest level of access within a Google Cloud organization. Granting this for a firewall rule update is a significant violation of the principle of least privilege.

By requesting Security Admin privileges, you are adhering to the least privilege principle by only requesting the specific permission needed to modify firewall rules without any unnecessary and excessive rights. You need to request it specifically from the Shared VPC Admin, as that's the responsible admin for this.

Further research on Google Cloud IAM roles can be found here:

Understanding Roles: <https://cloud.google.com/iam/docs/understanding-roles>

Predefined IAM Roles: https://cloud.google.com/iam/docs/understanding-roles#predefined_roles

Firewall IAM Roles: <https://cloud.google.com/vpc/docs/firewalls/iam-roles>

Question: 12

CertyIQ

You want to create a service in GCP using IPv6.
What should you do?

- A. Create the instance with the designated IPv6 address.
- B. Configure a TCP Proxy with the designated IPv6 address.
- C. Configure a global load balancer with the designated IPv6 address.
- D. Configure an internal load balancer with the designated IPv6 address.

Answer: C

Explanation:

The correct answer is **C. Configure a global load balancer with the designated IPv6 address.**

Here's the justification:

Google Cloud Platform (GCP) supports IPv6 for external facing services through its global load balancing offerings. To expose your service over IPv6, you need to place a global load balancer in front of it. Global load balancers, like HTTP(S) Load Balancing, can be configured with IPv6 external addresses, enabling clients using IPv6 to access your service. Options A, B, and D are not viable options for creating a service that is accessible via IPv6 to the general internet. Instances themselves (Option A) don't directly expose an external IPv6 address; you need a load balancer for that. TCP proxies (Option B) can work with IPv6, but they are not the primary method for serving general web traffic over IPv6. Internal load balancers (Option D) operate within a VPC network and don't have public IPv6 addresses. To make a service globally accessible with an IPv6 address, you need a global load balancer with an external facing IPv6 endpoint. Global load balancers distribute traffic across multiple backend instances and regions, providing high availability and scalability. This makes them the right choice when planning to create a public service that uses IPv6.

Authoritative links for further research:

Google Cloud Load Balancing: <https://cloud.google.com/load-balancing>

Configuring Global External HTTP(S) Load Balancing: <https://cloud.google.com/load-balancing/docs/https/setting-up-global-https-load-balancer>

IPv6 support for load balancers: <https://cloud.google.com/load-balancing/docs/ipv6>

Question: 13

CertyIQ

You want to deploy a VPN Gateway to connect your on-premises network to GCP. You are using a non BGP-capable on-premises VPN device. You want to minimize downtime and operational overhead when your network grows. The device supports only IKEv2, and you want to follow Google-recommended practices.
What should you do?

- A. "☐ Create a Cloud VPN instance. "☐ Create a policy-based VPN tunnel per subnet. "☐ Configure the appropriate local and remote traffic selectors to match your local and remote networks. "☐ Create the appropriate static routes.
- B. "☐ Create a Cloud VPN instance. "☐ Create a policy-based VPN tunnel. "☐ Configure the appropriate local and remote traffic selectors to match your local and remote networks. "☐ Configure the appropriate static routes.
- C. "☐ Create a Cloud VPN instance. "☐ Create a route-based VPN tunnel. "☐ Configure the appropriate local and

remote traffic selectors to match your local and remote networks. "❏ Configure the appropriate static routes.
D. "❏ Create a Cloud VPN instance. "❏ Create a route-based VPN tunnel. "❏ Configure the appropriate local and remote traffic selectors to 0.0.0.0/0. "❏ Configure the appropriate static routes.

Answer: B

Explanation:

The correct answer is **B**. Here's why:

The scenario requires connecting an on-premises network with a non-BGP VPN device to GCP, minimizing downtime, and following Google's best practices while using IKEv2. Option B correctly addresses these requirements.

Let's break down why other options are incorrect:

Option A: Creating a VPN tunnel per subnet is inefficient and doesn't scale well. It increases management overhead and complexity, directly contradicting the goal of minimizing operational overhead.

Option C: Route-based VPNs, while suitable for BGP, are not the appropriate choice for non-BGP devices. Route-based VPNs rely on routing protocols to exchange routes, which is not possible with a non-BGP capable device. Additionally, configuring traffic selectors to match local and remote network is not the best practice.

Option D: Setting traffic selectors to 0.0.0.0/0 is overly broad and a security risk, especially with static routes as it would send all outbound traffic from your network through the tunnel regardless of destination. Route-based VPN's is not the correct choice for non-BGP devices.

Option B is the ideal solution because:

1. **Cloud VPN Instance:** It correctly starts by creating a Cloud VPN gateway instance in GCP, essential for the VPN connection.
2. **Policy-Based VPN Tunnel:** It uses a policy-based VPN tunnel, which is the correct choice for non-BGP devices. Policy-based VPNs utilize traffic selectors to define which traffic should traverse the VPN tunnel. This approach is simple and ideal for non-BGP scenarios.
3. **Specific Traffic Selectors:** It correctly advises configuring local and remote traffic selectors. This limits the traffic passing over the VPN tunnel to only the necessary networks, enhancing security and avoiding unnecessary traffic.
4. **Static Routes:** With a policy-based tunnel, it is also necessary to configure the appropriate static routes for the traffic to pass in the tunnel.

In summary, Option B provides the best balance between functionality, security, and ease of management while adhering to Google's recommended practices for connecting on-premises networks to GCP using a non-BGP VPN device.

Authoritative Links:

Choosing a VPN option: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/choosing-vpn>

Policy-based VPN: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/policy-based-vpn>

Static Routes overview: <https://cloud.google.com/vpc/docs/routes>

Question: 14

CertyIQ

Your company just completed the acquisition of Altostrat (a current GCP customer). Each company has a separate organization in GCP and has implemented a custom DNS solution. Each organization will retain its current domain and host names until after a full transition and architectural review is done in one year.

These are the assumptions for both GCP environments.

"❏ Each organization has enabled full connectivity between all of its projects by using Shared VPC.

"Ç Both organizations strictly use the 10.0.0.0/8 address space for their instances, except for bastion hosts (for accessing the instances) and load balancers for serving web traffic.
"Ç There are no prefix overlaps between the two organizations.
"Ç Both organizations already have firewall rules that allow all inbound and outbound traffic from the 10.0.0.0/8 address space.
"Ç Neither organization has Interconnects to their on-premises environment.
You want to integrate networking and DNS infrastructure of both organizations as quickly as possible and with minimal downtime.
Which two steps should you take? (Choose two.)

- A. Provision Cloud Interconnect to connect both organizations together.
- B. Set up some variant of DNS forwarding and zone transfers in each organization.
- C. Connect VPCs in both organizations using Cloud VPN together with Cloud Router.
- D. Use Cloud DNS to create A records of all VMs and resources across all projects in both organizations.
- E. Create a third organization with a new host project, and attach all projects from your company and Altostrat to it using shared VPC.

Answer: BC

Explanation:

Okay, let's break down why options B and C are the correct choices for this scenario.

The core challenge is to quickly integrate two separate GCP organizations with existing, custom DNS solutions while maintaining minimal downtime and avoiding immediate architectural changes. Given the organizations already use Shared VPC and there are no IP overlaps in the 10.0.0.0/8 range, we can establish network connectivity and DNS resolution relatively simply.

Option **C**, connecting VPCs using **Cloud VPN and Cloud Router**, directly addresses the need for network integration. Cloud VPN establishes encrypted tunnels between the VPCs, and Cloud Router dynamically manages routes, enabling communication between resources in the two organizations' private address spaces without needing an Interconnect. This satisfies the need for connectivity without requiring major changes.

Option **B**, setting up **DNS forwarding and zone transfers**, tackles the DNS resolution challenge. Since both organizations use custom DNS, migrating to Cloud DNS and updating all records immediately would be complex and disruptive. Instead, setting up conditional forwarding in each organization's DNS server to forward queries for the other organization's domain to their DNS servers enables seamless resolution between the two environments. Zone transfers can help replicate data if necessary. This approach is low-impact, minimizes downtime, and maintains the existing DNS architectures.

Option **A** (Cloud Interconnect) is not a quick solution as it is typically a more complex process that involves dedicated physical connections and configuration. It is often used for on-premises connections rather than between GCP organizations. This does not align with the requirement to integrate quickly and with minimal downtime.

Option **D** (Cloud DNS A records) would require a major shift to Cloud DNS for both organizations and updates across many projects and is therefore not a minimal downtime solution. This also bypasses their current custom DNS, which is not aligned with the requirements for minimal impact.

Option **E** (creating a third organization) is an extreme measure that requires a complete overhaul of the existing project structure, which goes against the stated requirement of minimal downtime and integration as quickly as possible.

Therefore, the most suitable approach is to focus on establishing VPN tunnels and routing for network access (Option C) and forwarding between each organization's custom DNS for the time being (Option B) to avoid immediate changes.

Authoritative Links:

Cloud VPN: <https://cloud.google.com/vpn/docs/concepts/overview>

Cloud Router: <https://cloud.google.com/router/docs/concepts/overview>

Cloud DNS: <https://cloud.google.com/dns/docs/overview>

Question: 15

CertyIQ

Your on-premises data center has 2 routers connected to your Google Cloud environment through a VPN on each router. All applications are working correctly; however, all of the traffic is passing across a single VPN instead of being load-balanced across the 2 connections as desired.

During troubleshooting you find:

"Ç Each on-premises router is configured with a unique ASN.

"Ç Each on-premises router is configured with the same routes and priorities.

"Ç Both on-premises routers are configured with a VPN connected to a single Cloud Router.

"Ç BGP sessions are established between both on-premises routers and the Cloud Router.

"Ç Only 1 of the on-premises router's routes are being added to the routing table.

What is the most likely cause of this problem?

- A. The on-premises routers are configured with the same routes.
- B. A firewall is blocking the traffic across the second VPN connection.
- C. You do not have a load balancer to load-balance the network traffic.
- D. The ASNs being used on the on-premises routers are different.

Answer: D

Explanation:

The correct answer is D, that the ASNs being used on the on-premises routers are different. Here's why:

Google Cloud Router uses BGP (Border Gateway Protocol) to exchange routing information with on-premises routers. BGP relies on ASNs (Autonomous System Numbers) to uniquely identify networks. When a Cloud Router receives routes from multiple BGP peers (on-premises routers in this case) that have different ASNs, it assumes these are distinct paths to potentially different destinations. However, if multiple BGP sessions with different ASNs advertise the same routes (as stated in the question), Cloud Router will by default only use one of those paths (and install only one of those route's prefixes in its routing table), selecting a path based on best-path selection algorithm. It won't load-balance between them. This behavior is in line with how BGP normally functions.

To achieve load balancing across multiple VPN tunnels with Google Cloud Router, the on-premises routers must use the same ASN, presenting themselves as multiple paths from the same logical network. This allows the Google Cloud Router to see the VPN tunnels as equivalent paths for the same destination and distribute traffic accordingly. This is explained further in Google Cloud's documentation regarding dynamic routing and high-availability VPN configurations. The documentation also mentions that you can configure multi-path routing with equal cost paths.

Options A and B are incorrect. Option A states the routes are the same, which is actually required for load-balancing, not the cause of the issue. Option B suggests a firewall is blocking, but the question states one VPN is working, and only one route is being added to the routing table. Option C suggests needing a load balancer, but in this situation the load balancing is desired across the BGP sessions themselves (not a separate Google Cloud load balancer).

Authoritative Links:

Google Cloud Documentation - Cloud Router Overview: <https://cloud.google.com/router/docs/overview> - This page provides an overview of Cloud Router functionality and how it interacts with BGP.

Google Cloud Documentation - Creating Dynamic Routes using BGP: <https://cloud.google.com/network-connectivity/docs/router/how-to/create-dynamic-routes-using-bgp> - Information regarding BGP and dynamic

routing.

Google Cloud Documentation - BGP Router Overview: <https://cloud.google.com/network-connectivity/docs/router/concepts/bgp-overview> - Describes how BGP works with cloud router.

Question: 16

CertyIQ

You have ordered Dedicated Interconnect in the GCP Console and need to give the Letter of Authorization/Connecting Facility Assignment (LOA-CFA) to your cross-connect provider to complete the physical connection.

Which two actions can accomplish this? (Choose two.)

- A. Open a Cloud Support ticket under the Cloud Interconnect category.
- B. Download the LOA-CFA from the Hybrid Connectivity section of the GCP Console.
- C. Run `gcloud compute interconnects describe <interconnect>`.
- D. Check the email for the account of the NOC contact that you specified during the ordering process.
- E. Contact your cross-connect provider and inform them that Google automatically sent the LOA/CFA to them via email, and to complete the connection.

Answer: BD

Explanation:

Here's a detailed justification for why options B and D are the correct actions to obtain the LOA-CFA for a Dedicated Interconnect:

Option B: Download the LOA-CFA from the Hybrid Connectivity section of the GCP Console.

This is the primary method for obtaining the LOA-CFA. After ordering a Dedicated Interconnect in the GCP Console, Google generates the LOA-CFA, which is essential for the cross-connect provider to make the physical connection. The "Hybrid Connectivity" section within the GCP Console is the designated area where users can manage and retrieve information related to their interconnects, including the LOA-CFA. This document contains the necessary details about the Google side of the connection, such as the demarcation point and port information. Downloading it directly from the console ensures you have the correct, up-to-date version.

Option D: Check the email for the account of the NOC contact that you specified during the ordering process.

Google also sends the LOA-CFA to the email address specified for the Network Operations Center (NOC) contact during the Dedicated Interconnect ordering process. This acts as a notification and an alternative method to access the LOA-CFA. This email contains the document as an attachment, ensuring the relevant technical team also has immediate access to it. It serves as a backup and provides a direct notification mechanism.

Why other options are incorrect:

Option A: Open a Cloud Support ticket under the Cloud Interconnect category. While Google Cloud support is helpful, opening a ticket is not the standard or most efficient method to obtain the LOA-CFA. The document is made readily available through the console and email notifications.

Option C: Run `gcloud compute interconnects describe <interconnect>`. The `gcloud` command-line tool can provide information about the interconnect, but it does not directly output the LOA-CFA document. This command is used for retrieving information about the configuration but doesn't deliver the actual authorization document.

Option E: Contact your cross-connect provider and inform them that Google automatically sent the LOA/CFA to them via email, and to complete the connection. Google does not automatically send the LOA-

CFA to the cross-connect provider. The responsibility to provide the LOA-CFA to the provider falls on the customer requesting the interconnect.

In summary, downloading the LOA-CFA directly from the Hybrid Connectivity section in the GCP Console (B) and checking the NOC contact's email (D) are the two correct ways to retrieve the LOA-CFA necessary to complete the physical connection with the cross-connect provider.

Authoritative Links:

Dedicated Interconnect Overview: This Google Cloud documentation provides an overview of Dedicated Interconnect and the process involved.

Ordering Dedicated Interconnect: This page walks through ordering process and what is required from the customer

Question: 17

CertyIQ

Your company offers a popular gaming service. Your instances are deployed with private IP addresses, and external access is granted through a global load balancer. You believe you have identified a potential malicious actor, but aren't certain you have the correct client IP address. You want to identify this actor while minimizing disruption to your legitimate users.

What should you do?

- A. Create a Cloud Armor Policy rule that denies traffic and review necessary logs.
- B. Create a Cloud Armor Policy rule that denies traffic, enable preview mode, and review necessary logs.
- C. Create a VPC Firewall rule that denies traffic, enable logging and set enforcement to disabled, and review necessary logs.
- D. Create a VPC Firewall rule that denies traffic, enable logging and set enforcement to enabled, and review necessary logs.

Answer: B

Explanation:

The correct answer is **B. Create a Cloud Armor Policy rule that denies traffic, enable preview mode, and review necessary logs.**

Here's why:

Cloud Armor is Google Cloud's Web Application Firewall (WAF) service, designed for protecting applications from various threats at the edge of the network. It integrates with load balancers, making it ideal for the scenario described. Option B leverages Cloud Armor's preview mode, allowing administrators to simulate the impact of a rule before enforcing it. This is crucial for minimizing disruption to legitimate users while investigating the potential malicious actor. By using preview mode and enabling logging, administrators can review access logs and determine if the rule would block the desired IP address without impacting regular traffic. This approach provides a safe testing ground. VPC Firewall rules, on the other hand, are applied at the VM level and don't have the ability to use a preview mode. Additionally, they require more time to deploy and can impact all traffic across a subnet. Directly denying traffic, as proposed in options A and D, without testing could block legitimate users if the identified IP is incorrect. Option C's approach is more prudent in not immediately denying traffic, however, it does not have the benefit of preview mode, which cloud armor possesses, making B the most accurate answer. Cloud Armor's capabilities and its integration with load balancers make it the most appropriate tool for identifying potentially malicious traffic with minimal impact on valid users.

Authoritative Links:

Cloud Armor Overview: <https://cloud.google.com/armor/docs/overview>

Question: 18

Your company's web server administrator is migrating on-premises backend servers for an application to GCP. Libraries and configurations differ significantly across these backend servers. The migration to GCP will be lift-and-shift, and all requests to the servers will be served by a single network load balancer frontend.

You want to use a GCP-native solution when possible.

How should you deploy this service in GCP?

- A. Create a managed instance group from one of the images of the on-premises servers, and link this instance group to a target pool behind your load balancer.
- B. Create a target pool, add all backend instances to this target pool, and deploy the target pool behind your load balancer.
- C. Deploy a third-party virtual appliance as frontend to these servers that will accommodate the significant differences between these backend servers.
- D. Use GCP's ECMP capability to load-balance traffic to the backend servers by installing multiple equal-priority static routes to the backend servers.

Answer: B

Explanation:

The correct answer is **B. Create a target pool, add all backend instances to this target pool, and deploy the target pool behind your load balancer.**

Here's the justification:

Option B directly addresses the requirement for a single load balancer frontend distributing traffic across heterogeneous backend servers migrated "lift-and-shift". Target pools, in GCP's network load balancing context, are designed to group instances that receive traffic from the load balancer. They don't impose constraints on the underlying operating systems or configurations of the instances within the pool, fulfilling the "significant differences" criteria.

Option A, using managed instance groups (MIGs), is not suitable for this scenario since MIGs require homogeneous instances created from the same template. The differences in libraries and configurations across servers prevent utilizing MIGs effectively.

Option C, introducing a third-party virtual appliance, adds unnecessary complexity and management overhead. The goal is to utilize native GCP solutions where possible.

Option D, employing Equal-Cost Multi-Path (ECMP) routing, is not applicable for load balancing. ECMP routes are used to distribute traffic across multiple paths to the same destination, not multiple distinct servers behind a load balancer.

Therefore, using a target pool linked to the load balancer provides the simplest and most effective method for distributing traffic to the existing diverse set of backend server instances without requiring modifications to the migrated systems. This approach adheres to GCP-native best practices and minimizes operational overhead.

Authoritative Links:

Target Pools: <https://cloud.google.com/load-balancing/docs/target-pools>

Network Load Balancing: <https://cloud.google.com/load-balancing/docs/network>

Managed Instance Groups: <https://cloud.google.com/compute/docs/instance-groups>

ECMP Routing: <https://cloud.google.com/network-connectivity/docs/router/concepts/ecmp>

Question: 19

CertyIQ

You decide to set up Cloud NAT. After completing the configuration, you find that one of your instances is not using the Cloud NAT for outbound NAT.

What is the most likely cause of this problem?

- A. The instance has been configured with multiple interfaces.
- B. An external IP address has been configured on the instance.
- C. You have created static routes that use RFC1918 ranges.
- D. The instance is accessible by a load balancer external IP address.

Answer: B

Explanation:

The correct answer is **B. An external IP address has been configured on the instance.**

Here's why: Cloud NAT is designed to provide outbound internet connectivity for instances without public IP addresses by translating their private IP addresses to a shared public NAT IP. However, if an instance already possesses an external (public) IP address assigned directly to it, it will use that address for outbound communication and bypass Cloud NAT altogether. This is because having a public IP grants the instance direct internet access, rendering the NAT translation unnecessary. Cloud NAT's core function is to enable internet access for instances lacking a public IP, not to override existing public IP configurations. Therefore, if an instance isn't using Cloud NAT, a likely cause is that it has been configured with its own external IP. Options A, C, and D do not directly prevent an instance from using Cloud NAT. While multiple interfaces (A) might present complex routing situations, they do not inherently exclude NAT usage. Static routes (C), if correctly configured to route to the internet, would use NAT if the associated instance doesn't have a public IP. Likewise, access via an external load balancer (D) doesn't conflict with Cloud NAT; in fact, instances behind a load balancer would typically rely on NAT for egress.

Authoritative Links for further research:

Google Cloud NAT Documentation: <https://cloud.google.com/nat/docs/overview> (This is the main page for Cloud NAT and offers a lot of information. You can navigate the sidebar for more specific pages.)

Cloud NAT Concepts: <https://cloud.google.com/nat/docs/concepts> (Explains the functionality of cloud NAT and the types of traffic it handles.)

Cloud NAT limitations: <https://cloud.google.com/nat/docs/limitations> (Details the limitations of Cloud NAT, providing helpful context for troubleshooting)

Question: 20

CertyIQ

You want to set up two Cloud Routers so that one has an active Border Gateway Protocol (BGP) session, and the other one acts as a standby.

Which BGP attribute should you use on your on-premises router?

- A. AS-Path
- B. Community
- C. Local Preference
- D. Multi-exit Discriminator

Answer: D

Explanation:

The correct answer is **D. Multi-exit Discriminator (MED)**.

MED is the most appropriate BGP attribute for influencing inbound traffic routing from Google Cloud to your on-premises network when implementing an active/standby Cloud Router configuration. Specifically, you'd configure your on-premises router to advertise routes with a higher (or lower, depending on the desired behavior) MED value to the active Cloud Router and a different MED value to the standby Cloud Router.

Cloud Routers will propagate BGP routes received to Google's network with their MED values. Google's routing algorithm prioritizes routes based on MED: lower MED values indicate a preferred path (with some caveats). The Cloud Router with the lower MED advertised to Google becomes active. The standby Cloud Router would advertise routes with a higher MED value, making it the secondary path for traffic. The active Cloud Router thus receives lower MED updates, causing Google to send the traffic to that router and from that to the on-premises router.

Options A, B, and C are not the primary mechanisms for achieving active/standby failover in this context. AS-Path manipulation influences route selection based on the number of autonomous systems traversed, not path preference within the same AS. Community tags are primarily used for policy-based routing and don't influence the order of preference in BGP. Local Preference is an outbound attribute that dictates which router a network sends its traffic towards, not an inbound one from Google's perspective.

In summary, the MED attribute enables your on-premises router to signal to Google which Cloud Router should be used as the primary path for traffic, establishing the necessary active/standby routing behavior.

Authoritative Links:

Google Cloud Documentation - BGP routing options: <https://cloud.google.com/network-connectivity/docs/router/concepts/best-path-selection#med>

RFC 4271 - A Border Gateway Protocol 4 (BGP-4): <https://datatracker.ietf.org/doc/html/rfc4271> (See section 9.1.2 for MED description)

Question: 21**CertyIQ**

You are increasing your usage of Cloud VPN between on-premises and GCP, and you want to support more traffic than a single tunnel can handle. You want to increase the available bandwidth using Cloud VPN. What should you do?

- A. Double the MTU on your on-premises VPN gateway from 1460 bytes to 2920 bytes.
- B. Create two VPN tunnels on the same Cloud VPN gateway that point to the same destination VPN gateway IP address.
- C. Add a second on-premises VPN gateway with a different public IP address. Create a second tunnel on the existing Cloud VPN gateway that forwards the same IP range, but points at the new on-premises gateway IP.
- D. Add a second Cloud VPN gateway in a different region than the existing VPN gateway. Create a new tunnel on the second Cloud VPN gateway that forwards the same IP range, but points to the existing on-premises VPN gateway IP address.

Answer: C**Explanation:**

The correct answer is C. To increase bandwidth for Cloud VPN beyond a single tunnel's capacity, you need to implement multiple tunnels. Option C achieves this by adding a second on-premises VPN gateway with a different public IP address. Then, it creates a new tunnel on the existing Cloud VPN gateway, which forwards the same IP range but points to the new on-premises gateway. This effectively creates a second path for traffic, increasing the aggregate bandwidth. Option A is incorrect because increasing the MTU (Maximum

Transmission Unit) doesn't create additional bandwidth; it can optimize existing throughput but won't solve the limitation of a single tunnel. Option B is also incorrect because while creating multiple tunnels to the same destination IP can provide redundancy, it doesn't achieve increased bandwidth as both tunnels would likely utilize the same paths. Option D is incorrect since adding a VPN gateway in another region for the same IP ranges may create routing conflicts. Also, it would not contribute towards increasing bandwidth with the on-premises site, which is the core problem as stated. In essence, the correct strategy is to parallelize your VPN connections by adding a separate on-premises gateway and a corresponding tunnel on the GCP side. This approach allows you to distribute traffic across multiple tunnels, effectively multiplying your available bandwidth. Utilizing Cloud Router's dynamic routing features ensures consistent traffic balancing across these tunnels.

Authoritative Links for Further Research:

Google Cloud Documentation on Cloud VPN High Availability: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/high-availability-vpn>

Google Cloud Documentation on Cloud VPN Dynamic Routing: <https://cloud.google.com/network-connectivity/docs/router/how-to/dynamic-routing>

Question: 22

CertyIQ

You are disabling DNSSEC for one of your Cloud DNS-managed zones. You removed the DS records from your zone file, waited for them to expire from the cache, and disabled DNSSEC for the zone. You receive reports that DNSSEC validating resolves are unable to resolve names in your zone. What should you do?

- A. Update the TTL for the zone.
- B. Set the zone to the TRANSFER state.
- C. Disable DNSSEC at your domain registrar.
- D. Transfer ownership of the domain to a new registrar.

Answer: C

Explanation:

The correct answer is **C. Disable DNSSEC at your domain registrar**. Here's why:

When disabling DNSSEC for a Cloud DNS zone, the process involves more than just removing DS records within the zone itself. DNSSEC works by creating a chain of trust from the root zone down to your domain. Your domain registrar stores the Delegation Signer (DS) record, which acts as a crucial link in this chain, indicating that your zone is DNSSEC-enabled. Resolvers rely on this DS record to validate the authenticity of DNS responses from your zone.

Even after removing DS records from your zone file and disabling DNSSEC in Cloud DNS, the old DS record might still be cached by DNS resolvers. If the registrar still has the DS record, those resolvers will attempt DNSSEC validation, fail, and return errors. Consequently, they will not resolve names within your zone.

Therefore, to completely disable DNSSEC, you must remove the DS record at your domain registrar to break the chain of trust from the registrar's DNS servers downwards. This action signals to resolvers that your zone is no longer DNSSEC-enabled and they should stop attempting to validate it. Until this step is done, DNSSEC validating resolvers will continue failing to resolve domain names in your zone. This ensures that DNS lookups function correctly for users irrespective of whether they use validating or non-validating resolvers.

Option A, updating the TTL (Time To Live) for the zone, will influence how long resolvers cache records but won't directly address the core issue with DNSSEC validation. Options B and D are unrelated to this problem; setting the zone to the TRANSFER state is for zone transfers, and changing registrars isn't required for simply

disabling DNSSEC.

Authoritative Links for further research:

Google Cloud DNS Documentation on DNSSEC: <https://cloud.google.com/dns/docs/dnssec> - Specifically, look for sections related to managing and disabling DNSSEC.

ICANN - DNSSEC: <https://www.icann.org/resources/pages/dnssec-2012-02-25-en> - Provides general information about how DNSSEC works.

IETF RFC 4033 - DNS Security Introduction: <https://datatracker.ietf.org/doc/html/rfc4033> - The original document describing DNSSEC architecture.

Question: 23

CertyIQ

You have an application hosted on a Compute Engine virtual machine instance that cannot communicate with a resource outside of its subnet. When you review the flow and firewall logs, you do not see any denied traffic listed. During troubleshooting you find:

"Ç Flow logs are enabled for the VPC subnet, and all firewall rules are set to log.

"Ç The subnetwork logs are not excluded from Stackdriver.

"Ç The instance that is hosting the application can communicate outside the subnet.

"Ç Other instances within the subnet can communicate outside the subnet.

"Ç The external resource initiates communication.

What is the most likely cause of the missing log lines?

- A. The traffic is matching the expected ingress rule.
- B. The traffic is matching the expected egress rule.
- C. The traffic is not matching the expected ingress rule.
- D. The traffic is not matching the expected egress rule.

Answer: C

Explanation:

Here's a detailed justification for why the correct answer is **C. The traffic is not matching the expected ingress rule.**

The problem states that an application hosted on a Compute Engine VM cannot communicate with an external resource, despite other VMs in the same subnet working correctly. Flow logs are enabled and firewall rules are set to log, but no denied traffic is recorded. Crucially, the external resource initiates the communication. This means the relevant firewall rules to investigate are ingress rules, which control incoming traffic to the VM instance.

If traffic was being denied by an egress rule (option D), a log entry would be generated. The fact that there are no log entries implies that the traffic is not even reaching the point where a firewall rule can act on it. If traffic was matching an expected ingress or egress rule (options A and B), log entries would be generated (as stated in the problem).

The most plausible explanation is that the ingress firewall rules are not set up correctly to allow traffic to this specific instance. Since all VMs within the subnet, except this one, can communicate outbound, it implies that the egress rules (and potentially other configurations like the routing or network ACLs) are most likely correct. The missing log lines, together with the specific VM failure, and the fact that the external resource is initiating the connection, point strongly to the ingress rules as the culprit. A potential misconfiguration in the ingress rule could be not specifically allowing incoming traffic from the external resource's IP/port combination to the application's IP/port combination, or the ingress rule could be overly broad and not targeted at this specific instance. Therefore, the traffic is likely being dropped before it can match an applicable ingress rule and be logged, because either no applicable rule exists or one does exist, but with incorrect parameters.

Here are some authoritative links for further research:

VPC Firewall Rules: <https://cloud.google.com/vpc/docs/firewalls> - This documentation provides a comprehensive overview of VPC firewall rules in Google Cloud.

VPC Flow Logs: <https://cloud.google.com/vpc/docs/using-flow-logs> - This link details how VPC flow logs operate, including how to use them for troubleshooting network issues.

Compute Engine Networking: <https://cloud.google.com/compute/docs/networking> - This page discusses the general networking concepts related to Compute Engine instances.

Question: 24

CertyIQ

You have configured Cloud CDN using HTTP(S) load balancing as the origin for cacheable content. Compression is configured on the web servers, but responses served by Cloud CDN are not compressed.

What is the most likely cause of the problem?

- A. You have not configured compression in Cloud CDN.
- B. You have configured the web servers and Cloud CDN with different compression types.
- C. The web servers behind the load balancer are configured with different compression types.
- D. You have to configure the web servers to compress responses even if the request has a Via header.

Answer: D

Explanation:

The correct answer is D. Cloud CDN, by default, respects the Via header in requests. A Via header indicates the request has traversed a proxy or CDN. Web servers, upon receiving a request with a Via header, often assume that the requesting client already supports decompression and thus avoid compressing the response. If web servers are configured not to compress responses when Via headers are present, then Cloud CDN, which adds a Via header, will receive uncompressed content, and consequently serve uncompressed content to the end user. Option A is incorrect; Cloud CDN does not require separate compression configuration, it inherits compression from the origin server. Option B is incorrect; Cloud CDN uses the content served by the origin and does not recompress it. Option C is incorrect; while inconsistent compression configurations on origin servers can cause problems, this scenario specifically focuses on CDN behavior when a Via header is present. To ensure responses are compressed by the origin, web servers need to be configured to compress responses irrespective of the presence of a Via header. This often requires specific configuration changes in the web server configuration to override the default behavior that disables compression with Via headers.

For additional information, refer to these Google Cloud documentation links:

Cloud CDN overview: <https://cloud.google.com/cdn/docs/overview>

HTTP headers: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers>

Via Header: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Via>

Cloud CDN and Compression: While not directly stating the specific Via header behavior, understanding how Cloud CDN respects cache-control headers and origin response is important:

<https://cloud.google.com/cdn/docs/caching>

Question: 25

CertyIQ

You have a web application that is currently hosted in the us-central1 region. Users experience high latency when traveling in Asia. You've configured a network load balancer, but users have not experienced a performance improvement. You want to decrease the latency.

What should you do?

- A. Configure a policy-based route rule to prioritize the traffic.
- B. Configure an HTTP load balancer, and direct the traffic to it.
- C. Configure Dynamic Routing for the subnet hosting the application.
- D. Configure the TTL for the DNS zone to decrease the time between updates.

Answer: B

Explanation:

The correct answer is **B. Configure an HTTP load balancer, and direct the traffic to it.** Here's why:

The problem describes high latency for users in Asia accessing an application in us-central1. A network load balancer (NLB) operates at the transport layer (TCP/UDP), simply forwarding traffic to backend instances within the same region. It doesn't inherently improve latency across geographically distant users.

An HTTP(S) load balancer (specifically the Global External HTTP(S) Load Balancer) solves this issue by distributing traffic across multiple regions. By deploying backend instances closer to users in Asia (e.g., in asia-east1 or asia-northeast1), the HTTP(S) load balancer can route user requests to the nearest backend, significantly reducing latency.

Option A, policy-based routing, manages traffic flow within a network based on pre-defined policies; it won't reduce cross-continental latency. Option C, dynamic routing, optimizes paths between networks but doesn't address the issue of application proximity to end-users. Option D, adjusting DNS TTL, speeds up name resolution but doesn't impact the actual data transfer time, therefore won't resolve the latency. Only an HTTP(S) Load Balancer can strategically route users to backend instances physically nearer to them, thus lowering latency.

The key here is leveraging Google Cloud's global infrastructure to place backend resources closer to the users that consume it, and an HTTP(S) Load Balancer enables exactly this.

Further Research:

Google Cloud Load Balancing Overview: <https://cloud.google.com/load-balancing/docs/load-balancing-overview>

Global External HTTP(S) Load Balancer: <https://cloud.google.com/load-balancing/docs/https/>

Choosing a Load Balancer: <https://cloud.google.com/load-balancing/docs/choosing-load-balancer>

Question: 26

CertyIQ

You have an application running on Compute Engine that uses BigQuery to generate some results that are stored in Cloud Storage. You want to ensure that none of the application instances have external IP addresses. Which two methods can you use to accomplish this? (Choose two.)

- A. Enable Private Google Access on all the subnets.
- B. Enable Private Google Access on the VPC.
- C. Enable Private Services Access on the VPC.
- D. Create network peering between your VPC and BigQuery.
- E. Create a Cloud NAT, and route the application traffic via NAT gateway.

Answer: AE

Explanation:

The correct answer is **A and E.** Here's why:

Option A: Enable Private Google Access on all the subnets. Private Google Access allows instances without external IP addresses to access Google APIs and services (like BigQuery and Cloud Storage) using Google's private network. This means that communication happens within Google's infrastructure, avoiding the public internet. By enabling it on all relevant subnets, you ensure all application instances can reach these services without needing external IPs.<https://cloud.google.com/vpc/docs/private-google-access>

Option E: Create a Cloud NAT, and route the application traffic via NAT gateway. Cloud NAT provides a way for instances without external IP addresses to access the internet. While the question focused on access to BigQuery and Cloud Storage, which are Google services and handled by Private Google Access, Cloud NAT allows the application to connect outbound to external services. It achieves this by translating the internal private IP addresses to one or more external IPs provided by the NAT gateway. So, if the application needed to access other external services (besides Google APIs) then Cloud NAT will ensure that internal private IP addresses don't need to be exposed. This option addresses the case where accessing other external services beyond google APIs and Storage is needed.<https://cloud.google.com/nat/docs/overview>

Why the other options are incorrect:

B: Enable Private Google Access on the VPC. While Private Google Access is enabled within subnets, the VPC itself doesn't have the functionality. The configurations are done at subnet level.

C: Enable Private Services Access on the VPC. Private Services Access is used to privately connect to other services managed by Google or third-parties that are deployed within your VPC and not to Google APIs and services directly, so it's not needed to connect to BigQuery or Cloud Storage.<https://cloud.google.com/vpc/docs/private-services-access>

D: Create network peering between your VPC and BigQuery. Network peering establishes a connection between two VPC networks; in this case you would be peering with the Google internal network hosting the BigQuery service. This isn't necessary because Private Google Access already handles private connections to Google APIs.<https://cloud.google.com/vpc/docs/vpc-peering>

In Summary: By using Private Google Access on subnets and Cloud NAT, the application instances can securely communicate with Google services like BigQuery and Cloud Storage without requiring external IP addresses. Private Google Access handles the Google services communication while Cloud NAT allows secure outbound traffic for non google APIs and services. This combination helps maintain a more secure and controlled environment.

Question: 27

CertyIQ

You are designing a shared VPC architecture. Your network and security team has strict controls over which routes are exposed between departments. Your Production and Staging departments can communicate with each other, but only via specific networks. You want to follow Google-recommended practices. How should you design this topology?

- A. Create 2 shared VPCs within the shared VPC Host Project, and enable VPC peering between them. Use firewall rules to filter access between the specific networks.
- B. Create 2 shared VPCs within the shared VPC Host Project, and create a Cloud VPN/Cloud Router between them. Use Flexible Route Advertisement (FRA) to filter access between the specific networks.
- C. Create 2 shared VPCs within the shared VPC Service Project, and create a Cloud VPN/Cloud Router between them. Use Flexible Route Advertisement (FRA) to filter access between the specific networks.
- D. Create 1 VPC within the shared VPC Host Project, and share individual subnets with the Service Projects to filter access between the specific networks.

Answer: D

Explanation:

The correct answer is D because it aligns with Google's recommended practices for shared VPC and provides the necessary granularity for controlling inter-departmental network access. In a shared VPC setup, a single host project houses the network resources, which are then shared with service projects. Option D, utilizing one VPC in the host project and sharing specific subnets with service projects, allows for granular control at the subnet level. Different departments, such as Production and Staging, can be associated with distinct subnets, enabling selective communication. Firewall rules can be applied at the subnet level to further restrict traffic flows as needed. Options A, B, and C all involve multiple VPCs, either within the host or service projects. While these options are possible, they introduce unnecessary complexity. Peering or using VPN/Cloud Router for inter-VPC communication adds overhead and management challenges. The use of Flexible Route Advertisement (FRA) as suggested in B and C is not the appropriate mechanism for controlling access between subnets within a shared VPC; FRA is meant for dynamic route management in Cloud Router, and is not designed for the kind of granular subnet-level filtering required in this scenario. In contrast, option D utilizes the straightforward mechanism of subnet sharing to logically separate different network segments and applies familiar firewall rules to achieve the desired security and access control. This approach minimizes management overhead and stays consistent with the recommended Google best practices for shared VPC.

Authoritative Links for further research:

Shared VPC Overview: <https://cloud.google.com/vpc/docs/shared-vpc>

Shared VPC Concepts: <https://cloud.google.com/vpc/docs/shared-vpc-concepts>

Firewall Rules Overview: <https://cloud.google.com/vpc/docs/firewalls>

Question: 28

CertyIQ

You are adding steps to a working automation that uses a service account to authenticate. You need to drive the automation the ability to retrieve files from a Cloud Storage bucket. Your organization requires using the least privilege possible. What should you do?

- A. Grant the `compute.instanceAdmin` to your user account.
- B. Grant the `iam.serviceAccountUser` to your user account.
- C. Grant the read-only privilege to the service account for the Cloud Storage bucket.
- D. Grant the cloud-platform privilege to the service account for the Cloud Storage bucket.

Answer: C

Explanation:

The correct answer is **C. Grant the read-only privilege to the service account for the Cloud Storage bucket.**

Here's why:

The principle of least privilege dictates granting only the permissions necessary to perform a specific task. In this scenario, the automation needs to retrieve files from a Cloud Storage bucket. This operation solely requires read access; therefore, we must grant the minimum necessary privilege for this action.

Option A, granting `compute.instanceAdmin` to the user account, is incorrect because this role provides extensive control over Compute Engine instances, which is unrelated to accessing Cloud Storage. It also violates the least privilege principle by granting excessive permissions.

Option B, granting `iam.serviceAccountUser` to the user account, is also incorrect. This role allows impersonation of the service account, which isn't needed for the automation's purpose and over-privileges the user. The automation should directly use the service account's credentials rather than the user impersonating it.

Option D, granting cloud-platform privilege to the service account for the Cloud Storage bucket, is incorrect because cloud-platform is an overly broad role that grants access to all Google Cloud resources. This completely violates the least privilege principle, creating a significant security risk.

Option C correctly implements the principle of least privilege. By granting the service account read-only permissions on the specific Cloud Storage bucket, we restrict the automation to only performing the necessary retrieval function. This minimizes the attack surface and confines the potential damage if the service account is compromised.

Further Research:

Principle of Least Privilege: https://en.wikipedia.org/wiki/Principle_of_least_privilege

Cloud Storage IAM Permissions: <https://cloud.google.com/storage/docs/access-control/iam-permissions>

Understanding Roles: <https://cloud.google.com/iam/docs/understanding-roles>

Question: 29

CertyIQ

You converted an auto mode VPC network to custom mode. Since the conversion, some of your Cloud Deployment Manager templates are no longer working.

You want to resolve the problem.

What should you do?

- A. Apply an additional IAM role to the Google API's service account to allow custom mode networks.
- B. Update the VPC firewall to allow the Cloud Deployment Manager to access the custom mode networks.
- C. Explicitly reference the custom mode networks in the Cloud Armor whitelist.
- D. Explicitly reference the custom mode networks in the Deployment Manager templates.

Answer: D

Explanation:

The core issue arises from the change from auto mode to custom mode VPC networks. Auto mode networks automatically create subnets in each region, with predefined IP ranges. Cloud Deployment Manager templates, when designed for auto mode, often implicitly assume the existence and properties of these default subnets. When you switch to custom mode, these assumptions break down because you must now explicitly define the subnets. Therefore, the templates, which were previously working implicitly, now need to be updated to explicitly reference the specific subnets created in your custom mode network.

Option D directly addresses this by advocating for the modification of Deployment Manager templates to include specific references to the new custom mode networks and their subnets. This is the most logical and direct approach to the problem. Options A, B, and C are not relevant. IAM roles, firewall rules, and Cloud Armor whitelists are not directly related to this transition; they are more related to access control and security, while the core issue here is the network configuration being assumed by the templates.

Here's why each option is incorrect:

A: Apply an additional IAM role... IAM roles control what actions a service account can perform, not how subnets are discovered by infrastructure as code tools.

B: Update the VPC firewall... Firewalls control traffic flow. This issue is not related to traffic routing or access.

C: Explicitly reference the custom mode networks in the Cloud Armor whitelist... Cloud Armor whitelists control access to HTTP(S) load balancers, which is not directly related to Deployment Manager.

In essence, the templates need to be made aware of the custom network's structure, a configuration change addressed only by Option D.

Further research:

Google Cloud VPC Networks: <https://cloud.google.com/vpc/docs/vpc>

Cloud Deployment Manager: <https://cloud.google.com/deployment-manager/docs>

Auto mode vs. Custom mode VPC Networks: https://cloud.google.com/vpc/docs/vpc#auto-mode_and_custom-mode_networks

Question: 30

CertyIQ

You have recently been put in charge of managing identity and access management for your organization. You have several projects and want to use scripting and automation wherever possible. You want to grant the editor role to a project member.

Which two methods can you use to accomplish this? (Choose two.)

- A. GetIamPolicy() via REST API
- B. setIamPolicy() via REST API
- C. `gcloud pubsub add-iam-policy-binding Sprojectname --member user:Susername --role roles/editor`
- D. `gcloud projects add-iam-policy-binding Sprojectname --member user:Susername --role roles/editor`
- E. Enter an email address in the Add members field, and select the desired role from the drop-down menu in the GCP Console.

Answer: DE

Explanation:

The correct methods for programmatically granting the editor role to a project member are options D and E. Option D leverages the `gcloud` command-line tool, a powerful utility for managing Google Cloud resources. Specifically, `gcloud projects add-iam-policy-binding` directly modifies a project's IAM policy by adding a binding for the specified member (user) and role (roles/editor). This approach is ideal for automation and scripting within a Google Cloud environment. Option E describes the action of adding a member through the Google Cloud Platform (GCP) Console, a user interface to accomplish the same tasks. This is a non-scripted, manual method, but still a valid way of granting permissions. This provides a click-driven approach to managing IAM and is useful for cases when scripting is not required or desired.

Options A and B, which involve `GetIamPolicy()` and `setIamPolicy()` via the REST API, are also valid methods to modify IAM permissions programmatically, but they are more complex and require careful handling of the entire policy. They would involve reading the current policy, modifying it in a structured format (e.g., JSON), and then writing it back, which is more involved than the `add-iam-policy-binding` command. Option C incorrectly uses `gcloud pubsub` which is used for Pub/Sub topic related IAM, not general project-level permissions.

Therefore, options D and E are the most straightforward methods presented in the context of scripting (for option D) and using the interface directly (for E), to assign the editor role for a given project member.

Authoritative Links:

gcloud projects add-iam-policy-binding documentation:

<https://cloud.google.com/sdk/gcloud/reference/projects/add-iam-policy-binding>

IAM Overview: <https://cloud.google.com/iam/docs/overview>

Granting, Changing, and Revoking Access to Resources: <https://cloud.google.com/iam/docs/granting-changing-revoking-access>

Managing IAM with the GCP Console: <https://cloud.google.com/iam/docs/using-console>

Question: 31

You are using a 10-Gbps direct peering connection to Google together with the gsutil tool to upload files to Cloud Storage buckets from on-premises servers. The on-premises servers are 100 milliseconds away from the Google peering point. You notice that your uploads are not using the full 10-Gbps bandwidth available to you. You want to optimize the bandwidth utilization of the connection.

What should you do on your on-premises servers?

- A. Tune TCP parameters on the on-premises servers.
- B. Compress files using utilities like tar to reduce the size of data being sent.
- C. Remove the -m flag from the gsutil command to enable single-threaded transfers.
- D. Use the perfdiag parameter in your gsutil command to enable faster performance: `gsutil perfdiag gs://[BUCKET NAME]`.

Answer: A

Explanation:

The correct answer is **A. Tune TCP parameters on the on-premises servers.**

Here's the justification:

The observed underutilization of the 10 Gbps direct peering connection, despite having ample bandwidth, points to a likely TCP bottleneck. The 100ms round-trip time (RTT) between the on-premises servers and the Google peering point introduces latency, which can significantly impact TCP performance. TCP's congestion control mechanisms rely on acknowledgments (ACKs) to gauge network capacity. High latency leads to a delay in receiving ACKs, slowing down the rate at which TCP sends data. This is because the TCP window, which dictates the amount of data a sender can transmit before waiting for an ACK, might be too small.

Tuning TCP parameters, particularly the TCP send buffer size, allows for a larger TCP window. A larger window allows more data to be "in flight" and makes the connection less sensitive to higher RTT. This increase in the volume of in-flight data compensates for latency issues and utilizes more of the available bandwidth. Options B, C, and D do not directly address the TCP congestion control bottleneck. Compression (B) reduces data size but may not maximize bandwidth usage. Removing multi-threading (C) will decrease throughput by limiting transfer parallelism and perfdiag (D) analyzes performance but does not fix the problem. Addressing the TCP window is therefore the best solution.

For further reading on TCP performance and tuning, consult:

Google Cloud Documentation on Network Performance: <https://cloud.google.com/network-connectivity/docs/network-performance>

Linux Advanced Routing & Traffic Control: <https://lartc.org/> (specifically on TCP tuning)

RFC 7323 - TCP Extensions for High Performance: <https://datatracker.ietf.org/doc/html/rfc7323>

Question: 32

You work for a multinational enterprise that is moving to GCP.

These are the cloud requirements:

"☑ An on-premises data center located in the United States in Oregon and New York with Dedicated Interconnects connected to Cloud regions us-west1 (primary HQ) and us-east4 (backup)

"☑ Multiple regional offices in Europe and APAC

"☑ Regional data processing is required in europe-west1 and australia-southeast1

"☑ Centralized Network Administration Team

Your security and compliance team requires a virtual inline security appliance to perform L7 inspection for URL filtering. You want to deploy the appliance in us-west1.

What should you do?

A. "Ç Create 2 VPCs in a Shared VPC Host Project. "Ç Configure a 2-NIC instance in zone us-west1-a in the Host Project. "Ç Attach NIC0 in VPC #1 us-west1 subnet of the Host Project. "Ç Attach NIC1 in VPC #2 us-west1 subnet of the Host Project. "Ç Deploy the instance. "Ç Configure the necessary routes and firewall rules to pass traffic through the instance.

B. "Ç Create 2 VPCs in a Shared VPC Host Project. "Ç Configure a 2-NIC instance in zone us-west1-a in the Service Project. "Ç Attach NIC0 in VPC #1 us-west1 subnet of the Host Project. "Ç Attach NIC1 in VPC #2 us-west1 subnet of the Host Project. "Ç Deploy the instance. "Ç Configure the necessary routes and firewall rules to pass traffic through the instance.

C. "Ç Create 1 VPC in a Shared VPC Host Project. "Ç Configure a 2-NIC instance in zone us-west1-a in the Host Project. "Ç Attach NIC0 in us-west1 subnet of the Host Project. "Ç Attach NIC1 in us-west1 subnet of the Host Project "Ç Deploy the instance. "Ç Configure the necessary routes and firewall rules to pass traffic through the instance.

D. "Ç Create 1 VPC in a Shared VPC Service Project. "Ç Configure a 2-NIC instance in zone us-west1-a in the Service Project. "Ç Attach NIC0 in us-west1 subnet of the Service Project. "Ç Attach NIC1 in us-west1 subnet of the Service Project "Ç Deploy the instance. "Ç Configure the necessary routes and firewall rules to pass traffic through the instance.

Answer: A

Explanation:

The correct answer is A. Here's a detailed justification:

The scenario requires a centralized inline security appliance for L7 inspection within a Shared VPC environment. Shared VPC allows multiple projects to access a common VPC network, which is ideal for centralized network management. To achieve inline inspection, the security appliance needs to act as a gateway, intercepting and inspecting traffic before it reaches its destination. This necessitates having two network interfaces (NICs).

Option A correctly sets up the environment using two VPCs within the Shared VPC host project. The appliance, a 2-NIC instance, is also deployed within the host project. The NIC0 is attached to the first VPC and NIC1 is attached to the second VPC. This effectively creates a transit point for traffic. All traffic meant to enter or leave VPC #1 will traverse the security appliance, which allows it to perform the required L7 inspection. Option B incorrectly tries to deploy the appliance within a Service Project, violating the shared network model and preventing centralized control. Option C uses a single VPC, thus not allowing for the two network interface traffic filtering. Option D uses a Service Project for the VPC and the security appliance, violating the Shared VPC model and thus not allowing centralized network management.

The two VPC approach allows one VPC to serve as the "ingress" network where incoming traffic lands, and the other to serve as the "egress" network through which traffic leaves. By having the security appliance with interfaces in both, all traffic must pass through it. The instance resides within the Shared VPC host project allowing centralized administration and shared resources. Correct routing and firewall rules are necessary to direct traffic to the security appliance and allow it to pass only after inspection.

Therefore, Option A is the only correct approach that satisfies the requirements of Shared VPC and centralized security appliance for L7 inspection.

Authoritative Links:

Shared VPC Overview: <https://cloud.google.com/vpc/docs/shared-vpc>

Multi-NIC instances: <https://cloud.google.com/vpc/docs/create-multi-nic-vm>s

Network appliance architecture: <https://cloud.google.com/architecture/network-appliance-architectures-on-google-cloud>

You are designing a Google Kubernetes Engine (GKE) cluster for your organization. The current cluster size is expected to host 10 nodes, with 20 Pods per node and 150 services. Because of the migration of new services over the next 2 years, there is a planned growth for 100 nodes, 200 Pods per node, and 1500 services. You want to use VPC-native clusters with alias IP ranges, while minimizing address consumption. How should you design this topology?

- A. Create a subnet of size /25 with 2 secondary ranges of: /17 for Pods and /21 for Services. Create a VPC-native cluster and specify those ranges.
- B. Create a subnet of size /28 with 2 secondary ranges of: /24 for Pods and /24 for Services. Create a VPC-native cluster and specify those ranges. When the services are ready to be deployed, resize the subnets.
- C. Use gcloud container clusters create [CLUSTER NAME] --enable-ip-alias to create a VPC-native cluster.
- D. Use gcloud container clusters create [CLUSTER NAME] to create a VPC-native cluster.

Answer: A

Explanation:

Justification for Answer A:

Option A provides a correct configuration for a VPC-native GKE cluster utilizing alias IP ranges, optimizing for address space and scalability. VPC-native clusters assign Pod and Service IPs from secondary IP ranges within the VPC subnet, avoiding Network Address Translation (NAT). This eliminates performance bottlenecks associated with traditional IP address management and allows direct routing within the VPC.

Option A's subnet size of /25, representing 128 addresses, is sufficient for the cluster's initial node count of 10. It also leaves room for future growth. The primary IP range of the subnet will be used for the compute nodes. A /17 secondary range for pods can accommodate up to 131,072 addresses. With a target of 200 pods per node, across 100 nodes, 20,000 pod IPs will be required, well within the /17 range. The /21 secondary range for services allocates 2,048 addresses, also adequate for the projected 1500 services. Resizing subnets is a more complicated task compared to increasing the secondary range, so sizing them large enough from the start is the better strategy.

Options B, C, and D are incorrect. Option B's /28 subnet is insufficient for even the initial 10 nodes. The secondary IP range sizes are also far too small for the required number of Pods. Option C & D merely enable ip aliases and do not specify ranges which means defaults will be used that are not ideal for this scenario. Specifying the secondary ranges is necessary to efficiently manage the IP space for the expected pod and service growth.

By using appropriately sized secondary ranges, option A allows for future expansion while minimizing overall IP address consumption and reducing manual subnet changes. This approach perfectly aligns with best practices for deploying and scaling VPC-native GKE clusters.

Authoritative Links:

VPC-native clusters: <https://cloud.google.com/kubernetes-engine/docs/how-to/alias-ips>

IP Address Planning: <https://cloud.google.com/vpc/docs/vpc>

GKE Cluster Creation: <https://cloud.google.com/sdk/gcloud/reference/container/clusters/create>

Question: 34

CertyIQ

Your company has recently expanded their EMEA-based operations into APAC. Globally distributed users report that their SMTP and IMAP services are slow.

Your company requires end-to-end encryption, but you do not have access to the SSL certificates.

Which Google Cloud load balancer should you use?

- A. SSL proxy load balancer

- B. Network load balancer
- C. HTTPS load balancer
- D. TCP proxy load balancer

Answer: D

Explanation:

The correct answer is **D. TCP proxy load balancer**. Here's why:

The scenario requires end-to-end encryption for SMTP and IMAP services but explicitly states that the user does not have access to SSL certificates. This eliminates options that rely on the load balancer handling SSL termination. Both HTTPS load balancers (C) and SSL proxy load balancers (A) require access to SSL certificates for this purpose.

A Network Load Balancer (B) operates at the transport layer (Layer 4) and does not provide SSL capabilities nor does it inspect traffic; it simply forwards traffic based on IP protocol. It wouldn't address the end-to-end encryption requirement.

A TCP proxy load balancer, however, operates at the TCP layer (Layer 4) and forwards TCP traffic without requiring access to the content of the traffic. This allows the backend SMTP/IMAP servers to handle the SSL encryption and decryption using their own certificates. It does not decrypt traffic like the SSL Proxy Load Balancer or the HTTPS Load Balancer. The client connects with an encrypted connection (using TLS or SSL) to the backend server (SMTP/IMAP) through the TCP proxy.

Therefore, the TCP proxy load balancer provides the necessary load balancing without requiring the administrator to have access to SSL certificates, allowing for end-to-end encryption at the server level.

Authoritative Links:

Google Cloud Load Balancing Overview: <https://cloud.google.com/load-balancing/docs/load-balancing-overview>

TCP Proxy Load Balancer: <https://cloud.google.com/load-balancing/docs/tcp/>

SSL Proxy Load Balancer: <https://cloud.google.com/load-balancing/docs/ssl/>

HTTPS Load Balancer: <https://cloud.google.com/load-balancing/docs/https/>

Network Load Balancer: <https://cloud.google.com/load-balancing/docs/network/>

Question: 35

CertyIQ

Your company is working with a partner to provide a solution for a customer. Both your company and the partner organization are using GCP. There are applications in the partner's network that need access to some resources in your company's VPC. There is no CIDR overlap between the VPCs. Which two solutions can you implement to achieve the desired results without compromising the security? (Choose two.)

- A. VPC peering
- B. Shared VPC
- C. Cloud VPN
- D. Dedicated Interconnect
- E. Cloud NAT

Answer: AC

Explanation:

The correct answers are A (VPC peering) and C (Cloud VPN).

VPC Peering (A): VPC peering allows two VPC networks to connect privately, enabling resources in one network to communicate with resources in the other as if they were in the same network. Crucially, it establishes this connection without traffic traversing the public internet, enhancing security. Each VPC retains its own administrative control. Since the question states there's no CIDR overlap, peering is possible and a secure choice for direct communication between the two VPCs. This solution minimizes latency compared to other options because traffic routes directly between peered networks.

Cloud VPN (C): Cloud VPN establishes an encrypted IPsec tunnel between your VPC network and the partner's network, effectively extending your private network securely to theirs via the internet. This method ensures data confidentiality and integrity. It's a suitable option if you cannot or prefer not to peer directly, perhaps due to complexities in management. Cloud VPN uses standard security protocols, offering a well-established approach to secure network connectivity over the public internet.

Why other options are not ideal:

Shared VPC (B): While Shared VPC allows multiple projects to share a common VPC, it requires the partner's application to reside within your project which is not suitable based on the given use case.

Dedicated Interconnect (D): Dedicated Interconnect offers high-bandwidth, low-latency private connectivity, but is significantly more expensive and generally used for high-volume, long-term needs. It's overkill for this scenario and introduces unnecessary complexity.

Cloud NAT (E): Cloud NAT provides outbound internet connectivity for resources without public IP addresses, but doesn't enable communication between two VPCs directly. It is not relevant to this question.

In summary, VPC peering provides a direct and secure connection for inter-VPC communication without the public internet. Cloud VPN creates an encrypted tunnel for secure connectivity when direct peering is not preferred or possible. Both are suitable and secure solutions for the given use case.

Authoritative Links:

VPC Peering: <https://cloud.google.com/vpc/docs/vpc-peering>

Cloud VPN: <https://cloud.google.com/vpn/docs>

Question: 36

CertyIQ

You have a storage bucket that contains the following objects:

[1]
[1]
[1]
[1]

Cloud CDN is enabled on the storage bucket, and all four objects have been successfully cached. You want to remove the cached copies of all the objects with the prefix folder-a, using the minimum number of commands. What should you do?

- A. Add an appropriate lifecycle rule on the storage bucket.
- B. Issue a cache invalidation command with pattern /folder-a/*.
- C. Make sure that all the objects with prefix folder-a are not shared publicly.
- D. Disable Cloud CDN on the storage bucket. Wait 90 seconds. Re-enable Cloud CDN on the storage bucket.

Answer: B

Explanation:

The correct answer is **B. Issue a cache invalidation command with pattern /folder-a/*.**

Here's why:

Cloud CDN caches content to improve performance and reduce latency. When changes are made to the origin (storage bucket), the cached content needs to be updated to reflect these changes. Simply changing permissions or disabling/re-enabling CDN won't immediately remove cached content; it will eventually expire, but not in the prompt manner required.

Option A (Lifecycle Rule): Lifecycle rules primarily manage the lifecycle of objects stored within the bucket itself (e.g., deletion, archiving, versioning). They do not directly influence the Cloud CDN cache. While lifecycle rules might eventually delete the objects from the bucket, the CDN cache would still hold the outdated versions for some time.

Option C (Private Objects): Making objects private prevents access to the objects themselves but does not invalidate the CDN cache. The cached versions of these objects will still be served to users that have previously cached them. This is not the solution to explicitly clear cached objects.

Option D (Disable/Re-enable CDN): Disabling and re-enabling Cloud CDN will eventually clear the cache, but it's a time-consuming and disruptive process. It's not the minimum number of commands, and impacts performance. It also may not fully flush the cache on all edge locations.

Option B (Cache Invalidation): Cloud CDN provides a mechanism for invalidating cached content using a cache invalidation command. By specifying a pattern like `/folder-a/*`, we're instructing Cloud CDN to remove all cached objects with URLs starting with `/folder-a/` from its cache. This is the most efficient and direct way to achieve the required outcome. Cloud CDN invalidation is immediate. It will clear the cached objects at all edge locations and the clients will then fetch the latest version of the object.

Key Concepts:

Cloud CDN: A content delivery network that caches content closer to users to improve performance.

Cache Invalidation: The process of removing outdated content from a CDN cache.

URL Patterns: Used in cache invalidation commands to specify the objects or directories that need to be invalidated.

Authoritative Links:

[Cloud CDN Invalidation](#): Google Cloud documentation on invalidating Cloud CDN cache.

[Google Cloud CDN Overview](#): Overview of Google Cloud CDN.

Question: 37

CertyIQ

Your company is running out of network capacity to run a critical application in the on-premises data center. You want to migrate the application to GCP. You also want to ensure that the Security team does not lose their ability to monitor traffic to and from Compute Engine instances.

Which two products should you incorporate into the solution? (Choose two.)

- A. VPC flow logs
- B. Firewall logs
- C. Cloud Audit logs
- D. Stackdriver Trace
- E. Compute Engine instance system logs

Answer: AB

Explanation:

The correct answer is A and B: VPC Flow Logs and Firewall Logs.

VPC Flow Logs capture a record of network flows sent from and received by VM instances, including source/destination IP addresses, ports, protocols, and bytes transferred. This detailed flow information is crucial for security teams to analyze traffic patterns, identify anomalies, and ensure policy compliance.

<https://cloud.google.com/vpc/docs/flow-logs>

Firewall Logs, on the other hand, record attempts to connect to or from your Compute Engine instances which are dropped by firewall rules. This provides an insight into the effectiveness of your firewall rules and helps identify potential security threats trying to access your system. They complement Flow logs by showing blocked traffic in addition to allowed traffic. <https://cloud.google.com/vpc/docs/firewall-logs>

Cloud Audit Logs (C) primarily tracks administrative activities, such as API calls, not real-time network traffic, and is therefore less relevant for network monitoring. Stackdriver Trace (D) is used for performance analysis of application request flow, focusing on latency, and is not intended for general network traffic inspection. Compute Engine instance system logs (E) provide information about the operating system running within the instances but don't offer the detailed network traffic analysis needed for security monitoring. Thus, the combination of VPC flow logs and Firewall logs are essential to maintain the security monitoring of the migrated application in GCP.

Question: 38

CertyIQ

You want to apply a new Cloud Armor policy to an application that is deployed in Google Kubernetes Engine (GKE). You want to find out which target to use for your Cloud Armor policy. Which GKE resource should you use?

- A. GKE Node
- B. GKE Pod
- C. GKE Cluster
- D. GKE Ingress

Answer: D

Explanation:

The correct answer is **D. GKE Ingress**. Here's why:

Cloud Armor policies are designed to protect your applications from various threats by inspecting incoming traffic. They operate at the network edge, before requests reach your application's backend. In a GKE environment, the Ingress resource is the primary entry point for external traffic into your cluster. It acts as a load balancer and reverse proxy, directing incoming requests to the appropriate services and pods within the cluster.

Cloud Armor policies are directly associated with the load balancer that an Ingress resource creates. This allows you to apply the security policy to the traffic entering the cluster at the network edge. Options like GKE Node, GKE Pod, or the entire GKE Cluster are not suitable because they do not represent the entry point that load balancers or external traffic interacts with directly.

Therefore, when you want to find the correct target for a Cloud Armor policy in a GKE context, you need to focus on the resource that is responsible for handling ingress traffic and can be associated with a load balancer – which is the GKE Ingress.

For more information about Cloud Armor and its integration with GKE Ingress, you can refer to the official Google Cloud documentation:

Cloud Armor Overview: <https://cloud.google.com/armor/docs/overview>

Securing Ingress with Cloud Armor: https://cloud.google.com/kubernetes-engine/docs/how-to/ingress-features#cloud_armor

GKE Ingress: <https://cloud.google.com/kubernetes-engine/docs/concepts/ingress>

Question: 39

CertyIQ

You need to establish network connectivity between three Virtual Private Cloud networks, Sales, Marketing, and Finance, so that users can access resources in all three VPCs. You configure VPC peering between the Sales VPC and the Finance VPC. You also configure VPC peering between the Marketing VPC and the Finance VPC. After you complete the configuration, some users cannot connect to resources in the Sales VPC and the Marketing VPC. You want to resolve the problem.

What should you do?

- A. Configure VPC peering in a full mesh.
- B. Alter the routing table to resolve the asymmetric route.
- C. Create network tags to allow connectivity between all three VPCs.
- D. Delete the legacy network and recreate it to allow transitive peering.

Answer: A

Explanation:

The correct answer is **A. Configure VPC peering in a full mesh.**

Here's the justification:

VPC peering in Google Cloud does not support transitive routing. This means that if VPC A is peered with VPC B, and VPC B is peered with VPC C, VPC A and VPC C cannot communicate directly through VPC B. In the given scenario, the Sales VPC can communicate with the Finance VPC, and the Marketing VPC can communicate with the Finance VPC. However, Sales and Marketing VPCs cannot communicate with each other because their peering connections are only with the Finance VPC and not directly with each other. The users experiencing connection issues are likely trying to access resources between the Sales and Marketing VPCs.

To enable communication between all three VPCs, you need to establish peering connections in a full mesh configuration. This means each VPC should be peered directly with every other VPC. Therefore, in addition to the existing connections, you would need to create a peering connection between the Sales VPC and the Marketing VPC. This ensures all three VPCs are directly connected and can communicate with each other.

Option B is incorrect because routing table issues are not the root cause of transitive peering limitations, though routing table problems could occur if VPC networks had overlapping address ranges, but that was not mentioned in this question. Option C is also incorrect because Network Tags are for firewall rules and do not resolve the peering limitation. Option D is also incorrect because recreating the network as a legacy network, will not allow transitive peering.

Authoritative Links for further research:

VPC Network Peering Concepts: <https://cloud.google.com/vpc/docs/vpc-peering>

VPC Peering Limitations: <https://cloud.google.com/vpc/docs/vpc-peering#limitations> - Pay particular attention to "Transitive peering is not supported."

Question: 40

CertyIQ

You create multiple Compute Engine virtual machine instances to be used at TFTP servers.

Which type of load balancer should you use?

- A. HTTP(S) load balancer
- B. SSL proxy load balancer
- C. TCP proxy load balancer
- D. Network load balancer

Answer: D

Explanation:

The correct answer is **D. Network load balancer**. Here's why:

Network load balancers are ideal for non-HTTP traffic, operating at the transport layer (Layer 4 of the OSI model). TFTP (Trivial File Transfer Protocol) is a UDP-based protocol that doesn't rely on HTTP/HTTPS. Therefore, HTTP(S) load balancers (A) which are designed for web traffic, and SSL Proxy Load Balancers (B), built for encrypted TCP connections, are not suitable. TCP proxy load balancers (C) work with TCP protocols but aren't designed for UDP like TFTP.

Network Load Balancers handle raw IP traffic and can forward UDP traffic efficiently. They don't terminate connections or examine the payload, simply routing packets based on source and destination IPs and ports. This makes them highly performant and suitable for applications requiring low latency and high throughput, such as TFTP servers. Additionally, Network Load Balancers can handle a range of ports, making them adaptable to various network scenarios. For TFTP which uses port 69 and potentially other ports, this flexibility is ideal. The lack of payload inspection makes them cost-effective and provides better performance compared to proxy load balancers. They ensure availability by distributing incoming traffic across the TFTP server instances, preventing a single point of failure.

Authoritative Links:

Google Cloud Load Balancing Overview: <https://cloud.google.com/load-balancing/docs/load-balancing-overview>

Network Load Balancer: <https://cloud.google.com/load-balancing/docs/network/>

Choosing a Load Balancer: <https://cloud.google.com/load-balancing/docs/choosing-load-balancer>

Question: 41

CertyIQ

You want to configure load balancing for an internet-facing, standard voice-over-IP (VOIP) application. Which type of load balancer should you use?

- A. HTTP(S) load balancer
- B. Network load balancer
- C. Internal TCP/UDP load balancer
- D. TCP/SSL proxy load balancer

Answer: B

Explanation:

The correct answer is **B. Network load balancer**. Here's a detailed justification:

Network Load Balancers (NLBs) are ideal for non-HTTP(S) traffic, such as UDP and TCP, which are protocols crucial for VoIP applications. NLBs operate at the transport layer (Layer 4) of the OSI model, making them highly performant and capable of handling the real-time, low-latency requirements of voice communications. Unlike HTTP(S) load balancers, which are designed for web traffic, NLBs don't terminate connections or

inspect content, enabling them to process packets efficiently without introducing significant overhead. This is crucial for VoIP as it relies on constant, low-latency transmission of data. Internal TCP/UDP load balancers are not appropriate as they are designed for traffic within a VPC and not internet-facing traffic. TCP/SSL Proxy load balancers are more suited for TCP traffic that requires SSL termination at the load balancer, which is unnecessary for a standard VoIP application. Therefore, the Network Load Balancer provides the appropriate layer 4 load balancing solution for the given scenario.

Here are some links for further information:

Google Cloud Load Balancing Overview: <https://cloud.google.com/load-balancing/docs/load-balancing-overview>

Network Load Balancer Documentation: <https://cloud.google.com/load-balancing/docs/network>

Layer 4 Load Balancing: [https://en.wikipedia.org/wiki/Load_balancing_\(computing\)#Layer_4_load_balancing](https://en.wikipedia.org/wiki/Load_balancing_(computing)#Layer_4_load_balancing)

Question: 42

CertyIQ

You want to configure a NAT to perform address translation between your on-premises network blocks and GCP. Which NAT solution should you use?

- A. Cloud NAT
- B. An instance with IP forwarding enabled
- C. An instance configured with iptables DNAT rules
- D. An instance configured with iptables SNAT rules

Answer: C

Explanation:

The correct answer is **C. An instance configured with iptables DNAT rules.**

Here's the justification:

The scenario requires network address translation (NAT) between an on-premises network and Google Cloud Platform (GCP). Cloud NAT (option A) is designed for outbound internet traffic from GCP resources to the internet, not for translation between on-premises and GCP internal networks. While instance-based solutions (options B, C, and D) can achieve NAT, the specific type of translation needed dictates the correct approach.

Option B, an instance with IP forwarding enabled, only allows routing; it doesn't perform address translation. Therefore, it's not suitable for our NAT requirement.

Option C, an instance configured with iptables DNAT rules, is the best choice. DNAT (Destination NAT) translates the destination IP address. This is crucial because, for traffic originating from on-premises to GCP resources, the destination IP in the incoming packet needs to be translated from the on-premises address space to the corresponding GCP address. This translation is what allows on-premise traffic to reach and access the private GCP address range.

Option D, an instance configured with iptables SNAT rules, performs Source NAT, which translates the source IP address. While SNAT is used for traffic going out of the network (like for resources going to the internet, which Cloud NAT would handle), it's not the correct translation needed when on-premises initiate connections to GCP resources.

In short, DNAT on an instance allows you to present a different private IP address to the outside network, mapping to the actual IP address inside of GCP, essential for on-premises resources to access private GCP resources using NAT.

Here are some authoritative links for further research:

Iptables: <https://www.netfilter.org/documentation/>

Network Address Translation (NAT): <https://www.cloudflare.com/learning/network-layer/what-is-nat/>

Google Cloud NAT: <https://cloud.google.com/nat/docs/overview>

Differences between SNAT and DNAT: <https://www.fortinet.com/resources/cyberglossary/dnat-vs-snat>

Question: 43

CertyIQ

You need to ensure your personal SSH key works on every instance in your project. You want to accomplish this as efficiently as possible.

What should you do?

- A. Upload your public ssh key to the project Metadata.
- B. Upload your public ssh key to each instance Metadata.
- C. Create a custom Google Compute Engine image with your public ssh key embedded.
- D. Use gcloud compute ssh to automatically copy your public ssh key to the instance.

Answer: A

Explanation:

The most efficient way to ensure your personal SSH key works on every instance within a Google Cloud project is by uploading your public SSH key to the project's metadata (Option A). Project-wide metadata is inherited by all instances within the project. By adding the SSH key here, you avoid the need to configure each instance individually. This drastically reduces the time and effort required for key management, especially when dealing with numerous instances. Option B, uploading the key to individual instance metadata, is significantly more cumbersome and less scalable. Option C, creating a custom image, is overkill for this specific task and introduces unnecessary complexity. Finally, Option D, while gcloud compute ssh can be useful, it doesn't solve the requirement to have the key work on every instance automatically; it only automates connection with an existing key. Project metadata is specifically designed for such shared configurations and provides a centralized, manageable solution for SSH key deployment across the entire project. This aligns with best practices for efficient infrastructure management in cloud environments.

Authoritative Links:

Google Cloud Documentation - Adding and removing project-wide metadata:

<https://cloud.google.com/compute/docs/storing-retrieving-metadata#project-metadata>

Google Cloud Documentation - SSH keys: <https://cloud.google.com/compute/docs/connect/add-ssh-keys>

Question: 44

CertyIQ

In order to provide subnet level isolation, you want to force instance-A in one subnet to route through a security appliance, called instance-B, in another subnet.

What should you do?

- A. Create a more specific route than the system-generated subnet route, pointing the next hop to instance-B with no tag.
- B. Create a more specific route than the system-generated subnet route, pointing the next hop to instance-B with a tag applied to instance-A.
- C. Delete the system-generated subnet route and create a specific route to instance-B with a tag applied to instance-A.
- D. Move instance-B to another VPC and, using multi-NIC, connect instance-B's interface to instance-A's network. Configure the appropriate routes to force traffic through to instance-A.

Answer: B

Explanation:

The correct answer is B, creating a more specific route pointing to instance-B as the next hop, with a tag applied to instance-A. This approach leverages Google Cloud's custom static routes and tag-based routing.

Here's a breakdown:

Subnet Isolation: The goal is to isolate instance-A's traffic to route through instance-B, acting as a security appliance. This cannot be achieved solely via subnet boundaries.

System-Generated Subnet Routes: Google Cloud automatically creates routes for each subnet, directing traffic within that subnet. These routes are less specific than custom routes.

Custom Static Routes: Custom static routes, when configured, take precedence over system-generated routes due to longest prefix matching. This allows you to override the default behavior.

Next Hop: Specifying instance-B as the next hop directs traffic that matches the route to instance-B's IP address (acting as a router/firewall).

Tag-Based Routing: By applying a tag to instance-A, the custom route can be made specific to instance-A's traffic. Only instance-A's packets that match the prefix will be directed towards instance-B.

Why other options are incorrect:

Option A fails to target just the instance-A traffic, possibly affecting other instance traffic in the subnet.

Option C would break traffic flow within the subnet as the system route is required for local communication. Deleting a system-generated route isn't a standard way for traffic manipulation.

Option D is far more complex and not needed to achieve the required isolation, as it requires multiple VPCs and complex network configurations.

Key Concept: The ability to create custom routes based on tags and prefixes is fundamental to implementing complex networking requirements in Google Cloud.

Security Appliance: Instance-B acts as a middle-box inspecting all traffic from instance-A, enabling security policies and access controls.

Scalability: Tag-based routing offers a manageable way to scale policies across similar instances in a network.

By creating a more specific custom route with tag matching, one can successfully force traffic from instance-A to be routed through instance-B, achieving the necessary subnet-level isolation.

Authoritative Links:

Routes Overview: <https://cloud.google.com/vpc/docs/routes>

Using tags with routes: https://cloud.google.com/vpc/docs/routes#tagging_routes

Route priorities: https://cloud.google.com/vpc/docs/routes#route_priorities

Question: 45

CertyIQ

You create a Google Kubernetes Engine private cluster and want to use kubectl to get the status of the pods. In one of your instances you notice the master is not responding, even though the cluster is up and running. What should you do to solve the problem?

A. Assign a public IP address to the instance.

- B. Create a route to reach the Master, pointing to the default internet gateway.
- C. Create the appropriate firewall policy in the VPC to allow traffic from Master node IP address to the instance.
- D. Create the appropriate master authorized network entries to allow the instance to communicate to the master.

Answer: D

Explanation:

The correct solution is **D: Create the appropriate master authorized network entries to allow the instance to communicate to the master.**

Here's the justification:

Private GKE clusters, by design, isolate the master node from the public internet. This enhances security but introduces communication challenges. To interact with the master, clients like kubectl need authorized access. Option A, assigning a public IP, defeats the purpose of a private cluster. Option B, routing through the internet gateway, is unsuitable for private clusters and can expose the master to security risks. Option C, while related to network security, does not address the specific authorization mechanism required by GKE.

Master authorized networks specify which IP ranges can access the GKE master's API server. When an instance attempts to use kubectl to interact with the cluster, the master verifies if the instance's source IP is within these allowed ranges. If the instance is not in an authorized network, the master will reject the connection, leading to the "master not responding" issue. By adding the subnet or specific IP range where the instance resides to the master authorized networks, you grant it permission to access the API server. This ensures secure communication without exposing the master to the public internet. Therefore, modifying the master authorized networks is the correct action to allow kubectl to work from the instance.

This approach leverages GKE's built-in security features and avoids creating less secure configurations. Refer to Google Cloud documentation for more detailed explanations on private GKE clusters and master authorized networks:

[Private clusters - Google Kubernetes Engine \(GKE\) docs](#)

[Authorized networks - Google Kubernetes Engine \(GKE\) docs](#)

Question: 46

CertyIQ

Your company has a security team that manages firewalls and SSL certificates. It also has a networking team that manages the networking resources. The networking team needs to be able to read firewall rules, but should not be able to create, modify, or delete them.

How should you set up permissions for the networking team?

- A. Assign members of the networking team the compute.networkUser role.
- B. Assign members of the networking team the compute.networkAdmin role.
- C. Assign members of the networking team a custom role with only the compute.networks.* and the compute.firewalls.list permissions.
- D. Assign members of the networking team the compute.networkViewer role, and add the compute.networks.use permission.

Answer: B

Explanation:

The correct answer is **B. Assign members of the networking team the compute.networkAdmin role.**

Here's why:

The question specifies that the networking team needs to "read firewall rules," but cannot create, modify, or delete them. Let's analyze the options:

A. Assign members of the networking team the compute.networkUser role. This role is primarily intended for users who need to use existing networks but lack administrative privileges. It doesn't grant sufficient permissions to view firewall rules.

B. Assign members of the networking team the compute.networkAdmin role. The compute.networkAdmin role grants broad permissions over network resources, including reading firewall rules, viewing network configurations, and managing other network components. While this role has more permissions than strictly required for just reading firewall rules, it aligns with the broader responsibilities typically associated with a networking team in a collaborative environment. Crucially, it meets the requirement of reading firewall rules while not allowing them to create, modify, or delete them as that would require more specific firewall related roles.

C. Assign members of the networking team a custom role with only the compute.networks. and the compute.firewalls.list permissions. While a custom role could achieve the exact access needed, creating and maintaining a custom role adds complexity that isn't necessary when an existing predefined role, *compute.networkAdmin*, fits the requirement effectively. Also, *compute.networks.* permission does not grant view access to firewall rules.

D. Assign members of the networking team the compute.networkViewer role, and add the compute.networks.use permission. The compute.networkViewer role grants read-only access to network resources, but it does not include permissions to view firewall rules. Adding compute.networks.use is also not applicable to the requirement of viewing firewall rules.

Therefore, the compute.networkAdmin role is the most appropriate choice. It strikes a balance between providing the necessary permissions to read firewall rules, fulfilling the general duties of a networking team, while adhering to the limitations specified in the question. Predefined roles like compute.networkAdmin simplify role management by using Google's existing permission models. Using a more granular approach through a custom role would be overly complex and might introduce unnecessary complications.

Supporting Links:

Google Cloud IAM Roles: <https://cloud.google.com/iam/docs/understanding-roles>

Compute Engine IAM Roles: <https://cloud.google.com/compute/docs/access/iam> (specifically, the compute.networkAdmin role)

Predefined IAM roles: <https://cloud.google.com/iam/docs/predefined-roles>

Question: 47

CertyIQ

You have created an HTTP(S) load balanced service. You need to verify that your backend instances are responding properly.

How should you configure the health check?

- A. Set request-path to a specific URL used for health checking, and set proxy-header to PROXY_V1.
- B. Set request-path to a specific URL used for health checking, and set host to include a custom host header that identifies the health check.
- C. Set request-path to a specific URL used for health checking, and set response to a string that the backend service will always return in the response body.
- D. Set proxy-header to the default value, and set host to include a custom host header that identifies the health check.

Answer: C

Explanation:

Here's a detailed justification for why option C is the correct way to configure a health check for a Google Cloud HTTP(S) load balancer, along with explanations of why the other options are incorrect:

Correct Answer Justification (C):

Option C, "Set request-path to a specific URL used for health checking, and set response to a string that the backend service will always return in the response body," is the most appropriate because it utilizes the standard HTTP protocol and allows the load balancer to verify the health of backends by making a request to a specific, dedicated URL. The backend service, in turn, is expected to respond with a predefined string. This string matching mechanism confirms not only that the backend is reachable but also that the application is functioning as intended. The request-path specifies where on the backend server the load balancer sends the health check request. The response string makes sure that the load balancer is seeing the correct healthy response from the backend. This combination ensures the load balancer accurately determines the health of the backend.

Why Other Options are Incorrect:

Option A: Setting the proxy-header to PROXY_V1 is intended for forwarding the client's IP address and other connection information when using the PROXY protocol. While this can be useful for logging or other backend processing, it does not contribute to a health check's effectiveness. This protocol has nothing to do with validating backend health, and does not serve the load balancers health checking capabilities.

Option B: Setting the host header is useful for virtual hosting where a single server might host multiple websites. While it's good practice for regular requests, it's not the core requirement for validating health check responses. A custom host header will not tell the load balancer if the backend server is running and returning valid data. The key for health check success is the specific path and matching response string from the server.

Option D: Similar to option A, using the default proxy-header doesn't directly validate backend health. While the host header can be used, it is not the primary mechanism to confirm that the backend is responding appropriately. The core issue is that this option doesn't provide a check of application logic by specifying a path and response content.

Key Concepts:

Health Checks: Crucial for load balancers to determine which backend instances are healthy and ready to receive traffic. Without them, traffic might be sent to unresponsive instances, leading to service disruptions.

HTTP Health Checks: Use HTTP request/response mechanisms to probe backend instances, ensuring that they are not only reachable but also functioning correctly.

Load Balancer: Distributes network traffic across multiple backend servers to improve performance and availability.

Authoritative Links:

Google Cloud Load Balancing Health Checks: <https://cloud.google.com/load-balancing/docs/health-checks>
Configuring HTTP health checks: <https://cloud.google.com/load-balancing/docs/health-checks/configure-health-checks#http-health-checks>

In summary, option C allows for the load balancer to accurately confirm backend health using a specific URL and response string combination, making it the correct configuration for this health check scenario.

Question: 48

You need to give each member of your network operations team least-privilege access to create, modify, and

delete Cloud Interconnect VLAN attachments.
What should you do?

- A. Assign each user the editor role.
- B. Assign each user the compute.networkAdmin role.
- C. Give each user the following permissions only: compute.interconnectAttachments.create, compute.interconnectAttachments.get.
- D. Give each user the following permissions only: compute.interconnectAttachments.create, compute.interconnectAttachments.get, compute.routers.create, compute.routers.get, compute.routers.update.

Answer: B

Explanation:

The correct answer is **B. Assign each user the compute.networkAdmin role.**

Here's why: Creating, modifying, and deleting Cloud Interconnect VLAN attachments requires broad network administration permissions. While options C and D attempt to offer granular control, they fall short. compute.interconnectAttachments.create and compute.interconnectAttachments.get (as in options C and D) only allow for creating and viewing attachments, not modifying or deleting them. Additionally, option D's inclusion of router permissions is also insufficient as it doesn't cover the necessary interconnect attachment modification and deletion actions. Option A, assigning the 'editor' role, provides overly broad access, granting permissions beyond network administration and violating the principle of least privilege. The compute.networkAdmin role provides the necessary permissions to manage all aspects of network resources, including VLAN attachments, routers, and other network components. It strikes the right balance, offering sufficient power for the task without granting excessive privileges, aligning with best practices for security and role management. This role specifically includes the compute.interconnectAttachments.delete, compute.interconnectAttachments.update, and compute.interconnectAttachments.create permissions required for this use case.

For more details, refer to the following Google Cloud documentation:

Predefined IAM roles for Compute Engine: <https://cloud.google.com/iam/docs/understanding-roles#compute-engine-roles> - Specifically, explore the permissions granted to roles/compute.networkAdmin.
Cloud Interconnect permissions: <https://cloud.google.com/interconnect/docs/concepts/iam> - This outlines the specific permissions related to Cloud Interconnect resources.

Question: 49

CertyIQ

You have an application that is running in a managed instance group. Your development team has released an updated instance template which contains a new feature which was not heavily tested. You want to minimize impact to users if there is a bug in the new template.
How should you update your instances?

- A. Manually patch some of the instances, and then perform a rolling restart on the instance group.
- B. Using the new instance template, perform a rolling update across all instances in the instance group. Verify the new feature once the rollout completes.
- C. Deploy a new instance group and canary the updated template in that group. Verify the new feature in the new canary instance group, and then update the original instance group.
- D. Perform a canary update by starting a rolling update and specifying a target size for your instances to receive the new template. Verify the new feature on the canary instances, and then roll forward to the rest of the instances.

Answer: D

Explanation:

The best approach for updating instances with a potentially buggy template while minimizing user impact is **D: Perform a canary update by starting a rolling update and specifying a target size for your instances to receive the new template. Verify the new feature on the canary instances, and then roll forward to the rest of the instances.**

Here's why:

Canary Deployment: Option D employs a canary deployment strategy, a vital practice in software releases. It involves exposing the new template to a small subset of users or instances before a full rollout. This allows for early detection of issues and minimizes the blast radius if something goes wrong.

Rolling Update: Rolling updates are a safer way to deploy updates compared to a complete replacement of instances. By incrementally updating instances, the service remains available and minimizes downtime.

Target Size: By specifying a target size for the canary update, you can control how many instances are exposed to the new template. This gives you a "testbed" for the update without impacting the entire user base.

Verification: Thorough testing on the canary instances is crucial. If the new template exhibits any issues, you can halt the rollout, address the bugs, and avoid disrupting more users.

Roll Forward: If verification is successful, the remaining instances can then be updated, ensuring a stable rollout.

Avoids Manual Patching: Option A is not ideal because manual patching is prone to inconsistencies and is not scalable.

Option B exposes all instances to the risk of the untested update, which is directly against the core requirements.

Option C, while using canarying, introduces an entirely new managed instance group which has no traffic being routed to it and it does not address the question about updating the existing Managed Instance Group.

Relevant Concepts:

Managed Instance Groups (MIGs): MIGs are a service in Google Cloud Platform (GCP) that allows you to create and manage a group of identical virtual machines. They handle scaling, updating, and other management tasks.

Instance Template: An instance template defines the configuration of instances within a MIG, including the VM image, machine type, and other parameters.

Rolling Update: A method of deploying updates where instances are updated gradually, ensuring the service remains available.

Canary Deployment: A strategy that involves deploying a new version of an application to a small group of users or instances before rolling it out to everyone, allowing for testing and issue detection.

Authoritative Links:

Google Cloud Documentation on Updating MIGs: <https://cloud.google.com/compute/docs/instance-groups/creating-migs>

Google Cloud Documentation on Canary Deployments: <https://cloud.google.com/architecture/canary-deployments>

By implementing a canary update with a rolling update using MIGs, you can effectively balance the need for new features with the crucial goal of minimizing service disruption.

Question: 50

You have deployed a proof-of-concept application by manually placing instances in a single Compute Engine zone.

You are now moving the application to production, so you need to increase your application availability and ensure it can autoscale.

How should you provision your instances?

- A. Create a single managed instance group, specify the desired region, and select Multiple zones for the location.
- B. Create a managed instance group for each region, select Single zone for the location, and manually distribute instances across the zones in that region.
- C. Create an unmanaged instance group in a single zone, and then create an HTTP load balancer for the instance group.
- D. Create an unmanaged instance group for each zone, and manually distribute the instances across the desired zones.

Answer: A

Explanation:

The correct answer is **A: Create a single managed instance group, specify the desired region, and select Multiple zones for the location.**

Here's the justification:

Managed Instance Groups (MIGs) are the ideal choice for deploying applications that require high availability and autoscaling. By creating a single MIG and specifying a desired region with "Multiple zones," you achieve automatic distribution of instances across those zones within that region. This inherently provides redundancy, ensuring that if one zone experiences an outage, the application remains available in the other zones. This approach is superior to manual distribution which is error-prone and difficult to manage. The MIG handles instance creation, updates, deletion, and health monitoring automatically, reducing operational overhead. By deploying to multiple zones, the application achieves high availability while avoiding manual effort. Also, with auto-scaling capabilities of MIGs the application can scale in or scale out based on the usage metrics. Unlike unmanaged instance groups, MIGs seamlessly integrate with load balancers allowing for traffic distribution across healthy instances for fault tolerance. Options B, C, and D are incorrect because they involve manual management, single zone deployments, and lack the autoscaling capabilities provided by MIGs.

Relevant Concepts:

Managed Instance Groups (MIGs): Provide automated management of compute resources, including autoscaling and high availability. <https://cloud.google.com/compute/docs/instance-groups>

Regions and Zones: Regions are geographic locations, while zones are isolated locations within a region. Deploying across multiple zones provides high availability. <https://cloud.google.com/compute/docs/regions-zones>

Autoscaling: Automatically adjust the number of running instances based on demand, ensuring application performance and cost-effectiveness. <https://cloud.google.com/compute/docs/autoscaling>

Load Balancer: Distributes traffic across multiple instances to improve application performance and fault tolerance. <https://cloud.google.com/load-balancing>

Therefore, using a single regional MIG with multi-zone deployment is the best practice for achieving the stated requirements for increased availability and autoscaling of the production application.

Question: 51

CertyIQ

You have a storage bucket that contains two objects. Cloud CDN is enabled on the bucket, and both objects have been successfully cached. Now you want to make sure that one of the two objects will not be cached anymore, and will always be served to the internet directly from the origin.

What should you do?

- A. Ensure that the object you don't want to be cached anymore is not shared publicly.
- B. Create a new storage bucket, and move the object you don't want to be checked anymore inside it. Then edit the bucket setting and enable the private attribute.
- C. Add an appropriate lifecycle rule on the storage bucket containing the two objects.
- D. Add a Cache-Control entry with value private to the metadata of the object you don't want to be cached anymore. Invalidate all the previously cached copies.

Answer: D

Explanation:

The correct answer is **D. Add a Cache-Control entry with value private to the metadata of the object you don't want to be cached anymore. Invalidate all the previously cached copies.** This approach leverages standard HTTP caching mechanisms to control how Cloud CDN behaves. Specifically, the Cache-Control: private directive instructs intermediary caches, like Cloud CDN, that the response is specific to the requesting user and should not be stored in shared caches.

Here's a detailed breakdown:

Cache-Control Header: The Cache-Control header is a crucial part of HTTP caching. It allows origin servers to specify how intermediaries and browsers should handle responses, including whether they can be cached, for how long, and who is authorized to view the cached content.

private Directive: When the private directive is used in the Cache-Control header, it signifies that the response is intended for a single user and is not eligible for storage in shared caches like CDNs. This effectively prevents Cloud CDN from caching the specific object.

Object Metadata: In Google Cloud Storage, object metadata allows you to store key-value pairs related to an object. The Cache-Control header can be added as metadata to an object to influence its caching behavior.

Invalidation: Invalidating existing cached copies is important. Since the object was previously cached, simply changing the Cache-Control header won't remove it from the cache. Invalidation ensures that Cloud CDN fetches the updated object with the new Cache-Control setting, thereby enforcing the no-cache behavior.

Why other options are incorrect:

A. Ensuring the object is not shared publicly: This only impacts who can access the object, not whether a CDN will cache the object. Cloud CDN can cache publicly readable objects.

B. Creating a new bucket and making it private: This is overly complex and unnecessary for achieving the goal. This impacts object access permissions and bucket level configuration, but not specifically caching behaviors.

C. Adding a lifecycle rule: Lifecycle rules are primarily for managing object storage and deletion over time, not for controlling caching behavior.

Authoritative Links:

Google Cloud Storage Object Metadata: <https://cloud.google.com/storage/docs/metadata>

Cloud CDN Caching: <https://cloud.google.com/cdn/docs/caching>

Cache-Control Header: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Cache-Control>

In conclusion, using Cache-Control: private in object metadata and invalidating the cache is the most efficient and appropriate method to prevent a specific object from being cached by Cloud CDN, allowing it to be served directly from the origin for each request.

Question: 52

Your company offers a popular gaming service. Your instances are deployed with private IP addresses, and external access is granted through a global load balancer. You have recently engaged a traffic-scrubbing service and want to restrict your origin to allow connections only from the traffic-scrubbing service. What should you do?

- A. Create a Cloud Armor Security Policy that blocks all traffic except for the traffic-scrubbing service.
- B. Create a VPC Firewall rule that blocks all traffic except for the traffic-scrubbing service.
- C. Create a VPC Service Control Perimeter that blocks all traffic except for the traffic-scrubbing service.
- D. Create IPTables firewall rules that block all traffic except for the traffic-scrubbing service.

Answer: A

Explanation:

The correct answer is **A. Create a Cloud Armor Security Policy that blocks all traffic except for the traffic-scrubbing service.**

Here's the justification:

Cloud Armor is Google Cloud's Web Application Firewall (WAF) service, designed to protect applications and services from various threats, including unwanted traffic. Since the question specifies that external access is already managed through a global load balancer, Cloud Armor is the most appropriate tool. Cloud Armor security policies are applied at the load balancer level, allowing for granular control over incoming traffic based on source IP ranges, among other criteria. This precisely matches the requirement of restricting connections solely to the traffic-scrubbing service, which acts as a known source IP.

Option B, VPC Firewall rules, operate at the instance level within a VPC network. While useful for controlling internal traffic, VPC firewall rules are not the ideal choice for securing applications behind a global load balancer where filtering is required at the load balancer entry point. Option C, VPC Service Control Perimeters, focuses on data exfiltration prevention within Google Cloud projects and is not suitable for this traffic filtering scenario. Option D, using IPTables, would necessitate manual management across all involved instances, which is not scalable and is less manageable than centralizing at the load balancer via Cloud Armor. Cloud Armor also provides advanced features beyond basic IP filtering, such as protection against OWASP Top 10 vulnerabilities and geo-based access control. Therefore, leveraging Cloud Armor to allow traffic from the traffic-scrubbing service IP address range(s) and block all other traffic offers the best centralized and secure solution for this requirement.

Authoritative Links:

Cloud Armor Overview: <https://cloud.google.com/armor/docs/overview>

Cloud Armor Security Policies: <https://cloud.google.com/armor/docs/security-policy-concepts>

Load Balancing Overview: <https://cloud.google.com/load-balancing/docs/>

Question: 53

Your software team is developing an on-premises web application that requires direct connectivity to Compute Engine Instances in GCP using the RFC 1918 address space. You want to choose a connectivity solution from your on-premises environment to GCP, given these specifications:

- Your ISP is a Google Partner Interconnect provider.
- Your on-premises VPN device's internet uplink and downlink speeds are 10 Gbps.
- A test VPN connection between your on-premises gateway and GCP is performing at a maximum speed of 500 Mbps due to packet losses.
- Most of the data transfer will be from GCP to the on-premises environment.
- The application can burst up to 1.5 Gbps during peak transfers over the Interconnect.
- Cost and the complexity of the solution should be minimal.

How should you provision the connectivity solution?

- A. Provision a Partner Interconnect through your ISP.
- B. Provision a Dedicated Interconnect instead of a VPN.
- C. Create multiple VPN tunnels to account for the packet losses, and increase bandwidth using ECMP.
- D. Use network compression over your VPN to increase the amount of data you can send over your VPN.

Answer: A

Explanation:

The optimal solution is to provision a Partner Interconnect through your ISP (Option A). Here's why: The problem states that the ISP is already a Google Partner Interconnect provider, making this option readily available and simplifying the setup process. Partner Interconnect leverages a direct, private connection from the on-premises network through a partner's network to Google's network, bypassing the public internet. This provides greater reliability and consistent bandwidth compared to VPN solutions. The existing VPN is experiencing severe packet loss, limiting its throughput to a meager 500 Mbps, while the application needs to burst up to 1.5 Gbps. The 10 Gbps uplink speed of the on-premise VPN device is irrelevant here due to limitations within the VPN tunnel itself. A Partner Interconnect offers higher bandwidth options, which better suit the application's potential burst traffic of 1.5 Gbps. Using multiple VPN tunnels with ECMP (Option C) adds unnecessary complexity and may not fully resolve the underlying issues of packet loss and fluctuating VPN speeds, as these are not an inherent part of VPN design. Compressing network traffic over a VPN (Option D) may help a little, but it may not reach 1.5 Gbps and adds additional burden to the equipment. A dedicated interconnect (Option B) could be a viable alternative, but is more complex and typically more costly for the needs described in the scenario. Partner Interconnect prioritizes cost-effectiveness and ease of implementation for situations where a partner is already used as a carrier. Partner Interconnect also enables a more private, secure, reliable, and consistently performing connection.

Relevant Resources:

1. Google Cloud Partner Interconnect Overview: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/partner-interconnect>
2. Google Cloud VPN Overview: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/overview>

Question: 54

CertyIQ

Your company has just launched a new critical revenue-generating web application. You deployed the application for scalability using managed instance groups, autoscaling, and a network load balancer as frontend. One day, you notice severe bursty traffic that caused autoscaling to reach the maximum number of instances, and users of your application cannot complete transactions. After an investigation, you think it is a DDOS attack. You want to quickly restore user access to your application and allow successful transactions while minimizing cost. Which two steps should you take? (Choose two.)

- A. Use Cloud Armor to blacklist the attacker's IP addresses.
- B. Increase the maximum autoscaling backend to accommodate the severe bursty traffic.
- C. Create a global HTTP(s) load balancer and move your application backend to this load balancer.
- D. Shut down the entire application in GCP for a few hours. The attack will stop when the application is offline.
- E. SSH into the backend compute engine instances, and view the auth logs and syslogs to further understand the nature of the attack.

Answer: AC

Explanation:

Here's a detailed justification for why options A and C are the correct choices, and why the others are incorrect:

Correct Options:

A. Use Cloud Armor to blacklist the attacker's IP addresses. Cloud Armor is a web application firewall (WAF) service specifically designed to protect applications from attacks like DDoS. By quickly identifying and blacklisting the attacker's IP addresses, Cloud Armor can filter out the malicious traffic, allowing legitimate user traffic to reach the application. This action helps restore service availability and ensures that resources are not consumed by the attack traffic. This is a proactive measure that directly addresses the attack. [Cloud Armor Documentation](#)

C. Create a global HTTP(s) load balancer and move your application backend to this load balancer. Switching from a network load balancer to a global HTTP(s) load balancer provides several benefits during a DDoS attack. A global load balancer distributes traffic across multiple regions, enhancing resilience. It also has built-in DDoS protection capabilities through Google's global network infrastructure. This broader distribution of traffic makes it more challenging for attackers to overwhelm the application, and enables Google's infrastructure to absorb some of the attack traffic. Furthermore, it allows Cloud Armor rules to apply globally, which strengthens the protection. [Global Load Balancer Overview](#)

Incorrect Options:

B. Increase the maximum autoscaling backend to accommodate the severe bursty traffic. While autoscaling is beneficial for normal traffic fluctuations, solely relying on it during a significant DDoS attack is not ideal. The attacker can easily overwhelm the autoscaled instances before they spin up, causing potential cost overruns and potential service instability. Scaling up resources might make the application more resilient, but it's a reactive approach and more beneficial in the long-term rather than a short-term mitigation strategy.

D. Shut down the entire application in GCP for a few hours. The attack will stop when the application is offline. Shutting down the application is a drastic measure and results in severe service disruption and loss of revenue. This also provides a negative user experience. This approach is not a practical mitigation step. The attack may resume when the service is back online. The best approach is to mitigate an attack and stay available, rather than just removing the ability for an attack to be effective.

E. SSH into the backend compute engine instances, and view the auth logs and syslogs to further understand the nature of the attack. Although examining logs is crucial for post-attack analysis and learning, it's not the most effective short-term mitigation step when facing an ongoing DDoS attack that needs immediate mitigation. Analyzing logs would not restore application availability, rather it would slow down the response to resolving an outage. It's an important step for the investigation after the incident, not for real-time mitigation.

Question: 55

CertyIQ

You are creating a new application and require access to Cloud SQL from VPC instances without public IP addresses.

Which two actions should you take? (Choose two.)

- A. Activate the Service Networking API in your project.
- B. Activate the Cloud Datastore API in your project.
- C. Create a private connection to a service producer.
- D. Create a custom static route to allow the traffic to reach the Cloud SQL API.
- E. Enable Private Google Access.

Answer: AC

Explanation:

Here's the breakdown of why options A and C are the correct choices for enabling private access to Cloud SQL from VPC instances without public IP addresses:

Option A, "Activate the Service Networking API in your project," is crucial because it enables the necessary infrastructure for private connectivity between your VPC network and Google's managed services, like Cloud SQL. This API allows the establishment of private connections without exposing your instances to the public internet. It acts as a prerequisite for setting up private connections and managing network resources for service producers. [<https://cloud.google.com/vpc/docs/configure-private-service-access>]

Option C, "Create a private connection to a service producer," is the second essential step. This process specifically establishes a private link between your VPC network and the Cloud SQL service's network. Without this connection, instances in your VPC, lacking public IPs, would be unable to communicate with Cloud SQL. Private connection is established using Private Service Access, leveraging VPC peering. This ensures that all traffic remains within Google's network and avoids exposure to the public internet. [<https://cloud.google.com/vpc/docs/private-services-access>]

Option B, "Activate the Cloud Datastore API," is incorrect as it's unrelated to Cloud SQL connectivity. Cloud Datastore is a different database service with its own API. Option D, "Create a custom static route to allow the traffic to reach the Cloud SQL API," is not the correct approach for this scenario since Cloud SQL API is not an endpoint that requires routing and also static routing cannot solve connectivity between VPC networks. Lastly, option E, "Enable Private Google Access," is used for communication from VPC to Google APIs and doesn't apply here as Cloud SQL is a service producer, needing private connection.

In essence, the Service Networking API lays the foundation, and the private connection to the service producer (Cloud SQL) establishes the actual private communication path.

Question: 56

CertyIQ

You want to use Cloud Interconnect to connect your on-premises network to a GCP VPC. You cannot meet Google at one of its point-of-presence (POP) locations, and your on-premises router cannot run a Border Gateway Protocol (BGP) configuration.

Which connectivity model should you use?

- A. Direct Peering
- B. Dedicated Interconnect
- C. Partner Interconnect with a layer 2 partner
- D. Partner Interconnect with a layer 3 partner

Answer: D

Explanation:

The correct answer is **D. Partner Interconnect with a layer 3 partner**. Here's why:

The scenario outlines a need to connect an on-premises network to GCP using Cloud Interconnect, but with key limitations: no direct Google POP access and no BGP capability on the on-premises router.

Direct Peering (A) is unsuitable as it requires direct connection to a Google POP, which is explicitly stated as not feasible. Direct Peering is typically used for high-traffic, business-critical applications that require very low latency and do not involve transit networks.

Dedicated Interconnect (B) also requires a physical connection at a Google POP, thus failing the stated constraint. It's ideal for very high bandwidth use cases, but needs direct physical connection at the Google point of presence.

Partner Interconnect (C & D) is designed for situations where a direct connection to Google POP is not

available. It utilizes a service provider to bridge the connection. Both options C and D are valid options to use Partner Interconnect but option C is eliminated by the question text because it requires a layer 2 connection, which means it still needs BGP on your on-premise network.

The limitation regarding BGP incompatibility of the on-premise router is key. Layer 3 connections in Partner Interconnect rely on the partner's network to handle the BGP routing, effectively offloading that requirement from the on-premise network. Specifically, a **Layer 3 Partner Interconnect (D)** manages routing using BGP on the partner's side, which makes it the only valid option when the on-premise router cannot run BGP itself.

Therefore, **Partner Interconnect with a layer 3 partner (D)** is the best fit as it accommodates the geographical restriction by leveraging a service provider and circumvents the BGP limitation by offloading the routing responsibility.

Further Research:

[Google Cloud Interconnect Overview](#)

[Partner Interconnect](#)

[Dedicated Interconnect](#)

[Cloud Interconnect Layer 3 connections](#)

Question: 57

CertyIQ

You have configured a Compute Engine virtual machine instance as a NAT gateway. You execute the following command: `gcloud compute routes create no-ip-internet-route \`

```
--network custom-network1 \
--destination-range 0.0.0.0/0 \
--next-hop instance nat-gateway \
--next-hop instance-zone us-central1-a \
--tags no-ip --priority 800
```

You want existing instances to use the new NAT gateway.

Which command should you execute?

- A. `sudo sysctl -w net.ipv4.ip_forward=1`
- B. `gcloud compute instances add-tags [existing-instance] --tags no-ip`
- C. `gcloud builds submit --config=cloudbuild.waml --substitutions=TAG_NAME=no-ip`
- D. `gcloud compute instances create example-instance --network custom-network1 --subnet subnet-us-central --no-address --zone us-central1-a --image-family debian-9 --image-project debian-cloud --tags no-ip`

Answer: B

Explanation:

The correct answer is **B: `gcloud compute instances add-tags [existing-instance] --tags no-ip`**.

Here's the justification:

The provided `gcloud compute routes create` command establishes a custom route that directs traffic destined for the internet (0.0.0.0/0) through a specified Compute Engine instance acting as a NAT gateway. Crucially, this route is only applied to instances within the `custom-network1` network that have the tag `no-ip`. This tagging mechanism is the key to associating instances with this custom route. The command in option B adds the `no-ip` tag to existing instances, enabling them to start using the newly created NAT gateway. Option A (`sudo sysctl -w net.ipv4.ip_forward=1`) is relevant for enabling IP forwarding on the NAT gateway instance itself, not the instances that will use the gateway. Option C involves Cloud Builds which is unrelated to this scenario. Option D creates a new instance, which is not the goal of the question. Therefore, only Option B correctly modifies existing instances to utilize the custom route via the tag and the NAT gateway. Tagging is a powerful mechanism for targeted route application. <https://cloud.google.com/vpc/docs/routes>

Question: 58

CertyIQ

You need to configure a static route to an on-premises resource behind a Cloud VPN gateway that is configured for policy-based routing using the `gcloud` command.

Which next hop should you choose?

- A. The default internet gateway
- B. The IP address of the Cloud VPN gateway
- C. The name and region of the Cloud VPN tunnel
- D. The IP address of the instance on the remote side of the VPN tunnel

Answer: C

Explanation:

The correct answer is **C. The name and region of the Cloud VPN tunnel**. Here's why: When configuring a static route in Google Cloud to reach an on-premises network via a Cloud VPN using policy-based routing, you must specify the VPN tunnel as the next hop. This is because policy-based VPNs utilize selectors (like IP ranges) to determine which traffic should be routed through the VPN tunnel. Specifying the tunnel as the next hop directly associates the static route with that specific tunnel's policy.

Option A, the default internet gateway, is incorrect as it would route traffic to the public internet, bypassing the VPN connection entirely. Option B, the IP address of the Cloud VPN gateway, is also incorrect. While the gateway is crucial for the VPN connection, static routes must use the tunnel as the next hop, not the gateway's IP. The IP is part of the tunnel configuration but not directly used for routing. Option D, the IP address of an instance on the remote side, is incorrect; static routes point to a network gateway, not to specific machines behind the gateway.

By specifying the name and region of the Cloud VPN tunnel, you instruct the Google Cloud network to forward traffic matching the static route's destination through the configured tunnel to the on-premises network. This ensures that traffic is routed correctly through the VPN based on the defined policy. The command-line tool `gcloud` expects this identifier for tunnel-based routing when establishing a static route. This method aligns with Google Cloud's networking model and enables proper communication between cloud resources and on-premises networks. The tunnel name acts as a logical construct, allowing for efficient routing based on the underlying VPN configuration.

Authoritative Links:

Google Cloud Documentation on Creating Static Routes:

https://cloud.google.com/vpc/docs/routes#creating_static_routes

Google Cloud Documentation on Cloud VPN Overview:

<https://cloud.google.com/vpn/docs/concepts/overview>

Google Cloud Documentation on Policy Based VPN: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/policy-based-vpn>

Question: 59

CertyIQ

You need to enable Cloud CDN for all the objects inside a storage bucket. You want to ensure that all the object in the storage bucket can be served by the CDN.

What should you do in the GCP Console?

- A. Create a new cloud storage bucket, and then enable Cloud CDN on it.
- B. Create a new TCP load balancer, select the storage bucket as a backend, and then enable Cloud CDN on the backend.
- C. Create a new SSL proxy load balancer, select the storage bucket as a backend, and then enable Cloud CDN on the backend.
- D. Create a new HTTP load balancer, select the storage bucket as a backend, enable Cloud CDN on the backend, and make sure each object inside the storage bucket is shared publicly.

Answer: D

Explanation:

Here's a detailed justification for why option D is the correct answer:

Cloud CDN (Content Delivery Network) is designed to cache and serve content closer to users, reducing latency and improving performance. To use Cloud CDN with a Google Cloud Storage bucket, you must first make the storage bucket accessible through a load balancer. This is because Cloud CDN integrates with load balancers, using them as the entry point for requests before determining if the content can be served from the cache or needs to be fetched from the origin (the storage bucket in this case).

Option D correctly outlines the steps: creating an HTTP load balancer, setting the storage bucket as its backend, and enabling Cloud CDN on that backend. HTTP load balancers are designed for web traffic (HTTP and HTTPS), which is the common way content is accessed via a CDN. The crucial final step in option D is ensuring the objects inside the storage bucket are shared publicly. Cloud CDN cannot serve private objects directly; they must be publicly accessible to be cached and delivered.

Let's examine why the other options are incorrect:

Option A: Creating a new storage bucket and enabling Cloud CDN on it directly is not how Cloud CDN works. Cloud CDN does not directly integrate with storage buckets. It requires a load balancer in between to handle requests.

Option B: While TCP load balancers can distribute traffic, they operate at Layer 4 and are not suitable for HTTP/HTTPS content that Cloud CDN is designed for. Additionally, TCP load balancers do not integrate with Cloud CDN in the way required.

Option C: SSL proxy load balancers, like TCP load balancers, are not the correct type of load balancer to use for this purpose. While they do handle SSL/TLS, they are not specifically designed for the same functionality that an HTTP load balancer offers in concert with Cloud CDN. They do not directly support Cloud CDN.

Therefore, **Option D** is the only option that correctly enables Cloud CDN for objects in a storage bucket by utilizing an HTTP load balancer, setting the storage bucket as a backend, enabling CDN on the backend, and ensuring the objects are public.

Authoritative Links for further research:

Google Cloud CDN Overview: <https://cloud.google.com/cdn/docs/overview>

Setting up Cloud CDN with a Google Cloud Storage bucket: <https://cloud.google.com/cdn/docs/setting-up-cdn-with-gcs>

Google Cloud HTTP(S) Load Balancer: <https://cloud.google.com/load-balancing/docs/https/>

Cloud Storage Access Control: <https://cloud.google.com/storage/docs/access-control>

Question: 60

CertyIQ

Your company's Google Cloud-deployed, streaming application supports multiple languages. The application development team has asked you how they should support splitting audio and video traffic to different backend Google Cloud storage buckets. They want to use URL maps and minimize operational overhead. They are currently

using the following directory structure:

```
/fr/video  
/en/video  
/es/video  
/..video  
/fr/audio  
/en/audio  
/es/audio  
/..audio
```

Which solution should you recommend?

- A. Rearrange the directory structure, create a URL map and leverage a path rule such as `/video/*` and `/audio/*`.
- B. Rearrange the directory structure, create DNS hostname entries for video and audio and leverage a path rule such as `/video/*` and `/audio/*`.
- C. Leave the directory structure as-is, create a URL map and leverage a path rule such as `\[a-z] 2 \video` and `\[a-z] 2 \audio`.
- D. Leave the directory structure as-is, create a URL map and leverage a path rule such as `/*video` and `/*audio`.

Answer: A

Explanation:

The correct answer is A. Rearranging the directory structure to a more logical format, such as `/video/` and `/audio/`, directly under the root is the most efficient approach. This significantly simplifies path-based routing using URL maps. By employing path rules like `/video/` and `/audio/`, you can easily direct traffic destined for video content to one backend bucket and audio content to another. This structure aligns with the intended functionality of separating the different media types.

Option B is inefficient because relying on DNS entries for content type is redundant when you can use path-based routing directly, and introduces unnecessary complexity. Options C and D, while seemingly attempting to leverage the current directory, fail to provide clear separation. The structure `/[a-z] 2 /video` or `/*video` can cause conflicts if further directories are created, as a pattern of any 2 letters might lead to confusion. Also, these would require individual path rules for each language, creating unnecessary complexity.

Using URL maps with simple path rules offers a scalable and maintainable solution to routing traffic to different backend buckets. The primary goal is to minimize operational overhead and simplify the management of the application's routing. This straightforward structure makes it easy to add new languages or storage buckets in the future and keeps the URL mapping configuration clean. Utilizing cloud storage to store the media content and path-based routing through URL maps leverages core cloud concepts in a way that is simple, scalable, and cost-effective.

Further research can be found on Google Cloud's documentation about URL Maps and Content Routing:

[URL maps overview](#)

[Path matching](#)

Question: 61

CertyIQ

You want to establish a dedicated connection to Google that can access Cloud SQL via a public IP address and that does not require a third-party service provider.

Which connection type should you choose?

- A. Carrier Peering
- B. Direct Peering
- C. Dedicated Interconnect
- D. Partner Interconnect

Answer: B

Explanation:

The correct answer is **B. Direct Peering**. Here's why:

Direct Peering allows you to establish a direct network connection with Google's edge network at a Google point of presence (PoP) without relying on a third-party service provider. This means you can bypass the public internet for traffic destined to Google services, enhancing performance and security. The key requirements from the question are "dedicated connection to Google," "access Cloud SQL via a public IP address," and "does not require a third-party service provider". Direct Peering perfectly satisfies these. Cloud SQL instances can be accessed using their public IP addresses with Direct Peering, making this option suitable.

Carrier Peering (A) involves connecting through a third-party carrier and is therefore unsuitable. Dedicated Interconnect (C) also provides a direct connection but uses private IP addresses. While useful for hybrid setups and private APIs, it doesn't satisfy the requirement of accessing Cloud SQL over public IP addresses. Partner Interconnect (D) similarly uses a third-party provider, therefore not a solution without it. Direct Peering is specifically designed for direct public access to Google services, aligning with the scenario.

Further Research:

Google Cloud Direct Peering: <https://cloud.google.com/network-connectivity/docs/direct-peering/overview>

Google Cloud Interconnect Options: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/overview>

In summary, Direct Peering is the only option that meets all the specified criteria: a dedicated, direct connection to Google that doesn't involve a third-party and enables access to Cloud SQL via its public IP address.

Question: 62

CertyIQ

You are configuring a new instance of Cloud Router in your Organization's Google Cloud environment to allow connection across a new Dedicated Interconnect to your data center. Sales, Marketing, and IT each have a service project attached to the Organization's host project. Where should you create the Cloud Router instance?

- A. VPC network in all projects
- B. VPC network in the IT Project
- C. VPC network in the Host Project
- D. VPC network in the Sales, Marketing, and IT Projects

Answer: C

Explanation:

The correct location for the Cloud Router instance is the VPC network within the **Host Project (C)**. Cloud Routers are regional resources that facilitate dynamic routing using BGP (Border Gateway Protocol). They are integral to interconnecting a Google Cloud VPC network with external networks, such as a data center via Dedicated Interconnect. In a Shared VPC setup, where multiple service projects are attached to a single host project, the Cloud Router must reside within the host project's VPC network because the Dedicated Interconnect connection terminates in the host VPC.

Placing the Cloud Router within the service projects would not allow direct peering with the Dedicated Interconnect, as the interconnect's attachments and virtual interfaces are within the host project. The host project's VPC network is the central point for network management and connectivity for all associated service

projects in a Shared VPC environment. Cloud Routers manage routes across the entire shared VPC. While service projects use the same shared network, their resources and subnetworks within that network are still defined in the context of the host project. Therefore, the Cloud Router needs to be associated directly with the host project's VPC to manage the routes to the on-premises network via the Dedicated Interconnect, enabling connectivity for all service projects. The service project themselves can then share the routes via Shared VPC.

For detailed information regarding Shared VPC and Cloud Router, consult the following Google Cloud documentation:

Shared VPC overview: <https://cloud.google.com/vpc/docs/shared-vpc>

Cloud Router overview: <https://cloud.google.com/router/docs/overview>

Dedicated Interconnect Overview: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/dedicated-interconnect>

Question: 63

CertyIQ

You created a new VPC for your development team. You want to allow access to the resources in this VPC via SSH only.

How should you configure your firewall rules?

- A. Create two firewall rules: one to block all traffic with priority 0, and another to allow port 22 with priority 1000.
- B. Create two firewall rules: one to block all traffic with priority 65536, and another to allow port 3389 with priority 1000.
- C. Create a single firewall rule to allow port 22 with priority 1000.
- D. Create a single firewall rule to allow port 3389 with priority 1000.

Answer: C

Explanation:

The correct answer is C. Google Cloud's VPC firewall rules operate on an "allow" model; traffic is implicitly denied unless explicitly permitted by a rule. Therefore, you don't need a rule to explicitly block all traffic. Creating a single ingress (incoming) firewall rule with the action set to "allow," targeting the specific resources (e.g., using tags or network target), and specifying TCP port 22 (the standard SSH port) is sufficient. This rule allows SSH connections to your development VPC resources while implicitly blocking all other traffic by default. The priority of 1000 is arbitrary but needs to be set to a value within the permissible range. Lower values mean higher priority. Since no other rule is defined, a priority of 1000 works just fine.

Options A, B, and D are incorrect. Option A introduces an unnecessary rule with a priority 0 deny action, and this would block all traffic even though a later rule allows port 22, due to the processing order of firewall rules. Option B proposes to allow RDP port 3389, which is not SSH and a deny priority of 65536 isn't necessary. Option D incorrectly allows port 3389, again not SSH.

In summary, Google Cloud firewall's implicit deny combined with a single allow rule for port 22 provides the simplest and most effective method for securing SSH access to resources within a VPC, aligning with the principle of least privilege and the default-deny approach.

Authoritative Links:

VPC Firewall Rules Overview: <https://cloud.google.com/vpc/docs/firewalls>

Firewall Rule Priorities: <https://cloud.google.com/vpc/docs/firewalls#priorities>

Creating Firewall Rules: <https://cloud.google.com/vpc/docs/using-firewalls>

Question: 64

Your on-premises data center has 2 routers connected to your GCP through a VPN on each router. All applications are working correctly; however, all of the traffic is passing across a single VPN instead of being load-balanced across the 2 connections as desired.

During troubleshooting you find:

- "Ç Each on-premises router is configured with the same ASN.
- "Ç Each on-premises router is configured with the same routes and priorities.
- "Ç Both on-premises routers are configured with a VPN connected to a single Cloud Router.
- "Ç The VPN logs have no-proposal-chosen lines when the VPNs are connecting.
- "Ç BGP session is not established between one on-premises router and the Cloud Router.

What is the most likely cause of this problem?

- A. One of the VPN sessions is configured incorrectly.
- B. A firewall is blocking the traffic across the second VPN connection.
- C. You do not have a load balancer to load-balance the network traffic.
- D. BGP sessions are not established between both on-premises routers and the Cloud Router.

Answer: A

Explanation:

Okay, let's break down why option A, "One of the VPN sessions is configured incorrectly," is the most likely cause.

The problem description highlights several key issues: traffic is not load-balanced, only one VPN is being used, there are "no-proposal-chosen" errors in the VPN logs, and a BGP session is failing on one router. These points strongly indicate a misconfiguration in one of the VPN tunnels, rather than issues with routing or load balancing in general.

Here's why the other options are less likely:

B. A firewall is blocking the traffic across the second VPN connection: While firewalls can cause problems, the "no-proposal-chosen" error in the VPN logs suggests a fundamental incompatibility in the VPN tunnel configuration itself, before it even tries to pass traffic. A firewall would likely cause connection drops, not negotiation failures.

C. You do not have a load balancer to load-balance the network traffic: Cloud Router, as used with GCP VPNs, doesn't use a traditional load balancer for traffic distribution over multiple VPNs. Rather, it uses BGP to steer traffic based on routes learned from the on-premises routers. While a load balancer could be used in other scenarios, it is not relevant here.

D. BGP sessions are not established between both on-premises routers and the Cloud Router: The prompt states that the BGP session isn't established for one router not both. Although having a non-established BGP session is a problem, it doesn't explain why one connection is working and the other is not.

The "no-proposal-chosen" error means that the VPN tunnel negotiation is failing because the two sides (on-premises router and GCP) cannot agree on security parameters such as encryption algorithms, hashing algorithms, or Diffie-Hellman groups. This strongly indicates a misconfiguration on one end of the tunnel or that the proposed settings are not supported. The subsequent failure of BGP session on the same connection is a result of the tunnel failing to establish properly.

Therefore, option A is the most direct and logical cause for the symptoms. A misconfigured VPN tunnel on one end is sufficient to cause both the negotiation errors and the lack of traffic across the second VPN, and subsequent BGP failure.

Authoritative Links for Further Research:

Google Cloud VPN Documentation: <https://cloud.google.com/vpn/docs> - Provides a comprehensive overview

of Cloud VPN concepts and configurations.

Troubleshooting Cloud VPN: <https://cloud.google.com/network-connectivity/docs/vpn/troubleshooting> - Specific section on troubleshooting common problems with Cloud VPNs.

IKE and IPsec Configuration: https://en.wikipedia.org/wiki/Internet_Key_Exchange - For deeper understanding of the tunnel negotiation process and parameters involved.

Question: 65

CertyIQ

You need to define an address plan for a future new GKE cluster in your VPC. This will be a VPC native cluster, and the default Pod IP range allocation will be used. You must pre-provision all the needed VPC subnets and their respective IP address ranges before cluster creation. The cluster will initially have a single node, but it will be scaled to a maximum of three nodes if necessary. You want to allocate the minimum number of Pod IP addresses. Which subnet mask should you use for the Pod IP address range?

- A. /21
- B. /22
- C. /23
- D. /25

Answer: B

Explanation:

Here's a justification for why a /22 subnet mask is the correct choice for the Pod IP address range in this scenario:

A VPC-native GKE cluster allocates Pod IPs from a subnet. The default allocation strategy assigns a /24 subnet per node. Since the question specifies a maximum of three nodes, we need to account for three such /24 subnets. A /24 provides 256 addresses (2^8).

To determine the correct subnet for the total Pod IP range, we need to consolidate these node subnets. For three nodes, we need $3 * 256 = 768$ IP addresses. A /23 subnet offers 512 IPs (2^9), which is insufficient. Option A, /21 provides 2048 IPs, while it would certainly work it would be an overly large IP space allocation. A /25 provides 128 IPs, not even enough for a single node, so its not correct. A /22 subnet offers 1024 IP addresses (2^{10}). This is the smallest CIDR that accommodates the required 768 addresses for three nodes and meets the requirement for allocating the minimum necessary.

Therefore, a /22 subnet mask is the most efficient and appropriate choice for the Pod IP address range to ensure sufficient addresses for all future GKE nodes while minimizing IP address waste.

Authoritative Links:

GKE VPC-native documentation: <https://cloud.google.com/kubernetes-engine/docs/how-to/alias-ips>

VPC Subnetting concepts: <https://cloud.google.com/vpc/docs/vpc>

IP Address Calculation: <https://www.calculator.net/ip-subnet-calculator.html> (Use this to check the amount of IPs per subnet)

Question: 66

CertyIQ

You have created a firewall with rules that only allow traffic over HTTP, HTTPS, and SSH ports. While testing, you specifically try to reach the server over multiple ports and protocols; however, you do not see any denied connections in the firewall logs. You want to resolve the issue. What should you do?

- A. Enable logging on the default Deny Any Firewall Rule.
- B. Enable logging on the VM Instances that receive traffic.
- C. Create a logging sink forwarding all firewall logs with no filters.
- D. Create an explicit Deny Any rule and enable logging on the new rule.

Answer: D

Explanation:

The correct answer is **D. Create an explicit Deny Any rule and enable logging on the new rule.**

Here's why:

Firewall rules in Google Cloud's Virtual Private Cloud (VPC) networks operate using an implicit deny-all policy. This means that if no rule explicitly allows traffic, it's implicitly denied. However, implicit denials are not logged by default. The problem described indicates that while only specific ports are explicitly allowed, attempts to reach other ports aren't being logged, preventing proper troubleshooting.

Option A, enabling logging on the default deny-all rule, isn't possible because there isn't a configurable default rule. The implicit deny behavior is built-in but not loggable. Option B, enabling logging on the VM instances, would capture system-level logs, not firewall events which are essential for tracking network access control violations. Option C, while capturing all firewall logs, would not solve the problem if no traffic gets logged in the first place because of the implicit default deny.

Therefore, to specifically log denied traffic, you must explicitly create a new firewall rule that denies all traffic (using a low priority value, such as 65535 for outbound, and a high value, such as 0 for ingress) and then enable logging on that specific rule. This ensures that any traffic that doesn't match the allow rules will be matched by the explicit deny rule, generating logs that identify rejected connections and helping in debugging.

In essence, the implicit deny is a default behavior, while the explicit deny rule acts as a filter that catches otherwise untracked dropped traffic, enabling logging for debugging. Without the explicit deny rule with logging enabled, you will not see logs of traffic that falls outside explicitly allowed connections in the firewall rules, resulting in difficulty troubleshooting denied connections.

Further Reading:

Google Cloud Firewall Rules: <https://cloud.google.com/vpc/docs/firewalls>

Firewall Rules Logging: <https://cloud.google.com/vpc/docs/firewall-logging>

Question: 67

CertyIQ

In your company, two departments with separate GCP projects (code-dev and data-dev) in the same organization need to allow full cross-communication between all of their virtual machines in GCP. Each department has one VPC in its project and wants full control over their network. Neither department intends to recreate its existing computing resources. You want to implement a solution that minimizes cost. Which two steps should you take? (Choose two.)

- A. Connect both projects using Cloud VPN.
- B. Connect the VPCs in project code-dev and data-dev using VPC Network Peering.
- C. Enable Shared VPC in one project (e. g., code-dev), and make the second project (e. g., data-dev) a service project.
- D. Enable firewall rules to allow all ingress traffic from all subnets of project code-dev to all instances in project data-dev, and vice versa.
- E. Create a route in the code-dev project to the destination prefixes in project data-dev and use nexthop as the default gateway, and vice versa.

Answer: BD

Explanation:

Here's a detailed justification for choosing options B and D as the correct solution:

Option B: Connect the VPCs in project code-dev and data-dev using VPC Network Peering.

VPC Network Peering is the ideal choice for connecting VPCs across different projects within the same organization. It enables private IP address communication between the peered networks, without traversing the public internet. This is crucial for maintaining security and performance. It avoids the overhead of managing external IPs and allows for direct, low-latency communication that is not exposed to the public internet. Peering provides a direct path for traffic between the two networks, minimizing cost compared to solutions involving gateways or VPNs.

Option D: Enable firewall rules to allow all ingress traffic from all subnets of project code-dev to all instances in project data-dev, and vice versa.

While VPC peering establishes the connectivity, firewall rules are essential for allowing traffic flow between the peered networks. By creating ingress rules allowing traffic from all subnets in one VPC to all instances in the other, and vice versa, the desired full cross-communication is achieved. Without these rules, even with peering in place, traffic would be blocked by default firewall settings. This ensures that the VMs in each project can reach each other. Firewall rules are a fundamental aspect of network security and must be configured to allow communication after setting up network connectivity.

Why other options are incorrect:

Option A (Cloud VPN): Cloud VPN is used for connecting on-premises networks to GCP, or for cross-cloud connectivity, not for connecting VPCs within the same organization. It introduces unnecessary complexity and cost for this scenario.

Option C (Shared VPC): Shared VPC requires one project to be a host and the other a service project, giving control of the network to one department. The requirement specifies that each department should retain full control over their network. Therefore Shared VPC is not suitable.

Option E (Custom Routes with Default Gateway): This option is invalid because a default gateway can't be used as a next hop to reach private IP addresses in another VPC. Routes to private IP ranges need to use the peer connection. Creating routes using the default gateway for an internal target would not direct traffic to the other network and would be ineffective.

In summary, VPC Network Peering establishes the direct network path, and appropriate firewall rules allow traffic to flow freely between the VPCs. This combination fulfills the requirements of full cross-communication, cost-effectiveness, and decentralized network control.

Authoritative Links:

VPC Network Peering: <https://cloud.google.com/vpc/docs/vpc-peering>

VPC Firewall Rules: <https://cloud.google.com/vpc/docs/firewalls>

Question: 68

CertyIQ

You need to create a GKE cluster in an existing VPC that is accessible from on-premises. You must meet the following requirements:

- IP ranges for pods and services must be as small as possible.
- The nodes and the master must not be reachable from the internet.
- You must be able to use kubectl commands from on-premises subnets to manage the cluster.

How should you create the GKE cluster?

- A. "Create a private cluster that uses VPC advanced routes. "Set the pod and service ranges as /24. "Set up a network proxy to access the master.
- B. "Create a VPC-native GKE cluster using GKE-managed IP ranges. "Set the pod IP range as /21 and service IP range as /24. "Set up a network proxy to access the master.
- C. "Create a VPC-native GKE cluster using user-managed IP ranges. "Enable a GKE cluster network policy, set the pod and service ranges as /24. "Set up a network proxy to access the master. "Enable master authorized networks.
- D. "Create a VPC-native GKE cluster using user-managed IP ranges. "Enable privateEndpoint on the cluster master. "Set the pod and service ranges as /24. "Set up a network proxy to access the master. "Enable master authorized networks.

Answer: D

Explanation:

Here's a breakdown of why option D is the correct approach for creating the GKE cluster based on the requirements:

VPC-Native Cluster: The requirement to use the existing VPC necessitates creating a VPC-native cluster. This type of cluster leverages the VPC network for its pod and service IP addresses, avoiding complex networking setups. (Reference: <https://cloud.google.com/kubernetes-engine/docs/how-to/alias-ips>)

User-Managed IP Ranges: The requirement for small IP ranges for pods and services implies using user-managed IP ranges, not GKE-managed. This gives the user fine-grained control over IP address allocation, enabling smaller subnets (e.g. /24). (Reference: <https://cloud.google.com/kubernetes-engine/docs/how-to/flexible-pod-cidr>)

Private Endpoint for Master: To prevent internet reachability to the master, enabling the privateEndpoint option is crucial. This exposes the master only on the private network, fulfilling the security requirement. (Reference: <https://cloud.google.com/kubernetes-engine/docs/how-to/private-clusters>)

Small IP Ranges: Setting pod and service ranges as /24 ensures each has a smaller block of IP addresses (256 addresses each), fulfilling the requirement of minimized IP ranges.

Network Proxy for Master Access: Although the cluster's master has a private endpoint, to reach it from on-premises networks for kubectl access, a network proxy, like a bastion host or a VPN/Cloud Interconnect connection, becomes necessary. This is because private GKE endpoints are reachable within the VPC network by default, but require a connection to the network for access from on-premises.

Master Authorized Networks: Enabling master authorized networks allows specifying on-premises subnets to access the GKE master's control plane. This allows you to configure which IPs or CIDRs can access the master. (Reference: <https://cloud.google.com/kubernetes-engine/docs/how-to/authorized-networks>)

In summary, option D addresses all specific requirements: it creates a VPC-native cluster with user-managed IP ranges for small pod/service subnets, secures the master with a private endpoint, establishes a secure access mechanism from on-premises through a network proxy, and specifically allows authorized on-premises subnets to reach the master. Other options fail to achieve all of these constraints simultaneously.

Question: 69

CertyIQ

You are creating an instance group and need to create a new health check for HTTP(s) load balancing. Which two methods can you use to accomplish this? (Choose two.)

- A. Create a new health check using the gcloud command line tool.
- B. Create a new health check using the VPC Network section in the GCP Console.

- C. Create a new health check, or select an existing one, when you complete the load balancer's backend configuration in the GCP Console.
- D. Create a new legacy health check using the gcloud command line tool.
- E. Create a new legacy health check using the Health checks section in the GCP Console.

Answer: AC

Explanation:

Let's analyze why options A and C are the correct methods for creating a new health check during instance group creation for HTTP(s) load balancing. Option A, creating a health check using the gcloud command-line tool, is a direct and flexible way to manage Google Cloud resources. The gcloud tool provides granular control over resource configurations, including health checks. It's suitable for both scripted deployments and manual configurations. Option C aligns with the typical workflow of setting up load balancing in the Google Cloud Console. When configuring the backend service of a load balancer, the console prompts you to either select an existing health check or create a new one, thereby directly linking the health check with the service's backend instances, an instance group. This integrates the health check creation process within the broader load balancing configuration.

Options B, D, and E are incorrect. Option B, using the VPC network section, is not where health checks are configured. Health checks are resource specific to load balancing and are created independently. Similarly, option D mentioning creation of "legacy" health check via gcloud is deprecated. Google recommends using the newer health check system. Option E refers to legacy health checks which are no longer encouraged; the current best practice is to create non legacy health checks from the load balancing section itself or via the command line as demonstrated with gcloud in option A. Hence option A and C are the valid responses.

Supporting Links:

gcloud compute health-checks create: <https://cloud.google.com/sdk/gcloud/reference/compute/health-checks/create>

Creating a health check for load balancing: <https://cloud.google.com/load-balancing/docs/health-checks/create-hc>

Health Checks Overview: <https://cloud.google.com/load-balancing/docs/health-checks>

Question: 70

CertyIQ

You are in the early stages of planning a migration to GCP. You want to test the functionality of your hybrid cloud design before you start to implement it in production. The design includes services running on a Compute Engine Virtual Machine instance that need to communicate to on-premises servers using private IP addresses. The on-premises servers have connectivity to the internet, but you have not yet established any Cloud Interconnect connections. You want to choose the lowest cost method of enabling connectivity between your instance and on-premises servers and complete the test in 24 hours. Which connectivity method should you choose?

- A. Cloud VPN
- B. 50-Mbps Partner VLAN attachment
- C. Dedicated Interconnect with a single VLAN attachment
- D. Dedicated Interconnect, but don't provision any VLAN attachments

Answer: A

Explanation:

The correct answer is A, Cloud VPN. Here's why:

The scenario emphasizes speed and cost-effectiveness for a short-term (24-hour) test of hybrid connectivity before production implementation. Cloud VPN offers the quickest setup time and the lowest cost for this purpose. It establishes an IPsec VPN tunnel over the public internet between the Compute Engine instance in GCP and the on-premises environment. This allows communication via private IP addresses as required by the scenario.

Option B, a 50-Mbps Partner VLAN attachment, involves a physical connection via a partner network and would take significantly longer than 24 hours to provision, making it unsuitable for a quick test. Partner Interconnect is also more expensive than Cloud VPN.

Options C and D, involving Dedicated Interconnect, are both even more complex and time-consuming than Partner Interconnect because they require physical cross connects between your premises and Google's network, which would make them too costly and infeasible for a 24-hour test. Furthermore, option D includes not provisioning VLAN attachments rendering it useless for connectivity.

In summary, Cloud VPN is ideal for quickly testing hybrid connectivity using private IPs due to its low cost, rapid deployment, and ability to leverage the existing internet connection. It is designed for such use cases where speed and cost take priority over bandwidth and latency.

Relevant Resources:

Cloud VPN Overview: <https://cloud.google.com/vpn/docs/concepts/overview>

Choosing a Network Connectivity Product: <https://cloud.google.com/network-connectivity/docs/choose-network-connectivity>

Cloud Interconnect Overview: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/overview>

Question: 71

CertyIQ

You want to implement an IPSec tunnel between your on-premises network and a VPC via Cloud VPN. You need to restrict reachability over the tunnel to specific local subnets, and you do not have a device capable of speaking Border Gateway Protocol (BGP).

Which routing option should you choose?

- A. Dynamic routing using Cloud Router
- B. Route-based routing using default traffic selectors
- C. Policy-based routing using a custom local traffic selector
- D. Policy-based routing using the default local traffic selector

Answer: C

Explanation:

The correct answer is **C. Policy-based routing using a custom local traffic selector**. Here's why:

The scenario requires an IPSec tunnel with restricted reachability to specific subnets, and the on-premises device doesn't support BGP. Dynamic routing (A) with Cloud Router relies on BGP to exchange routes, which isn't viable in this case. Route-based routing (B) with default traffic selectors automatically creates tunnels based on subnet ranges within your VPC. This approach does not allow restricting to specific on-premises subnets, as it's designed for connecting entire subnets. Similarly, Policy-based routing using the default local traffic selector (D) covers the entire network and does not allow for precise reachability control.

Policy-based routing, in contrast, allows for the configuration of specific local traffic selectors. These selectors define which source and destination IP ranges (e.g., on-premises subnets) are permitted to communicate through the tunnel. By creating a custom local traffic selector, you can accurately specify the

allowed subnets, thus fulfilling the requirement of limiting reachability over the tunnel to particular local subnets without BGP. This method offers granular control and is the ideal choice for scenarios like this. You define the selector with the on-premise subnets, and the remote one is the VPC subnets.

Authoritative Links for further research:

Cloud VPN Overview: <https://cloud.google.com/vpn/docs/concepts/overview>

Choosing a routing option: <https://cloud.google.com/vpn/docs/concepts/choosing-networks-routing>

Policy-based VPN: <https://cloud.google.com/vpn/docs/how-to/configuring-policy-based-vpn>

IPSec Traffic Selectors: <https://www.fortinet.com/resources/cyberglossary/traffic-selector> (General definition of traffic selector - although not google specific, the concept is the same)

Question: 72

CertyIQ

You have enabled HTTP(S) load balancing for your application, and your application developers have reported that HTTP(S) requests are not being distributed correctly to your Compute Engine Virtual Machine instances. You want to find data about how the request are being distributed.

Which two methods can accomplish this? (Choose two.)

- A. On the Load Balancer details page of the GCP Console, click on the Monitoring tab, select your backend service, and look at the graphs.
- B. In Stackdriver Error Reporting, look for any unacknowledged errors for the Cloud Load Balancers service.
- C. In Stackdriver Monitoring, select Resources > Metrics Explorer and search for https/request_bytes_count metric.
- D. In Stackdriver Monitoring, select Resources > Google Cloud Load Balancers and review the Key Metrics graphs in the dashboard.
- E. In Stackdriver Monitoring, create a new dashboard and track the https/backend_request_count metric for the load balancer.

Answer: AE

Explanation:

Here's a detailed justification for why options A and E are correct, and why the others are incorrect:

Correct Options:

A. On the Load Balancer details page of the GCP Console, click on the Monitoring tab, select your backend service, and look at the graphs. The GCP Console provides a user-friendly interface to monitor the performance of your load balancers. The monitoring tab for a load balancer directly displays key metrics related to its performance and traffic distribution to its backend services. These graphs visualize the request flow, response times, and other relevant parameters, allowing quick identification of traffic distribution issues. You can view details for specific backend services to see how requests are routed.

Link: <https://cloud.google.com/load-balancing/docs/monitoring/using-cloud-monitoring>

E. In Stackdriver Monitoring, create a new dashboard and track the https/backend_request_count metric for the load balancer. Stackdriver Monitoring (now part of Google Cloud Observability) allows you to create customized dashboards to track specific metrics. The https/backend_request_count metric directly measures the number of requests received by the backend instances from the load balancer. This provides direct insight into traffic distribution, enabling you to identify imbalances and troubleshoot routing issues. By creating a custom dashboard you can tailor your monitoring to focus specifically on this metric for easier analysis.

Link: <https://cloud.google.com/monitoring/charts/metrics>

Incorrect Options:

B. In Stackdriver Error Reporting, look for any unacknowledged errors for the Cloud Load Balancers service. Stackdriver Error Reporting primarily focuses on capturing and aggregating application-level errors. While load balancers may log errors, this service is not the appropriate location to directly track how requests are distributed. Errors could indicate that something went wrong, but won't necessarily reveal traffic distribution issues.

C. In Stackdriver Monitoring, select Resources > Metrics Explorer and search for `https/request_bytes_count` metric. The `https/request_bytes_count` metric focuses on the volume of data transferred in requests. While this is a useful metric for understanding traffic load, it doesn't specifically illustrate how the requests are distributed among the backend instances.

D. In Stackdriver Monitoring, select Resources > Google Cloud Load Balancers and review the Key Metrics graphs in the dashboard. While Stackdriver Monitoring's default dashboard for Google Cloud Load Balancers provides helpful overview metrics, it doesn't provide specific details about the per-backend request distribution out of the box. Custom dashboards and specific metrics (like E) provide better insights in this situation.

In summary, options A and E provide methods focused on viewing actual request distribution across the backend services. Option A offers quick visuals via the GCP console and option E provides more customizable detail via a custom metrics dashboard. The other options address related but less specific issues, and thus are not the best approach to troubleshooting request distribution.

Question: 73

CertyIQ

You want to use Partner Interconnect to connect your on-premises network with your VPC. You already have an Interconnect partner.
What should you first?

- A. Log in to your partner's portal and request the VLAN attachment there.
- B. Ask your Interconnect partner to provision a physical connection to Google.
- C. Create a Partner Interconnect type VLAN attachment in the GCP Console and retrieve the pairing key.
- D. Run `gcloud compute interconnect attachments partner update <attachment> / --region <region> --admin-enabled`.

Answer: C

Explanation:

The correct first step when establishing a Partner Interconnect connection to Google Cloud is to create a Partner Interconnect type VLAN attachment within the GCP Console. This action generates a unique pairing key, essential for the subsequent configuration with the partner. Option A is incorrect because the VLAN attachment needs to exist in GCP first. Option B is premature; a physical connection is needed, but the GCP side needs the VLAN attachment first to establish a logical connection. Option D is also incorrect as it relates to updating an existing attachment, not initiating a new one.

The pairing key acts as a shared secret, linking the Google Cloud VLAN attachment to the physical connection being provisioned by the partner. This step initiates the logical connection process within Google's infrastructure, enabling your partner to match their physical connection to your virtual resource. Therefore, creating the VLAN attachment and retrieving the pairing key are foundational to establishing a Partner Interconnect. Without this, your partner cannot correctly associate their physical infrastructure with your Google Cloud VPC.

Here's a breakdown:

1. **VLAN Attachment Creation:** Establishes the logical endpoint in GCP for the connection.
2. **Pairing Key Generation:** A secure identifier linking your GCP environment to the partner's physical

connection.

3. **Partner Configuration:** The partner uses the pairing key to configure their infrastructure to connect to the GCP endpoint.

Therefore, the chronological order is critical: logical resource creation, pairing, then physical connection linking to that resource.

For further details, refer to these authoritative Google Cloud documentation links:

Partner Interconnect Overview: This page provides a high-level overview of Partner Interconnect.

Ordering Partner Interconnect: This page details the steps involved in ordering a Partner Interconnect connection.

Creating a Partner Interconnect Attachment: This document specifically outlines the process for creating a Partner Interconnect VLAN attachment and retrieving the pairing key.

Question: 74

CertyIQ

You need to centralize the Identity and Access Management permissions and email distribution for the WebServices Team as efficiently as possible. What should you do?

- A. Create a Google Group for the WebServices Team.
- B. Create a G Suite Domain for the WebServices Team.
- C. Create a new Cloud Identity Domain for the WebServices Team.
- D. Create a new Custom Role for all members of the WebServices Team.

Answer: A

Explanation:

The most efficient way to centralize IAM permissions and email distribution for a team is by using a Google Group. Google Groups act as a single point of management for a collection of users. By adding the WebServices Team members to a Google Group, you can grant permissions to the group rather than individual users. This greatly simplifies permission management and ensures consistent access control. Furthermore, Google Groups come with built-in email distribution functionality, automatically sending messages to all members of the group. This achieves both the goals stated in the scenario – centralizing IAM and email distribution.

Creating a new G Suite or Cloud Identity domain would add unnecessary complexity, expense, and administrative overhead. These solutions are intended for managing separate organizations or identities rather than a single team. Custom Roles provide permission assignment at a more granular level and are not suitable for team management and email distribution.

Therefore, utilizing Google Groups is the most efficient and straightforward approach for managing a team's permissions and email communication within an existing organizational context. It aligns perfectly with the requirements of centralizing management with minimal overhead.

<https://support.google.com/groups/answer/2464926?hl=en><https://cloud.google.com/iam/docs/overview>

Question: 75

CertyIQ

You are using the gcloud command line tool to create a new custom role in a project by copying a predefined role. You receive this error message:

INVALID_ARGUMENT: Permission resource manager.projects.list is not valid
What should you do?

- A. Add the resource manager.projects.get permission, and try again.
- B. Try again with a different role with a new name but the same permissions.
- C. Remove the resource manager.projects.list permission, and try again.
- D. Add the resource manager.projects.setIamPolicy permission, and try again.

Answer: C

Explanation:

The error "INVALID_ARGUMENT: Permission resource manager.projects.list is not valid" when creating a custom role indicates that you're trying to include a permission that cannot be directly assigned to a custom role. The resource manager.projects.list permission is a special permission often used in broader organizational access scenarios, not specific project-level custom roles. Custom roles are designed for granular control within a specific project, and the list permission typically applies to enumerating a collection (like projects) rather than operating on a specific resource.

Option A, adding resource manager.projects.get, is incorrect; it doesn't address the core issue of the invalid list permission. Option B, trying with a different role name, is irrelevant as the problem is the permission itself, not the role name. Option D, adding resource manager.projects.setIamPolicy, is also incorrect; this permission is for setting IAM policies, not for being included in a custom role's base permissions.

Option C is the correct solution. You must remove the resource manager.projects.list permission from the source role before copying it to your custom role. Custom roles should be limited to permissions within their project scope, and the list permission violates this principle. Therefore, by removing the problematic permission, you are creating a role that aligns with the expected granularity and is within IAM rules.

Relevant links for further research:

[Understanding custom roles](#)

[IAM permissions](#)

[gcloud iam roles create](#)

Question: 76

CertyIQ

One instance in your VPC is configured to run with a private IP address only. You want to ensure that even if this instance is deleted, its current private IP address will not be automatically assigned to a different instance. In the GCP Console, what should you do?

- A. Assign a public IP address to the instance.
- B. Assign a new reserved internal IP address to the instance.
- C. Change the instance's current internal IP address to static.
- D. Add custom metadata to the instance with key internal-address and value reserved.

Answer: C

Explanation:

The correct answer is C: Change the instance's current internal IP address to static. Here's why:

The problem requires ensuring a private IP address remains exclusively associated with a specific resource, even if that resource (the instance) is deleted. By default, internal IP addresses are allocated dynamically to instances within a VPC. When an instance is deleted, its dynamic IP is released and can be reassigned. Making

the IP address static prevents this reassignment.

Option A, assigning a public IP, doesn't address the core problem of keeping the internal IP reserved, as public IPs are for external communication. Option B, assigning a new reserved internal IP, would effectively allocate a new static IP but not reserve the existing private IP that the question refers to. Option D, adding custom metadata, is irrelevant to IP address management within GCP networking. Metadata is about providing information about an instance, not configuring network behavior.

Changing an instance's current internal IP to static makes that specific private IP address persist within the project, even after the instance is deleted. If a new instance needs that IP, you'll have to explicitly select that static internal IP. This achieves the desired outcome: the original instance's private IP address is now reserved and won't be automatically assigned elsewhere.

For further information on static internal IPs, refer to the official Google Cloud documentation:

Reserving a static internal IP address: <https://cloud.google.com/vpc/docs/configure-static-ip-address>

IP addresses: <https://cloud.google.com/vpc/docs/vpc> (Search within the page for "Internal IP Addresses")

These resources from Google Cloud provide detailed information on how internal and external IPs are managed, as well as how to reserve IPs.

Question: 77

CertyIQ

After a network change window one of your company's applications stops working. The application uses an on-premises database server that no longer receives any traffic from the application. The database server IP address is 10.2.1.25. You examine the change request, and the only change is that 3 additional VPC subnets were created. The new VPC subnets created are 10.1.0.0/16, 10.2.0.0/16, and 10.3.1.0/24/ The on-premises router is advertising 10.0.0.0/8.

What is the most likely cause of this problem?

- A. The less specific VPC subnet route is taking priority.
- B. The more specific VPC subnet route is taking priority.
- C. The on-premises router is not advertising a route for the database server.
- D. A cloud firewall rule that blocks traffic to the on-premises database server was created during the change.

Answer: B

Explanation:

The problem stems from how route selection works within a network, particularly regarding route specificity. The on-premises router advertises the broad 10.0.0.0/8 route. The newly created VPC subnets introduce routes for 10.1.0.0/16, 10.2.0.0/16, and 10.3.1.0/24. Because 10.2.1.25 falls within the 10.2.0.0/16 range, this route becomes more specific than the on-premises 10.0.0.0/8 route. Route selection algorithms prioritize the most specific route. Therefore, traffic intended for the database server at 10.2.1.25 will now be routed within the VPC following the 10.2.0.0/16 route instead of exiting to on-premises through the 10.0.0.0/8 route. This causes the application's traffic to not reach the database server on-premises. Options A, C and D are not the most likely scenario here given the situation. Option A is incorrect as we are considering the most specific route. Option C cannot be true as the router is advertising the 10.0.0.0/8 route that would cover the 10.2.1.25 ip. Option D is incorrect as the problem described is related to routing and not firewall rules.

For further research, see:

Google Cloud VPC Routing: <https://cloud.google.com/vpc/docs/routes>

Subnet Routing: <https://cloud.google.com/vpc/docs/using-routes>

IP Prefix Length: <https://en.wikipedia.org/wiki/Subnetwork>

Question: 78

CertyIQ

You need to create a new VPC network that allows instances to have IP addresses in both the 10.1.1.0/24 network and the 172.16.45.0/24 network. What should you do?

- A. Configure global load balancing to point 172.16.45.0/24 to the correct instance.
- B. Create unique DNS records for each service that sends traffic to the desired IP address.
- C. Configure an alias-IP range of 172.16.45.0/24 on the virtual instances within the VPC subnet of 10.1.1.0/24.
- D. Use VPC peering to allow traffic to route between the 10.1.0.0/24 network and the 172.16.45.0/24 network.

Answer: C

Explanation:

The correct answer is **C. Configure an alias-IP range of 172.16.45.0/24 on the virtual instances within the VPC subnet of 10.1.1.0/24.**

This approach directly addresses the requirement of instances having IP addresses from both specified ranges within a single VPC network. Alias IP ranges allow you to assign secondary IP addresses from different subnets to the same network interface of a virtual machine. By assigning the 172.16.45.0/24 range as an alias IP range to instances primarily within the 10.1.1.0/24 subnet, you effectively allow instances to have interfaces in both IP ranges without needing multiple network interfaces.

Option A, using global load balancing, is not relevant to assigning IP addresses to instances. Global load balancing distributes traffic across multiple regions, not individual IP ranges within a VPC. Similarly, Option B, DNS records, map domain names to IP addresses; they do not configure IP addresses for instances within a VPC. Option D, VPC peering, connects two separate VPC networks. While peering could enable communication between separate networks, it doesn't allow instances to directly use multiple IP ranges within the same network, which is the core requirement.

Alias IPs provide a mechanism to achieve logical separation while still residing within the same underlying subnet. This avoids the complexities of managing multiple interfaces. You are expanding the logical addressing of the instance without adding more physical NICs.

For further research, consult the official Google Cloud documentation on Alias IP Ranges:

<https://cloud.google.com/vpc/docs/alias-ip>

Question: 79

CertyIQ

You are deploying a global external TCP load balancing solution and want to preserve the source IP address of the original layer 3 payload. Which type of load balancer should you use?

- A. HTTP(S) load balancer
- B. Network load balancer
- C. Internal load balancer
- D. TCP/SSL proxy load balancer

Answer: D

Explanation:

The correct answer is **D. TCP/SSL proxy load balancer**. This choice is appropriate because TCP/SSL proxy load balancers are designed to preserve the source IP address of the client when forwarding traffic to backend instances. They accomplish this by using the Proxy Protocol, which adds a header containing the client's original IP to the TCP connection. Unlike HTTP(S) load balancers, which operate at the application layer and terminate the TCP connection, a TCP/SSL proxy load balancer forwards the connection transparently. Network load balancers work at layer 4 and typically don't preserve source IPs; they often replace them with the load balancer's IP, which impacts connection tracking and visibility on the backend. Internal load balancers are for internal traffic within a VPC, not external global traffic. HTTP(S) load balancers function at layer 7 and terminate TCP connections; therefore, they cannot preserve original IP addresses in the way required in the scenario.

The specific requirement of preserving the original layer 3 source IP for backend systems in this global, external configuration directly aligns with the functionality of a TCP/SSL proxy load balancer which utilizes the proxy protocol to achieve this. Therefore, TCP/SSL Proxy load balancers address the need to retain client information for logging, security, and application logic dependent on the original source address. It is crucial for applications that require true client IP data, rather than just the load balancer's IP address.

For more detailed information, consider reviewing these resources:

Google Cloud documentation on TCP Proxy Load Balancers: <https://cloud.google.com/load-balancing/docs/tcp/>

Google Cloud documentation on the Proxy Protocol: <https://cloud.google.com/load-balancing/docs/proxy-protocol>

Google Cloud documentation on different load balancing options: <https://cloud.google.com/load-balancing/docs/load-balancing-overview>

Question: 80

CertyIQ

Your company has a single Virtual Private Cloud (VPC) network deployed in Google Cloud with access from your on-premises network using Cloud Interconnect. You must configure access only to Google APIs and services that are supported by VPC Service Controls through hybrid connectivity with a service level agreement (SLA) in place. What should you do?

- A. Configure the existing Cloud Routers to advertise the Google API's public virtual IP addresses.
- B. Use Private Google Access for on-premises hosts with restricted.googleapis.com virtual IP addresses.
- C. Configure the existing Cloud Routers to advertise a default route, and use Cloud NAT to translate traffic from your on-premises network.
- D. Add Direct Peering links, and use them for connectivity to Google APIs that use public virtual IP addresses.

Answer: B

Explanation:

The correct answer is **B. Use Private Google Access for on-premises hosts with restricted.googleapis.com virtual IP addresses**.

Here's the justification:

The scenario requires secure and SLA-backed access to Google APIs and services supported by VPC Service Controls from an on-premises network via Cloud Interconnect. Private Google Access for on-premises hosts, specifically utilizing restricted.googleapis.com, fulfills this requirement. This configuration routes API traffic from on-premises through the Cloud Interconnect connection and the VPC network, bypassing the public internet. The restricted.googleapis.com domain ensures only VPC Service Controls-compliant APIs are accessible, enforcing security policies. This method avoids the complexities and potential security risks

associated with advertising public IP ranges (Option A), relying on Cloud NAT (Option C) for on-premises traffic, or setting up separate Direct Peering links (Option D), which aren't suited for this specific need of private access to supported APIs. Private Google Access is the recommended and most secure approach for accessing Google services over a private network with an SLA. Furthermore, using `restricted.googleapis.com` ensures alignment with VPC Service Controls requirements.

Here are some links for further research:

Private Google Access for on-premises hosts: <https://cloud.google.com/vpc/docs/private-access-options#private-google-access>

VPC Service Controls: <https://cloud.google.com/vpc-service-controls>

restricted.googleapis.com: <https://cloud.google.com/vpc/docs/private-access-options#restricted-api>

Cloud Interconnect: <https://cloud.google.com/interconnect>

Question: 81

CertyIQ

Your company's security team tends to use managed services when possible. You need to build a dashboard to show the number of deny hits that occur against configured firewall rules without increasing operational overhead. What should you do?

- A. Configure Firewall Rules Logging. Use Firewall Insights to display the number of hits.
- B. Configure Firewall Rules Logging. View the logs in Cloud Logging, and create a custom dashboard in Cloud Monitoring to display the number of hits.
- C. Configure a firewall appliance from the Google Cloud Marketplace. Route all traffic through this appliance, and apply the firewall rules at this layer. Use the firewall appliance to display the number of hits.
- D. Configure Packet Mirroring on the VPC. Apply a filter with an IP address list of the Denied Firewall rules. Configure an intrusion detection system (IDS) appliance as the receiver to display the number of hits.

Answer: A

Explanation:

The correct answer is **A. Configure Firewall Rules Logging. Use Firewall Insights to display the number of hits.**

Here's why:

Firewall Rules Logging in Google Cloud Platform (GCP) automatically captures details about connections that match your firewall rules, including both allow and deny actions. This is a managed service, aligning with the requirement to minimize operational overhead. Firewall Insights is a powerful tool built directly on top of these logs, offering a pre-built dashboard specifically for visualizing firewall rule activity, including deny hits. This eliminates the need to build custom dashboards or process logs manually. Option B, while also leveraging Firewall Rules Logging, requires additional work to set up a custom dashboard in Cloud Monitoring, increasing operational effort and contradicting the goal. Options C and D introduce complexities with network appliances and packet mirroring, adding management burden and not focusing on leveraging managed services. The packet mirroring approach in D, in particular, is overkill for simply tracking denied traffic. Packet mirroring adds complexity since it duplicates traffic, potentially impacting performance and involving more configuration. Furthermore, IDS appliance setup also adds to complexity. Thus, A is the most efficient approach for fulfilling requirements, leveraging GCP native and managed logging and analytics solutions.

Authoritative links:

Firewall Rules Logging: <https://cloud.google.com/vpc/docs/firewall-rules-logging>

Firewall Insights: <https://cloud.google.com/network-intelligence-center/docs/firewall-insights>

Question: 82

You are configuring your Google Cloud environment to connect to your on-premises network. Your configuration must be able to reach Cloud Storage APIs and your Google Kubernetes Engine nodes across your private Cloud Interconnect network. You have already configured a Cloud Router with your Interconnect VLAN attachments. You now need to set up the appropriate router advertisement configuration on the Cloud Router. What should you do?

- A. Configure the route advertisement to the default setting.
- B. On the on-premises router, configure a static route for the storage API virtual IP address which points to the Cloud Router's link-local IP address.
- C. Configure the route advertisement to the custom setting, and manually add prefix 199.36.153.8/30 to the list of advertisements. Leave all other options as their default settings.
- D. Configure the route advertisement to the custom setting, and manually add prefix 199.36.153.8/30 to the list of advertisements. Advertise all visible subnets to the Cloud Router.

Answer: D

Explanation:

Justification for Answer D:

The correct answer is **D**, which involves configuring a custom route advertisement on the Cloud Router to include the Google Cloud API range (199.36.153.8/30) and all visible subnets. Here's why:

1. **Access to Google Cloud APIs:** Google services like Cloud Storage are accessed via a virtual IP (VIP) address range, specifically 199.36.153.8/30. This range needs to be advertised to the on-premises network through BGP so that on-premises systems can route traffic correctly to these APIs using the private interconnect.
2. **Private Access for GKE:** GKE nodes, while in the Google Cloud VPC, require routes from the on-prem network via Cloud Router for internal communication. Simply configuring a default route advertisement would not work because the Cloud Router only advertises internal subnets by default; external services IP addresses are not.
3. **Custom Route Advertisement:** The requirement of accessing both Google APIs and GKE nodes necessitates a custom advertisement. A default setting in Cloud Router would not include the external range. Custom advertisement allows for this flexibility.
4. **Advertising Visible Subnets:** In addition to the Google API range, the Cloud Router also needs to advertise the VPC's subnets for the on-prem network to reach the GKE nodes. This is what makes option D the correct choice as it advertises all visible subnets.
5. **Why other options are incorrect:** Option A is incorrect because it does not accommodate the requirement to reach the API IP range. Option B is incorrect as it's not a matter of setting a static route on the on-premises router; the on-prem router learns routes via BGP advertisements from the Cloud Router. Option C is incorrect because it does not also advertise visible subnets to enable GKE internal traffic to flow.

In summary, option D is the most complete and correct configuration because it ensures that both Google APIs and the GKE nodes are reachable from the on-premises network, all whilst routing through the private interconnect.

Authoritative Links for Further Research:

Google Cloud Interconnect Overview: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/overview>

Cloud Router Overview: <https://cloud.google.com/network-connectivity/docs/router/concepts/overview>

Custom Route Advertisements: <https://cloud.google.com/network-connectivity/docs/router/how->

Question: 83

CertyIQ

You are configuring load balancing for a standard three-tier (web, application, and database) application. You have configured an external HTTP(S) load balancer for the web servers. You need to configure load balancing for the application tier of servers. What should you do?

- A. Configure a forwarding rule on the existing load balancer for the application tier.
- B. Configure equal cost multi-path routing on the application servers.
- C. Configure a new internal HTTP(S) load balancer for the application tier.
- D. Configure a URL map on the existing load balancer to route traffic to the application tier.

Answer: C

Explanation:

The correct answer is C: Configure a new internal HTTP(S) load balancer for the application tier. Here's why:

The scenario describes a three-tier application where the web tier is already load-balanced externally. The application tier, being internal, requires its own load balancing solution. Option A, configuring a forwarding rule on the existing external load balancer, is incorrect because the application tier is typically not directly exposed to the internet. Option D, configuring a URL map on the existing external load balancer, would also expose the internal application tier, violating security best practices. Option B, using equal-cost multi-path (ECMP) routing, is more about path selection within a network and not a load balancing solution.

Internal HTTP(S) load balancers are designed for distributing traffic within a Virtual Private Cloud (VPC) network. They enable high availability and scalability for internal services. They provide a single entry point for application servers, regardless of their location within the VPC, while offering features such as health checks, session persistence, and traffic management. This approach decouples the web tier from the application tier, allowing independent scaling and management. Consequently, using an internal load balancer isolates the application tier from external access and provides the desired load-balancing functionality. The external load balancer will pass requests to the application tier via the internal load balancer.

Relevant Resources:

Google Cloud Load Balancing Overview: <https://cloud.google.com/load-balancing/docs/load-balancing-overview>

Internal HTTP(S) Load Balancing: <https://cloud.google.com/load-balancing/docs/internal>

Choosing a load balancer: <https://cloud.google.com/load-balancing/docs/choosing-load-balancer>

Question: 84

CertyIQ

Your organization has a new security policy that requires you to monitor all egress traffic payloads from your virtual machines in region us-west2. You deployed an intrusion detection system (IDS) virtual appliance in the same region to meet the new policy. You now need to integrate the IDS into the environment to monitor all egress traffic payloads from us-west2. What should you do?

- A. Enable firewall logging, and forward all filtered egress firewall logs to the IDS.
- B. Enable VPC Flow Logs. Create a sink in Cloud Logging to send filtered egress VPC Flow Logs to the IDS.
- C. Create an internal TCP/UDP load balancer for Packet Mirroring, and add a packet mirroring policy filter for egress traffic.

D. Create an internal HTTP(S) load balancer for Packet Mirroring, and add a packet mirroring policy filter for egress traffic.

Answer: C

Explanation:

The correct answer is **C. Create an internal TCP/UDP load balancer for Packet Mirroring, and add a packet mirroring policy filter for egress traffic.**

Here's why:

The primary requirement is to monitor the payload of all egress traffic, not just connection metadata. Options A and B, involving firewall logging and VPC Flow Logs, respectively, only provide information about the connections themselves (e.g., source/destination IPs, ports) but not the actual data transmitted. They are useful for network activity analysis, but not for deep packet inspection required by an IDS.

Packet Mirroring is a feature designed for exactly this purpose: copying network packets for analysis by tools like IDS. It replicates network traffic from a source (virtual machine instances in this case) to a destination (the IDS appliance). An internal TCP/UDP load balancer (not HTTP(S) as in option D) is necessary to act as a stable endpoint to receive the mirrored packets from the mirrored instances. Mirroring policies provide granular control over which traffic is copied, and in this case, the policy would filter for egress traffic. An internal TCP/UDP load balancer is suitable because mirroring works at the network layer (L3/L4), where TCP and UDP operate and not at the application layer (L7) where HTTP(S) works.

Using an internal TCP/UDP load balancer with Packet Mirroring provides a scalable, robust, and efficient way to forward all egress traffic for deeper analysis by the IDS, satisfying the requirements of the new security policy. Therefore, option C is the most appropriate and accurate solution.

Relevant links:

Packet Mirroring documentation: <https://cloud.google.com/vpc/docs/packet-mirroring>

Internal TCP/UDP Load Balancer documentation: <https://cloud.google.com/load-balancing/docs/internal>

VPC Flow Logs documentation: <https://cloud.google.com/vpc/docs/flow-logs>

Firewall Rules Logging: <https://cloud.google.com/firewall/docs/logging/usage>

Question: 85

CertyIQ

You are developing an HTTP API hosted on a Compute Engine virtual machine instance that must be invoked only by multiple clients within the same Virtual Private Cloud (VPC). You want clients to be able to get the IP address of the service. What should you do?

A. Reserve a static external IP address and assign it to an HTTP(S) load balancing service's forwarding rule. Clients should use this IP address to connect to the service.

B. Ensure that clients use Compute Engine internal DNS by connecting to the instance name with the url `https://[INSTANCE_NAME].[ZONE].c.[PROJECT_ID].internal/`.

C. Reserve a static external IP address and assign it to an HTTP(S) load balancing service's forwarding rule. Then, define an A record in Cloud DNS. Clients should use the name of the A record to connect to the service.

D. Ensure that clients use Compute Engine internal DNS by connecting to the instance name with the url `https://[API_NAME]/[API_VERSION]/`.

Answer: B

Explanation:

Justification for Answer B:

The core requirement is for clients within the same VPC to access an HTTP API hosted on a Compute Engine instance using its IP address, without exposing it to the public internet. Option B leverages Compute Engine's internal DNS, a built-in service that automatically resolves instance names to their internal IP addresses within a VPC network. This means clients within the VPC can resolve the instance's internal IP simply by knowing its name, zone, and project ID, forming a URL like `https://[INSTANCE_NAME].[ZONE].c.[PROJECT_ID].internal/`. No public IP or complex DNS setup is required.

Internal DNS is private and only accessible within the VPC, fulfilling the requirement to keep the API private. It simplifies access and configuration, as each instance automatically registers with internal DNS. Clients can communicate securely and directly with the instance over the VPC's private network, avoiding external traffic routes. This approach is cost-effective as it does not necessitate public IPs or load balancers.

Option A, while viable for public access, unnecessarily exposes the service to the internet with a static external IP. Similarly, Option C introduces complexity with Cloud DNS configuration, which is not required for internal VPC access. Option D suggests a non-standard URL format, mixing API name/version with instance URLs, which does not leverage internal DNS in the intended manner.

Therefore, Option B directly addresses the requirements by utilizing built-in, efficient, and secure internal DNS for private service access within the VPC, making it the correct solution.

Authoritative Links:

Google Cloud Documentation on Internal DNS: <https://cloud.google.com/compute/docs/internal-dns>

Google Cloud Documentation on VPC Networks: <https://cloud.google.com/vpc/docs/>

Google Cloud Documentation on Compute Engine Instances:

<https://cloud.google.com/compute/docs/instances/>

Question: 86

CertyIQ

You recently deployed Cloud VPN to connect your on-premises data center to Google Cloud. You need to monitor the usage of this VPN and set up alerts in case traffic exceeds the maximum allowed. You need to be able to quickly decide whether to add extra links or move to a Dedicated Interconnect. What should you do?

- A. In the Network Intelligence Center, check for the number of packet drops on the VPN.
- B. In the Google Cloud Console, use Monitoring Query Language to create a custom alert for bandwidth utilization.
- C. In the Monitoring section of the Google Cloud Console, use the Dashboard section to select a default dashboard for VPN usage.
- D. In the VPN section of the Google Cloud Console, select the VPN under hybrid connectivity, and then select monitoring to display utilization on the dashboard.

Answer: B

Explanation:

The correct answer is **B. In the Google Cloud Console, use Monitoring Query Language to create a custom alert for bandwidth utilization.**

Here's why:

Option B directly addresses the problem's requirements. It leverages Google Cloud Monitoring's powerful capability to create custom alerts based on specific metrics. Bandwidth utilization is a critical metric for monitoring VPN usage and capacity. By using Monitoring Query Language (MQL), you can define precise thresholds for alert triggers, enabling proactive responses to excessive traffic. For example, you could set an alert to fire when the average VPN bandwidth usage exceeds 80% of its capacity over a 5-minute interval.

Option A, while useful for identifying issues, doesn't directly address bandwidth utilization. Packet drops could indicate various problems, not just capacity limits. Option C offers pre-defined dashboards, which might lack the granularity and specific alert configurations required for informed decisions about expanding capacity. Option D provides basic VPN monitoring but lacks the custom alerting functionality crucial for proactive capacity management.

The problem requires monitoring and setting alerts for bandwidth utilization specifically, and option B is the only solution that provides the necessary control to create custom alerts based on a defined threshold. This allows for quick detection of capacity issues and data-driven decisions on whether to add more VPN tunnels or switch to a Dedicated Interconnect.

Authoritative Links:

Google Cloud Monitoring: <https://cloud.google.com/monitoring>

Monitoring Query Language (MQL): <https://cloud.google.com/monitoring/mql>

Cloud VPN Monitoring: <https://cloud.google.com/network-connectivity/docs/vpn/how-to/monitor-vpn>

Cloud VPN Metrics: https://cloud.google.com/monitoring/api/metrics_gcp#gcp-vpn (Search for vpn related metrics like sent_bytes_count, received_bytes_count, etc.)

Question: 87

CertyIQ

You have applications running in the us-west1 and us-east1 regions. You want to build a highly available VPN that provides 99.99% availability to connect your applications from your project to the cloud services provided by your partner's project while minimizing the amount of infrastructure required. Your partner's services are also in the us-west1 and us-east1 regions. You want to implement the simplest solution. What should you do?

- A. Create one Cloud Router and one HA VPN gateway in each region of your VPC and your partner's VPC. Connect your VPN gateways to the partner's gateways. Enable global dynamic routing in each VPC.
- B. Create one Cloud Router and one HA VPN gateway in the us-west1 region of your VPC. Create one OpenVPN Access Server in each region of your partner's VPC. Connect your VPN gateway to your partner's servers.
- C. Create one OpenVPN Access Server in each region of your VPC and your partner's VPC. Connect your servers to the partner's servers.
- D. Create one Cloud Router and one HA VPN gateway in the us-west1 region of your VPC and your partner's VPC. Connect your VPN gateways to the partner's gateways with a pair of tunnels. Enable global dynamic routing in each VPC.

Answer: D

Explanation:

The correct answer is **D** because it provides the simplest, most cost-effective, and highly available solution using Google Cloud's managed services, HA VPN.

Option D proposes creating a single Cloud Router and HA VPN gateway in one region (us-west1) for both your VPC and your partner's VPC. This avoids unnecessary complexity and cost of setting up infrastructure in each region. The HA VPN gateway, by design, is resilient and provides 99.99% availability. Connecting the gateways with a pair of tunnels enhances redundancy. Finally, enabling global dynamic routing ensures traffic can be routed through the active tunnel, and if one fails, the other automatically takes over. This configuration aligns with the requirement of minimizing infrastructure while maintaining high availability.

Option A, while achieving high availability, creates more infrastructure than necessary by deploying in both regions. Options B and C introduce OpenVPN, which requires more management effort, patching, and monitoring and is not a managed service. They also won't provide the 99.99% SLA of HA VPN. Option C further deviates from the requested solution by suggesting OpenVPN for both VPCs, contradicting the requirement to connect to the partner's project via VPN. HA VPN, being a managed Google Cloud service,

requires less administrative overhead and is the simplest approach to satisfy the requirements for a highly available connection.

Authoritative Links:

Google Cloud HA VPN: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/ha-vpn>

Google Cloud Router: <https://cloud.google.com/network-connectivity/docs/router/concepts/overview>

Google Cloud Dynamic Routing: <https://cloud.google.com/network-connectivity/docs/router/concepts/dynamic-routing>

Question: 88

CertyIQ

You need to create the network infrastructure to deploy a highly available web application in the us-east1 and us-west1 regions. The application runs on Compute Engine instances, and it does not require the use of a database. You want to follow Google-recommended practices. What should you do?

- A. Create one VPC with one subnet in each region.
Create a regional network load balancer in each region with a static IP address.
Enable Cloud CDN on the load balancers.
Create an A record in Cloud DNS with both IP addresses for the load balancers.
- B. Create one VPC with one subnet in each region.
Create a global load balancer with a static IP address.
Enable Cloud CDN and Google Cloud Armor on the load balancer.
Create an A record using the IP address of the load balancer in Cloud DNS.
- C. Create one VPC in each region, and peer both VPCs.
Create a global load balancer.
Enable Cloud CDN on the load balancer.
Create a CNAME for the load balancer in Cloud DNS.
- D. Create one VPC with one subnet in each region.
Create an HTTP(S) load balancer with a static IP address.
Choose the standard tier for the network.
Enable Cloud CDN on the load balancer.
Create a CNAME record using the load balancer's IP address in Cloud DNS.

Answer: B

Explanation:

The correct answer is **B** because it aligns with Google's best practices for deploying highly available, multi-regional web applications on Compute Engine. Here's a detailed breakdown:

Single VPC with Multi-Regional Subnets: Creating one Virtual Private Cloud (VPC) network with subnets in both us-east1 and us-west1 provides a unified network infrastructure, simplifying management and ensuring seamless communication between resources across regions. This avoids the complexity of peering separate VPCs (as in option C).

Global Load Balancer: Using a global load balancer is crucial for high availability and traffic management across regions. A global load balancer, specifically the HTTP(S) Load Balancer, automatically directs traffic to the closest healthy backend instance, regardless of region, which is a key factor for availability and user experience. This is superior to using regional load balancers (as in option A).

Static IP Address for the Global Load Balancer: A static IP address ensures that the load balancer's IP address remains consistent, even if underlying infrastructure changes, which simplifies DNS management and external accessibility.

Cloud CDN and Google Cloud Armor: Enabling Cloud CDN (Content Delivery Network) caches content closer to users, reducing latency and improving application performance. Cloud Armor provides security features

like WAF (Web Application Firewall) protecting against threats. Both contribute to the application's availability, performance and security.

DNS Configuration: Using an A record in Cloud DNS for the load balancer's static IP address is the correct approach. This directs domain traffic to the global load balancer. This avoids using CNAME directly to the load balancer because using CNAME to a load balancer is not advised.

Option A is flawed as regional load balancers lack the global traffic distribution needed for high availability across multiple regions. Option C is unnecessarily complex with VPC peering and doesn't use the appropriate DNS configuration. Option D incorrectly uses the standard network tier which is not optimal for global traffic management, and uses a CNAME record incorrectly. Thus, only option B adheres to best practices.

Authoritative Links:

Google Cloud Load Balancing Overview: <https://cloud.google.com/load-balancing/docs/load-balancing-overview>

Google Cloud VPC Network Overview: <https://cloud.google.com/vpc/docs/vpc>

Cloud CDN Overview: <https://cloud.google.com/cdn/docs/overview>

Google Cloud Armor Overview: <https://cloud.google.com/armor/docs/overview>

Cloud DNS Overview: <https://cloud.google.com/dns/docs/overview>

Question: 89

CertyIQ

You are the network administrator responsible for hybrid connectivity at your organization. Your developer team wants to use Cloud SQL in the us-west1 region in your Shared VPC. You configured a Dedicated Interconnect connection and a Cloud Router in us-west1, and the connectivity between your Shared VPC and on-premises data center is working as expected. You just created the private services access connection required for Cloud SQL using the reserved IP address range and default settings. However, your developers cannot access the Cloud SQL instance from on-premises. You want to resolve the issue. What should you do?

- A.1. Modify the VPC Network Peering connection used for Cloud SQL, and enable the import and export of routes.
- 2. Create a custom route advertisement in your Cloud Router to advertise the Cloud SQL IP address range.
- B.1. Change the VPC routing mode to global.
- 2. Create a custom route advertisement in your Cloud Router to advertise the Cloud SQL IP address range.
- C.1. Create an additional Cloud Router in us-west2.
- 2. Create a new Border Gateway Protocol (BGP) peering connection to your on-premises data center.
- 3. Modify the VPC Network Peering connection used for Cloud SQL, and enable the import and export of routes.
- D.1. Change the VPC routing mode to global.
- 2. Modify the VPC Network Peering connection used for Cloud SQL, and enable the import and export of routes.

Answer: A

Explanation:

The core issue is that on-premises resources cannot reach the Cloud SQL instance, which is accessed through private services access and uses a peered VPC. Private services access involves a VPC peering connection between your Shared VPC and Google's service producer VPC. By default, these peering connections do not automatically exchange custom routes.

Option A correctly addresses this issue. First, **modifying the VPC Network Peering connection to enable import and export of routes (Step 1)** is crucial. This allows your Shared VPC to learn the routes to the Cloud SQL IP range from the service producer's VPC and vice versa. Without this, your on-premises network will not know how to reach the Cloud SQL instances, and the response from the SQL instance won't find a return path to your on-premises resources.

Second, **creating a custom route advertisement in the Cloud Router to advertise the Cloud SQL IP address range (Step 2)** ensures that your on-premises network, which connects to Google Cloud via the Dedicated Interconnect and Cloud Router, learns how to reach that Cloud SQL range. This route advertisement propagates the required IP range to your BGP-speaking on-premise routers via BGP. Thus establishing the full path of communications from your on-premises resources to the Cloud SQL instance.

Option B is incorrect because changing the routing mode to global (Step 1) does not solve the core problem of route exchange with the peering connection. Global routing would mainly affect subnet-based routing within the VPC itself. It is also redundant since dedicated interconnect traffic by default follows global dynamic routing. Option C is incorrect because creating another Cloud Router in us-west2 (Step 1) is not necessary as the SQL instance resides in us-west1 and does not improve connectivity. Option D is incorrect because changing the VPC routing mode to global (Step 1) does not address the problem and is not required.

In summary, the correct approach is to enable route exchange through the peering connection and explicitly advertise the Cloud SQL IP range via the Cloud Router. This will establish the necessary routing for on-premises resources to reach the Cloud SQL instance.

Authoritative Links:

VPC Network Peering: <https://cloud.google.com/vpc/docs/vpc-peering>

Private Services Access: <https://cloud.google.com/vpc/docs/private-services-access>

Cloud Router Custom Route Advertisements: <https://cloud.google.com/router/docs/how-to/advertisement-options>

Question: 90

CertyIQ

Your company has separate Virtual Private Cloud (VPC) networks in a single region for two departments: Sales and Finance. The Sales department's VPC network already has connectivity to on-premises locations using HA VPN, and you have confirmed that the subnet ranges do not overlap. You plan to peer both VPC networks to use the same HA tunnels for on-premises connectivity, while providing internet connectivity for the Google Cloud workloads through Cloud NAT. Internet access from the on-premises locations should not flow through Google Cloud. You need to propagate all routes between the Finance department and on-premises locations. What should you do?

- A. Peer the two VPCs, and use the default configuration for the Cloud Routers.
- B. Peer the two VPCs, and use Cloud Router's custom route advertisements to announce the peered VPC network ranges to the on-premises locations.
- C. Peer the two VPCs. Configure VPC Network Peering to export custom routes from Sales and import custom routes on Finance's VPC network. Use Cloud Router's custom route advertisements to announce a default route to the on-premises locations.
- D. Peer the two VPCs. Configure VPC Network Peering to export custom routes from Sales and import custom routes on Finance's VPC network. Use Cloud Router's custom route advertisements to announce the peered VPC network ranges to the on-premises locations.

Answer: D

Explanation:

Okay, let's break down why option D is the correct solution for this network engineering problem.

The core requirement is to enable communication between the Finance VPC and on-premises locations via the existing HA VPN connection established in the Sales VPC, while also ensuring internet access for both VPCs through Cloud NAT and preventing on-premises internet traffic from routing through Google Cloud.

Here's why option D works:

1. **VPC Peering:** Peering the Sales and Finance VPCs establishes a private connection between them,

allowing resources in both VPCs to communicate. This is fundamental for the Finance VPC to access the Sales VPC's VPN connection.

2. **Export/Import Custom Routes:** By enabling "export custom routes" in the Sales VPC peering configuration and "import custom routes" in the Finance VPC peering configuration, routes learned by the Sales VPC's Cloud Router (including the on-premises routes learned over the VPN) are propagated to the Finance VPC. This allows Finance resources to reach on-premises locations.
3. **Cloud Router Custom Route Advertisement:** Option D correctly dictates the use of custom route advertisements on the Sales VPC's Cloud Router. It should advertise the peered Finance VPC network ranges to the on-premises locations via the HA VPN. This is crucial for directing traffic destined for the Finance VPC from the on-premises network to the appropriate gateway.

Option A fails because it does not propagate the routes between the networks. The default configuration for Cloud Router would not allow on-premises locations to learn about the Finance VPC network. Option B is incorrect because it assumes that the on-premises should learn the peered VPC network. Option C does not address the on-premises knowing about the Finance network; it proposes a default route for on-premises, which we do not need based on the requirements.

In summary, option D ensures that: (1) the Finance VPC can reach on-premises through the Sales VPC's HA VPN; (2) that the Finance VPC and the Sales VPC know how to route traffic to the other; (3) on-premises locations know how to route traffic to Finance networks and (4) internet traffic for GCP workloads utilizes Cloud NAT.

Authoritative Links for further research:

VPC Network Peering: <https://cloud.google.com/vpc/docs/vpc-peering>

Cloud Router Custom Route Advertisements: <https://cloud.google.com/router/docs/how-to/advertised-routes>

HA VPN: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/overview>

Cloud NAT: <https://cloud.google.com/nat/docs/overview>

Question: 91

CertyIQ

You recently noticed a recurring daily spike in network usage in your Google Cloud project. You need to identify the virtual machine (VM) instances and type of traffic causing the spike in traffic utilization while minimizing the cost and management overhead required. What should you do?

- A.Enable VPC Flow Logs and send the output to BigQuery for analysis.
- B.Enable Firewall Rules Logging for all allowed traffic and send the output to BigQuery for analysis.
- C.Configure Packet Mirroring to send all traffic to a VM. Use Wireshark on the VM to identify traffic utilization for each VM in the VPC.
- D.Deploy a third-party network appliance and configure it as the default gateway. Use the third-party network appliance to identify users with high network traffic.

Answer: A

Explanation:

The best approach to identify the source of the network usage spike with minimal cost and management overhead is **A. Enable VPC Flow Logs and send the output to BigQuery for analysis.**

Here's why:

VPC Flow Logs: VPC Flow Logs record a sample of network flows sent from and received by VM instances. This gives insights into network traffic without capturing the packet data itself, reducing the storage

requirements and cost. <https://cloud.google.com/vpc/docs/using-flow-logs>

BigQuery: BigQuery is a cost-effective, fully managed, serverless data warehouse that allows you to analyze large datasets quickly. It is ideal for analyzing the VPC Flow Logs data to identify the VMs and traffic types contributing most to the spike. <https://cloud.google.com/bigquery/docs>

Cost Optimization: By sampling network flows instead of capturing every packet, VPC Flow Logs minimize the storage and processing costs. BigQuery's pay-as-you-go pricing model ensures that you only pay for the queries you run.

Management Overhead: Both VPC Flow Logs and BigQuery are managed services, minimizing the operational overhead required for setup and maintenance.

Here's why the other options are less suitable:

B. Enable Firewall Rules Logging: While firewall rules logging can provide some information, it only logs traffic that matches firewall rules (allowed or denied) and doesn't offer detailed insights into the traffic flow or usage patterns like VPC Flow Logs do. It also incurs costs from logging all allowed traffic, potentially higher than VPC Flow Logs.

C. Configure Packet Mirroring: Packet mirroring duplicates network traffic and sends it to a specific VM for analysis. While effective for detailed packet inspection, it's costly and resource-intensive as it requires maintaining a dedicated VM and processing all mirrored traffic. This is an overkill approach for merely identifying VMs causing a spike.

D. Deploy a third-party network appliance: Deploying a third-party network appliance introduces additional complexity, management overhead, and cost. A virtual appliance can be effective, but isn't required for this scenario; a cloud-native solution utilizing VPC Flow Logs provides the needed information in a simpler and more cost-effective manner.

In summary, VPC Flow Logs combined with BigQuery provides the most efficient and cost-effective way to analyze network traffic and pinpoint the source of the usage spike. This solution leverages Google Cloud's managed services to reduce operational overhead while providing the necessary insights.

Question: 92

CertyIQ

You need to enable Private Google Access for use by some subnets within your Virtual Private Cloud (VPC). Your security team set up the VPC to send all internet-bound traffic back to the on-premises data center for inspection before egressing to the internet, and is also implementing VPC Service Controls in the environment for API-level security control. You have already enabled the subnets for Private Google Access. What configuration changes should you make to enable Private Google Access while adhering to your security team's requirements?

- A.1. Create a private DNS zone with a CNAME record for *.googleapis.com to restricted.googleapis.com, with an A record pointing to Google's restricted API address range.
- 2. Create a custom route that points Google's restricted API address range to the default internet gateway as the next hop.
- B.1. Create a private DNS zone with a CNAME record for *.googleapis.com to restricted.googleapis.com, with an A record pointing to Google's restricted API address range.
- 2. Change the custom route that points the default route (0/0) to the default internet gateway as the next hop.
- C.1. Create a private DNS zone with a CNAME record for *.googleapis.com to private.googleapis.com, with an A record pointing to Google's private API address range.
- 2. Change the custom route that points the default route (0/0) to the default internet gateway as the next hop.
- D.1. Create a private DNS zone with a CNAME record for *.googleapis.com to private.googleapis.com, with an A record pointing to Google's private API address range.
- 2. Create a custom route that points Google's private API address range to the default internet gateway as the next hop.

Answer: A

Explanation:

The correct answer is **A**. Here's why:

Private Google Access allows instances without external IP addresses to reach Google APIs and services. However, the security requirement mandates that all internet-bound traffic, including API traffic, is routed through the on-premises data center for inspection. This creates a conflict as Private Google Access typically bypasses user-defined routes.

To address this, we need to force traffic to Google's restricted API address range (which is treated as internet-bound traffic) through the user-defined default route (which is directed towards the on-premises firewall). This is achieved using a combination of DNS resolution and custom routing.

First, a private DNS zone is created with a CNAME record for *.googleapis.com pointing to restricted.googleapis.com. This redirects all API requests to Google's restricted API endpoints. An A record is then created for restricted.googleapis.com, resolving it to Google's reserved IP address range for restricted API access. This ensures all traffic to Google APIs uses a specific, dedicated range.

Second, we need to create a custom route that specifies that the restricted API range must go through the default internet gateway. This sends the traffic to your on-premises firewall, which is designed to inspect and forward the traffic to the internet. This satisfies the security requirement by ensuring all API traffic is inspected. This is a different route than the default 0/0 route to the default internet gateway, which may send standard internet traffic.

Option B is incorrect because modifying the default 0/0 route would send ALL traffic to the default internet gateway, not only the specific Google API traffic. Options C and D use the incorrect private.googleapis.com which is not used when sending traffic via an on-premises gateway for security. Additionally, D makes the routing configuration incorrect.

Key Concepts: Private Google Access, User-Defined Routes, Private DNS Zones, Restricted Google API Endpoints.

Relevant Documentation:

Private Google Access: <https://cloud.google.com/vpc/docs/private-access-options>

Using Private Google Access: <https://cloud.google.com/vpc/docs/configure-private-google-access>

Restricted Google API IP range: <https://cloud.google.com/vpc/docs/configure-private-google-access#configuring-custom-routes>

Question: 93

CertyIQ

You have deployed an HTTP(s) load balancer, but health checks to port 80 on the Compute Engine virtual machine instance are failing, and no traffic is sent to your instances. You want to resolve the problem. Which commands should you run?

A. `gcloud compute instances add-access-config instance-1`

B. `gcloud compute firewall-rules create allow-lb --network load-balancer --allow tcp --destination-ranges 130.211.0.0/22,35.191.0.0/16 --direction EGRESS`

C. `gcloud compute firewall-rules create allow-lb --network load-balancer --allow tcp --source-ranges 130.211.0.0/22,35.191.0.0/16 --direction INGRESS`

D. `gcloud compute health-checks update http health-check --unhealthy-threshold 10`

Answer: C

Explanation:

The issue is that the health checks from the Google Cloud HTTP(s) load balancer to the Compute Engine

instances are failing, preventing traffic from reaching the instances. This failure typically indicates a network connectivity or firewall rule problem. Option C correctly addresses this issue by creating a firewall rule to allow traffic from the Google Cloud health check ranges. Specifically, `gcloud compute firewall-rules create allow-lb` creates a new firewall rule named "allow-lb". The `--network` load-balancer flag specifies the network where this rule applies. The `--allow tcp` flag allows TCP traffic, essential for HTTP(s). The crucial part is `--source-ranges 130.211.0.0/22,35.191.0.0/16`, which permits incoming traffic from the IP ranges that Google's health checks originate from. The `--direction INGRESS` flag means the rule allows incoming traffic to the instances. The other options are not relevant to this problem. Option A addresses external IP assignment to the instance, not health check failures. Option B creates an egress rule which is opposite to what's needed here as health checks are ingress connections. Option D modifies the health check parameters instead of fixing network connectivity. Therefore, option C is the most effective solution, as it permits health checks to reach the backend instances. The health check process requires the source range from Google cloud to be allowed access to the backend instances on the configured health check port.

Authoritative Links:

Google Cloud Documentation on Health Checks: <https://cloud.google.com/load-balancing/docs/health-checks>

Google Cloud Documentation on Firewall Rules: <https://cloud.google.com/vpc/docs/firewalls>

Google Cloud IP Ranges for Health Checks: <https://cloud.google.com/load-balancing/docs/health-checks#source-ips>

Question: 94

CertyIQ

You deployed a hub-and-spoke architecture in your Google Cloud environment that uses VPC Network Peering to connect the spokes to the hub. For security reasons, you deployed a private Google Kubernetes Engine (GKE) cluster in one of the spoke projects with a private endpoint for the control plane. You configured authorized networks to be the subnet range where the GKE nodes are deployed. When you attempt to reach the GKE control plane from a different spoke project, you cannot access it. You need to allow access to the GKE control plane from the other spoke projects. What should you do?

- A. Add a firewall rule that allows port 443 from the other spoke projects.
- B. Enable Private Google Access on the subnet where the GKE nodes are deployed.
- C. Configure the authorized networks to be the subnet ranges of the other spoke projects.
- D. Deploy a proxy in the spoke project where the GKE nodes are deployed and connect to the control plane through the proxy.

Answer: C

Explanation:

The correct answer is **C. Configure the authorized networks to be the subnet ranges of the other spoke projects.**

Here's the justification:

Private GKE clusters with private endpoints restrict access to the control plane to only authorized networks. Initially, these authorized networks were configured to be solely the subnet where the GKE nodes reside. This configuration is why accessing the control plane from other spoke projects fails; their subnet ranges were not permitted.

To enable access from other spoke projects, you must expand the authorized networks to include the IP address ranges of those spoke subnets. By explicitly adding these subnet ranges to the GKE control plane's authorized networks, you grant permissions for resources in those networks to communicate with the GKE control plane. This direct approach avoids the complexities of implementing proxies (Option D) or fiddling with

firewall rules (Option A). Private Google Access (Option B) facilitates private connectivity to Google APIs, which is not directly relevant for control plane access within your peered network. The focus should remain on properly authorizing network sources, which is directly addressed by modifying the authorized networks list.

Further research:

GKE Authorized Networks Documentation: Provides details on how to configure authorized networks for GKE private clusters.

VPC Network Peering Overview: Explains the functionality of VPC peering, important to understand the context of the question.

Question: 95

CertyIQ

You recently deployed your application in Google Cloud. You need to verify your Google Cloud network configuration before deploying your on-premises workloads. You want to confirm that your Google Cloud network configuration allows traffic to flow from your cloud resources to your on-premises network. This validation should also analyze and diagnose potential failure points in your Google Cloud network configurations without sending any data plane test traffic. What should you do?

- A. Use Network Intelligence Center's Connectivity Tests.
- B. Enable Packet Mirroring on your application and send test traffic.
- C. Use Network Intelligence Center's Network Topology visualizations.
- D. Enable VPC Flow Logs and send test traffic.

Answer: C

Explanation:

The correct answer is **C. Use Network Intelligence Center's Network Topology visualizations.**

Here's why:

Network Topology in Google Cloud's Network Intelligence Center is designed to provide a visual representation of your network architecture, including connections between VPCs, subnets, and on-premises networks (via Cloud Interconnect or VPN). It analyzes your network configurations, not data plane traffic, to depict how traffic should flow based on defined routes, firewalls, and other networking rules. This visualization helps you to understand the intended path of traffic and pinpoint potential misconfigurations that could prevent connectivity from your cloud resources to your on-premises network, all without generating any actual traffic. This aligns directly with the problem's need to validate configuration and identify failure points without sending data.

Option A, Connectivity Tests, while valuable, involves creating and running probes that send data packets, violating the requirement to avoid sending data plane traffic. Option B, Packet Mirroring, is for traffic analysis by creating copies of packets and requires live traffic, which again contradicts the question's parameters. Similarly, option D, VPC Flow Logs, collects information about network traffic that has already traversed your network, thus requiring actual traffic and not addressing the need for configuration analysis without data flow.

Network Topology provides a static, configuration-based analysis of your network, precisely what's needed to diagnose potential issues before on-premises workloads are migrated. It helps confirm the intended path of traffic without relying on active connections, providing insights into potential blockages based on existing configurations.

Authoritative Links:

Google Cloud Network Intelligence Center - Network Topology: <https://cloud.google.com/network->

Question: 96

In your Google Cloud organization, you have two folders: Dev and Prod. You want a scalable and consistent way to enforce the following firewall rules for all virtual machines (VMs) with minimal cost:

- Port 8080 should always be open for VMs in the projects in the Dev folder.
- Any traffic to port 8080 should be denied for all VMs in your projects in the Prod folder.

What should you do?

- A.Create and associate a firewall policy with the Dev folder with a rule to open port 8080. Create and associate a firewall policy with the Prod folder with a rule to deny traffic to port 8080.
- B.Create a Shared VPC for the Dev projects and a Shared VPC for the Prod projects. Create a VPC firewall rule to open port 8080 in the Shared VPC for Dev. Create a firewall rule to deny traffic to port 8080 in the Shared VPC for Prod. Deploy VMs to those Shared VPCs.
- C.In all VPCs for the Dev projects, create a VPC firewall rule to open port 8080. In all VPCs for the Prod projects, create a VPC firewall rule to deny traffic to port 8080.
- D.Use Anthos Config Connector to enforce a security policy to open port 8080 on the Dev VMs and deny traffic to port 8080 on the Prod VMs.

Answer: A

Explanation:

The correct answer is A because it leverages Google Cloud's hierarchical firewall policies for efficient and scalable management. Firewall policies, associated at the folder level, are inherited by all projects and VMs within that folder. This avoids the need for repetitive rule creation in individual VPC networks. Option A directly addresses the requirement to enforce different firewall rules for Dev and Prod folders with minimal effort by configuring policies at the folder level which automatically propagate down.

Option B, using Shared VPCs, is overly complex for the task. It introduces unnecessary architectural changes (shared VPC configuration, which is not mentioned in question requirements) and is less efficient compared to simply using firewall policies. While Shared VPCs can be helpful, they're not the right solution here where the goal is merely to establish firewall rules.

Option C, manually creating firewall rules in each VPC, is not scalable and prone to human error. As the number of VPCs and projects grow, this approach becomes extremely difficult to manage consistently. This is not ideal as it requires individual management across VPCs and is not organizationally scalable.

Option D, using Anthos Config Connector, is an over-engineered solution. While Config Connector is powerful, using it to manage basic firewall rules is unnecessary. It adds unnecessary complexity and overhead when firewall policies are a more straightforward and direct fit. Firewall policies are a simpler, quicker and native solution within GCP itself.

Folder level firewall policies provide a centralized and scalable way to manage network security. They ensure consistent rule enforcement across all resources within their scope. In summary, option A offers the simplest, most scalable, and least costly approach to implementing the specified firewall rules by using native GCP features such as firewall policies. This aligns with best practices and the principle of least privilege by providing a way to manage these network rules at the highest necessary level in the resource hierarchy.<https://cloud.google.com/vpc/docs/firewall-policies><https://cloud.google.com/vpc/docs/firewalls>

Question: 97

You need to configure the Border Gateway Protocol (BGP) session for a VPN tunnel you just created between two Google Cloud VPCs, 10.1.0.0/16 and 172.16.0.0/16. You have a Cloud Router (router-1) in the 10.1.0.0/16 network and a second Cloud Router (router-2) in the 172.16.0.0/16 network. Which configuration should you use for the BGP session?

A.

Router	BGP Interface Name	BGP IP	BGP Peer IP	Peer ASN
router-1	if-tunnel-a-to-b-if-0	169.254.0.254	169.254.0.254	65502
router-2	if-tunnel-b-to-a-if-0	169.254.0.254	169.254.0.254	65501

B.

Router	BGP Interface Name	BGP IP	BGP Peer IP	Peer ASN
router-1	if-tunnel-a-to-b-if-0	10.1.0.1	172.16.0.1	15052
router-2	if-tunnel-b-to-a-if-0	172.16.0.1	10.1.0.1	15501

C.

Router	BGP Interface Name	BGP IP	BGP Peer IP	Peer ASN
router-1	if-tunnel-a-to-b-if-0	169.254.20.1	169.254.20.2	65002
router-2	if-tunnel-b-to-a-if-0	169.254.20.2	169.254.20.1	65001

D.

Router	BGP Interface Name	BGP IP	BGP Peer IP	Peer ASN
router-1	if-tunnel-a-to-b-if-0	172.16.0.254	10.1.0.254	16552
router-2	if-tunnel-b-to-a-if-0	10.1.0.254	172.16.0.254	16551

Answer: C

Explanation:

A) wrong because peer address need be different B and D) wrong because google asn range is 64512-65534, 4200000000-4294967294. C) 100% sure.

Question: 98

Your company's on-premises network is connected to a VPC using a Cloud VPN tunnel. You have a static route of 0.0.0.0/0 with the VPN tunnel as its next hop defined in the VPC. All internet bound traffic currently passes through the on-premises network. You configured Cloud NAT to translate the primary IP addresses of Compute Engine instances in one region. Traffic from those instances will now reach the internet directly from their VPC and not from the on-premises network. Traffic from the virtual machines (VMs) is not translating addresses as expected. What should you do?

A. Lower the TCP Established Connection Idle Timeout for the NAT gateway.

B. Add firewall rules that allow ingress and egress of the external NAT IP address, have a target tag that is on the Compute Engine instances, and have a priority value higher than the priority value of the default route to the VPN gateway.

C. Add a default static route to the VPC with the default internet gateway as the next hop, the network tag associated with the Compute Engine instances, and a higher priority than the priority of the default route to the VPN tunnel.

D. Increase the default min-ports-per-vm setting for the Cloud NAT gateway.

Answer: C

Explanation:

The issue is that traffic from Compute Engine instances, expected to use Cloud NAT for internet access, is still being routed through the on-premises network via the VPN tunnel. This occurs because a static route with a 0.0.0.0/0 destination (the default route) and the VPN tunnel as the next hop takes precedence over the implicit Cloud NAT behavior. Cloud NAT only translates addresses when no other route covers the destination.

Option C correctly addresses the problem by suggesting the addition of another static route. This new route is a default route (0.0.0.0/0) pointing to the default internet gateway (which is the implicit next hop for internet-bound traffic within the VPC), but is specifically targeted using a network tag that is applied to the Compute Engine instances. The route also has a higher priority than the existing VPN route. This prioritizes the Cloud NAT path for VMs with the specific tag, allowing traffic to flow through Cloud NAT.

Here's how this works:

1. **Route Order:** VPC routes are evaluated based on the best match and priority.
2. **Specific vs General:** More specific routes take precedence (e.g., 10.0.0.0/24 is more specific than 10.0.0.0/8). If two routes have the same destination, the one with a higher priority (lower number) is chosen.
3. **Tag-Based Routing:** Routes can target specific network tags, which allows for a more granular control of routing behavior.
4. **Implicit Default:** By default, if no route matches, a VM will egress the VPC network through the default internet gateway.
5. **Resolving the Conflict:** When the VM has the tag, the new route will be the chosen path. This enables the VM to send internet traffic directly through the Cloud NAT. When there's no tag, it will take the default route which is through the on-premises VPN tunnel.

Option A is incorrect because the TCP timeout won't affect the routing. Option B is incorrect because firewall rules don't affect the routing decision. Option D is incorrect because Cloud NAT's min-ports setting does not relate to the route selection.

Authoritative Links:

VPC Route Overview: <https://cloud.google.com/vpc/docs/routes>

Cloud NAT: <https://cloud.google.com/nat/docs/overview>

Using Tags: <https://cloud.google.com/vpc/docs/tags-firewall-rules>

Question: 99

CertyIQ

You are designing a Partner Interconnect hybrid cloud connectivity solution with geo-redundancy across two metropolitan areas. You want to follow Google-recommended practices to set up the following region/metro pairs:

- (region 1/metro 1)
- (region 2/metro 2)

What should you do?

- A. Create a Cloud Router in region 1 with two VLAN attachments connected to metro1-zone1-x.
Create a Cloud Router in region 2 with two VLAN attachments connected to metro1-zone2-x.
- B. Create a Cloud Router in region 1 with one VLAN attachment connected to metro1-zone1-x.
Create a Cloud Router in region 2 with two VLAN attachments connected to metro2-zone2-x.
- C. Create a Cloud Router in region 1 with one VLAN attachment connected to metro1-zone2-x.
Create a Cloud Router in region 2 with one VLAN attachment connected to metro2-zone2-x.
- D. Create a Cloud Router in region 1 with one VLAN attachment connected to metro1-zone1-x and one VLAN attachment connected to metro1-zone2-x.
Create a Cloud Router in region 2 with one VLAN attachment connected to metro2-zone1-x and one VLAN

attachment to metro2-zone2-x.

Answer: D

Explanation:

The correct answer is **D**. Here's a detailed justification:

Option D aligns with Google's best practices for achieving high availability and geo-redundancy with Partner Interconnect. The goal is to establish resilient connections between your on-premises network and your Google Cloud environment across two separate metropolitan areas (metros).

Each metro location provides a point of presence (PoP) where your on-premises network connects to Google's network. Within each metro, there are multiple availability zones (e.g., zone1-x and zone2-x). To achieve redundancy within each metro, you should create at least two VLAN attachments connecting to different zones.

By creating a Cloud Router in each region (region 1 and region 2), you enable dynamic routing via BGP (Border Gateway Protocol). Each Cloud Router is then associated with two VLAN attachments. Critically, these two VLAN attachments are located in different availability zones within the same metro. This allows for redundancy in case one zone experiences issues.

The solution utilizes two regions (region 1/metro 1 and region 2/metro 2). This strategy provides geo-redundancy, ensuring that traffic can be routed even if a whole metropolitan area experiences a disruption. Each region also has redundancy within the metro itself with 2 VLAN attachments in different zones.

Option A is incorrect because it attempts to connect both attachments in region 2 to the same metro, which doesn't provide geo-redundancy. Option B is incorrect as it only connects one VLAN attachment in region 1 and doesn't provide redundancy within that metro, and also connects both attachments in region 2 to one metro. Option C is incorrect because it provides only one VLAN attachment in each metro and doesn't provide redundancy within each metro.

In summary, option D correctly implements the required redundancy and geographical diversity across two metros and availability zones, aligning with Google's recommended practices.

Authoritative links for further research:

Partner Interconnect Overview: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/partner-interconnect>

Partner Interconnect Topology: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/topology>

Cloud Router Overview: <https://cloud.google.com/network-connectivity/docs/router/concepts/overview>

High Availability for Partner Interconnect: <https://cloud.google.com/network-connectivity/docs/interconnect/how-to/configure-high-availability>

Question: 100

CertyIQ

You are designing the network architecture for your organization. Your organization has three developer teams: Web, App, and Database. All of the developer teams require access to Compute Engine instances to perform their critical tasks. You are part of a small network and security team that needs to provide network access to the developers. You need to maintain centralized control over network resources, including subnets, routes, and firewalls. You want to minimize operational overhead. How should you design this topology?

A. Configure a host project with a Shared VPC. Create service projects for Web, App, and Database.

B. Configure one VPC for Web, one VPC for App, and one VPC for Database. Configure HA VPN between each VPC.

C. Configure three Shared VPC host projects, each with a service project: one for Web, one for App, and one for Database.

D. Configure one VPC for Web, one VPC for App, and one VPC for Database. Use VPC Network Peering to connect all VPCs in a full mesh.

Answer: A

Explanation:

The correct answer is **A: Configure a host project with a Shared VPC. Create service projects for Web, App, and Database.** This approach aligns best with the requirements of centralized control, minimized operational overhead, and providing network access to multiple development teams. Shared VPC allows a single VPC network to be shared across multiple projects within an organization. The host project holds the VPC network, subnets, routes, and firewalls, enabling centralized management by the network and security team. Service projects, assigned to each development team (Web, App, and Database), can then create and manage their Compute Engine instances, while inheriting the centrally managed network resources. This reduces operational overhead since network configurations are only handled in one place and ensures consistent network policies across the organization. By decoupling network management from application management, you achieve a clear separation of responsibilities. Options B, C and D involve independent VPCs or complex network peering which make centralized management and policy control difficult and increase operational overhead. Option C introduces unnecessary complexity by using multiple host projects and would also hinder the central control of network resources. Option D, while achieving connectivity, creates an N*N problem, making management of multiple peering connections complex.

Authoritative Links:

Shared VPC Overview: <https://cloud.google.com/vpc/docs/shared-vpc>

Shared VPC Concepts: <https://cloud.google.com/vpc/docs/shared-vpc-concepts>

Choosing a VPC topology: <https://cloud.google.com/architecture/vpc-design>

Question: 101

CertyIQ

Your company has 10 separate Virtual Private Cloud (VPC) networks, with one VPC per project in a single region in Google Cloud. Your security team requires each VPC network to have private connectivity to the main on-premises location via a Partner Interconnect connection in the same region. To optimize cost and operations, the same connectivity must be shared with all projects. You must ensure that all traffic between different projects, on-premises locations, and the internet can be inspected using the same third-party appliances. What should you do?

A. Configure the third-party appliances with multiple interfaces and specific Partner Interconnect VLAN attachments per project. Create the relevant routes on the third-party appliances and VPC networks.

B. Configure the third-party appliances with multiple interfaces, with each interface connected to a separate VPC network. Create separate VPC networks for on-premises and internet connectivity. Create the relevant routes on the third-party appliances and VPC networks.

C. Consolidate all existing projects' subnetworks into a single VPC. Create separate VPC networks for on-premises and internet connectivity. Configure the third-party appliances with multiple interfaces, with each interface connected to a separate VPC network. Create the relevant routes on the third-party appliances and VPC networks.

D. Configure the third-party appliances with multiple interfaces. Create a hub VPC network for all projects, and create separate VPC networks for on-premises and internet connectivity. Create the relevant routes on the third-party appliances and VPC networks. Use VPC Network Peering to connect all projects' VPC networks to the hub VPC. Export custom routes from the hub VPC and import on all projects' VPC networks.

Answer: D

Explanation:

The correct answer is **D**. Here's why:

Option D utilizes a hub-and-spoke architecture, a common best practice for centralized network management and security inspection in cloud environments. A central "hub" VPC acts as the transit point for all traffic. Each of the 10 existing VPCs (the "spokes") connects to the hub VPC via VPC Network Peering. This peering allows network traffic to flow between them as if they were a single network. Importantly, with VPC Network Peering custom routes are exchanged and can be configured to allow traffic flow through the centralized hub. The hub VPC also contains separate subnets for on-premises (connected via Partner Interconnect) and internet connectivity (likely through Cloud NAT or a similar service). The third-party appliances, also located in the hub VPC, have interfaces connected to all three types of subnets (spoke, on-prem, internet), allowing them to inspect and process all traffic traversing through the network. This configuration fulfills the requirement to inspect all traffic using the same appliances and shared connectivity and isolates each project into their own VPC.

Option A fails to centralize security inspection and manage network connectivity efficiently as the third-party appliances become specific to each project. Option B does not leverage the benefits of a hub-and-spoke model and instead requires a separate VPC for each project, complicating management. Option C is impractical as consolidating all projects' subnets into a single VPC creates a significant single point of failure and management complexity.

Key Concepts:

Hub-and-Spoke Topology: A centralized network architecture where a central hub network connects to multiple spoke networks. This design simplifies network management, security, and shared services.

VPC Network Peering: A service that connects two VPC networks, allowing them to communicate with each other using private IP addresses.

Custom Routes: Allow for granular control of traffic flow within a VPC network, enabling traffic redirection to the third-party appliances.

Partner Interconnect: Provides a dedicated, private connection between on-premises networks and Google Cloud.

Authoritative Links:

VPC Network Peering: <https://cloud.google.com/vpc/docs/vpc-peering>

Hub and Spoke Network Design: <https://cloud.google.com/architecture/networking/creating-hub-and-spoke-networks>

Custom Routes: <https://cloud.google.com/vpc/docs/routes>

Question: 102

CertyIQ

You have just deployed your infrastructure on Google Cloud. You now need to configure the DNS to meet the following requirements:

- Your on-premises resources should resolve your Google Cloud zones.
- Your Google Cloud resources should resolve your on-premises zones.
- You need the ability to resolve ".internal" zones provisioned by Google Cloud.

What should you do?

A. Configure an outbound server policy, and set your alternative name server to be your on-premises DNS resolver. Configure your on-premises DNS resolver to forward Google Cloud zone queries to Google's public DNS 8.8.8.8.

B. Configure both an inbound server policy and outbound DNS forwarding zones with the target as the on-premises DNS resolver. Configure your on-premises DNS resolver to forward Google Cloud zone queries to Google Cloud's DNS resolver.

C. Configure an outbound DNS server policy, and set your alternative name server to be your on-premises DNS

resolver. Configure your on-premises DNS resolver to forward Google Cloud zone queries to Google Cloud's DNS resolver.

D. Configure Cloud DNS to DNS peer with your on-premises DNS resolver. Configure your on-premises DNS resolver to forward Google Cloud zone queries to Google's public DNS 8.8.8.8.

Answer: B

Explanation:

The correct answer is B, which involves configuring both inbound and outbound DNS forwarding. To understand why, let's break down the requirements and how each part of the solution addresses them.

Requirement 1: On-premises resources resolving Google Cloud zones. This necessitates a mechanism for on-premises DNS resolvers to forward queries for Google Cloud-hosted domains to Google Cloud DNS. An inbound server policy on Google Cloud allows on-premises DNS servers to forward queries to the cloud. We configure this with the target as the on-premises DNS resolver, so the queries can reach it.

Requirement 2: Google Cloud resources resolving on-premises zones. Similarly, this requires Google Cloud DNS to forward requests for on-premises domains to the on-premises DNS resolvers. An outbound DNS forwarding zone configured within Cloud DNS is used here to forward queries to on-premises DNS. The target for this zone is the IP address of the on-premises DNS resolver.

Requirement 3: Resolving ".internal" zones provisioned by Google Cloud. Google Cloud automatically resolves these internal zones, so no additional configuration is needed on either the on-premises or the cloud side.

Option B correctly configures both inbound and outbound forwarding policies. This ensures bidirectional DNS resolution between on-premises and Google Cloud environments. The on-premises DNS server also needs to be configured to forward queries for the Google Cloud zones to Google Cloud's internal DNS resolver; this is how Google Cloud resolves the Google Cloud zones with their respective IP. This is why alternative A or D will not work. Option C only addresses the resolution of on-prem zones by the Google Cloud and does not allow for on-prem systems to resolve resources in the Google Cloud.

In contrast, option A is incorrect because forwarding Google Cloud zone queries to Google's public DNS (8.8.8.8) would not resolve private Google Cloud zones. Option C lacks an inbound policy to allow on-prem resolutions of Google Cloud zones. Option D uses DNS peering which is not the right way to solve this problem and also uses a public DNS resolver.

Authoritative Links for further research:

Google Cloud DNS Overview: <https://cloud.google.com/dns/docs/overview>

DNS forwarding with inbound server policy: <https://cloud.google.com/dns/docs/using-dns-forwarding#inbound>

DNS forwarding with outbound forwarding zones: <https://cloud.google.com/dns/docs/using-dns-forwarding#outbound>

Cloud DNS server policies: <https://cloud.google.com/dns/docs/policies>

Question: 103

CertyIQ

Your organization uses a hub-and-spoke architecture with critical Compute Engine instances in your Virtual Private Clouds (VPCs). You are responsible for the design of Cloud DNS in Google Cloud. You need to be able to resolve Cloud DNS private zones from your on-premises data center and enable on-premises name resolution from your hub-and-spoke VPC design. What should you do?

- A.1. Configure a private DNS zone in the hub VPC, and configure DNS forwarding to the on-premises server.
2. Configure DNS peering from the spoke VPCs to the hub VPC.

- B.1. Configure a DNS policy in the hub VPC to allow inbound query forwarding from the spoke VPCs.
- 2. Configure the spoke VPCs with a private zone, and set up DNS peering to the hub VPC.
- C.1. Configure a DNS policy in the spoke VPCs, and configure your on-premises DNS as an alternate DNS server.
- 2. Configure the hub VPC with a private zone, and set up DNS peering to each of the spoke VPCs.
- D.1. Configure a DNS policy in the hub VPC, and configure the on-premises DNS as an alternate DNS server.
- 2. Configure the spoke VPCs with a private zone, and set up DNS peering to the hub VPC.

Answer: B

Explanation:

Here's a detailed justification for why option B is the correct solution, along with relevant concepts and links:

Option B correctly addresses both requirements: resolving Cloud DNS private zones from on-premises and enabling on-premises name resolution from the hub-and-spoke VPCs.

Step 1: Hub VPC Inbound Forwarding

Configuring a DNS policy in the hub VPC to allow inbound query forwarding from the spoke VPCs enables the hub to receive DNS queries from the spokes. This is essential for the spokes to be able to resolve resources within the hub or on-premises if configured.

This leverages Google Cloud's Inbound DNS Forwarding Policy feature, enabling private DNS zones in the spokes to be resolved via the hub VPC's DNS resolvers.

Step 2: Spoke VPCs and DNS Peering

Each spoke VPC is configured with its own private DNS zone. This supports localized naming within each spoke.

Setting up DNS peering from the spoke VPCs to the hub VPC allows the spokes to send DNS queries to the hub's DNS resolvers for name resolution outside of their own private zones. This centralizes DNS resolution through the hub.

This peering allows for the resolution of names in the hub's private zone, as well as on-premises names once forwarding is configured on the hub.

Why other options are incorrect:

Option A: Configuring DNS forwarding in the hub VPC to on-premises (as the primary step) is backwards. The hub needs to receive DNS queries from the spokes first, then forward to on-premises if required.

Option C: Configuring a DNS policy on spokes and using on-premises as an alternate server does not address hub-to-spoke DNS resolution and lacks clarity in terms of resolving hub zone names in spoke VPCs.

Option D: This is similar to option A, where forwarding on the hub to on-prem without first receiving DNS requests from the spokes creates an indirect configuration.

Key Concepts:

Hub-and-Spoke Architecture: A network topology where a central hub VPC connects to multiple spoke VPCs.

Cloud DNS Private Zones: Used for managing DNS records within a Google Cloud VPC network.

DNS Peering: A mechanism that allows one VPC to query another VPC's Cloud DNS resolvers.

DNS Forwarding: Directing DNS queries to specified resolvers like on-prem DNS servers.

Inbound DNS Forwarding Policy: A policy that allows Google Cloud to receive DNS queries from on-prem or peered VPCs in order to resolve a private DNS zone.

Authoritative Links:

Cloud DNS overview: <https://cloud.google.com/dns/docs/overview>

DNS Peering: <https://cloud.google.com/dns/docs/using-dns-peering>

Inbound Forwarding: <https://cloud.google.com/dns/docs/using-forwarding-policies>

In summary, Option B creates a flow where DNS queries from the spoke VPCs are handled by the hub VPC and onward to on-premises if needed. This structure allows for both the resolution of names within the Google cloud infrastructure and within on-premises networks.

Question: 104

CertyIQ

You have a Cloud Storage bucket in Google Cloud project XYZ. The bucket contains sensitive data. You need to design a solution to ensure that only instances belonging to VPCs under project XYZ can access the data stored in this Cloud Storage bucket. What should you do?

- A. Configure Private Google Access to privately access the Cloud Storage service using private IP addresses.
- B. Configure a VPC Service Controls perimeter around project XYZ, and include storage.googleapis.com as a restricted service in the service perimeter.
- C. Configure Cloud Storage with projectPrivate Access Control List (ACL) that gives permission to the project team based on their roles.
- D. Configure Private Service Connect to privately access Cloud Storage from all VPCs under project XYZ.

Answer: B

Explanation:

The correct answer is **B. Configure a VPC Service Controls perimeter around project XYZ, and include storage.googleapis.com as a restricted service in the service perimeter.**

Here's why:

VPC Service Controls (VPC SC) provides a security perimeter around Google Cloud resources, enabling you to restrict access based on the identity of the client accessing the data and the network they are using. By creating a perimeter around project XYZ and including storage.googleapis.com as a restricted service, you ensure that only requests originating from within that perimeter (i.e., from VPCs within project XYZ) can access the Cloud Storage bucket, regardless of the source's IP address or authentication credentials. This effectively prevents unauthorized access from outside project XYZ.

Option A, Private Google Access, only enables private access to Google APIs from within a VPC network, but doesn't restrict which networks can access Cloud Storage. It doesn't provide the strong perimeter security required to limit access based on the originating project. Option C, project-private ACLs, only control access based on user/service account roles, not the network from which the access is initiated. It does not prevent users from accessing storage from a network external to project XYZ if the user/service account has correct permissions. Option D, Private Service Connect, establishes private endpoints to Google services, but it also doesn't enforce network-based perimeter security to restrict access to project-specific VPCs like VPC SC does.

VPC SC provides a crucial defense in depth mechanism, moving beyond just user authentication and focusing on network location as a security control. This is necessary for protecting highly sensitive data. For more details, refer to the official Google Cloud documentation:

VPC Service Controls Overview: <https://cloud.google.com/vpc-service-controls/docs/overview>

VPC Service Controls Service Perimeters: <https://cloud.google.com/vpc-service-controls/docs/service-perimeters>

Question: 105

CertyIQ

You are maintaining a Shared VPC in a host project. Several departments within your company have infrastructure

in different service projects attached to the Shared VPC and use Identity and Access Management (IAM) permissions to manage the cloud resources in those projects. VPC Network Peering is also set up between the Shared VPC and a common services VPC that is not in a service project. Several users are experiencing failed connectivity between certain instances in different Shared VPC service projects and between certain instances and the internet. You need to validate the network configuration to identify whether a misconfiguration is the root cause of the problem. What should you do?

- A. Review the VPC audit logs in Cloud Logging for the affected instances.
- B. Use Secure Shell (SSH) to connect to the affected Compute Engine instances, and run a series of PING tests to the other affected endpoints and the 8.8.8.8 IPv4 address.
- C. Run Connectivity Tests from Network Intelligence Center to check connectivity between the affected endpoints in your network and the internet.
- D. Enable VPC Flow Logs for all VPCs, and review the logs in Cloud Logging for the affected instances.

Answer: C

Explanation:

The correct answer is **C. Run Connectivity Tests from Network Intelligence Center to check connectivity between the affected endpoints in your network and the internet.**

Here's why:

Connectivity Tests within the Network Intelligence Center are specifically designed to diagnose and troubleshoot network connectivity issues. They allow you to verify reachability between various points within your Google Cloud environment and the internet, including scenarios involving Shared VPCs, peered networks, and different projects. This is crucial in a complex setup like the one described where connectivity problems exist across service projects and potentially towards the internet. These tests go beyond simple pings; they analyze the entire network path, including firewall rules, routing tables, and VPC peering configurations. This provides detailed information about where connectivity fails, helping pinpoint the source of the problem. The tool displays a visualization of the network path which provides insights into any misconfigured firewalls, routes, or peering issues that might be impeding traffic.

Option A is not ideal because while VPC audit logs can be helpful for understanding configuration changes or actions, they don't directly diagnose current connectivity failures. Similarly, option D, VPC flow logs, provide information about existing traffic flow but don't actively probe or test connectivity issues like the Network Intelligence Center does. They are useful for identifying traffic patterns, but not for pinpointing configuration issues blocking traffic. Finally, option B, SSH and ping, while a basic troubleshooting technique, only checks reachability at the instance level and provides limited visibility into the path within a complex multi-project VPC environment; it's not sufficient to trace the path or identify the source of complex network misconfigurations such as those with peering, routing, and firewall settings.

Therefore, using Connectivity Tests provides a targeted, comprehensive solution tailored for identifying the root cause of complex network problems in Google Cloud environments, making it the most efficient and effective approach for this situation.

Authoritative Links:

[Connectivity Tests Overview](#)
[Using Connectivity Tests](#)
[Shared VPC Overview](#)

Question: 106

CertyIQ

Your organization has Compute Engine instances in us-east1, us-west2, and us-central1. Your organization also has an existing Cloud Interconnect physical connection in the East Coast of the United States with a single VLAN

attachment and Cloud Router in us-east1. You need to provide a design with high availability and ensure that if a region goes down, you still have access to all your other Virtual Private Cloud (VPC) subnets. You need to accomplish this in the most cost-effective manner possible. What should you do?

- A.1. Configure your VPC routing in regional mode.
2. Add an additional Cloud Interconnect VLAN attachment in the us-east1 region, and configure a Cloud Router in us-east1.
- B.1. Configure your VPC routing in global mode.
2. Add an additional Cloud Interconnect VLAN attachment in the us-east1 region, and configure a Cloud Router in us-east1.
- C.1. Configure your VPC routing in global mode.
2. Add an additional Cloud Interconnect VLAN attachment in the us-west2 region, and configure a Cloud Router in us-west2.
- D.1. Configure your VPC routing in regional mode.
2. Add additional Cloud Interconnect VLAN attachments in the us-west2 and us-central1 regions, and configure Cloud Routers in us-west2 and us-central1.

Answer: C

Explanation:

The correct answer is C because it provides the most cost-effective high-availability solution for connecting on-premises resources to multiple Google Cloud regions via Cloud Interconnect. Global routing is crucial (step 1) as it enables dynamic route exchange between on-premises and all VPC subnets regardless of the region, meaning connectivity persists even if a region goes down. With regional routing, routes are only exchanged within a specific region, hindering connectivity to other regions in case of failure. Adding a second VLAN attachment in a different region (us-west2), along with its associated Cloud Router (step 2), provides the needed redundancy and failover capabilities. If the east coast connection or Cloud Router fails, the traffic will be routed through the west coast attachment seamlessly. Option A is incorrect because regional routing does not allow for cross-region connectivity. Option B is incorrect because although using global routing is correct, adding the second connection in the same region doesn't provide true high availability. Option D is incorrect due to the high cost and unnecessary complexity that comes from adding VLAN attachments and routers in every region.

Key Concepts:

Global Routing: Allows dynamic routing exchange across all subnets in a VPC regardless of region.

Regional Routing: Limits route exchange to subnets within a specific region.

Cloud Interconnect VLAN Attachment: Connects an on-premises network to a VPC network via a physical Cloud Interconnect.

Cloud Router: Manages dynamic routing between a VPC network and an on-premises network using BGP.

Authoritative Links:

VPC Network routing options: <https://cloud.google.com/vpc/docs/routes>

Cloud Interconnect Overview: <https://cloud.google.com/interconnect/docs/concepts/overview>

Cloud Router Overview: <https://cloud.google.com/router/docs/overview>

Question: 107

CertyIQ

You recently configured Google Cloud Armor security policies to manage traffic to your application. You discover that Google Cloud Armor is incorrectly blocking some traffic to your application. You need to identify the web application firewall (WAF) rule that is incorrectly blocking traffic. What should you do?

- A.Enable firewall logs, and view the logs in Firewall Insights.
- B.Enable HTTP(S) Load Balancing logging with sampling rate equal to 1, and view the logs in Cloud Logging.

C.Enable VPC Flow Logs, and view the logs in Cloud Logging.

D.Enable Google Cloud Armor audit logs, and view the logs on the Activity page in the Google Cloud Console.

Answer: B

Explanation:

The correct answer is **B. Enable HTTP(S) Load Balancing logging with sampling rate equal to 1, and view the logs in Cloud Logging.** Here's why:

Google Cloud Armor is a Web Application Firewall (WAF) that operates at the edge of Google's network, protecting applications behind HTTP(S) Load Balancers. To troubleshoot why Cloud Armor is incorrectly blocking traffic, you need detailed logs of the requests that are being evaluated by Cloud Armor rules. HTTP(S) Load Balancer logging, when set to a sampling rate of 1, captures every request that passes through the load balancer, including the Cloud Armor policy evaluation results (specifically, which rule triggered a blocking action). This detailed log is written to Cloud Logging, which is the central log management service in Google Cloud. By examining these logs, you can pinpoint the specific Cloud Armor rule that is incorrectly blocking requests by analyzing the logs for fields like `securityPolicy.matchedRule` and `securityPolicy.action`.

Option A, enabling firewall logs and viewing them in Firewall Insights, is incorrect because firewall logs relate to VPC firewall rules, not to the WAF capabilities of Cloud Armor. Option C, VPC Flow Logs, captures network traffic at the virtual machine instance level but doesn't provide details about the WAF evaluation. Finally, option D, Google Cloud Armor audit logs, primarily track administrative actions (like policy updates) and won't provide the granular request-level details needed to diagnose blocked traffic due to specific WAF rules. Therefore, HTTP(S) Load Balancing logs with a sampling rate of 1 is the most appropriate method for diagnosing the issue.

Authoritative Links:

Google Cloud Load Balancing Logging: <https://cloud.google.com/load-balancing/docs/logging>

Google Cloud Armor logging: <https://cloud.google.com/armor/docs/logging>

Question: 108

CertyIQ

You are the Organization Admin for your company. One of your engineers is responsible for setting up multiple host projects across multiple folders and sharing subnets with service projects. You need to enable the engineer's Identity and Access Management (IAM) configuration to complete their task in the fewest number of steps. What should you do?

A.Set up the engineer with Compute Shared VPC Admin IAM role at the folder level.

B.Set up the engineer with Compute Shared VPC Admin IAM role at the organization level.

C.Set up the engineer with Compute Shared VPC Admin IAM role and Project IAM Admin role at the folder level.

D.Set up the engineer with Compute Shared VPC Admin IAM role and Project IAM Admin role at the organization level.

Answer: B

Explanation:

The correct answer is **B. Set up the engineer with Compute Shared VPC Admin IAM role at the organization level.**

Here's a detailed justification:

The scenario requires an engineer to manage Shared VPC across multiple folders and projects. Shared VPC

allows resources from service projects to use a network defined in a host project. The Compute Shared VPC Admin role is specifically designed for managing Shared VPC configurations, including enabling host projects and attaching service projects.

Granting this role at the **organization level (B)** is the most efficient approach as it allows the engineer to perform Shared VPC tasks across all folders and projects within the organization without needing separate permissions at each folder level. This adheres to the principle of least privilege while still enabling the required actions.

Option A, setting the role at the folder level, would necessitate granting the role multiple times for each folder the engineer needs to manage, which is less efficient and more cumbersome. Options C and D include the Project IAM Admin role, which is unnecessary for the engineer's described task. The engineer does not require the ability to manage IAM policies on individual projects, they only require the ability to manage the shared vpc itself. This would be an example of over privileging, which should be avoided. Granting permissions at the organization level (B) allows the engineer to control all Shared VPC configurations throughout the organization, streamlining their work, and following the principle of least privilege by only granting permissions needed for the task.

Authoritative Links:

Shared VPC Roles: <https://cloud.google.com/vpc/docs/shared-vpc#roles>

IAM best practices: <https://cloud.google.com/iam/docs/best-practices>

Understanding IAM: <https://cloud.google.com/iam/docs/overview>

Question: 109

CertyIQ

You recently deployed Compute Engine instances in regions us-west1 and us-east1 in a Virtual Private Cloud (VPC) with default routing configurations. Your company security policy mandates that virtual machines (VMs) must not have public IP addresses attached to them. You need to allow your instances to fetch updates from the internet while preventing external access. What should you do?

- A. Create a Cloud NAT gateway and Cloud Router in both us-west1 and us-east1.
- B. Create a single global Cloud NAT gateway and global Cloud Router in the VPC.
- C. Change the instances' network interface external IP address from None to Ephemeral.
- D. Create a firewall rule that allows egress to destination 0.0.0.0/0.

Answer: A

Explanation:

Here's a breakdown of why option A is the correct solution and why the other options are incorrect:

Justification for Option A (Create a Cloud NAT gateway and Cloud Router in both us-west1 and us-east1):

Cloud NAT (Network Address Translation) allows Compute Engine instances without public IP addresses to access the internet. It works by translating the private IP addresses of your VMs to a public IP address provided by Cloud NAT, enabling outbound connections. Critically, it does not allow inbound connections originating from the internet to the VMs. Because the requirement is that VMs in both us-west1 and us-east1 need internet access, a Cloud NAT gateway, paired with a Cloud Router, is needed in each region. Cloud Router is essential for Cloud NAT to function; it dynamically manages the routing of traffic. Cloud NAT is not global; it is a regional service, thus requiring an independent implementation in each relevant region. This option fulfills the security policy requirement of no public IP addresses and allows for internet access for software updates.

Why other options are incorrect:

B. Create a single global Cloud NAT gateway and global Cloud Router in the VPC: Cloud NAT is a regional service, not global. Therefore, there is no such thing as a global Cloud NAT gateway or Cloud Router. This option is fundamentally incorrect.

C. Change the instances' network interface external IP address from None to Ephemeral: Assigning an ephemeral (or static) public IP address to the VM interface directly violates the security policy stating that VMs must not have public IP addresses. This option fails to fulfill a key requirement.

D. Create a firewall rule that allows egress to destination 0.0.0.0/0: While a firewall rule allowing egress to 0.0.0.0/0 is necessary, it doesn't, by itself, provide the VMs a way to access the internet when they don't have public IP addresses. It just permits the traffic if it can somehow get there. This option doesn't solve the core issue of private VMs needing internet access, thus requiring Cloud NAT.

Authoritative Links:

Cloud NAT Overview: <https://cloud.google.com/nat/docs/overview>

Cloud NAT Concepts: <https://cloud.google.com/nat/docs/concepts>

Cloud Router Overview: <https://cloud.google.com/router/docs/overview>

Compute Engine Network Overview: <https://cloud.google.com/compute/docs/networks-and-firewall>

Question: 110

CertyIQ

You are designing a new global application using Compute Engine instances that will be exposed by a global HTTP(S) load balancer. You need to secure your application from distributed denial-of-service and application layer (layer 7) attacks. What should you do?

- A. Configure VPC Service Controls and create a secure perimeter. Define fine-grained perimeter controls and enforce that security posture across your Google Cloud services and projects.
- B. Configure a Google Cloud Armor security policy in your project, and attach it to the backend service to secure the application.
- C. Configure VPC firewall rules to protect the Compute Engine instances against distributed denial-of-service attacks.
- D. Configure hierarchical firewall rules for the global HTTP(S) load balancer public IP address at the organization level.

Answer: B

Explanation:

The correct answer is **B. Configure a Google Cloud Armor security policy in your project, and attach it to the backend service to secure the application.**

Here's why:

Google Cloud Armor is a Web Application Firewall (WAF) service specifically designed to protect applications exposed through Google Cloud load balancers (including HTTP(S) load balancers) from various threats. It offers features like:

DDoS protection: Cloud Armor mitigates distributed denial-of-service attacks by identifying and blocking malicious traffic patterns.

Application Layer (L7) Protection: It provides rules to defend against common web attacks like SQL injection, cross-site scripting (XSS), and other OWASP Top 10 vulnerabilities.

Custom Rules: You can create custom rules based on specific traffic characteristics, allowing you to tailor security to your application's needs.

Integration with Load Balancers: Cloud Armor policies are directly attached to the backend service of your HTTP(S) load balancer, ensuring that traffic is inspected before reaching your application servers.

Option A, VPC Service Controls, primarily focuses on creating a secure perimeter around Google Cloud resources and data by restricting access based on identity and network location. It doesn't provide the granular, application-layer protection that's required for web applications. While important for overall security, it's not the primary solution for the specific threat model.

Option C, VPC firewall rules, operates at the network layer (L3/L4) and primarily controls traffic based on IP addresses, protocols, and ports. While useful for general network security, they are insufficient for mitigating sophisticated L7 attacks that target application logic and request parameters.

Option D, hierarchical firewall rules for the load balancer IP at the organization level, is also inadequate. Firewall rules are primarily for controlling network traffic and are not designed to inspect the application-layer traffic that Google Cloud Armor is designed to handle. They can not provide the necessary protection against attacks at the layer 7.

Therefore, configuring a Google Cloud Armor security policy is the most appropriate solution for securing the application against DDoS and application-layer attacks when using a global HTTP(S) load balancer.

Authoritative Links:

Google Cloud Armor Overview: <https://cloud.google.com/armor/docs/overview>

Google Cloud Armor Security Policies: <https://cloud.google.com/armor/docs/security-policy-concepts>

VPC Service Controls: <https://cloud.google.com/vpc-service-controls/docs>

VPC Firewall Rules: <https://cloud.google.com/vpc/docs/firewalls>

Question: 111

CertyIQ

Your organization's security policy requires that all internet-bound traffic return to your on-premises data center through HA VPN tunnels before egressing to the internet, while allowing virtual machines (VMs) to leverage private Google APIs using private virtual IP addresses 199.36.153.4/30. You need to configure the routes to enable these traffic flows. What should you do?

- A. Configure a custom route 0.0.0.0/0 with a priority of 500 whose next hop is the default internet gateway. Configure another custom route 199.36.153.4/30 with priority of 1000 whose next hop is the VPN tunnel back to the on-premises data center.
- B. Configure a custom route 0.0.0.0/0 with a priority of 1000 whose next hop is the internet gateway. Configure another custom route 199.36.153.4/30 with a priority of 500 whose next hop is the VPN tunnel back to the on-premises data center.
- C. Announce a 0.0.0.0/0 route from your on-premises router with a MED of 1000. Configure a custom route 199.36.153.4/30 with a priority of 1000 whose next hop is the default internet gateway.
- D. Announce a 0.0.0.0/0 route from your on-premises router with a MED of 500. Configure another custom route 199.36.153.4/30 with a priority of 1000 whose next hop is the VPN tunnel back to the on-premises data center.

Answer: C

Explanation:

The correct answer is C. This solution effectively routes internet-bound traffic through the on-premises data center while allowing access to private Google APIs via their respective virtual IP addresses. Option C leverages Border Gateway Protocol (BGP) route advertisement for the 0.0.0.0/0 (default) route from the on-premises router. By assigning a lower MED (Multi-Exit Discriminator) of 1000, we ensure that this route is preferred by Google Cloud's routing system. This forces all outbound internet traffic from Google Cloud to be directed towards the on-premises VPN tunnel. Simultaneously, a custom static route is configured within Google Cloud for the private Google API address range (199.36.153.4/30). This route has a priority of 1000 and specifies the default internet gateway as the next hop. Therefore, API traffic will directly reach Google via the public internet instead of the VPN tunnel. Options A and B are incorrect because they incorrectly prioritize routes. A assigns a higher priority to the internet gateway for 0.0.0.0/0 and B does it for the private API range,

which would lead to traffic going directly to the internet. Option D fails because it does not use the correct next hop for the private api range and also implies routing all 199.36.153.4/30 through the VPN tunnel. Using BGP from the on-premises router with the correct MED gives preference to the on-prem tunnel.

Relevant concepts include:

BGP (Border Gateway Protocol): A routing protocol used to exchange routing information between different networks, including on-premises and cloud networks. <https://cloud.google.com/network-connectivity/docs/router/concepts/bgp>

MED (Multi-Exit Discriminator): An attribute used by BGP to influence the path selection. Lower MED values indicate a more preferred path.

Custom Routes: Manually defined routes within Google Cloud VPC that override the default routing behavior. <https://cloud.google.com/vpc/docs/routes>

Default Internet Gateway: Google Cloud's gateway for connecting resources to the internet.

HA VPN: A Google Cloud service for creating highly available VPN tunnels. <https://cloud.google.com/network-connectivity/docs/vpn/concepts/overview>

Private Google Access: Allows Google Cloud resources to reach Google APIs privately. <https://cloud.google.com/vpc/docs/private-access-options>

By combining BGP route advertisements and custom static routes, you can achieve the desired traffic routing according to the organization's security requirements.

Question: 112

CertyIQ

Your company has defined a resource hierarchy that includes a parent folder with subfolders for each department. Each department defines their respective project and VPC in the assigned folder and has the appropriate permissions to create Google Cloud firewall rules. The VPCs should not allow traffic to flow between them. You need to block all traffic from any source, including other VPCs, and delegate only the intra-VPC firewall rules to the respective departments. What should you do?

- A. Create a VPC firewall rule in each VPC to block traffic from any source, with priority 0.
- B. Create a VPC firewall rule in each VPC to block traffic from any source, with priority 1000.
- C. Create two hierarchical firewall policies per department's folder with two rules in each: a high-priority rule that matches traffic from the private CIDRs assigned to the respective VPC and sets the action to allow, and another lower-priority rule that blocks traffic from any other source.
- D. Create two hierarchical firewall policies per department's folder with two rules in each: a high-priority rule that matches traffic from the private CIDRs assigned to the respective VPC and sets the action to `goto_next`, and another lower-priority rule that blocks traffic from any other source.

Answer: D

Explanation:

Okay, let's break down why option D is the correct solution for this scenario, focusing on hierarchical firewall policies and the `goto_next` action.

The core requirement is to isolate department VPCs, preventing cross-VPC traffic while allowing departments control over intra-VPC firewall rules. Standard VPC firewall rules, as in options A and B, are defined at the VPC level and cannot effectively enforce a company-wide policy of isolation at the folder level, nor do they offer the control needed for allowing intra-VPC communication per department. They can only block traffic with a specified priority.

Hierarchical firewall policies, on the other hand, are designed to be applied at the folder or organization level, offering a centralized approach to defining and enforcing policies across multiple VPCs within that scope. This directly addresses the need for organizational control over VPC isolation.

Option C is closer to correct than A or B as it utilizes hierarchical firewall policies, however, it introduces a problem where each department will ultimately allow any traffic originating from their own CIDR, not just the traffic within their VPC. There needs to be another lower priority rule to block the rest, but this is not enough.

Option D correctly implements a two-rule hierarchical policy approach:

1. **High-Priority "goto_next" Rule:** This rule, applied at each department's folder, matches traffic from the private CIDRs assigned to that specific VPC. Importantly, it uses the `goto_next` action rather than `allow`. `goto_next` doesn't permit traffic but passes the firewall rule evaluation to the next policy or VPC rule. This allows the department specific firewall rules to be considered for the matching traffic, but only if the traffic originates within the same VPC.
2. **Lower-Priority Block Rule:** This rule catches any traffic not matched by the high priority rule and blocks it. Since the high-priority rule only matches traffic originating from within the VPC's CIDR, this effectively blocks all cross-VPC traffic as well as traffic coming from outside the VPC's CIDR range. This rule is crucial for preventing cross-VPC communication.

By employing `goto_next` and a lower-priority block policy rule, option D achieves the required isolation and delegation of control. The organizational-level policy establishes the cross-VPC block, and the departments retain control over intra-VPC rules, as those rules are evaluated because of the use of `goto_next`.

Authoritative Links:

Hierarchical Firewall Policies: <https://cloud.google.com/vpc/docs/firewall-policies>

goto_next Action: https://cloud.google.com/vpc/docs/firewall-policies#rule_evaluation

Question: 113

CertyIQ

You have two Google Cloud projects in a perimeter to prevent data exfiltration. You need to move a third project inside the perimeter; however, the move could negatively impact the existing environment. You need to validate the impact of the change. What should you do?

- A. Enable Firewall Rules Logging inside the third project.
- B. Modify the existing VPC Service Controls policy to include the new project in dry run mode.
- C. Monitor the Resource Manager audit logs inside the perimeter.
- D. Enable VPC Flow Logs inside the third project, and monitor the logs for negative impact.

Answer: B

Explanation:

The correct answer is **B. Modify the existing VPC Service Controls policy to include the new project in dry run mode.**

Here's why:

VPC Service Controls (VPC SC) is a critical tool for establishing security perimeters around Google Cloud resources, preventing data exfiltration by limiting access to services within the perimeter. When adding a new project, especially when potential impacts are a concern, the 'dry run' mode offers a crucial safety net. It allows you to observe how the new project would interact with the current VPC SC policy without actually enforcing the policy on that project.

Option A, enabling Firewall Rules Logging, is helpful for monitoring network traffic but doesn't address the perimeter's impact. Similarly, Option D, enabling VPC Flow Logs, while useful for flow analysis, also doesn't proactively test the impact of adding a project to the VPC SC perimeter. Option C, monitoring Resource Manager audit logs, primarily tracks changes to resources and won't highlight potential conflicts or

unintended access blocks caused by the expanded perimeter.

By modifying the existing VPC Service Controls policy to include the new project in dry-run mode, you get detailed insights into what actions would be blocked or allowed if the policy were fully enforced. This includes logs showing violations and allowed access based on the modified policy. Dry run enables validation of the new project's interaction with the existing perimeter setup. This is vital because misconfiguration can lead to application downtime and significant operational disruptions. After validating the changes, the dry-run can be removed from the policy to fully enforce the change.

Relevant Cloud Computing Concepts:

VPC Service Controls (VPC SC): A security feature that enables you to establish a perimeter around your Google Cloud resources to prevent unauthorized access and data exfiltration.

Dry Run Mode: A feature in many security tools that allows you to test the effect of a policy change without actually enforcing it. This helps understand potential impacts before implementation.

Security Perimeters: Boundaries established to control access to sensitive resources.

Authoritative Links:

[VPC Service Controls Documentation](#): Official documentation for VPC Service Controls on Google Cloud.

[VPC Service Controls dry run mode](#): Specific documentation on using the dry-run mode.

Question: 114

CertyIQ

You are configuring an HA VPN connection between your Virtual Private Cloud (VPC) and on-premises network. The VPN gateway is named VPN_GATEWAY_1. You need to restrict VPN tunnels created in the project to only connect to your on-premises VPN public IP address: 203.0.113.1/32. What should you do?

- A. Configure a firewall rule accepting 203.0.113.1/32, and set a target tag equal to VPN_GATEWAY_1.
- B. Configure the Resource Manager constraint `constraints/compute.restrictVpnPeerIPs` to use an `allowList` consisting of only the 203.0.113.1/32 address.
- C. Configure a Google Cloud Armor security policy, and create a policy rule to allow 203.0.113.1/32.
- D. Configure an access control list on the peer VPN gateway to deny all traffic except 203.0.113.1/32, and attach it to the primary external interface.

Answer: B

Explanation:

The correct answer is **B. Configure the Resource Manager constraint `constraints/compute.restrictVpnPeerIPs` to use an `allowList` consisting of only the 203.0.113.1/32 address.**

This is the recommended approach because it directly addresses the requirement of restricting VPN tunnel creation within the project. The `constraints/compute.restrictVpnPeerIPs` constraint is a powerful tool within Google Cloud's Resource Manager service specifically designed for governing the creation of VPN tunnels. By setting this constraint with an `allowList` containing only the desired on-premises IP (203.0.113.1/32), you enforce that only VPN tunnels connecting to this specific IP are permitted. This constraint operates at the organization, folder, or project level, providing centralized control over resource creation. This approach ensures that even if users attempt to configure tunnels to other external IPs, the constraint will block the creation process, preventing unwanted connections and bolstering security. This is the most efficient and granular method to enforce this specific policy. Options A, C, and D address different aspects of networking or security but don't directly restrict VPN tunnel creation based on peer IP addresses within the context of Google Cloud. Option A is for firewall rules for instances; option C is for protecting web services. Option D, while valid, is about controlling traffic at the on-premises VPN gateway and is not within the scope of Google Cloud configuration.

Further research:

Resource Manager Constraints: <https://cloud.google.com/resource-manager/docs/organization-policy/constraints-list>

compute.restrictVpnPeerIPs Constraint: Specifically search for this constraint in the above list

Cloud VPN Overview: <https://cloud.google.com/vpn/docs/concepts/overview>

Question: 115

CertyIQ

Your company has recently installed a Cloud VPN tunnel between your on-premises data center and your Google Cloud Virtual Private Cloud (VPC). You need to configure access to the Cloud Functions API for your on-premises servers. The configuration must meet the following requirements:

- Certain data must stay in the project where it is stored and not be exfiltrated to other projects.
- Traffic from servers in your data center with RFC 1918 addresses do not use the internet to access Google Cloud APIs.
- All DNS resolution must be done on-premises.
- The solution should only provide access to APIs that are compatible with VPC Service Controls.

What should you do?

- A.1. Create an A record for private.googleapis.com using the 199.36.153.8/30 address range.
2. Create a CNAME record for *.googleapis.com that points to the A record.
3. Configure your on-premises routers to use the Cloud VPN tunnel as the next hop for the addresses you used in the A record.
4. Remove the default internet gateway from the VPC where your Cloud VPN tunnel terminates.
- B.1. Create an A record for restricted.googleapis.com using the 199.36.153.4/30 address range.
2. Create a CNAME record for *.googleapis.com that points to the A record.
3. Configure your on-premises routers to use the Cloud VPN tunnel as the next hop for the addresses you used in the A record.
4. Configure your on-premises firewalls to allow traffic to the restricted.googleapis.com addresses.
- C.1. Create an A record for restricted.googleapis.com using the 199.36.153.4/30 address range.
2. Create a CNAME record for *.googleapis.com that points to the A record.
3. Configure your on-premises routers to use the Cloud VPN tunnel as the next hop for the addresses you used in the A record.
4. Remove the default internet gateway from the VPC where your Cloud VPN tunnel terminates.
- D.1. Create an A record for private.googleapis.com using the 199.36.153.8/30 address range.
2. Create a CNAME record for *.googleapis.com that points to the A record.
3. Configure your on-premises routers to use the Cloud VPN tunnel as the next hop for the addresses you used in the A record.
4. Configure your on-premises firewalls to allow traffic to the private.googleapis.com addresses.

Answer: B

Explanation:

Here's a detailed justification for why option B is the correct answer:

Option B correctly addresses all the requirements laid out in the scenario. It leverages restricted.googleapis.com, which is specifically designed for private API access compatible with VPC Service Controls. This directly satisfies the requirement that access should be limited to VPC Service Controls compatible APIs and prevent data exfiltration as the traffic is destined to the restricted IP range. The use of the 199.36.153.4/30 address range is correct for this purpose and aligns with Google's documented private access configurations. By creating an A record for restricted.googleapis.com and a CNAME record pointing all API requests (*.googleapis.com) to the restricted domain, all API traffic gets directed to this private endpoint. Configuring on-premises routers to use the VPN tunnel as the next hop for traffic to the restricted IP range ensures that RFC 1918 addresses from the data center do not use the public internet to access Google Cloud APIs as the route is confined to the VPN tunnel. Lastly, configuring on-premises firewalls to allow traffic to

restricted.googleapis.com ensures the network path has no blocks.

Option A is incorrect because it uses private.googleapis.com which is intended for private access within Google's network, and might not provide the required level of VPC Service Controls support. Option C is incorrect because while it correctly uses restricted.googleapis.com, it incorrectly removes the internet gateway from the VPC. This is not required and the question does not require modification to VPC default network settings. Option D is incorrect because it also uses private.googleapis.com and adds firewall rules which is not necessary. The most important consideration to satisfy the question's requirements is to ensure all API traffic is directed to restricted.googleapis.com which provides the necessary private API access.

Key Concepts:

restricted.googleapis.com: A Google API endpoint designed for private access, compatible with VPC Service Controls.

VPC Service Controls: A Google Cloud feature that allows you to define security perimeters around your resources, limiting data exfiltration.

RFC 1918 Addresses: Private IP addresses used within private networks, such as data centers.

CNAME Record: A DNS record that creates an alias of one domain name to another domain name.

A Record: A DNS record that maps a hostname to an IP address.

Authoritative Links:

Private Google Access Options: <https://cloud.google.com/vpc/docs/private-access-options>

Using Private Google Access: <https://cloud.google.com/vpc/docs/configure-private-google-access>

VPC Service Controls: <https://cloud.google.com/vpc-service-controls/docs>

Question: 116

CertyIQ

You need to configure a Google Kubernetes Engine (GKE) cluster. The initial deployment should have 5 nodes with the potential to scale to 10 nodes. The maximum number of Pods per node is 8. The number of services could grow from 100 to up to 1024. How should you design the IP schema to optimally meet this requirement?

- A. Configure a /28 primary IP address range for the node IP addresses. Configure a /25 secondary IP range for the Pods. Configure a /22 secondary IP range for the Services.
- B. Configure a /28 primary IP address range for the node IP addresses. Configure a /25 secondary IP range for the Pods. Configure a /21 secondary IP range for the Services.
- C. Configure a /28 primary IP address range for the node IP addresses. Configure a /28 secondary IP range for the Pods. Configure a /21 secondary IP range for the Services.
- D. Configure a /28 primary IP address range for the node IP addresses. Configure a /24 secondary IP range for the Pods. Configure a /22 secondary IP range for the Services.

Answer: D

Explanation:

Here's a breakdown of why option D is the correct answer, along with supporting concepts:

Node IP Addresses: A /28 subnet provides 16 IP addresses, 13 of which are usable. Given the requirement of 5 nodes initially scaling to 10, a /28 subnet provides sufficient addresses with some room for future growth, which is why it is used for Node IP Addresses.

Pod IP Addresses: Each node can have a maximum of 8 pods. At a maximum of 10 nodes, that could be a total of 80 Pods. The key reason a /24 subnet is selected is because a /24 subnet gives us 256 IP addresses, and Kubernetes is using a concept called IP-per-pod, which means that each pod is given a unique IP address. 80 IP addresses are needed at a maximum to handle all of the Pods, meaning a /24 is chosen because it has

ample space for future growth and prevents IP exhaustion. A /25 would give us 128 addresses, which while sufficient for this use case does not leave ample room for growth compared to the chosen /24 subnet.

Service IP Addresses: Services in Kubernetes use ClusterIPs, which are internal IP addresses. The question specified a growth from 100 to 1024 services. A /22 subnet provides 1024 usable IPs which is exactly the range we need for our services. This number of IP addresses is required because each service needs a unique cluster IP. **Why other options are incorrect:**

A, B, and C: all fail to adequately handle the pod subnet. Specifically, they choose either a /25 or a /28 for pods, both of which are too small to handle the maximum pods across the 10 nodes, with room for future growth.

Key Concepts:

Subnetting: Dividing IP networks into smaller, more manageable subnets to efficiently allocate IP addresses.

CIDR Notation (e.g., /28, /24): Used to specify the size of an IP address range.

IP-per-pod: A GKE configuration where each pod has its own unique IP address within the cluster.

Kubernetes Services: A way to expose applications within a Kubernetes cluster via a virtual IP address.

Authoritative Links:

Google Cloud VPC Subnets: <https://cloud.google.com/vpc/docs/vpc>

GKE Networking Overview: <https://cloud.google.com/kubernetes-engine/docs/concepts/networking>

Kubernetes IP Addresses: <https://kubernetes.io/docs/concepts/services-networking/service/>

GKE IP-per-Pod : <https://cloud.google.com/kubernetes-engine/docs/how-to/alias-ips>

In summary, Option D's subnet choices for nodes, pods, and services provide the optimal balance of current requirements with space for future growth, adhering to the IP-per-pod model used by GKE and providing sufficient addresses for the services being used.

Question: 117

CertyIQ

You are migrating a three-tier application architecture from on-premises to Google Cloud. As a first step in the migration, you want to create a new Virtual Private Cloud (VPC) with an external HTTP(S) load balancer. This load balancer will forward traffic back to the on-premises compute resources that run the presentation tier. You need to stop malicious traffic from entering your VPC and consuming resources at the edge, so you must configure this policy to filter IP addresses and stop cross-site scripting (XSS) attacks. What should you do?

- A. Create a Google Cloud Armor policy, and apply it to a backend service that uses an unmanaged instance group backend.
- B. Create a hierarchical firewall ruleset, and apply it to the VPC's parent organization resource node.
- C. Create a Google Cloud Armor policy, and apply it to a backend service that uses an internet network endpoint group (NEG) backend.
- D. Create a VPC firewall ruleset, and apply it to all instances in unmanaged instance groups.

Answer: C

Explanation:

The correct answer is **C. Create a Google Cloud Armor policy, and apply it to a backend service that uses an internet network endpoint group (NEG) backend.**

Here's why:

The scenario requires a way to filter malicious traffic at the edge of the Google Cloud VPC, specifically targeting IP addresses and preventing XSS attacks, before it reaches the on-premises presentation tier.

Google Cloud Armor is specifically designed for this purpose. It's a Web Application Firewall (WAF) service that provides defense against various web-based attacks.

Option A is incorrect because while Cloud Armor can be applied to a backend service, unmanaged instance groups do not represent a connection point to on-premises resources. An unmanaged instance group is for managing compute instances within Google Cloud.

Option B is incorrect because while hierarchical firewall rules are useful for managing access control within a VPC organization, they do not provide the specific filtering capabilities needed for web application security like IP allow/deny lists and WAF rules. They are more focused on network layer access control.

Option D is incorrect because VPC firewall rules operate at the instance level and are not suitable for filtering malicious HTTP/S traffic at the edge of the network. They also lack the specific application security features like XSS protection that Cloud Armor offers.

Option C correctly leverages the capabilities of Google Cloud Armor. When applying Cloud Armor, we must attach it to a backend service of the load balancer. The crucial point here is that we're routing traffic to on-premises resources. For this, we use an internet network endpoint group (NEG) as the backend. An internet NEG allows you to direct traffic from the Google Cloud load balancer to endpoints located outside of Google Cloud, like your on-premises presentation tier. By applying the Cloud Armor policy to this backend service, all traffic directed towards the on-premises resources first passes through the WAF layer, which will filter malicious IP addresses and protect against XSS attacks as per the requirement. Cloud Armor offers features such as predefined rules, custom rules, and IP allow/deny lists for fine-grained control.

Authoritative Links:

Google Cloud Armor: <https://cloud.google.com/armor>

Network Endpoint Groups (NEGs): <https://cloud.google.com/load-balancing/docs/negs>

Internet NEGs: <https://cloud.google.com/load-balancing/docs/negs/internet-negs>

Question: 118

CertyIQ

You just finished your company's migration to Google Cloud and configured an architecture with 3 Virtual Private Cloud (VPC) networks: one for Sales, one for Finance, and one for Engineering. Every VPC contains over 100 Compute Engine instances, and now developers using instances in the Sales VPC and the Finance VPC require private connectivity between each other. You need to allow communication between Sales and Finance without compromising performance or security. What should you do?

- A. Configure an HA VPN gateway between the Finance VPC and the Sales VPC.
- B. Configure the instances that require communication between each other with an external IP address.
- C. Create a VPC Network Peering connection between the Finance VPC and the Sales VPC.
- D. Configure Cloud NAT and a Cloud Router in the Sales and Finance VPCs.

Answer: C

Explanation:

The correct answer is C: Create a VPC Network Peering connection between the Finance VPC and the Sales VPC.

VPC Network Peering enables private connectivity between two VPC networks, regardless of whether they belong to the same project or organization. It allows for direct IP address communication within the RFC 1918 private IP address space, avoiding the need to traverse the public internet. This approach directly satisfies the requirement for private connectivity between the Sales and Finance VPCs without compromising performance or security.

Option A, configuring an HA VPN gateway, is typically used for connecting on-premises networks to Google

Cloud or connecting two VPCs over the public internet. While secure, it introduces latency and complexity that aren't needed for internal VPC communication within Google Cloud. Option B, using external IP addresses, directly violates the security requirement by exposing instances to the public internet. This would also negate the benefits of having separate VPCs for security. Option D, using Cloud NAT, is designed to enable instances without external IPs to reach the internet; it does not facilitate communication between VPCs.

VPC Network Peering offers superior performance compared to VPN since there are no gateway overheads and traffic travels over Google's internal network. It's also more secure than exposing instances with public IP addresses. Furthermore, since peering uses private IP addressing, it follows the principle of least privilege, only permitting communication that is required and avoiding the need for complex network address translation rules.

Authoritative links:

Google Cloud VPC Network Peering Documentation: <https://cloud.google.com/vpc/docs/vpc-peering>

Google Cloud VPC Overview: <https://cloud.google.com/vpc/docs/overview>

Question: 119

CertyIQ

You have provisioned a Partner Interconnect connection to extend connectivity from your on-premises data center to Google Cloud. You need to configure a Cloud Router and create a VLAN attachment to connect to resources inside your VPC. You need to configure an Autonomous System number (ASN) to use with the associated Cloud Router and create the VLAN attachment.

What should you do?

- A. Use a 4-byte private ASN 4200000000-4294967294.
- B. Use a 2-byte private ASN 64512-65535.
- C. Use a public Google ASN 15169.
- D. Use a public Google ASN 16550.

Answer: D

Explanation:

Here's a detailed justification for why option D is the correct choice, focusing on ASN requirements for Google Cloud Partner Interconnect:

When configuring a Cloud Router for a Partner Interconnect connection, you must specify a Border Gateway Protocol (BGP) Autonomous System Number (ASN) to establish routing adjacencies. Google Cloud mandates the use of a public ASN for Partner Interconnect connections, specifically in the range of 16550 to 65534. Option D (public ASN 16550) falls squarely within this permitted range. Options A and B, on the other hand, suggest private ASNs. While private ASNs are used in other networking scenarios, they are explicitly disallowed with Partner Interconnect. Using a private ASN for the Cloud Router would lead to BGP peering failures. Option C proposes a public ASN that is also not allowed for customer use (it is Google's ASN). It is essential to use an appropriate ASN from the Google-defined public range to ensure successful BGP session establishment. The ASN, along with the peering IP addresses, helps in identifying the peer routers and facilitates routing information exchange. The specific ASN 16550 is a valid option among the available Google Public ASNs. A correct ASN is mandatory for proper BGP setup and therefore, effective connectivity between your on-premises infrastructure and Google Cloud. Using a wrong ASN will result in the failure of the BGP peering and will not allow the VLAN attachment to function correctly.

Authoritative links for further research:

Google Cloud Documentation on Router ASN: <https://cloud.google.com/network-connectivity/docs/router/concepts/asn>

Google Cloud Documentation on Partner Interconnect: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/partner-interconnect>

RFC 1930, Guidelines for creation, selection, and registration of an Autonomous System (AS): <https://datatracker.ietf.org/doc/html/rfc1930> (Although RFC 1930 discusses ASNs generally, it does not specify Google's specific implementation for GCP, which is found in the links above).

Question: 120

CertyIQ

You are configuring a new application that will be exposed behind an external load balancer with both IPv4 and IPv6 addresses and support TCP pass-through on port 443. You will have backends in two regions: us-west1 and us-east1. You want to serve the content with the lowest possible latency while ensuring high availability and autoscaling. Which configuration should you use?

- A. Use global SSL Proxy Load Balancing with backends in both regions.
- B. Use global TCP Proxy Load Balancing with backends in both regions.
- C. Use global external HTTP(S) Load Balancing with backends in both regions.
- D. Use Network Load Balancing in both regions, and use DNS-based load balancing to direct traffic to the closest region.

Answer: B

Explanation:

The correct answer is B, using a global TCP Proxy Load Balancer with backends in both regions. Here's why:

The question specifies TCP pass-through on port 443, which is a key indicator for needing a TCP-based load balancer, not HTTP(S) or SSL. HTTP(S) load balancers terminate SSL/TLS connections at the load balancer itself, preventing TCP pass-through. Global load balancing is required for routing traffic to the closest backend (us-west1 or us-east1) for minimum latency, and ensures high availability across regions. Network Load Balancing (NLB), option D, operates at layer 4 and can provide TCP load balancing but does not inherently provide global load balancing. NLB in each region plus DNS based routing would require custom configurations and is less optimal than the built-in global capabilities of GCP load balancers. In addition, using DNS-based load balancing for latency can be inaccurate due to DNS caching and does not react to backend health as quickly or dynamically as a managed load balancer. TCP Proxy Load Balancing handles TCP connections directly. It does not terminate TLS, fulfilling the TCP pass-through requirement. Global TCP Proxy Load Balancing provides the flexibility to configure autoscaling and load balancing between backend resources spread across the two regions. This approach offers the desired low latency by directing users to the nearest healthy backend based on their location, leveraging Google's global network. Option A, SSL Proxy Load Balancing, would terminate SSL and therefore not support TCP passthrough, making it incorrect for the stated requirements. Option C is for HTTP(S) traffic and therefore not suitable.

Authoritative Links:

Google Cloud Load Balancing Overview: <https://cloud.google.com/load-balancing/docs/load-balancing-overview>

TCP Proxy Load Balancing: <https://cloud.google.com/load-balancing/docs/tcp/>

Choosing a Load Balancer: <https://cloud.google.com/load-balancing/docs/choosing-load-balancer>

Question: 121

CertyIQ

In your project my-project, you have two subnets in a Virtual Private Cloud (VPC): subnet-a with IP range

10.128.0.0/20 and subnet-b with IP range 172.16.0.0/24. You need to deploy database servers in subnet-a. You will also deploy the application servers and web servers in subnet-b. You want to configure firewall rules that only allow database traffic from the application servers to the database servers. What should you do?

A. Create network tag app-server and service account Add the tag to the application servers, and associate the service account with the database servers. Run the following command: `gcloud compute firewall-rules create app-db-firewall-rule \`

```
--action allow \
--direction ingress \
--rules top:3306 \
--source-tags app-server \
--target-service-accounts [email protected]
project.iam.gserviceaccount.com
```

B. Create service accounts and Associate service account sa-app with the application servers, and associate the service account sa-db with the database servers. Run the following command: `gcloud compute firewall-rules create app-db-firewall-rule`

```
--allow TCP:3306 \
--source-service-accounts [email protected]
demo.iam.gserviceaccount.com \
--target-service-accounts [email protected]
project.iam.gserviceaccount.com
```

C. Create service accounts and Associate the service account sa-app with the application servers, and associate the service account sa-db with the database servers. Run the following command: `gcloud compute firewall-rules create app-db-firewall-rule`

```
--allow TCP:3306 \
--source-ranges 10.128.0.0/20 \
--source-service-accounts [email protected]
project.iam.gserviceaccount.com \
--target-service-accounts [email protected]
project.iam.gserviceaccount.com
```

D. Create network tags app-server and db-server. Add the app-server tag to the application servers, and add the db-server tag to the database servers. Run the following command: `gcloud compute firewall-rules create app-db-firewall-rule \`

```
--action allow \
--direction ingress \
--rules tcp:3306 \
--source-ranges 10.128.0.0/20 \
--source-tags app-server \
--target-tags db-server
```

Answer: B

Explanation:

Here's a breakdown of why option B is the correct solution and why the others are not:

Justification for Option B:

Option B correctly leverages service accounts for granular firewall control, aligning with Google Cloud's security best practices. It creates two service accounts: sa-app for application servers and sa-db for database servers. By associating these service accounts with their respective servers, we can use them as identifiers in firewall rules. This ensures that only traffic originating from the application servers (identified by sa-app) is permitted to reach the database servers (identified by sa-db) on port 3306 (commonly used for MySQL). This approach is more secure than relying solely on IP ranges or network tags as it decouples the firewall rules from the underlying network configuration or the instance tags and instead relies on identity. This is a best practice and supports least privilege principles.

The command `gcloud compute firewall-rules create app-db-firewall-rule --allow TCP:3306 --source-service-accounts [email protected]demo.iam.gserviceaccount.com --target-service-accounts [email protected]project.iam.gserviceaccount.com` establishes an ingress firewall rule, allowing TCP traffic on port 3306 only from instances with service account sa-app to instances with service account sa-db.

Why other options are incorrect:

Option A: This option uses a combination of network tags and service accounts. While network tags can be used for firewall rules, they are not as secure as service accounts, as they can be applied or removed from a server without any authorization. Furthermore, the command does not specify TCP protocol. Instead, it refers to "top:3306". This is incorrect. Additionally, the service account is the one for the project, not for the database server itself, which should be independent.

Option C: While it uses service accounts correctly, it incorrectly combines service accounts and source IP ranges for access control. This introduces unnecessary complexity and defeats the purpose of using service accounts for finer-grained access control, plus is less secure. Source ranges are typically not recommended and should be avoided unless explicitly required.

Option D: This option uses network tags for both source and target, which is less secure and doesn't provide the identity-based access control as service accounts. Furthermore, the specified source range 10.128.0.0/20, which is the entire source subnet, would allow any server in that subnet to potentially reach the databases, defeating the purpose.

Key Concepts:

Service Accounts: Provide an identity for workloads rather than users, enabling secure access to Google Cloud resources.

Firewall Rules: Control network traffic into and out of virtual machine instances.

Ingress Rules: Apply to traffic coming into a virtual network.

Least Privilege: Granting only the necessary permissions for a task.

Authoritative Links:

Google Cloud VPC Firewall Rules: <https://cloud.google.com/vpc/docs/firewalls>

Google Cloud Service Accounts: <https://cloud.google.com/iam/docs/service-accounts>

Creating Firewall Rules with Service Accounts: <https://cloud.google.com/vpc/docs/firewalls#sa-firewall>

Question: 122

CertyIQ

You are planning a large application deployment in Google Cloud that includes on-premises connectivity. The application requires direct connectivity between workloads in all regions and on-premises locations without address translation, but all RFC 1918 ranges are already in use in the on-premises locations. What should you do?

- A. Use multiple VPC networks with a transit network using VPC Network Peering.
- B. Use overlapping RFC 1918 ranges with multiple isolated VPC networks.
- C. Use overlapping RFC 1918 ranges with multiple isolated VPC networks and Cloud NAT.
- D. Use non-RFC 1918 ranges with a single global VPC.

Answer: D

Explanation:

The correct solution is **D: Use non-RFC 1918 ranges with a single global VPC.**

Here's the justification:

The core requirement is direct connectivity without NAT between Google Cloud regions and on-premises, with a caveat that all RFC 1918 (private) IP ranges are already utilized on-premises. This means we cannot use the standard private IP ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) in our VPC network.

Option A, using multiple VPCs and peering, doesn't address the IP overlap problem. Peering wouldn't allow overlapping ranges, and still requires address space to connect all the VPCs. Option B introduces overlapping

RFC 1918 ranges, which would cause routing conflicts and make direct connectivity impossible without address translation. Option C also fails for the same reason as B. Furthermore, NAT will break the required direct connectivity requirement.

Option D elegantly resolves the issue by leveraging non-RFC 1918 ranges (publicly routable IP addresses) for the Google Cloud VPC. Because these ranges are not used on-premises, they eliminate the overlap. Utilizing a single global VPC ensures that the non-RFC 1918 addresses are globally accessible across all regions. This, combined with Cloud Interconnect or Cloud VPN, allows on-premises resources to reach Google Cloud workloads using their unique addresses without any NAT. Therefore, workloads in all regions and on-premises have direct, routable connectivity, fulfilling all requirements.

Key Concepts:

RFC 1918: Defines private IP address ranges.

VPC (Virtual Private Cloud): A logically isolated private network within Google Cloud.

Global VPC: A VPC that spans multiple regions, allowing seamless connectivity within it.

Non-RFC 1918 Addresses: Publicly routable IP addresses, essential for this scenario to avoid conflicts.

Cloud Interconnect/Cloud VPN: Services that establish a secure connection between on-premises locations and Google Cloud.

Authoritative Links:

VPC networks overview: <https://cloud.google.com/vpc/docs/vpc>

IP address ranges: <https://cloud.google.com/vpc/docs/vpc#ipranges>

Cloud Interconnect: <https://cloud.google.com/interconnect/docs/overview>

Cloud VPN: <https://cloud.google.com/vpn/docs/overview>

Question: 123

CertyIQ

Your company's security team wants to limit the type of inbound traffic that can reach your web servers to protect against security threats. You need to configure the firewall rules on the web servers within your Virtual Private Cloud (VPC) to handle HTTP and HTTPS web traffic for TCP only. What should you do?

- A. Create an allow on match ingress firewall rule with the target tag "web-server" to allow all IP addresses for TCP port 80.
- B. Create an allow on match egress firewall rule with the target tag "web-server" to allow all IP addresses for TCP port 80.
- C. Create an allow on match ingress firewall rule with the target tag "web-server" to allow all IP addresses for TCP ports 80 and 443.
- D. Create an allow on match egress firewall rule with the target tag "web-server" to allow web server IP addresses for TCP ports 80 and 443.

Answer: C

Explanation:

The correct answer is C because it accurately addresses the requirement of allowing only HTTP and HTTPS traffic (TCP ports 80 and 443) to web servers within a VPC. Let's break down why:

Ingress vs. Egress: The question specifies inbound traffic, therefore, an **ingress** rule is required. Egress rules control outbound traffic. Options B and D incorrectly use egress rules.

Allow Rule: Security best practices involve a default deny approach. Thus, we need to create an **allow** rule to explicitly permit the desired traffic.

Target Tag: The question implies that multiple web servers exist, meaning we can apply a tag (e.g., "web-

server") to all instances to simplify rule management. Applying the rule to tagged instances ensures that the firewall applies to all web servers simultaneously.

TCP Ports 80 and 443: HTTP (port 80) and HTTPS (port 443) are the standard ports for web traffic, as mentioned. To fulfill the requirement of handling both, the rule must allow traffic on both ports.

All IP Addresses: The question does not mention a specific source for allowed traffic. The rule allows all IP addresses. This means any client on the internet can access the server on port 80 and 443, which aligns with standard web server usage. This broad allowance might be restricted in a real-world scenario by limiting it to only specific sources or VPC.

Option A is incorrect because it only allows traffic on port 80, missing the requirement for HTTPS traffic on port 443. Options B and D are incorrect because they use egress rules, which control outbound traffic.

Authoritative links:

Google Cloud Firewall Rules: <https://cloud.google.com/vpc/docs/firewalls>

Firewall Rule Concepts: <https://cloud.google.com/vpc/docs/firewall-concepts>

Using Tags in Firewall Rules: <https://cloud.google.com/vpc/docs/firewalls#tags>

Question: 124

CertyIQ

You successfully provisioned a single Dedicated Interconnect. The physical connection is at a colocation facility closest to us-west2. Seventy-five percent of your workloads are in us-east4, and the remaining twenty-five percent of your workloads are in us-central1. All workloads have the same network traffic profile. You need to minimize data transfer costs when deploying VLAN attachments. What should you do?

- A. Keep the existing Dedicated interconnect. Deploy a VLAN attachment to a Cloud Router in us-west2, and use VPC global routing to access workloads in us-east4 and us-central1.
- B. Keep the existing Dedicated Interconnect. Deploy a VLAN attachment to a Cloud Router in us-east4, and deploy another VLAN attachment to a Cloud Router in us-central1.
- C. Order a new Dedicated Interconnect for a colocation facility closest to us-east4, and use VPC global routing to access workloads in us-central1.
- D. Order a new Dedicated Interconnect for a colocation facility closest to us-central1, and use VPC global routing to access workloads in us-east4.

Answer: B

Explanation:

The optimal solution for minimizing data transfer costs in this scenario is **B**. Here's why:

Dedicated Interconnect connections, while providing high bandwidth and low latency, incur egress data transfer costs when data traverses between regions. Choosing a single Dedicated Interconnect location and then accessing resources in other regions will increase cross-region traffic and costs. Since 75% of your workload is in us-east4 and only 25% in us-central1, having a VLAN attachment in the location of the largest workload, us-east4, is the most efficient start. Creating an additional VLAN attachment to a Cloud Router in us-central1 handles the remaining traffic without incurring long distance transfers. This distributes the ingress traffic efficiently from the on-premises location to the closest GCP locations for a more cost effective approach.

Option A is not optimal. While it uses a single Dedicated Interconnect, sending 100% of traffic through us-west2 and then relying on VPC global routing to reach us-east4 and us-central1 incurs cross-region egress costs for the majority (75%) of the traffic traveling to us-east4. Options C and D are costly as ordering additional Dedicated Interconnects significantly increases infrastructure costs when a single connection already exists. Furthermore, option C would be less cost efficient as the majority of the workload is in us-east4

and having the Dedicated Interconnect in us-east4 is the best option to reduce cross-region transfers, and the same logic goes for option D where the majority workload is not in us-central1.

Therefore, option B strategically places VLAN attachments in the regions where most of the resources reside, minimizing cross-region data transfer, thus reducing costs.

Authoritative links for further research:

Google Cloud Interconnect Overview: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/overview>

Google Cloud Interconnect Pricing: <https://cloud.google.com/network-connectivity/interconnect/pricing>

Google Cloud Network Topology: <https://cloud.google.com/vpc/docs/vpc>

Question: 125

CertyIQ

You are designing a hybrid cloud environment. Your Google Cloud environment is interconnected with your on-premises network using HA VPN and Cloud Router in a central transit hub VPC. The Cloud Router is configured with the default settings. Your on-premises DNS server is located at 192.168.20.88. You need to ensure that your Compute Engine resources in multiple spoke VPCs can resolve on-premises private hostnames using the domain corp.altostrat.com while also resolving Google Cloud hostnames. You want to follow Google-recommended practices. What should you do?

- A.1. Create a private forwarding zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com that points to 192.168.20.88. Associate the zone with the hub VPC.
- 2. Create a private peering zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com associated with the spoke VPCs, with the hub VPC as the target.
- 3. Set a custom route advertisement on the Cloud Router for 35.199.192.0/19.
- 4. Configure VPC peering in the spoke VPCs to peer with the hub VPC.
- B.1. Create a private forwarding zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com that points to 192.168.20.88.
- 2. Associate the zone with the hub VPC. Create a private peering zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com associated with the spoke VPCs, with the hub VPC as the target.
- 3. Set a custom route advertisement on the Cloud Router for 35.199.192.0/19.
- C.1. Create a private forwarding zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com that points to 192.168.20.88. Associate the zone with the hub VPC.
- 2. Create a private peering zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com associated with the spoke VPCs, with the hub VPC as the target.
- 3. Set a custom route advertisement on the Cloud Router for 35.199.192.0/19.
- 4. Create a hub-and-spoke VPN deployment in each spoke VPC to connect back to the on-premises network directly.
- D.1. Create a private forwarding zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com that points to 192.168.20.88. Associate the zone with the hub VPC.
- 2. Create a private peering zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com associated with the spoke VPCs, with the hub VPC as the target.
- 3. Set a custom route advertisement on the Cloud Router for 35.199.192.0/19.
- 4. Create a hub and spoke VPN deployment in each spoke VPC to connect back to the hub VPC.

Answer: A

Explanation:

The correct answer is **A**. Here's a breakdown of why and how it satisfies the requirements:

The goal is to enable Compute Engine instances in multiple spoke VPCs to resolve both on-premises private hostnames (under corp.altostrat.com) and Google Cloud hostnames, following best practices. Option A achieves this efficiently using Cloud DNS and VPC Peering.

1. **Private Forwarding Zone in Hub VPC:** Creating a private forwarding zone in Cloud DNS for corp.altostrat.com pointing to the on-premises DNS server (192.168.20.88) within the hub VPC allows

the hub to resolve on-prem hostnames. This centralizes DNS resolution for on-premises resources.
<https://cloud.google.com/dns/docs/zones/forwarding-zones>

2. **Private Peering Zone in Spoke VPCs:** By creating a private peering zone for corp.altostrat.com in each spoke VPC, pointing to the hub VPC, the spoke VPCs delegate DNS resolution for the corp.altostrat.com domain to the hub, where the forwarding zone exists. This eliminates the need for individual forwarding configurations in each spoke, promoting scalability and management.
<https://cloud.google.com/dns/docs/zones/peering-zones>
3. **Custom Route Advertisement:** The Cloud Router already automatically advertises the 35.199.192.0/19 range used for Cloud DNS resolver addresses. However, Explicitly advertising this is a good practice for robustness. If you have non-default routing setup. This ensures on-premises network knows how to route traffic to the Cloud DNS resolvers. <https://cloud.google.com/network-connectivity/docs/router/how-to/advertise-custom-ranges>
4. **VPC Peering:** Establishing VPC peering between spoke VPCs and the hub VPC is crucial for network communication between these environments. This allows the spoke VPCs to access the hub's private forwarding zone for on-premises host resolution, as well as utilize the shared hybrid network infrastructure. <https://cloud.google.com/vpc/docs/vpc-peering>

Option B misses the need for explicit VPC peering for proper communication between spoke and hub. Option C incorrectly suggests using hub-and-spoke VPNs when VPC peering is more efficient for private communication. Option D incorrectly suggests hub-and-spoke VPNs and repeats the same mistake with the custom route advertisement as C. They both suggest creating VPNs in a situation where they are not necessary and greatly increase complexity.

Question: 126

CertyIQ

You have the following firewall ruleset applied to all instances in your Virtual Private Cloud (VPC):

Direction	Action	Address range	Port	Priority
egress	deny	192.0.2.0/24	80	100
egress	deny	198.51.100.0/24	80	200
ingress	allow	203.0.113.0/24	80	300

You need to update the firewall rule to add the following rule to the ruleset:

Direction	Action	Address range	Port	Logging
egress	deny	192.0.2.42/32	80	true

You are using a new user account. You must assign the appropriate identity and Access Management (IAM) user roles to this new user account before updating the firewall rule. The new user account must be able to apply the update and view firewall logs. What should you do?

- A. Assign the compute.securityAdmin and logging.viewer rule to the new user account. Apply the new firewall rule with a priority of 50.
- B. Assign the compute.securityAdmin and logging.bucketWriter role to the new user account. Apply the new firewall rule with a priority of 150.
- C. Assign the compute.orgSecurityPolicyAdmin and logging.viewer role to the new user account. Apply the new firewall rule with a priority of 50.

D.Assign the compute.orgSecurityPolicyAdmin and logging.bucketWriter role to the new user account. Apply the new firewall rule with a priority of 150.

Answer: A

Explanation:

A. Assign the compute.securityAdmin and logging.viewer rule to the new user account. Apply the new firewall rule with a priority of 50.

<https://cloud.google.com/logging/docs/access-control>

Question: 127

CertyIQ

Your organization has a single project that contains multiple Virtual Private Clouds (VPCs). You need to secure API access to your Cloud Storage buckets and BigQuery datasets by allowing API access only from resources in your corporate public networks. What should you do?

- A.Create an access context policy that allows your VPC and corporate public network IP ranges, and then attach the policy to Cloud Storage and BigQuery.
- B.Create a VPC Service Controls perimeter for your project with an access context policy that allows your corporate public network IP ranges.
- C.Create a firewall rule to block API access to Cloud Storage and BigQuery from unauthorized networks.
- D.Create a VPC Service Controls perimeter for each VPC with an access context policy that allows your corporate public network IP ranges.

Answer: B

Explanation:

The correct answer is **B: Create a VPC Service Controls perimeter for your project with an access context policy that allows your corporate public network IP ranges.**

Here's why: VPC Service Controls (VPC SC) is designed to establish a security perimeter around your Google Cloud resources, specifically preventing unauthorized data exfiltration. This is crucial when you need to restrict access to sensitive services like Cloud Storage and BigQuery. Instead of relying solely on network-based firewalls, VPC SC offers a context-aware approach.

An access context policy, part of VPC SC, lets you define which requests are allowed based on various attributes such as IP address ranges, user identity, and device posture. In this scenario, you want to restrict access from your corporate public networks only. By applying this policy at the project level, you can establish a single security boundary that protects all resources within that project, regardless of which VPC they reside in. This is more manageable than creating perimeters for every VPC, as suggested in option D.

Option A, while seemingly related to access context policies, doesn't leverage the perimeter protection offered by VPC SC. The access context policy would not enforce a boundary to prevent unauthorized access. Option C, firewalls, manage network traffic flow; they are inadequate for controlling API access at the project level, especially with the context-aware criteria.

Thus, creating a VPC Service Controls perimeter at the project level with a corresponding access context policy limiting to your corporate public network IP ranges is the most effective way to secure API access to Cloud Storage and BigQuery in the specified scenario.

Authoritative Links:

VPC Service Controls Overview: <https://cloud.google.com/vpc-service-controls/docs/overview>

Question: 128

CertyIQ

Your company has provisioned 2000 virtual machines (VMs) in the private subnet of your Virtual Private Cloud (VPC) in the us-east1 region. You need to configure each VM to have a minimum of 128 TCP connections to a public repository so that users can download software updates and packages over the internet. You need to implement a Cloud NAT gateway so that the VMs are able to perform outbound NAT to the internet. You must ensure that all VMs can simultaneously connect to the public repository and download software updates and packages. Which two methods can you use to accomplish this? (Choose two.)

- A. Configure the NAT gateway in manual allocation mode, allocate 2 NAT IP addresses, and update the minimum number of ports per VM to 256.
- B. Create a second Cloud NAT gateway with the default minimum number of ports configured per VM to 64.
- C. Use the default Cloud NAT gateway's NAT proxy to dynamically scale using a single NAT IP address.
- D. Use the default Cloud NAT gateway to automatically scale to the required number of NAT IP addresses, and update the minimum number of ports per VM to 128.
- E. Configure the NAT gateway in manual allocation mode, allocate 4 NAT IP addresses, and update the minimum number of ports per VM to 128.

Answer: DE

Explanation:

Here's a detailed justification for why options D and E are the correct methods to enable 2000 VMs with 128 TCP connections each to a public repository using Cloud NAT:

Option D: Correct

Cloud NAT dynamically scales the number of NAT IP addresses as needed. By utilizing the default Cloud NAT gateway's automatic scaling, you eliminate the need to manually manage IP address allocations. Crucially, you must increase the minimum number of ports per VM to 128, which directly ensures that each VM can establish the required number of concurrent connections. The default of 64 ports is insufficient in this case. This dynamic scaling behavior, combined with the correctly configured minimum ports, addresses the core requirement of simultaneous connections.

Option E: Correct

Configuring the NAT gateway in manual allocation mode allows you to control the number of NAT IPs. For 2000 VMs requiring 128 simultaneous connections, 4 IP addresses are a good starting point. However, it's critical to also increase the minimum number of ports per VM to 128. This ensures each VM can establish its required 128 connections. While manual allocation requires more planning upfront, it provides a predictable resource pool for outbound connections.

Why other options are incorrect:

Option A: Allocating only 2 NAT IP addresses and setting 256 minimum ports per VM is insufficient. The number of NAT IPs is the primary resource that limits outbound concurrent connections. 2 IPs is highly likely to cause connection exhaustion under load of 2000 VMs, even with 256 ports.

Option B: Creating a second NAT gateway with default ports per VM is redundant and inefficient. While it adds capacity, it doesn't guarantee each VM will have 128 concurrent connections and does not resolve the issue at hand. You should adjust the port range on the initial gateway instead.

Option C: While the default Cloud NAT proxy does scale, using only a single NAT IP address will limit the amount of total concurrent connections. With 2000 VMs requiring 128 connections each, a single NAT IP won't scale to provide that many concurrent connections.

Cloud Computing Concepts:

Cloud NAT: Network Address Translation service in Google Cloud, enabling VMs in private subnets to connect to the internet.

NAT IP Address: Public IP address used to translate a private IP when communicating to the Internet

NAT Ports: Ports on a NAT IP address used for specific connections. Limiting the number of ports per VM will limit the concurrent connections it can hold

Automatic Scaling: A Cloud NAT feature that automatically allocates and deallocates NAT IPs based on usage.

Manual Allocation: A Cloud NAT feature where the user controls the number of NAT IPs.

Authoritative Links:

Cloud NAT Overview: <https://cloud.google.com/nat/docs/overview>

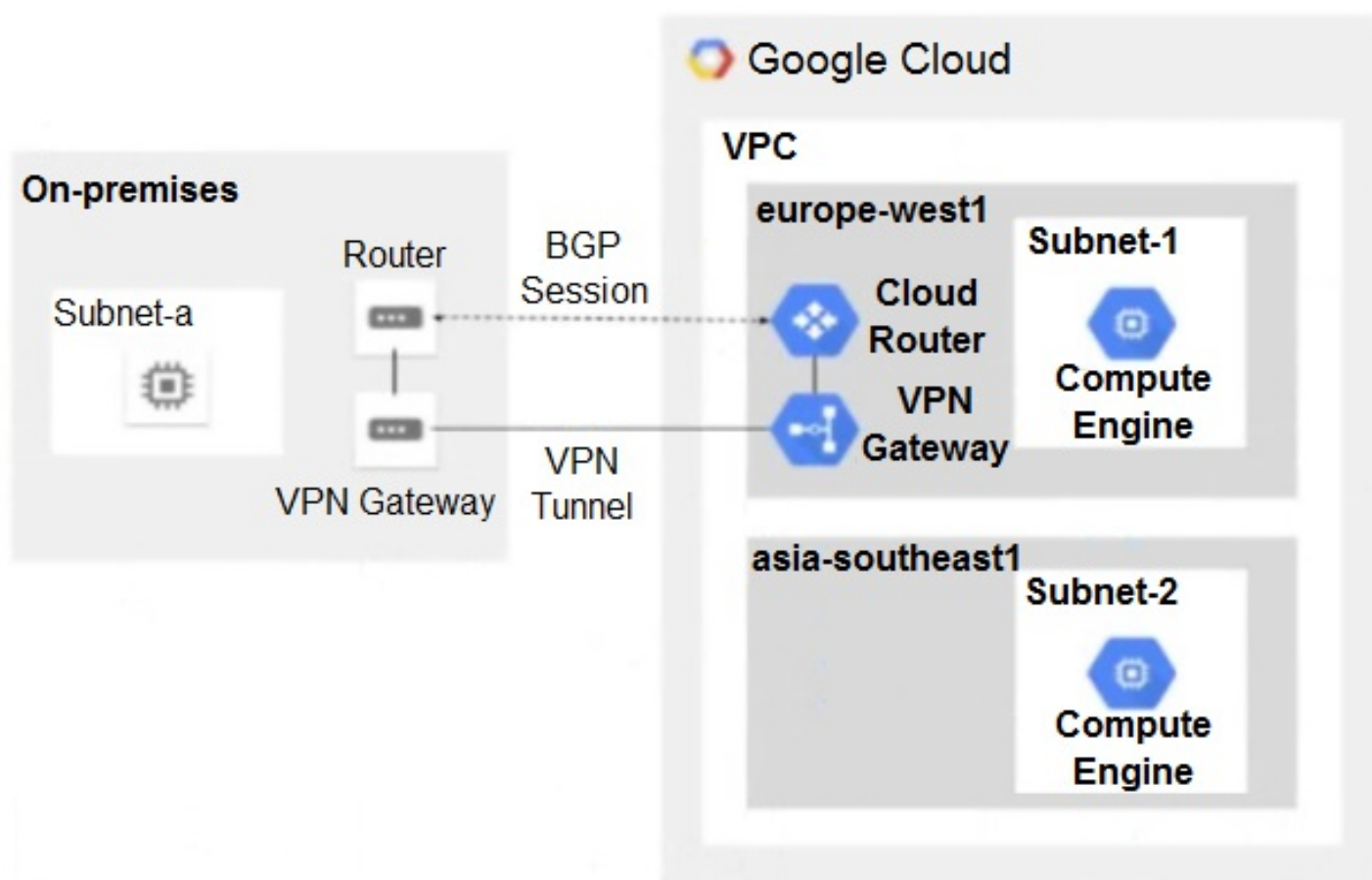
Cloud NAT Configuration: <https://cloud.google.com/nat/docs/using/configure-nat>

Cloud NAT Port Allocation: <https://cloud.google.com/nat/docs/ports>

Question: 129

CertyIQ

You have the following routing design. You discover that Compute Engine instances in Subnet-2 in the asia-southeast1 region cannot communicate with compute resources on-premises. What should you do?



- A. Configure a custom route advertisement on the Cloud Router.
- B. Enable IP forwarding in the asia-southeast1 region.
- C. Change the VPC dynamic routing mode to Global.
- D. Add a second Border Gateway Protocol (BGP) session to the Cloud Router.

Answer: C

Explanation:

A) wrong, by default cloud router sent vpc cidr address. B) wrong, ip forwarding is setup to interface only, not region. C) correct, need adjust dynamic routing to global. D) wrong, no make sense.

Question: 130**CertyIQ**

You are designing a hybrid cloud environment for your organization. Your Google Cloud environment is interconnected with your on-premises network using Cloud HA VPN and Cloud Router. The Cloud Router is configured with the default settings. Your on-premises DNS server is located at 192.168.20.88 and is protected by a firewall, and your Compute Engine resources are located at 10.204.0.0/24. Your Compute Engine resources need to resolve on-premises private hostnames using the domain corp.altostrat.com while still resolving Google Cloud hostnames. You want to follow Google-recommended practices. What should you do?

- A.1. Create a private forwarding zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com that points to 192.168.20.88.
- 2. Configure your on-premises firewall to accept traffic from 10.204.0.0/24.
- 3. Set a custom route advertisement on the Cloud Router for 10.204.0.0/24
- B.1. Create a private forwarding zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com that points to 192.168.20.88.
- 2. Configure your on-premises firewall to accept traffic from 35.199.192.0/19
- 3. Set a custom route advertisement on the Cloud Router for 35.199.192.0/19.
- C.1. Create a private forwarding zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com that points to 192.168.20.88.
- 2. Configure your on-premises firewall to accept traffic from 10.204.0.0/24.
- 3. Modify the /etc/resolv.conf file on your Compute Engine instances to point to 192.168.20.88
- D.1. Create a private zone in Cloud DNS for 'corp.altostrat.com' called corp-altostrat-com.
- 2. Configure DNS Server Policies and create a policy with Alternate DNS servers to 192.168.20.88.
- 3. Configure your on-premises firewall to accept traffic from 35.199.192.0/19.
- 4. Set a custom route advertisement on the Cloud Router for 35.199.192.0/19.

Answer: B**Explanation:**

Option B is the correct solution because it addresses the core requirements of hybrid DNS resolution while adhering to Google's best practices. Creating a private forwarding zone in Cloud DNS for corp.altostrat.com and pointing it to the on-premises DNS server (192.168.20.88) allows Compute Engine instances to resolve internal hostnames in that domain. Critically, Cloud DNS uses a specific range (35.199.192.0/19) for forwarding traffic, not the Compute Engine's IP range (10.204.0.0/24). Hence, the on-premises firewall must accept traffic from 35.199.192.0/19 for DNS queries to reach the on-premises server. Finally, advertising 35.199.192.0/19 via Cloud Router ensures that on-premises networks can reach Cloud DNS forwarders. This approach avoids directly modifying instance configurations (like /etc/resolv.conf) and uses the recommended private forwarding zones for DNS resolution, integrating hybrid DNS within Cloud DNS and on-premises DNS. Other options either miss key aspects of Google Cloud DNS implementation or make inappropriate changes to compute engine instances. Option A incorrectly tries to use the compute engine's ip range, and is missing the advertisement for 35.199.192.0/19. Option C also uses the wrong IP range for firewall configuration and proposes incorrect /etc/resolv.conf configuration. Option D misses the need for a forwarding zone and instead attempts to configure DNS server policies, which are not suited for this scenario and still doesn't solve the problem of routing the correct IPs for DNS requests.

Authoritative Links:

Cloud DNS Private Zones: <https://cloud.google.com/dns/docs/private-zones>

Cloud DNS Forwarding Zones: <https://cloud.google.com/dns/docs/forwarding>

Cloud Router Custom Route Advertisements: <https://cloud.google.com/network->

Question: 131

CertyIQ

Your company has a single Virtual Private Cloud (VPC) network deployed in Google Cloud with on-premises connectivity already in place. You are deploying a new application using Google Kubernetes Engine (GKE), which must be accessible only from the same VPC network and on-premises locations. You must ensure that the GKE control plane is exposed to a predefined list of on-premises subnets through private connectivity only. What should you do?

- A. Create a GKE private cluster with a private endpoint for the control plane. Configure VPC Networking Peering export/import routes and custom route advertisements on the Cloud Routers. Configure authorized networks to specify the desired on-premises subnets.
- B. Create a GKE private cluster with a public endpoint for the control plane. Configure VPC Networking Peering export/import routes and custom route advertisements on the Cloud Routers.
- C. Create a GKE private cluster with a private endpoint for the control plane. Configure authorized networks to specify the desired on-premises subnets.
- D. Create a GKE public cluster. Configure authorized networks to specify the desired on-premises subnets.

Answer: A

Explanation:

The correct answer is A because it aligns with the requirements of providing secure, private access to a GKE cluster's control plane from both within the VPC and from specified on-premises networks. Option A involves creating a GKE private cluster, which is essential for keeping the control plane endpoint private, ensuring it's not directly exposed to the public internet. The option specifies using a private endpoint for the control plane, which means the control plane's API server is only reachable through internal IP addresses within the VPC. This isolates the control plane from the outside world. To allow access from on-premises networks, we need to use VPC Network Peering for route exchange, this will be configured for each network that needs to be accessed. Additionally, custom route advertisements on Cloud Routers allow for the on-prem networks to advertise their subnets for proper routing. Furthermore, the important part of the solution is to configure authorized networks on the cluster, ensuring only the desired on-premises subnets can interact with the GKE control plane. This combination of private cluster, private endpoint, VPC peering, route advertisements, and authorized networks satisfies all requirements while adhering to best practices for security and network segmentation. Options B, C, and D fall short on various aspects. Option B suggests a public endpoint, which is directly contrary to the requirement for private connectivity. Option C omits the critical step of route advertisement for on-premises networks to be reachable, and while it correctly uses a private endpoint, it wouldn't work for access from on-premises without route configuration. Option D proposes a public cluster, again contradicting the private connectivity requirement.

Further research can be conducted using the following authoritative links:

GKE Private Clusters: <https://cloud.google.com/kubernetes-engine/docs/how-to/private-clusters>

VPC Network Peering: <https://cloud.google.com/vpc/docs/vpc-peering>

Cloud Router Custom Route Advertisements: <https://cloud.google.com/router/docs/how-to/advertising-custom-ips>

GKE Authorized Networks: <https://cloud.google.com/kubernetes-engine/docs/how-to/authorized-networks>

Question: 132

CertyIQ

You built a web application with several containerized microservices. You want to run those microservices on Cloud Run. You must also ensure that the services are highly available to your customers with low latency. What should

you do?

- A. Deploy the Cloud Run services to multiple availability zones. Create a global TCP load balancer. Add the Cloud Run endpoints to its backend service.
- B. Deploy the Cloud Run services to multiple regions. Create serverless network endpoint groups (NEGs) that point to the services. Create a global HTTPS load balancer, and attach the serverless NEGs as backend services of the load balancer.
- C. Deploy the Cloud Run services to multiple availability zones. Create Cloud Endpoints that point to the services. Create a global HTTPS load balancer, and attach the Cloud Endpoints to its backend
- D. Deploy the Cloud Run services to multiple regions. Configure a round-robin A record in Cloud DNS.

Answer: B

Explanation:

The correct answer is B. Deploying Cloud Run services across multiple regions ensures high availability by distributing the application across physically separate locations. This mitigates the risk of a regional outage impacting the entire application. Serverless Network Endpoint Groups (NEGs) are specifically designed to connect load balancers to serverless platforms like Cloud Run, providing the necessary abstraction for load balancing traffic. A global HTTPS load balancer is essential for directing user requests to the nearest healthy instance of the service, minimizing latency and providing a consistent experience worldwide. Using serverless NEGs with a global load balancer allows for efficient traffic management and automatic scaling of the application based on demand. Option A is incorrect because availability zones within a region are still susceptible to a broader regional failure. Option C incorrectly utilizes Cloud Endpoints, which are designed for API management rather than direct load balancing of Cloud Run services. Option D using round-robin DNS can cause inconsistent and inefficient routing patterns and lacks the health check capabilities of a load balancer. Round-robin DNS can also be ineffective with caching and can lead to uneven traffic distribution. Thus, option B provides the optimal solution for global high availability and low latency using Google Cloud's capabilities.

Authoritative Links:

Cloud Run Deployment: <https://cloud.google.com/run/docs/deploying>

Serverless NEGs: <https://cloud.google.com/load-balancing/docs/negs/serverless-neg>

Global HTTP(S) Load Balancer: <https://cloud.google.com/load-balancing/docs/https>

Regions and Zones: <https://cloud.google.com/compute/docs/regions-zones>

Question: 133

CertyIQ

You have an HA VPN connection with two tunnels running in active/passive mode between your Virtual Private Cloud (VPC) and on-premises network. Traffic over the connection has recently increased from 1 gigabit per second (Gbps) to 4 Gbps, and you notice that packets are being dropped. You need to configure your VPN connection to Google Cloud to support 4 Gbps. What should you do?

- A. Configure the remote autonomous system number (ASN) to 4096.
- B. Configure a second Cloud Router to scale bandwidth in and out of the VPC.
- C. Configure the maximum transmission unit (MTU) to its highest supported value.
- D. Configure a second set of active/passive VPN tunnels.

Answer: D

Explanation:

The correct answer is **D. Configure a second set of active/passive VPN tunnels.**

Here's why: HA VPN tunnels in Google Cloud have a per-tunnel throughput limit. A single HA VPN tunnel can

typically handle around 1.5 Gbps of traffic. When traffic increases to 4 Gbps, a single active tunnel will be insufficient, leading to packet drops as observed. While active/passive mode provides redundancy, it doesn't increase the total capacity of the connection. To achieve 4 Gbps throughput, adding more VPN tunnels is necessary. By configuring a second set of active/passive VPN tunnels, you effectively double the bandwidth capacity available for the connection. The Cloud Routers will distribute the traffic across the available tunnels, allowing you to reach the desired 4 Gbps bandwidth. Option A is incorrect because the remote ASN is not directly related to bandwidth capacity; it's an identifier for routing purposes. Option B is incorrect since while a second Cloud Router provides redundancy, it doesn't increase per-tunnel bandwidth. Option C, changing the MTU, can sometimes improve performance, but it won't address the fundamental problem of bandwidth exceeding a single tunnel's capacity; MTU is mostly concerned with packet fragmentation and size. The solution involves scaling horizontally by introducing new tunnels rather than attempting to change the configuration of existing infrastructure.

For further research, you can consult Google Cloud's documentation on:

HA VPN: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/ha-vpn>

Cloud Router: <https://cloud.google.com/network-connectivity/docs/router/concepts/overview>

VPN Tunnel Limits: (While this isn't explicitly stated, understanding concepts behind HA VPN helps understand the limits)

Question: 134

CertyIQ

You recently deployed two network virtual appliances in us-central1. Your network appliances provide connectivity to your on-premises network, 10.0.0.0/8. You need to configure the routing for your Virtual Private Cloud (VPC). Your design must meet the following requirements:

- All access to your on-premises network must go through the network virtual appliances.
- Allow on-premises access in the event of a single network virtual appliance failure.
- Both network virtual appliances must be used simultaneously.

Which method should you use to accomplish this?

- A. Configure two routes for 10.0.0.0/8 with different priorities, each pointing to separate network virtual appliances.
- B. Configure an internal HTTP(S) load balancer with the two network virtual appliances as backends. Configure a route for 10.0.0.0/8 with the internal HTTP(S) load balancer as the next hop.
- C. Configure a network load balancer for the two network virtual appliances. Configure a route for 10.0.0.0/8 with the network load balancer as the next hop.
- D. Configure an internal TCP/UDP load balancer with the two network virtual appliances as backends. Configure a route for 10.0.0.0/8 with the internal load balancer as the next hop.

Answer: D

Explanation:

Okay, let's break down why option D is the correct approach for this scenario.

The core requirement is to route traffic destined for the on-premises network (10.0.0.0/8) through two network virtual appliances (NVAs) in an active-active manner, ensuring failover in case one NVA goes down. Option A is unsuitable because using different priorities for routes won't achieve simultaneous usage; the lower priority route is only used during failover. Options B and C are unsuitable because HTTP(S) Load Balancer and Network Load Balancers are designed for web and TCP/UDP based traffic respectively and do not handle arbitrary traffic forwarding as the given scenario demands.

Option D, utilizing an internal TCP/UDP load balancer, is the best solution. This type of load balancer distributes TCP and UDP traffic across its backend instances, which in this case, are the two NVAs. A single

route with the internal TCP/UDP load balancer as the next hop is then configured for the 10.0.0.0/8 destination. This setup ensures that traffic is actively distributed across both NVAs, using them simultaneously. If one NVA becomes unhealthy, the load balancer will automatically redirect traffic to the remaining healthy NVA. This provides the desired active-active usage with seamless failover in the event of an appliance failure.

In essence, the internal TCP/UDP load balancer acts as an intelligent router, distributing traffic based on the health of the NVAs while keeping the routing simple.

Here are some authoritative links for further research:

Google Cloud Internal TCP/UDP Load Balancer: <https://cloud.google.com/load-balancing/docs/internal/>
Google Cloud Routing: <https://cloud.google.com/vpc/docs/routes>
Network Virtual Appliance (NVA) patterns: <https://cloud.google.com/architecture/network-virtual-appliances-patterns>

Question: 135

CertyIQ

You are responsible for enabling Private Google Access for the virtual machine (VM) instances in your Virtual Private Cloud (VPC) to access Google APIs. All VM instances have only a private IP address and need to access Cloud Storage. You need to ensure that all VM traffic is routed back to your on-premises data center for traffic scrubbing via your existing Cloud Interconnect connection. However, VM traffic to Google APIs should remain in the VPC. What should you do?

- A.1. Delete the default route in your VPC.
- 2. Create a private Cloud DNS zone for googleapis.com, create a CNAME for *.googleapis.com to restricted googleapis.com, and create an A record for restricted googleapis.com that resolves to the addresses in 199.36.153.4/30.
- 3. Create a static route in your VPC for the range 199.36.153.4/30 with the default internet gateway as the next hop.
- B.1. Delete the default route in your VPC and configure your on-premises router to advertise 0.0.0.0/0 via Border Gateway Protocol (BGP).
- 2. Create a public Cloud DNS zone with a CNAME for *.google.com to private googleapis.com, create a CNAME for * googleapis.com to private googleapis.com, and create an A record for Private googleapis.com that resolves to the addresses in 199.36.153.8/30.
- 3. Create a static route in your VPC for the range 199.36.153.8/30 with the default internet gateway as the next hop.
- C.1. Configure your on-premises router to advertise 0.0.0.0/0 via Border Gateway Protocol (BGP) with a lower priority (MED) than the default VPC route.
- 2. Create a private Cloud DNS zone for googleapis.com, create a CNAME for * googieapis.com to private googleapis.com, and create an A record for private.googleapis.com that resolves to the addresses in 199.36.153.8/30.
- 3. Create a static route in your VPC for the range 199.36.153.8/30 with the default internet gateway as the next hop.
- D.1. Delete the default route in your VPC and configure your on-premises router to advertise 0.0.0.0/0 via Border Gateway Protocol (BGP).
- 2. Create a private Cloud DNS zone for googleapis.com, create a CNAME for * googieapis.com to Private googleapis.com, and create an A record for private.googleapis.com that resolves to the addresses in 199.36.153.8/30.
- 3. Create a static route in your VPC for the range 199.36.153.8/30 with the default internet gateway as the next hop.

Answer: D

Explanation:

The correct answer is **D**. Here's why:

The primary goal is to force all VM traffic to traverse the on-premises data center for inspection, except for

Google API traffic which should remain within the VPC. This requires careful route manipulation and DNS resolution.

Option D correctly addresses this:

1. **Deleting the Default Route & On-Premises Route Advertisement:** Removing the default route in the VPC ensures that no traffic is routed directly to the internet. By advertising 0.0.0.0/0 via BGP from the on-premises router, all traffic, except specifically defined routes, is forced to go through the on-premises data center due to BGP's route selection mechanism. This aligns with the requirement of scrubbing all VM traffic before it exits the VPC, since the BGP advertised route will have a higher priority (lower MED) for this traffic than the default Google VPC route.
2. **Private DNS Zone & Resolution:** A private DNS zone is created for googleapis.com along with a CNAME that directs *.googleapis.com to private.googleapis.com. The A record for private.googleapis.com resolves to 199.36.153.8/30, which is the range of Google's Private Google Access IPs. This ensures that when a VM tries to resolve a Google API endpoint, it receives a private IP address, keeping the traffic within the VPC.
3. **Static Route to Google APIs:** A static route is created within the VPC for the 199.36.153.8/30 range with the default internet gateway as the next hop. This explicitly routes traffic destined for Google APIs (which is now resolving to an IP in the range of 199.36.153.8/30) directly to Google, bypassing the on-premises data center. This ensures that API traffic stays within the Google infrastructure as desired.

Why other options are incorrect:

Option A: Creates a default route with the internet gateway, which will route Google API traffic to the internet instead of keeping it in the VPC

Option B: Creates a public DNS zone which does not align with best practices. The use of private googleapis.com in this option is not correctly formatted. Furthermore, the default Google VPC route will take priority over the BGP advertised route for all traffic

Option C: The lack of deleting the default route means the default Google VPC route will take priority over the BGP advertised route for all traffic. Furthermore, the use of a lower priority MED for the on-premise advert will mean that all traffic is routed through the default VPC route, and not to on-prem.

Authoritative Links:

Private Google Access: <https://cloud.google.com/vpc/docs/private-google-access>

Cloud DNS Private Zones: <https://cloud.google.com/dns/docs/private-zones>

Cloud Router BGP: <https://cloud.google.com/network-connectivity/docs/router/concepts/bgp>

Static Routes: <https://cloud.google.com/vpc/docs/routes#staticroutes>

Question: 136

CertyIQ

You are designing a hub-and-spoke network architecture for your company's cloud-based environment. You need to make sure that all spokes are peered with the hub. The spokes must use the hub's virtual appliance for internet access. The virtual appliance is configured in high-availability mode with two instances using an internal load balancer with IP address 10.0.0.5. What should you do?

- A.1. Create a default route in the hub VPC that points to IP address 10.0.0.5.
 2. Delete the default internet gateway route in the hub VPC, and create a new higher-priority route that is tagged only to the appliances with a next hop of the default internet gateway.
 3. Export the custom routes in the hub.
 4. Import the custom routes in the spokes.
- B.1. Create a default route in the hub VPC that points to IP address 10.0.0.5.
 2. Delete the default internet gateway route in the hub VPC, and create a new higher-priority route that is tagged only to the appliances with a next hop of the default internet gateway.

3. Export the custom routes in the hub. Import the custom routes in the spokes.
 4. Delete the default internet gateway route of the spokes.
- C.1. Create two default routes in the hub VPC that point to the next hop instances of the virtual appliances.
2. Delete the default internet gateway route in the hub VPC, and create a new higher-priority route that is tagged only to the appliances with a next hop of the default internet gateway.
3. Export the custom routes in the hub. Import the custom routes in the spokes.
- D.1. Create a default route in the hub VPC that points to IP address 10.0.0.5.
2. Delete the default internet gateway route in the hub VPC, and create a new higher-priority route that is tagged only to the appliances with a next hop of the default internet gateway.
3. Create a new route in the spoke VPC that points to IP address 10.0.0.5.

Answer: B

Explanation:

The correct answer is **B**. Here's why:

This scenario requires routing traffic from spoke VPCs to the internet through a virtual appliance located in the hub VPC. The key is to use custom route exchange via VPC Network Peering.

Step-by-step justification:

1. **Create a default route in the hub VPC that points to IP address 10.0.0.5:** This step establishes the default path for traffic originating from within the hub VPC destined for the internet through the internal load balancer that sits in front of the virtual appliances. This ensures that even traffic within the hub is sent to the appliance first and from there to the internet.
2. **Delete the default internet gateway route in the hub VPC, and create a new higher-priority route that is tagged only to the appliances with a next hop of the default internet gateway:** This step is crucial for ensuring that only traffic from the appliance instances are routed to the internet gateway. When a route is tagged for a particular network or tag only the routes matching that tag will go out to the next hop internet gateway, other traffic will match the default route set to 10.0.0.5. The purpose of removing the old gateway and then creating one with higher priority (lower number) is to ensure that the tag for the appliance is matched when it leaves the appliance.
3. **Export the custom routes in the hub:** By exporting custom routes in the hub VPC, the default route to 10.0.0.5 and the tagged route to the internet gateway will be advertised to the peered spoke VPCs. This allows the spokes to utilize the hub's appliance for internet access.
4. **Import the custom routes in the spokes:** Importing the exported routes from the hub VPC in the spoke VPCs ensures that the spokes are aware of the default route via the hub VPC's internal load balancer.
5. **Delete the default internet gateway route of the spokes:** The default internet gateway route needs to be removed from the spokes, as it will bypass the hub appliance. After removing this route, the only path to internet will be the one imported from the hub VPC.

Why other options are incorrect:

Option A is missing step 4, the deletion of the default internet gateway route in the spokes which is crucial for forcing traffic through the appliance.

Option C is incorrect because it suggests creating two default routes pointing to individual appliance instances. This bypasses the internal load balancer (10.0.0.5) designed for high availability, and it is not the correct way to use an internal load balancer. The instances behind the internal load balancer will also not have internet connectivity if the gateway is removed and tagged for the virtual appliance only.

Option D is incomplete. It creates a default route, a more specific route using tags, but doesn't correctly configure the spokes to send traffic through the hub.

Relevant Cloud Concepts:

VPC Network Peering: Enables private network connectivity between VPC networks.

Custom Routes: Allows for customized traffic routing within a VPC.

Route Export/Import: Facilitates route sharing between peered VPC networks.

Internal Load Balancer: Distributes traffic across multiple instances within a VPC, ensuring high availability.

Route Tagging: Allows you to match the specific routes to go out a specific next hop based on the origin network or instance.

Authoritative Links:

Google Cloud VPC Route Overview: <https://cloud.google.com/vpc/docs/routes>

Google Cloud VPC Network Peering: <https://cloud.google.com/vpc/docs/vpc-peering>

Google Cloud Load Balancing: <https://cloud.google.com/load-balancing>

Question: 137

CertyIQ

You configured Cloud VPN with dynamic routing via Border Gateway Protocol (BGP). You added a custom route to advertise a network that is reachable over the VPN tunnel. However, the on-premises clients still cannot reach the network over the VPN tunnel. You need to examine the logs in Cloud Logging to confirm that the appropriate routers are being advertised over the VPN tunnel. Which filter should you use in Cloud Logging to examine the logs?

A.resource.type= "gce_router"

B.resource.type= "gce_network_region"

C.resource.type= "vpn_tunnel"

D.resource.type= "vpn_gateway"

Answer: A

Explanation:

The correct answer is **A. resource.type="gce_router"**. Here's why:

The scenario involves a Cloud VPN with dynamic routing using BGP. BGP routes are managed by Cloud Routers in Google Cloud. These routers are responsible for advertising the custom routes to your on-premises network. Therefore, to examine logs related to BGP route advertisements, specifically those related to custom routes over a VPN tunnel, we need to focus on logs emitted by these Cloud Routers.

Cloud Logging uses a structured format where each log entry is associated with a specific resource type. The resource.type field allows filtering based on the source of the log. gce_router is the specific resource type for Cloud Routers. This means log entries related to BGP updates, route learning, and advertising will be tagged with this resource type. Options B, C and D are not related to the route processing. gce_network_region refers to a Google Cloud region and not the router responsible for advertising the routes; vpn_tunnel focuses on the tunnel itself, but not on the route management; while vpn_gateway concerns the endpoint, not the route exchange. Filtering on the gce_router resource type will provide the most relevant information for troubleshooting BGP route advertisements.

By using resource.type="gce_router" in Cloud Logging, you can effectively filter out irrelevant logs and focus directly on the router logs, thus isolating the source of the issue with the advertised custom routes. Examining these logs will help determine if the custom route was successfully advertised, or if there are any issues with route propagation. This enables debugging route problems specific to Google Cloud's Cloud Router.

Authoritative Links:

Cloud Logging Resource Types: <https://cloud.google.com/logging/docs/reference/v2/resource-list>

Cloud Router Overview: <https://cloud.google.com/router/docs/overview>

Cloud VPN Dynamic Routing: <https://cloud.google.com/vpn/docs/concepts/dynamic-routing>

Question: 138

CertyIQ

Your company has a single Virtual Private Cloud (VPC) network deployed in Google Cloud with access from on-premises locations using Cloud Interconnect connections. Your company must be able to send traffic to Cloud Storage only through the Interconnect links while accessing other Google APIs and services over the public internet. What should you do?

- A. Use the default public domains for all Google APIs and services.
- B. Use Private Service Connect to access Cloud Storage, and use the default public domains for all other Google APIs and services.
- C. Use Private Google Access, with restricted.googleapis.com virtual IP addresses for Cloud Storage and private.googleapis.com for all other Google APIs and services.
- D. Use Private Google Access, with private.googleapis.com virtual IP addresses for Cloud Storage and restricted.googleapis.com virtual IP addresses for all other Google APIs and services.

Answer: B

Explanation:

The correct answer is **B: Use Private Service Connect to access Cloud Storage, and use the default public domains for all other Google APIs and services.**

Here's the justification:

The requirement is to route Cloud Storage traffic exclusively through the Cloud Interconnect links while allowing other Google API traffic over the public internet. This necessitates a mechanism to selectively direct traffic.

Private Service Connect (PSC) allows you to establish private connections to Google services, including Cloud Storage, by creating private endpoints within your VPC. These endpoints are reachable only through your internal network, enforcing the required path for Cloud Storage over the Interconnect.

For other Google APIs, utilizing the default public domains directs traffic over the public internet, fulfilling the second requirement. There's no need to redirect them through interconnect because they are specifically designated to use the public internet.

Option A would route all traffic through the public internet, failing to meet the requirement for Cloud Storage. Options C and D utilize Private Google Access, which provides access to Google APIs via internal IPs, but not specifically over interconnect, therefore not a solution for this case. While restricted.googleapis.com can control access to specific services, it would also direct the other API traffic over private IP which doesn't match the requirement to reach them via public internet. Private Service Connect offers granularity and isolation of network path that Private Google Access lacks.

In summary, PSC provides the necessary control to ensure Cloud Storage traffic traverses the Interconnect, while using public domains for other APIs delivers them through public internet, precisely fulfilling the problem statement's conditions.

Authoritative Links:

Private Service Connect: <https://cloud.google.com/vpc/docs/private-service-connect>

Private Google Access: <https://cloud.google.com/vpc/docs/private-google-access>

Cloud Interconnect: <https://cloud.google.com/interconnect/docs/overview>

Question: 139

Your organization has a Google Cloud Virtual Private Cloud (VPC) with subnets in us-east1, us-west4, and europe-west4 that use the default VPC configuration. Employees in a branch office in Europe need to access the resources in the VPC using HA VPN. You configured the HA VPN associated with the Google Cloud VPC for your organization with a Cloud Router deployed in europe-west4. You need to ensure that the users in the branch office can quickly and easily access all resources in the VPC. What should you do?

- A. Create custom advertised routes for each subnet.
- B. Configure each subnet's VPN connections to use Cloud VPN to connect to the branch office.
- C. Configure the VPC dynamic routing mode to Global.
- D. Set the advertised routes to Global for the Cloud Router.

Answer: C

Explanation:

The correct answer is **C. Configure the VPC dynamic routing mode to Global.**

Here's why:

The scenario requires branch office users in Europe to access resources across all subnets within the VPC (us-east1, us-west4, and europe-west4) via an HA VPN tunnel connected to a Cloud Router in europe-west4. By default, VPC networks use regional dynamic routing. This means that routes learned by the Cloud Router are only propagated within its region (europe-west4 in this case). For users in the branch office to reach resources in us-east1 and us-west4, the routes to those subnets need to be propagated globally.

Option C directly addresses this limitation by changing the VPC dynamic routing mode to Global. With Global routing, the Cloud Router learns routes from all subnets within the VPC, regardless of the region. These routes are then advertised to the on-premises network via BGP over the HA VPN tunnel. This enables the branch office users to access resources across all regions within the VPC using a single HA VPN setup and the Cloud Router in europe-west4.

Option A, creating custom advertised routes for each subnet, is unnecessarily complex and can be error-prone. Option B, configuring each subnet's VPN, would be significantly more expensive and difficult to manage. Option D, setting advertised routes to Global for the Cloud Router, isn't directly supported; it is the VPC's dynamic routing mode that needs to be global.

Therefore, configuring the VPC dynamic routing mode to Global (Option C) is the most efficient and effective solution, allowing seamless access to resources in all regions via the single HA VPN tunnel, while avoiding unnecessary complexity and cost.

Authoritative Links:

Google Cloud Documentation on VPC Dynamic Routing:

https://cloud.google.com/vpc/docs/routes#dynamic_routing

Google Cloud Documentation on Cloud Router: <https://cloud.google.com/router/docs/concepts/overview>

Google Cloud Documentation on HA VPN: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/overview>

Question: 140

Your organization uses a Shared VPC architecture with a host project and three service projects. You have Compute Engine instances that reside in the service projects. You have critical workloads in your on-premises data center. You need to ensure that the Google Cloud instances can resolve on-premises hostnames via the Dedicated

Interconnect you deployed to establish hybrid connectivity. What should you do?

- A.1. Create a Cloud DNS private forwarding zone in the host project of the Shared VPC that forwards the private zone to the on-premises DNS servers.
- 2. In your Cloud Router, add a custom route advertisement for the IP 35.199.192.0/19 to the on-premises environment.
- B.1. Create a Cloud DNS private forwarding zone in the host project of the Shared VPC that forwards the Private zone to the on-premises DNS servers.
- 2. In your Cloud Router, add a custom route advertisement for the IP 169.254 169.254 to the on-premises environment.
- C.1. Configure a Cloud DNS private zone in the host project of the Shared VPC.
- 2. Set up DNS forwarding to your Google Cloud private zone on your on-premises DNS servers to point to the inbound forwarder IP address in your host project
- 3. In your Cloud Router, add a custom route advertisement for the IP 169.254 169 254 to the on-premises environment.
- D.1. Configure a Cloud DNS private zone in the host project of the Shared VPC.
- 2. Set up DNS forwarding to your Google Cloud private zone on your on-premises DNS servers to point to the inbound forwarder IP address in your host project.
- 3. Configure a DNS policy in the Shared VPC to allow inbound query forwarding with your on-premises DNS server as the alternative DNS server.

Answer: A

Explanation:

The correct answer is A because it accurately addresses the requirement for on-premises hostname resolution from Google Cloud instances in a Shared VPC environment. Creating a Cloud DNS private forwarding zone in the host project, as described in step 1 of option A, is the essential first step. This zone is configured to forward DNS queries for the on-premises domain to the designated on-premises DNS servers. This effectively allows the Google Cloud instances to leverage the on-premises DNS infrastructure for name resolution. Step 2 of option A is also critical. While not directly related to DNS resolution, it addresses a potential issue related to hybrid connectivity. The 35.199.192.0/19 range is used by Google for health checks for cloud interconnect. By adding this custom route advertisement in cloud router we are allowing health check to reach on-prem and detect that route is available. This can influence failover behavior. Option B is incorrect because it suggests using the 169.254.169.254, this IP is the metadata server for VMs and not related to Cloud Interconnects for health checks. Option C and D are incorrect because they propose configuring a private zone and setting up DNS forwarding on on-prem dns server, which implies that the on premise dns is going to resolve the GCP internal names, which is not required. They also suggest advertising 169.254.169.254 IP range which is not required. Overall option A is correct because it sets up the forwarding in the host project to leverage on premise DNS servers for name resolution and also adds the required network route for health checks.

Authoritative links for further research:

Cloud DNS Private Zones: <https://cloud.google.com/dns/docs/private-zones>

Cloud DNS Forwarding: <https://cloud.google.com/dns/docs/forwarding>

Shared VPC: <https://cloud.google.com/vpc/docs/shared-vpc>

Cloud Router Custom Advertisements: <https://cloud.google.com/router/docs/custom-advertisements>

Google Cloud Interconnect IP ranges: <https://cloud.google.com/vpc/docs/special-ip-ranges>

Question: 141

CertyIQ

Your organization is implementing a new security policy to control how firewall rules are applied to control flows between virtual machines (VMs). Using Google-recommended practices, you need to set up a firewall rule to enforce strict control of traffic between VM A and VM B. You must ensure that communications flow only from VM A to VM B within the VPC, and no other communication paths are allowed. No other firewall rules exist in the VPC. Which firewall rule should you configure to allow only this communication path?

A.Firewall rule direction: ingress

Action: allow -

Target: VM B service account -

Source ranges: VM A service account

Priority: 1000

B.Firewall rule direction: ingress

Action: allow -

Target: specific VM B tag -

Source ranges: VM A tag and VM A source IP address

Priority: 1000

C.Firewall rule direction: ingress

Action: allow -

Target: VM A service account -

Source ranges: VM B service account and VM B source IP address

Priority: 100

D.Firewall rule direction: ingress

Action: allow -

Target: specific VM A tag -

Source ranges: VM B tag and VM B source IP address

Priority: 100

Answer: A

Explanation:

The correct answer is A because it aligns with Google Cloud's best practices for configuring firewall rules to achieve granular network control. Option A creates an ingress (incoming) rule that explicitly allows traffic to reach VM B (the target). It employs VM B's service account as the target, ensuring the rule only applies to that specific instance. Crucially, it restricts the source to VM A's service account, effectively allowing traffic only from VM A. This approach uses service accounts for identity management, a recommended practice for enhanced security over IP-based filtering or tags. The rule's action is set to 'allow,' permitting the desired communication. The priority of 1000 is standard, allowing for other rules to take precedence if needed. In contrast, Option B improperly mixes tags and IP addresses, a less robust approach compared to service accounts. Options C and D wrongly apply the target, making them ineffective for allowing traffic to VM B. Option C also incorrectly sets up the source. Using service accounts offers a more secure and dynamic approach, since IP addresses could change, and tags are less specific than identities. Therefore, option A is the only option that achieves a secure and specific method for allowing traffic from VM A to VM B.

Further research on firewall rule configuration and service accounts within Google Cloud can be found at:

VPC firewall rules overview: <https://cloud.google.com/vpc/docs/firewalls>

Using service accounts with firewall rules: <https://cloud.google.com/vpc/docs/firewall-rules-service-accounts>

Best practices for firewall rules: <https://cloud.google.com/architecture/security-best-practices-vpc-firewalls>

Question: 142

CertyIQ

You have configured a service on Google Cloud that connects to an on-premises service via a Dedicated Interconnect. Users are reporting recent connectivity issues. You need to determine whether the traffic is being dropped because of firewall rules or a routing decision. What should you do?

- A. Use the Network Intelligence Center Connectivity Tests to test the connectivity between the VPC and the on-premises network.
- B. Use Network Intelligence Center Network Topology to check the traffic flow, and replay the traffic from the time period when the connectivity issue occurred.
- C. Configure VPC Flow Logs. Review the logs by filtering on the source and destination.
- D. Configure a Compute Engine instance on the same VPC as the service running on Google Cloud to run a traceroute targeted at the on-premises service.

Answer: A

Explanation:

Option A, using Network Intelligence Center Connectivity Tests, is the most effective approach for diagnosing connectivity issues between a VPC and an on-premises network via Dedicated Interconnect. These tests simulate traffic flow along a specified path and provide detailed results, including whether the traffic was allowed or blocked by firewall rules or routing configurations. This targeted approach quickly isolates the problem area. Option B, using Network Topology, provides a visual representation of the network but does not offer specific details on packet drop reasons or replay functionalities. Option C, VPC Flow Logs, provides valuable insights into traffic flow but requires post-analysis, and logs do not explicitly state the exact reason for packet drops due to firewall rules or routing decisions; it needs interpretation. Option D, a traceroute from a Compute Engine instance, is useful for testing reachability but does not pinpoint the root cause at a granular level, such as routing or firewall rules, within the Google Cloud Network environment. Connectivity Tests are designed explicitly for end-to-end troubleshooting, whereas other tools offer a view that is not tailored for root cause identification of network path issues. Therefore, Network Intelligence Center Connectivity Tests provides immediate, focused, and actionable data for network connectivity problems, and it is a more efficient option for determining whether firewall rules or routing decisions are causing the dropped traffic.

Authoritative Links:

Google Cloud Network Intelligence Center - Connectivity Tests: <https://cloud.google.com/network-intelligence-center/docs/connectivity-tests/concepts>

Google Cloud Dedicated Interconnect: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/dedicated-interconnect>

Google Cloud VPC Firewall Rules: <https://cloud.google.com/vpc/docs/firewalls>

Google Cloud VPC Routing: <https://cloud.google.com/vpc/docs/routes>

Question: 143

CertyIQ

You are configuring a new HTTP application that will be exposed externally behind both IPv4 and IPv6 virtual IP addresses, using ports 80, 8080, and 443. You will have backends in two regions: us-west1 and us-east1. You want to serve the content with the lowest-possible latency while ensuring high availability and autoscaling, and create native content-based rules using the HTTP hostname and request path. The IP addresses of the clients that connect to the load balancer need to be visible to the backends. Which configuration should you use?

- A. Use Network Load Balancing
- B. Use TCP Proxy Load Balancing with PROXY protocol enabled
- C. Use External HTTP(S) Load Balancing with URL Maps and custom headers
- D. Use External HTTP(S) Load Balancing with URL Maps and an X-Forwarded-For header

Answer: D

Explanation:

The correct answer is **D: Use External HTTP(S) Load Balancing with URL Maps and an X-Forwarded-For**

header. This choice effectively addresses all requirements outlined. External HTTP(S) Load Balancing is specifically designed for handling HTTP(S) traffic, supporting both IPv4 and IPv6, and distributing traffic across multiple backends in different regions, thus ensuring high availability and low latency through geographic proximity. URL Maps allow for sophisticated, content-based routing rules based on hostnames and request paths, fulfilling the requirement for native, flexible routing. Moreover, External HTTP(S) Load Balancers natively support autoscaling, adjusting resources automatically to handle traffic fluctuations. Critically, these load balancers add an X-Forwarded-For header to requests, preserving the original client IP address, which is necessary for visibility on the backend servers. Options A and B are unsuitable. Network Load Balancing (A) operates at Layer 4 (TCP/UDP) and does not support content-based routing or X-Forwarded-For headers natively. TCP Proxy Load Balancing with PROXY protocol enabled (B) works at TCP Layer and, while supporting client IP visibility with the PROXY protocol, does not handle HTTP(S) traffic directly or support URL mapping like HTTP(S) load balancing. Custom headers (C), while possible, are not the standard or most efficient way to pass client IPs and are also not part of the default configurations needed, making X-Forwarded-For the standard method and thus answer D the best option.

Here are some authoritative links for further research:

Google Cloud Load Balancing Overview: <https://cloud.google.com/load-balancing/docs/load-balancing-overview>

External HTTP(S) Load Balancing: <https://cloud.google.com/load-balancing/docs/https/>

URL Maps: <https://cloud.google.com/load-balancing/docs/url-maps>

X-Forwarded-For Header: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Forwarded-For>

Question: 144

CertyIQ

You need to define an address plan for a future new Google Kubernetes Engine (GKE) cluster in your Virtual Private Cloud (VPC). This will be a VPC-native cluster, and the default Pod IP range allocation will be used. You must pre-provision all the needed VPC subnets and their respective IP address ranges before cluster creation. The cluster will initially have a single node, but it will be scaled to a maximum of three nodes if necessary. You want to allocate the minimum number of Pod IP addresses. Which subnet mask should you use for the Pod IP address range?

- A./21
- B./22
- C./23
- D./25

Answer: B

Explanation:

Here's a breakdown of why a /22 subnet mask is the correct choice for the Pod IP address range in this scenario:

The core requirement is to minimize the allocated IP addresses while supporting a future maximum of three nodes in the VPC-native GKE cluster. VPC-native clusters, by default, allocate Pod IP addresses from a subnet separate from the node IPs. Each node gets a /24 subnet for its Pods.

A /21 subnet offers 2048 IP addresses, which is far more than required for three nodes. A /23 subnet provides 512 IPs, still more than required. A /25 subnet would provide only 128 IPs, which would be insufficient as each node would need at least a /24 for its Pods.

A /22 subnet offers 1024 IP addresses. With the default allocation of a /24 subnet per node, this is enough to support three nodes ($3 \times 256 = 768$ IPs) with room to spare for future pods, if needed, while minimizing address usage. Using a /22 avoids over-allocating address space, optimizing IP address utilization and

preventing exhaustion within the VPC. The recommended sizing provides flexibility without excessive waste. A /22 subnet provides the perfect balance between meeting the scalability requirement (3 nodes) and keeping the IP space efficient.

For further research, refer to the following documentation:

GKE VPC-native (alias IP) clusters: <https://cloud.google.com/kubernetes-engine/docs/how-to/alias-ips>

GKE IP address management: <https://cloud.google.com/kubernetes-engine/docs/concepts/alias-ips>

Subnetting Basics: <https://www.cloudflare.com/learning/network-layer/what-is-subnetting/>

In summary, the /22 mask efficiently caters to the requirements, using the least number of IP addresses for the given scenario.

Question: 145

CertyIQ

You are responsible for designing a new connectivity solution for your organization's enterprise network to access and use Google Workspace. You have an existing Shared VPC with Compute Engine instances in us-west1. Currently, you access Google Workspace via your service provider's internet access. You want to set up a direct connection between your network and Google. What should you do?

- A. Order a Dedicated Interconnect connection in the same metropolitan area. Create a VLAN attachment, a Cloud Router in us-west1, and a Border Gateway Protocol (BGP) session between your Cloud Router and your router.
- B. Order a Direct Peering connection in the same metropolitan area. Configure a Border Gateway Protocol (BGP) session between Google and your router.
- C. Configure HA VPN in us-west1. Configure a Border Gateway Protocol (BGP) session between your Cloud Router and your on-premises data center.
- D. Order a Carrier Peering connection in the same metropolitan area. Configure a Border Gateway Protocol (BGP) session between Google and your router.

Answer: B

Explanation:

The correct answer is **B**, opting for a Direct Peering connection. This solution aligns best with the requirements of establishing a direct connection for accessing Google Workspace from an organization's enterprise network, given the existing infrastructure in us-west1.

Here's why:

Direct Peering Purpose: Direct Peering is designed for directly connecting to Google's network for accessing Google Workspace and other Google services, offering improved performance, reliability, and potentially cost savings for high-bandwidth traffic compared to using public internet access. This meets the requirement of "a direct connection between your network and Google".

Location Requirement: The scenario does not require a specific connectivity method to the VPC, rather direct access to Google services, therefore a connection in the same metropolitan area is adequate and required for Direct Peering.

BGP for Routing: BGP is the industry-standard protocol for routing between autonomous systems, making it the appropriate method for establishing connectivity and exchanging routes between Google and the organization's network.

Other Options Disqualified:

A (Dedicated Interconnect): While also a direct connection, it's primarily aimed at connecting to the VPC, not specifically for Google Workspace. Dedicated Interconnect requires a VLAN attachment and a Cloud Router within the VPC, which is not the core need in this question.

C (HA VPN): HA VPN uses IPsec over the public internet, not a direct connection. It is for access to a VPC, not Google Workspace.

D (Carrier Peering): Carrier peering is similar to direct peering, but is generally used for connecting to Google using a service provider and can add additional costs and latency compared to direct.

Therefore, Direct Peering (option B) is the most efficient and suitable solution for directly connecting to Google's network for Google Workspace access.

Authoritative Links:

Google Cloud Direct Peering: <https://cloud.google.com/network-connectivity/docs/direct-peering>

Google Cloud Connectivity Options: <https://cloud.google.com/network-connectivity/docs/overview#connectivity>

Question: 146

CertyIQ

You suspect that one of the virtual machines (VMs) in your default Virtual Private Cloud (VPC) is under a denial-of-service attack. You need to analyze the incoming traffic for the VM to understand where the traffic is coming from. What should you do?

- A. Enable Data Access audit logs of the VPC. Analyze the logs and get the source IP addresses from the subnetworks.get field.
- B. Enable VPC Flow Logs for the subnet. Analyze the logs and get the source IP addresses from the connection field.
- C. Enable VPC Flow Logs for the VPC. Analyze the logs and get the source IP addresses from the src_location field.
- D. Enable Data Access audit logs of the subnet. Analyze the logs and get the source IP addresses from the networks.get field.

Answer: B

Explanation:

The correct answer is **B. Enable VPC Flow Logs for the subnet. Analyze the logs and get the source IP addresses from the connection field.** Here's why:

VPC Flow Logs are designed to capture network traffic entering and exiting your VPC networks and subnets. They record details about IP traffic flowing between virtual machines (VMs) and other network endpoints. For identifying the source of a potential denial-of-service (DoS) attack, VPC Flow Logs are ideal because they provide detailed information about the connections, including the source and destination IP addresses, ports, protocols, and the action (accepted or denied).

Specifically, the 'connection' field within VPC Flow Logs contains the essential source IP address, allowing you to pinpoint the origin of the traffic bombarding your VM. By analyzing these logs, you can identify patterns and pinpoint which sources are sending excessive traffic, indicative of a DoS attack.

Option A is incorrect because Data Access audit logs are primarily concerned with logging API calls made to GCP services, not network traffic flows within your VPC. They don't provide the granular IP-level connection details needed for a DoS investigation. Options C and D are incorrect because 'src_location' field is not part of VPC Flow Logs; and similarly, Data Access audit logs do not have a 'networks.get' field associated with network flow data.

Therefore, enabling VPC Flow Logs at the subnet level and analyzing the connection fields offers the most direct and accurate way to understand incoming traffic and identify the source IP addresses associated with a potential denial-of-service attack.

Authoritative Links for further research:

VPC Flow Logs Overview: <https://cloud.google.com/vpc/docs/flow-logs>

Using VPC Flow Logs: <https://cloud.google.com/vpc/docs/using-flow-logs>

VPC Flow Logs fields: https://cloud.google.com/vpc/docs/flow-logs#log_record

Question: 147

CertyIQ

You are responsible for configuring firewall policies for your company in Google Cloud. Your security team has a strict set of requirements that must be met to configure firewall rules.

- Always allow Secure Shell (SSH) from your corporate IP address.
- Restrict SSH access from all other IP addresses.

There are multiple projects and VPCs in your Google Cloud organization. You need to ensure that other VPC firewall rules cannot bypass the security team's requirements. What should you do?

- A.1. Configure a hierarchical firewall policy to the organization node to allow TCP port 22 for your corporate IP address with priority 0.
- 2. Configure a hierarchical firewall policy to the organization node to deny TCP port 22 for all IP addresses with priority 1.
- B.1. Configure a VPC firewall rule to allow TCP port 22 for your corporate IP address with priority 0.
- 2. Configure a VPC firewall rule to deny TCP port 22 for all IP addresses with priority 1.
- C.1. Configure a VPC firewall rule to allow TCP port 22 for your corporate IP address with priority 1.
- 2. Configure a VPC firewall rule to deny TCP port 22 for all IP addresses with priority 0.
- D.1. Configure a hierarchical firewall policy to the organization node to allow TCP port 22 for your corporate IP address with priority 1
- 2. Configure a hierarchical firewall policy to the organization node to deny TCP port 22 for all IP addresses with priority 0.

Answer: A

Explanation:

The correct answer is A because it leverages hierarchical firewall policies at the organization level to enforce security requirements consistently across all projects and VPCs. Hierarchical firewall policies are evaluated before VPC firewall rules, providing a central point for enforcing organization-wide security mandates. Option A first establishes an allow rule for SSH (TCP port 22) originating from the corporate IP address with the highest priority (0), ensuring this specific access is always permitted. Then, it sets a deny rule for SSH for all other IP addresses with a slightly lower priority (1). Since lower numerical values indicate higher priority, this structure guarantees that the corporate IP can connect via SSH, and all other traffic will be blocked, effectively enforcing the security team's requirements. Option B incorrectly uses VPC firewall rules, which are evaluated after hierarchical rules. This would not prevent other projects from circumventing security policies, and wouldn't provide centralized control. Options C and D have priority issues. In Option C, the deny rule has a higher priority than the allow rule, which will block all traffic, even from the corporate IP address. Option D also has a priority error; the deny rule is at the highest priority (0), and therefore overrides the intended allow rule (1). Therefore, Option A is the most effective and secure way to implement the necessary security standards using Google Cloud's hierarchical firewall policies.

Supporting Concepts:

Hierarchical Firewall Policies: These policies are evaluated before VPC firewall rules and are inherited by all projects and VPCs within the organization, establishing a centralized method for security management.

Priority: In Google Cloud firewall rules, lower priority numbers take precedence. A rule with priority 0 will be evaluated before a rule with priority 1.

Centralized Security: Enforcing security at the organization level allows consistent security controls across

the entire organization and avoids the risk of a project misconfiguration circumventing policies.

Authoritative Links:

Hierarchical firewall policies: <https://cloud.google.com/vpc/docs/firewall-policies>

Firewall rules overview: <https://cloud.google.com/vpc/docs/firewalls>

Firewall rule priorities: <https://cloud.google.com/vpc/docs/firewalls#priority>

Question: 148

CertyIQ

You are designing a new application that has backends internally exposed on port 800. The application will be exposed externally using both IPv4 and IPv6 via TCP on port 700. You want to ensure high availability for this application. What should you do?

- A. Create a network load balancer that used backend services containing one instance group with two instances.
- B. Create a network load balancer that uses a target pool backend with two instances.
- C. Create a TCP proxy that uses a zonal network endpoint group containing one instance.
- D. Create a TCP proxy that uses backend services containing an instance group with two instances.

Answer: A

Explanation:

The correct answer is A because it leverages a network load balancer, which is designed for TCP and UDP traffic, making it suitable for the stated requirement of exposing the application via TCP on port 700. Network load balancers operate at the transport layer (Layer 4), providing high performance and low latency by directly forwarding traffic to backend instances. Backend services, which are a core component of load balancers, enable health checks, session affinity (if needed), and routing rules. Using an instance group with multiple instances (two, as stated) ensures high availability by distributing traffic across healthy backends and automatically handling instance failures. This approach aligns with the prompt's HA requirement as if one instance fails, traffic can be routed to the other without impacting the external facing service.

Option B is incorrect because target pools are a legacy feature and do not support backend services or health checks, limiting their capability to fulfill the high availability and health monitoring requirements. Option C is incorrect because a TCP proxy, though suitable for TCP traffic, uses zonal network endpoint groups which do not inherently provide high availability and is therefore not the ideal HA solution. Also, a proxy needs a global forwarding rule. Option D, while using backend services and instance groups, utilizes a proxy load balancer, which is meant for more complex routing and is not ideal for simple TCP forwarding where performance is key. In contrast, a network load balancer offers a simpler, more efficient solution for directing TCP traffic directly to backend instances, without the added overhead of a proxy. This approach directly fulfills the high availability design needs.

Supporting Documentation:

Google Cloud Network Load Balancer: <https://cloud.google.com/load-balancing/docs/network>

Google Cloud Backend Services: <https://cloud.google.com/load-balancing/docs/backend-service>

Google Cloud Instance Groups: <https://cloud.google.com/compute/docs/instance-groups>

Comparison of Load Balancers: <https://cloud.google.com/load-balancing/docs/choosing-load-balancer>

Question: 149

CertyIQ

You work for a university that is migrating to Google Cloud.

These are the cloud requirements:

- On-premises connectivity with 10 Gbps
- Lowest latency access to the cloud
- Centralized Networking Administration Team

New departments are asking for on-premises connectivity to their projects. You want to deploy the most cost-efficient interconnect solution for connecting the campus to Google Cloud.

What should you do?

- A. Use Shared VPC, and deploy the VLAN attachments and Dedicated Interconnect in the host project.
- B. Use Shared VPC, and deploy the VLAN attachments in the service projects. Connect the VLAN attachment to the Shared VPC's host project.
- C. Use standalone projects, and deploy the VLAN attachments in the individual projects. Connect the VLAN attachment to the standalone projects' Dedicated Interconnects.
- D. Use standalone projects and deploy the VLAN attachments and Dedicated Interconnects in each of the individual projects.

Answer: A

Explanation:

The correct answer is A, utilizing Shared VPC and deploying VLAN attachments and Dedicated Interconnect in the host project. This approach aligns with several requirements. Firstly, Shared VPC allows a central Networking Administration Team to manage network resources, including the Dedicated Interconnect, within a designated host project. This ensures consistent policies and control. Secondly, having the Dedicated Interconnect in the host project allows all service projects, managed under the Shared VPC, to leverage the high-bandwidth 10 Gbps connection. The VLAN attachments, also residing in the host project, are the logical links for service projects to connect to this Dedicated Interconnect. This centralized setup promotes cost efficiency by avoiding redundant interconnects and simplifies network administration. Deploying Dedicated Interconnect in each individual project (options C and D) would lead to significantly higher costs and management overhead as each department would need its own interconnect. Placing VLAN attachments in service projects (option B) complicates the connectivity model as they must still connect to the single Dedicated Interconnect in the host project. The centralized approach is the most efficient for connectivity and administrative ease.

Further research on Shared VPC:

[Google Cloud Documentation on Shared VPC](#)

[Google Cloud Documentation on Dedicated Interconnect](#)

Question: 150

CertyIQ

You have several microservices running in a private subnet in an existing Virtual Private Cloud (VPC). You need to create additional serverless services that use Cloud Run and Cloud Functions to access the microservices. The network traffic volume between your serverless services and private microservices is low. However, each serverless service must be able to communicate with any of your microservices. You want to implement a solution that minimizes cost. What should you do?

- A. Deploy your serverless services to the serverless VPC. Peer the serverless service VPC to the existing VPC. Configure firewall rules to allow traffic between the serverless services and your existing microservices.
- B. Create a serverless VPC access connector for each serverless service. Configure the connectors to allow traffic between the serverless services and your existing microservices.
- C. Deploy your serverless services to the existing VPC. Configure firewall rules to allow traffic between the serverless services and your existing microservices.
- D. Create a serverless VPC access connector. Configure the serverless service to use the connector for

communication to the microservices.

Answer: D

Explanation:

Here's a detailed justification for why option D is the correct answer:

Option D, "Create a serverless VPC access connector. Configure the serverless service to use the connector for communication to the microservices," is the most cost-effective and efficient solution for connecting serverless services (Cloud Run and Cloud Functions) to microservices in a private VPC, given the low network traffic volume.

Serverless VPC Access connectors allow serverless environments to communicate with resources in a VPC network. Unlike VPC peering, which creates a full network connection, connectors enable access through a proxy, which is more suitable for lower traffic scenarios and is a simpler configuration. Creating a single connector allows all serverless services to leverage it, reducing complexity and resource consumption compared to having individual connectors for each service (as in option B). This also avoids the overhead and potential increased latency associated with peering different VPCs as proposed in option A.

Option C, deploying serverless services directly into the existing VPC, would not utilize the advantages of serverless computing. These services would still require virtual machines or other infrastructure, negating the serverless model, and would have scaling and management complexity.

In summary, a single Serverless VPC Access connector offers the necessary connection while minimizing costs and operational overhead. It provides a proxy connection for all serverless services, handling low-volume traffic effectively to the existing microservices.

Authoritative links for further research:

Serverless VPC Access Overview: <https://cloud.google.com/vpc/docs/configure-serverless-vpc-access>

Choosing a Network Connectivity Option: <https://cloud.google.com/architecture/network-connectivity-options>

Understanding Serverless VPC Access: <https://cloud.google.com/serverless-vpc-access/docs/overview>

Question: 151

CertyIQ

You have provisioned a Dedicated Interconnect connection of 20 Gbps with a VLAN attachment of 10 Gbps. You recently noticed a steady increase in ingress traffic on the Interconnect connection from the on-premises data center. You need to ensure that your end users can achieve the full 20 Gbps throughput as quickly as possible. Which two methods can you use to accomplish this? (Choose two.)

- A. Configure an additional VLAN attachment of 10 Gbps in another region. Configure the on-premises router to advertise routes with the same multi-exit discriminator (MED).
- B. Configure an additional VLAN attachment of 10 Gbps in the same region. Configure the on-premises router to advertise routes with the same multi-exit discriminator (MED).
- C. From the Google Cloud Console, modify the bandwidth of the VLAN attachment to 20 Gbps.
- D. From the Google Cloud Console, request a new Dedicated Interconnect connection of 20 Gbps, and configure a VLAN attachment of 10 Gbps.
- E. Configure Link Aggregation Control Protocol (LACP) on the on-premises router to use the 20-Gbps Dedicated Interconnect connection.

Answer: BC

Explanation:

Here's a detailed justification for why options B and C are the correct methods to achieve 20 Gbps throughput on a Dedicated Interconnect, along with supporting concepts and links:

Option B is correct: Creating an additional 10 Gbps VLAN attachment in the same region allows you to utilize the full 20 Gbps capacity of your Dedicated Interconnect. The key here is using the same Multi-Exit Discriminator (MED) value when advertising routes from your on-premises router. The MED tells Google Cloud which path is preferred for traffic leaving your on-premises network. By using the same MED value across both attachments, Google Cloud's BGP router can load balance traffic across both attachments to make full use of available bandwidth up to 20 Gbps. This effectively creates an active-active path configuration for increased throughput. This is a fundamental networking concept for optimizing resource utilization.

Option C is correct: The fastest way to get all bandwidth on one vlan is by increasing the size of it. Google allows you to request an increased size for an existing attachment.

Why other options are incorrect:

Option A: While adding a VLAN attachment in a different region can increase overall capacity, it does not solve the problem of achieving 20 Gbps on a single connection for the purposes of this question. This would be more applicable for geo-redundancy or disaster recovery. Additionally, the multi-exit discriminator helps with path selection but does not help with total capacity.

Option D: Requesting a new 20 Gbps Dedicated Interconnect and configuring a 10 Gbps attachment is redundant and does not utilize the original 20 Gbps connection to its fullest. This option is more about scaling and not to achieve the requested throughput.

Option E: Link Aggregation Control Protocol (LACP) is not applicable for Dedicated Interconnect VLAN attachments with Google Cloud. LACP is used for link aggregation at Layer 2 on your own infrastructure. While you can bundle physical links between the on-prem router and your cloud provider, this is not applicable to Google. Link Aggregation can be done on the premise side if this is a limitation.

Key Concepts:

Dedicated Interconnect: A physical connection between your on-premises network and Google's network for low-latency, high-throughput connectivity.

VLAN Attachment: A logical connection over a Dedicated Interconnect, used to carry network traffic.

Multi-Exit Discriminator (MED): A BGP attribute used to influence routing path selection when multiple paths exist to reach the same destination.

BGP (Border Gateway Protocol): The routing protocol used to exchange routing information between Google Cloud and on-premises routers.

Active-Active Load Balancing: Using multiple paths simultaneously to distribute traffic and increase throughput.

Authoritative Links:

Google Cloud Interconnect Overview: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/overview>

Google Cloud Dedicated Interconnect VLAN Attachments: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/vlan-attachments>

Google Cloud BGP Configuration: <https://cloud.google.com/network-connectivity/docs/interconnect/how-to/configure-interconnect-bgp>

In summary, to achieve the full 20 Gbps throughput, you should leverage the existing Dedicated Interconnect by adding a VLAN attachment in the same region and making sure to keep your MED the same for all your VLAN attachments, or simply modify the existing attachment bandwidth via the console to the 20 Gbps you have available.

Question: 152

Your company has a Virtual Private Cloud (VPC) with two Dedicated Interconnect connections in two different regions: us-west1 and us-east1. Each Dedicated Interconnect connection is attached to a Cloud Router in its respective region by a VLAN attachment. You need to configure a high availability failover path. By default, all ingress traffic from the on-premises environment should flow to the VPC using the us-west1 connection. If us-west1 is unavailable, you want traffic to be rerouted to us-east1. How should you configure the multi-exit discriminator (MED) values to enable this failover path?

- A. Use regional routing. Set the us-east1 Cloud Router to a base priority of 100, and set the us-west1 Cloud Router to a base priority of 1
- B. Use global routing. Set the us-east1 Cloud Router to a base priority of 100, and set the us-west1 Cloud Router to a base priority of 1
- C. Use regional routing. Set the us-east1 Cloud Router to a base priority of 1000, and set the us-west1 Cloud Router to a base priority of 1
- D. Use global routing. Set the us-east1 Cloud Router to a base priority of 1000, and set the us-west1 Cloud Router to a base priority of 1

Answer: D

Explanation:

The correct answer is D because it leverages the Multi-Exit Discriminator (MED) attribute within Border Gateway Protocol (BGP) to influence the preferred path for incoming traffic. MED is used to signal a preferred route to a BGP peer (in this case, the on-premises router). With global routing, routes learned in one region are propagated to all other regions within the same VPC network, ensuring consistent reachability. Setting the MED value for the us-west1 Cloud Router to a lower value (implicitly 0 since a priority isn't provided in the answer choice and a lower MED is preferred) and the MED for the us-east1 Cloud Router to 1000 establishes us-west1 as the preferred path. In the event of a failure at us-west1, the on-premises router will automatically switch to us-east1 due to its lower MED. Option A & C are incorrect because MED is the core component for influencing BGP selection amongst the connections. Option B is wrong as MED can only be influenced at the BGP session level not the route level and Global routing allows for reachability of routes across different regions and doesn't impact path selection. This ensures seamless failover without manual intervention. MED values are relative, and a higher number means lower preference.

For further research on Cloud Router's functionality with BGP and MED attributes, please refer to the following official Google Cloud documentation:

[Cloud Router Overview](#)

[BGP Routing Options](#)

[Multi-Exit Discriminator \(MED\)](#) (external reference, as MED is a BGP standard)

Question: 153

You have the following private Google Kubernetes Engine (GKE) cluster deployment:

```

gcloud container clusters describe customer-1-cluster --zone us-central1-c
...

clusterIpv4Cidr: 192.168.36.0/24
endpoint: 192.168.38.2
ipAllocationPolicy:
  clusterIpv4Cidr: 192.168.36.0/24
  clusterIpv4CidrBlock: 192.168.36.0/24
  clusterSecondaryRangeName: customer-1-pods
  servicesIpv4Cidr: 192.168.37.0/24
  servicesIp4CidrBlock: 192.168.37.0/24
  servicesSecondaryRangeName: customer-1-svc
  useIpAliases: true
...

masterAuthorizedNetworksConfig:
...

privateClusterConfig:
  enablePrivateEndpoint: true
  enablePrivateNodes: true
  masterIpv4CidrBlock: 192.168.38.0/28
  privateEndpoint: 192.168.38.2
  publicEndpoint: 35.224.37.17
...

servicesIpv4Cidr: 192.162.37.0/24
...

subnetwork: customer-1-nodes
zone: us-central1-c

```

You have a virtual machine (VM) deployed in the same VPC in the subnetwork kubernetes-management with internal IP address 192.168.40.2/24 and no external IP address assigned. You need to communicate with the cluster master using kubectl. What should you do?

- A. Add the network 192.168.40.0/24 to the masterAuthorizedNetworksConfig. Configure kubectl to communicate with the endpoint 192.168.38.2.
- B. Add the network 192.168.38.0/28 to the masterAuthorizedNetworksConfig. Configure kubectl to communicate with the endpoint 192.168.38.2
- C. Add the network 192.168.36.0/24 to the masterAuthorizedNetworksConfig. Configure kubectl to communicate with the endpoint 192.168.38.2
- D. Add an external IP address to the VM, and add this IP address in the masterAuthorizedNetworksConfig. Configure kubectl to communicate with the endpoint 35.224.37.17.

Answer: A

Explanation:

- A. Add the network 192.168.40.0/24 to the masterAuthorizedNetworksConfig. Configure kubectl to

communicate with the endpoint 192.168.38.2.

Reference:

<https://cloud.google.com/kubernetes-engine/docs/how-to/authorized-networks>

Question: 154

CertyIQ

Your company's logo is published as an image file across multiple websites that are hosted by your company. You have implemented Cloud CDN; however, you want to improve the performance of the cache hit ratio associated with this image file. What should you do?

- A. Configure custom cache keys for the backend service that holds the image file, and clear the Host and Protocol checkboxes.
- B. Configure the default time to live (TTL) as 0 for the image file.
- C. Configure versioned URLs for each domain to serve users the image file before the cache entry expires.
- D. Configure Cloud Storage as a custom origin backend to host the image file, and select multi-region as the location type.

Answer: A

Explanation:

Here's a detailed justification for why option A is the correct solution and why the other options are incorrect:

Correct Answer: A. Configure custom cache keys for the backend service that holds the image file, and clear the Host and Protocol checkboxes.

Justification:

Cloud CDN relies on cache keys to determine whether a request can be served from the cache or needs to be forwarded to the origin. By default, Cloud CDN uses the full request URL (including protocol, host, and path) as the cache key. When an image is served from multiple websites, each with its own domain name (host), each website will have its own cache entry for the same image file. This leads to multiple cache entries, reduced cache hit ratio, and more origin requests.

Configuring custom cache keys allows us to **normalize the key** across different domains pointing to the same image. By clearing the Host and Protocol checkboxes, Cloud CDN will only use the path portion of the URL when generating the cache key. This means that requests for the logo from different domains will now map to a single cache entry, dramatically increasing the cache hit ratio and improving performance by reducing the need to fetch from the origin server. This enhances the efficiency of CDN by serving cached content to multiple website visitors from diverse domains.

Why other options are incorrect:

B. Configure the default time to live (TTL) as 0 for the image file: Setting the TTL to 0 would effectively disable caching, meaning each request would need to go back to the origin server. This would defeat the purpose of using Cloud CDN and significantly worsen performance.

C. Configure versioned URLs for each domain to serve users the image file before the cache entry expires: This would involve generating different URLs for each website, effectively preventing any caching optimization because these are all treated as different assets. While versioning is a good strategy for cache invalidation in some scenarios when a file changes, it would be counterproductive here because our goal is to serve the same image with better caching.

D. Configure Cloud Storage as a custom origin backend to host the image file, and select multi-region as

the location type: While Cloud Storage is a valid option for origin storage, it doesn't solve the cache hit ratio problem on its own. Moreover, while multi-region storage is good practice for availability, choosing it doesn't directly affect cache hit ratio. If Cloud Storage was being used as a CDN backend in this scenario without customized keys, the issues with different website domains having separate cache entries would still persist.

Authoritative Links:

Google Cloud CDN Cache Keys: <https://cloud.google.com/cdn/docs/cache-keys>

Google Cloud CDN Overview: <https://cloud.google.com/cdn/docs/overview>

Google Cloud CDN Using Custom Cache Keys: <https://cloud.google.com/cdn/docs/using-cache-keys>

In summary, customizing cache keys is crucial for optimizing caching when the same content is accessed through different URLs, which is the key issue highlighted in the given scenario.

Question: 155

CertyIQ

Your company recently migrated to Google Cloud in a single region. You configured separate Virtual Private Cloud (VPC) networks for two departments: Department A and Department B. Department A has requested access to resources that are part of Department B's VPC. You need to configure the traffic from private IP addresses to flow between the VPCs using multi-NIC virtual machines (VMs) to meet security requirements. Your configuration also must:

- Support both TCP and UDP protocols
- Provide fully automated failover
- Include health-checks
- Require minimal manual intervention in the client VMs

Which approach should you take?

- A.Create the VMs in the same zone, and configure static routes with IP addresses as next hops.
- B.Create the VMs in different zones, and configure static routes with instance names as next hops.
- C.Create an instance template and a managed instance group. Configure a single internal load balancer, and define a custom static route with the internal TCP/UDP load balancer as the next hop.
- D.Create an instance template and a managed instance group. Configure two separate internal TCP/UDP load balancers for each protocol (TCP/UDP), and configure the client VMs to use the internal load balancers' virtual IP addresses.

Answer: C

Explanation:

Okay, let's break down why option C is the correct solution and why the others fall short.

Why Option C is Correct:

Option C leverages a managed instance group (MIG) and a single internal TCP/UDP load balancer, which perfectly addresses all requirements. Here's how:

1. **Multi-NIC VMs (Implicit):** The use of an internal load balancer implies that intermediary VMs are handling traffic forwarding, effectively acting as multi-NIC routers. The internal load balancer sends traffic to one or more instances in the managed instance group which have interfaces in the networks for both departments
2. **Private IP Connectivity:** Internal load balancers operate within VPCs, ensuring private IP communication between the networks.
3. **TCP and UDP Support:** A single internal TCP/UDP load balancer (as opposed to separate load balancers for each protocol) handles both TCP and UDP traffic, eliminating the need to manage two

separate load balancing configurations. It supports both TCP and UDP, as stated in the official documentation.

4. **Automated Failover:** MIGs provide automatic failover. If a VM within the group fails, the MIG will automatically recreate and reincorporate a new VM. The load balancer seamlessly reroutes traffic to the remaining healthy VMs.
5. **Health Checks:** Load balancers include health checks. Traffic is only routed to VMs that are deemed healthy by the configured checks.
6. **Minimal Client Intervention:** Clients within Department A's VPC simply need to use the internal load balancer's IP address as their next hop. This setup minimizes changes to client VMs.
7. **Custom Static Route:** You create a custom static route in Department A's VPC pointing the destination range for Department B's VPC to the internal load balancer's IP address as the next hop. This route ensures traffic is sent to the load balancer.

Why Other Options Are Incorrect:

Option A: Using static routes with IP addresses as next hops doesn't provide automatic failover. If the VM used as the next hop fails, traffic will be lost until a manual intervention is made to change the route. Furthermore, it's not aligned with the best practices of using managed instance groups for scaling and availability.

Option B: This option attempts to improve on the first by using instance names instead of static IP addresses, but this still does not provide automated failover or health checks. Using instance names as next-hop targets is not supported in static routes within VPCs.

Option D: While using internal TCP/UDP load balancers for each protocol provides the requested functionality, a single TCP/UDP load balancer is all that is required to meet all the requirements. Using two load balancers would add extra complexity and costs, which the problem statement is trying to avoid.

Authoritative Links:

Internal TCP/UDP Load Balancing: <https://cloud.google.com/load-balancing/docs/internal/>

Managed Instance Groups: <https://cloud.google.com/compute/docs/instance-groups/>

VPC Routes: <https://cloud.google.com/vpc/docs/routes>

In conclusion, option C provides the most efficient, scalable, and resilient solution using a managed instance group and a single internal TCP/UDP load balancer, while also minimizing manual intervention and providing automated failover through health checks.

Question: 156

CertyIQ

You are designing an IP address scheme for new private Google Kubernetes Engine (GKE) clusters. Due to IP address exhaustion of the RFC 1918 address space in your enterprise, you plan to use privately used public IP space for the new clusters. You want to follow Google-recommended practices. What should you do after designing your IP scheme?

- A. Create the minimum usable RFC 1918 primary and secondary subnet IP ranges for the clusters. Re-use the secondary address range for the pods across multiple private GKE clusters.
- B. Create the minimum usable RFC 1918 primary and secondary subnet IP ranges for the clusters. Re-use the secondary address range for the services across multiple private GKE clusters.
- C. Create privately used public IP primary and secondary subnet ranges for the clusters. Create a private GKE cluster with the following options selected: `--enable-ip-alias` and `--enable-private-nodes`.
- D. Create privately used public IP primary and secondary subnet ranges for the clusters. Create a private GKE cluster with the following options selected: `--disable-default-snat`, `--enable-ip-alias`, and `--enable-private-nodes`.

Answer: C

Explanation:

The correct answer is **C**. Here's why:

Option C aligns with Google's best practices for using privately used public IP addresses in GKE. Since RFC 1918 space is exhausted, using privately used public IP space is necessary. Option C suggests creating primary and secondary subnet ranges from this public IP space. When creating the GKE cluster, using the `--enable-ip-alias` flag is crucial because it allows pods to have routable IP addresses, improving network performance and simplifying management. The `--enable-private-nodes` option ensures that nodes are not exposed to the public internet, which is essential for security when using publicly routable IP addresses in a private context. The combination of these configurations makes the GKE setup secure and fully operational using private nodes and Alias IPs within the chosen privately used public IP space.

Option A is incorrect because it attempts to use RFC 1918 addresses when the problem states there is exhaustion. Additionally, reusing secondary ranges for pods across multiple clusters will result in IP address conflicts and network routing problems.

Option B suffers from the same exhaustion issue as Option A and proposes reusing secondary ranges for services across multiple clusters. This practice will also result in network conflicts.

Option D is also incorrect despite using the proper IP ranges and enabling `ip-alias` and `private-nodes`. The `--disable-default-snat` flag is not the recommended option for this specific use case when using privately used public IP ranges. It's a specific choice made to allow GKE control of source NAT and implies a more complex use case, not for general use of private GKE clusters. While `--disable-default-snat` is a valid option, it's not relevant to the problem description and is generally not recommended, especially when using privately used public IP ranges.

In summary, option C ensures proper IP range usage, utilizes Alias IPs for optimized pod networking and keeps the nodes private using private nodes.

Authoritative Links:

Google Cloud Documentation - Alias IP Ranges: <https://cloud.google.com/kubernetes-engine/docs/how-to/alias-ips>

Google Cloud Documentation - Private Clusters: <https://cloud.google.com/kubernetes-engine/docs/how-to/private-clusters>

Google Cloud Documentation - `--disable-default-snat`:
<https://cloud.google.com/sdk/gcloud/reference/container/clusters/create#--disable-default-snat>

Question: 157**CertyIQ**

You want Cloud CDN to serve the `https://www.example.com/images/spacetime.png` static image file that is hosted in a private Cloud Storage bucket. You are using the `USE_ORIGIN_HEADERS` cache mode. You receive an HTTP 403 error when opening the file in your browser, and you see that the HTTP response has a `Cache-Control: private, max-age=0` header. How should you correct this issue?

- A. Enable negative caching for the backend bucket.
- B. Change the cache mode to Force cache all content.
- C. Configure a Cloud Storage bucket permission that gives allUsers the Storage Legacy Object Reader role.
- D. Increase the default time-to-live (TTL) for the backend service.

Answer: C**Explanation:**

The core issue is that Cloud CDN cannot access the private Cloud Storage object because it lacks the necessary permissions, leading to a 403 error. The Cache-Control: private, max-age=0 header further indicates that the object was not intended for caching by intermediate servers.

Option C addresses this directly. By granting the allUsers principal the Storage Legacy Object Reader role on the bucket, we make the objects publicly readable. This allows Cloud CDN, acting as a user, to retrieve the file for caching and delivery. It's essential for Cloud CDN to have read access to the origin to function correctly.

Option A is incorrect because negative caching deals with caching error responses, not with accessing content that is forbidden. Option B is not a solution; it forces caching without addressing the underlying access control problem. This would only cache the 403 response. Option D would affect how long the cached object is stored but will not resolve the initial issue of not being able to access the object from the bucket.

The "USE_ORIGIN_HEADERS" cache mode means that Cloud CDN respects the cache headers sent from the origin. This requires that the origin server (Cloud Storage) allows Cloud CDN to access it in the first place. With proper permissions in place, the caching mechanism defined in the origin (if applicable) will be followed. If no cache header is sent by Cloud Storage, Cloud CDN uses its default TTL.

Therefore, making the objects publicly readable through IAM permissions (Option C) is the most effective solution to allow Cloud CDN to serve the content while also considering the use of origin headers.

Relevant Resources:

Cloud CDN overview: <https://cloud.google.com/cdn/docs/overview>

Cloud CDN cache modes: <https://cloud.google.com/cdn/docs/cache-modes>

Cloud Storage permissions: <https://cloud.google.com/storage/docs/access-control/iam-permissions>

Cloud Storage Roles: <https://cloud.google.com/storage/docs/access-control/iam-roles>

Question: 158

CertyIQ

You are deploying an application that runs on Compute Engine instances. You need to determine how to expose your application to a new customer. You must ensure that your application meets the following requirements:

- Maps multiple existing reserved external IP addresses to the instance
- Processes IP Encapsulating Security Payload (ESP) traffic

What should you do?

- A. Configure a target pool, and create protocol forwarding rules for each external IP address.
- B. Configure a backend service, and create an external network load balancer for each external IP address.
- C. Configure a target instance, and create a protocol forwarding rule for each external IP address to be mapped to the instance.
- D. Configure the Compute Engine instances' network interface external IP address from None to Ephemeral. Add as many external IP addresses as required.

Answer: A

Explanation:

The correct answer is **A. Configure a target pool, and create protocol forwarding rules for each external IP address.**

Here's the justification:

Target Pools: Target pools are designed to direct traffic to groups of virtual machines. This makes them suitable for managing multiple instances behind a single external IP.

Protocol Forwarding Rules: Protocol forwarding rules are essential for mapping external IP addresses to a specific protocol and port on your instances. They are designed to handle non-HTTP(S) traffic, like ESP, making them ideal for this scenario where IPsec is required. Unlike load balancers which are more application aware, forwarding rules operate at the network layer allowing for raw IP protocol forwarding.

Mapping Multiple IPs: To fulfill the requirement of mapping "multiple existing reserved external IP addresses", you need a mechanism that allows for configuration of multiple mappings to the same set of instances. This is accomplished with a forwarding rule per external IP address.

ESP Traffic: ESP traffic, part of IPsec, operates at the network layer and requires direct protocol forwarding, which is exactly what protocol forwarding rules do. Load Balancers are optimized for HTTP(s) and don't handle network-level protocol forwarding such as ESP without extensive configuration workarounds.

Option B is incorrect because external network load balancers are designed for load balancing TCP/UDP traffic at layer 4 and do not directly map reserved IPs to specific instances. External network load balancers do not support ESP traffic in the way forwarding rules do. Option C is incorrect because target instances can only handle a single instance, not a pool of them. Furthermore, it also lacks features to manage the forwarding of multiple external IPs and is not suitable for ESP traffic. Option D is wrong because assigning ephemeral IPs directly to the instances' network interfaces does not allow multiple external IPs mapping and direct ESP handling.

In summary, using a target pool allows for a group of instances to be managed as a destination and protocol forwarding rules provide the ability to map multiple reserved external IPs directly to these instances, handling network level traffic such as ESP, which accurately meets the scenario requirements.

Authoritative Links for Further Research:

Google Cloud Documentation on Protocol Forwarding: <https://cloud.google.com/load-balancing/docs/protocol-forwarding>

Google Cloud Documentation on Target Pools: <https://cloud.google.com/compute/docs/load-balancing/target-pools>

Google Cloud Documentation on External Network Load Balancing: <https://cloud.google.com/load-balancing/docs/network/>

Question: 159

CertyIQ

Your product team has web servers running on both us-east1 and us-west1 regions in the prod-servers project. Your security team plans to install an intrusion detection system (IDS) in their own Google Cloud project to inspect the incoming network traffic. What should you do?

- A. Create a new project and a VPC for the security team.
Peer the new VPC with the web servers' VPC in the prod-servers project.
Create an internal load balancer and the IDS system in both us-east1 and us-west1.
Enable Packet Mirroring, and create packet mirroring policies inside the new project.
- C. Create a host project and a Shared VPC for the security team.
Make prod-servers a service project, and relocate the web servers to shared subnets in both regions.
Enable IP forwarding on all the web servers.
Create the IDS system in a non-shared subnet of us-east1 or us-west1.
Configure the web servers to forward the packets to the IDS system.
- C. Create a new project and a VPC for the security team.
Peer the new VPC with the web servers' VPC in the prod-servers project.
Enable IP forwarding on all the web servers.
Install the IDS system in both us-east1 and us-west1.
Configure the web servers to forward the packets to the IDS system.
- D. Create a host project and a Shared VPC for the security team.
Make prod-servers a service project, and relocate the web servers to shared subnets in both regions.
Create an internal load balancer and the IDS system in a subnet in either us-east1 or us-west1.
Enable Packet Mirroring, and create a packet mirroring policy inside the host project.

Answer: A

Explanation:

The correct answer is **A**. Here's why:

Option A provides a secure and scalable solution using Google Cloud's recommended practices. By creating a separate project and VPC for the security team, you enforce strong isolation and separation of concerns. Peering the security VPC with the prod-servers VPC allows secure network connectivity between the two environments without exposing traffic to the public internet. Utilizing internal load balancers in each region (us-east1 and us-west1) ensures high availability and resilience for the IDS system. Packet Mirroring is the crucial feature, enabling the redirection of network traffic from the web servers to the IDS without requiring changes to the web servers themselves, ensuring minimal performance overhead and maintaining simplicity. The mirror policy resides in the security project because that is where the inspection is occurring, which also keeps the project roles well aligned. The traffic can be mirrored to multiple internal load balancers, which can then send the mirrored traffic to the IDS for inspection.

Option B is incorrect because using a Shared VPC for this purpose is not the best approach. Moving all the web servers to a Shared VPC is not required to inspect the traffic.

Option C is incorrect as it suggests enabling IP forwarding and configuring web servers to forward packets, a process that introduces complexity, increased server load, and potential security vulnerabilities. This manual forwarding mechanism is also prone to misconfiguration and would not be as efficient as Packet Mirroring.

Option D is incorrect because, while it correctly uses a shared VPC for better isolation, and uses packet mirroring, it locates the load balancer and IDS in only one region, which contradicts the statement of both us-east1 and us-west1 regions.

Authoritative links for further research:

VPC Network Peering: <https://cloud.google.com/vpc/docs/vpc-peering>

Internal Load Balancing: <https://cloud.google.com/load-balancing/docs/internal>

Packet Mirroring: <https://cloud.google.com/vpc/docs/packet-mirroring>

Shared VPC: <https://cloud.google.com/vpc/docs/shared-vpc>

Question: 160

CertyIQ

You are in the process of deploying an internal HTTP(S) load balancer for your web server virtual machine (VM) instances. What two prerequisite tasks must be completed before creating the load balancer? (Choose two.)

- A. Choose a region.
- B. Create firewall rules for health checks.
- C. Reserve a static IP address for the load balancer.
- D. Determine the subnet mask for a proxy-only subnet.
- E. Determine the subnet mask for Serverless VPC Access.

Answer: AD

Explanation:

Let's break down why options A and D are the necessary prerequisites for setting up an internal HTTP(S) load balancer in Google Cloud.

Option A, "Choose a region," is crucial because an internal load balancer operates within a specific region. All associated resources, like the backend VM instances and the load balancer itself, must reside within the same

region for proper communication and functionality. This regional scope ensures low-latency connections for your internal services. Without specifying a region, the deployment would be impossible.

Option D, "Determine the subnet mask for a proxy-only subnet," is also essential. Internal load balancers, particularly HTTP(S) ones, require a dedicated proxy-only subnet within the chosen region. This subnet is solely for the load balancer's proxies, which handle traffic distribution. The subnet mask dictates the size and address space of this dedicated subnet, and it must be planned carefully to avoid address conflicts and ensure sufficient capacity for the proxies. Therefore, identifying the subnet mask for the proxy-only subnet before creation is vital.

Options B, C, and E are not immediate prerequisites, though they might be needed later. Firewall rules (option B) are needed to allow health checks, but the load balancer can be created before firewall rules are defined and can still be set up afterward. Similarly, reserving a static IP address (option C) can be useful and is generally recommended but is not a requirement for the load balancer's initial creation. Option E, concerning Serverless VPC Access, is unrelated to internal load balancing, and only applies to serverless services, not the VM backend.

In summary, choosing a region and determining the subnet mask for the proxy-only subnet are the two absolute necessities before an internal HTTP(S) load balancer can be created.

Authoritative Links:

Google Cloud Load Balancing Overview: <https://cloud.google.com/load-balancing/docs/load-balancing-overview>

Internal HTTP(S) Load Balancing: <https://cloud.google.com/load-balancing/docs/internal>

Proxy-only subnets: <https://cloud.google.com/vpc/docs/configure-proxy-only-subnet>

Question: 161

CertyIQ

Your team is developing an application that will be used by consumers all over the world. Currently, the application sits behind a global external application load balancer. You need to protect the application from potential application-level attacks. What should you do?

- A.Enable Cloud CDN on the backend service.
- B.Create multiple firewall deny rules to block malicious users, and apply them to the global external application load balancer.
- C.Create a Google Cloud Armor security policy with web application firewall rules, and apply the security policy to the backend service
- D.Create a VPC Service Controls perimeter with the global external application load balancer as the protected service, and apply it to the backend service.

Answer: C

Explanation:

The correct answer is **C. Create a Google Cloud Armor security policy with web application firewall rules, and apply the security policy to the backend service.**

Here's the justification:

Google Cloud Armor is a Web Application Firewall (WAF) service specifically designed to protect applications from layer 7 (application-level) attacks such as SQL injection, cross-site scripting (XSS), and other common web exploits. It provides pre-configured and customizable rules to detect and mitigate these threats. Since the scenario requires protection against application-level attacks for a globally accessible application, Cloud Armor is the ideal choice.

Option A, enabling Cloud CDN, focuses on caching content to improve performance and reduce latency, not on security. While CDN can reduce traffic to the origin, it doesn't actively protect against application-level attacks. Option B, creating firewall deny rules, operates at the network layer (layer 3/4) and is not effective against application-specific attacks. Firewall rules would primarily block traffic based on IP addresses and ports, not on the content of HTTP requests. Option D, using VPC Service Controls, creates a security perimeter around your cloud resources, focusing on data exfiltration prevention rather than actively blocking web attacks. It's not designed to be a WAF.

Therefore, using a Google Cloud Armor policy directly addresses the need for application-level security by inspecting incoming requests and blocking malicious patterns. Cloud Armor policies are associated with backend services, seamlessly integrating with the existing global external application load balancer setup. This setup offers centralized and consistent security enforcement, crucial for globally distributed applications.

Authoritative Links:

Google Cloud Armor Overview: <https://cloud.google.com/armor/docs/overview>

Web Application Firewall (WAF) Definition: <https://owasp.org/www-project-top-ten/> (Although OWASP is not directly Google Cloud specific, it's an industry standard for Web Security.)

Google Cloud Load Balancing Documentation: <https://cloud.google.com/load-balancing/docs/>

Question: 162

CertyIQ

You are a network administrator at your company planning a migration to Google Cloud and you need to finish the migration as quickly as possible. To ease the transition, you decided to use the same architecture as your on-premises network: a hub-and-spoke model. Your on-premises architecture consists of over 50 spokes. Each spoke does not have connectivity to the other spokes, and all traffic is sent through the hub for security reasons. You need to ensure that the Google Cloud architecture matches your on-premises architecture. You want to implement a solution that minimizes management overhead and cost, and uses default networking quotas and limits. What should you do?

- A. Connect all the spokes to the hub with Cloud VPN.
- B. Connect all the spokes to the hub with VPC Network Peering.
- C. Connect all the spokes to the hub with Cloud VPN. Use a third-party network appliance as a default gateway to prevent connectivity between the spokes.
- D. Connect all the spokes to the hub with VPC Network Peering. Use a third-party network appliance as a default gateway to prevent connectivity between the spokes.

Answer: D

Explanation:

Here's a breakdown of why option D is the correct solution, with supporting justifications:

Why Option D is Best:

Option D, "Connect all the spokes to the hub with VPC Network Peering. Use a third-party network appliance as a default gateway to prevent connectivity between the spokes," best aligns with the requirements due to a combination of cost-effectiveness, minimal management overhead, and adherence to the desired hub-and-spoke isolation:

1. **VPC Network Peering:** This is the primary method for connecting VPC networks within Google Cloud. It allows direct, private connectivity between VPC networks, providing high bandwidth and low latency without traffic leaving the Google network. This is more performant and often less expensive than Cloud VPN for internal connectivity. It's ideal for establishing the hub-and-spoke network topology within Google Cloud.

2. **Hub and Spoke Model:** Using VPC peering between each spoke VPC and the hub VPC recreates the existing architecture with minimal adjustments.
3. **Third-Party Network Appliance:** The key requirement is to maintain the isolation between spokes, similar to the on-premise setup where traffic is forced through the hub. VPC peering natively allows all connected networks to directly communicate with each other. Introducing a third-party network appliance (like a virtual firewall) as the default gateway in the hub network allows controlling all traffic between spokes and enforce the hub-and-spoke topology. This appliance acts as a centralized control point, enforcing that all inter-spoke traffic is routed through the hub for inspection and isolation policies. This maintains the existing security architecture.

Google Cloud Network Appliance Solutions

4. **Minimizing Management Overhead:** VPC peering is a managed service, reducing the operational overhead compared to managing VPN connections. This removes the overhead of maintaining VPN tunnels and reduces the overall complexity of the solution. Additionally, centralising management of security policies through a virtual appliance simplifies maintenance and enforcement.
5. **Cost-Effectiveness:** VPC peering generally has lower cost than Cloud VPN for intra-Google Cloud traffic. Also, this solution maintains the existing design with minimal adjustments making it more cost effective in the long run.

Why Other Options are Less Suitable:

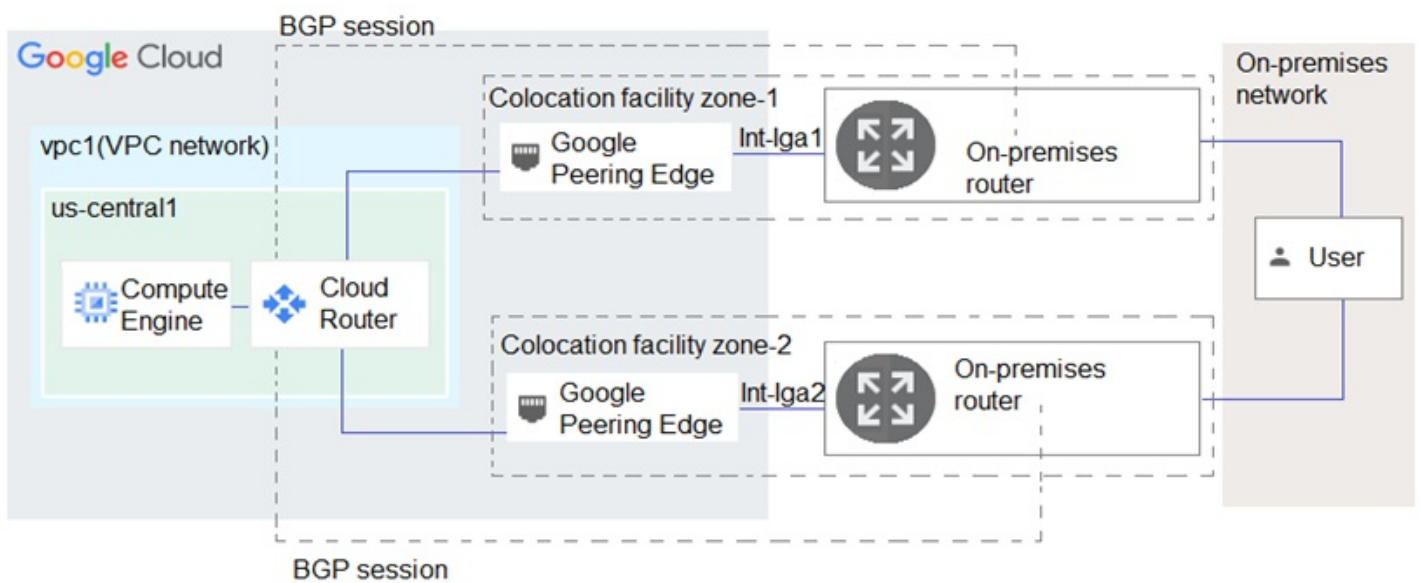
Option A and C: Using Cloud VPN for all connections would be more expensive, add management overhead due to VPN tunnel configuration and maintenance, and will add latency for private communications. VPN is a good solution to connect your on-premise datacenter to the VPC, but not for connecting within VPCs.

Option B: While using VPC peering is correct for the connections, it lacks the enforcement of the hub-and-spoke traffic flow as all peered networks can communicate with each other directly.

Question: 163

You have the networking configuration shown in the diagram. Two VLAN attachments associated with two Dedicated Interconnect connections terminate on the same Cloud Router (mycloudrouter). The Interconnect connections terminate on two separate on-premises routers. You advertise the same prefixes from the Border Gateway Protocol (BGP) sessions associated with each of the VLAN attachments.

You notice an asymmetric traffic flow between the two Interconnect connections. Which of the following actions should you take to troubleshoot the asymmetric traffic flow?



- A. From the Google Cloud console, navigate to Cloud Logging to view VPC Flow Logs and review the results.
- B. From the Cloud CLI, run `gcloud compute --project PROJECT_ID routers get-status mycloudrouter --region REGION` and review the results.
- C. From the Google Cloud console, navigate to the Hybrid Connectivity, select the Cloud Router, and view BGP sessions.
- D. From the Cloud CLI, run `gcloud compute routers describe mycloudrouter --region REGION` and review the results.

Answer: B

Explanation:

From the Cloud CLI, run `gcloud compute --project PROJECT_ID routers get-status mycloudrouter --region REGION` and review the results.

Question: 164

CertyIQ

You are responsible for designing a new connectivity solution between your organization's on-premises data center and your Google Cloud Virtual Private Cloud (VPC) network. Currently, there is no end-to-end connectivity. You must ensure a service level agreement (SLA) of 99.99% availability. What should you do?

- A. Use one Dedicated Interconnect connection in a single metropolitan area. Configure one Cloud Router and enable global routing in the VPC.
- B. Use a Direct Peering connection between your on-premises data center and Google Cloud. Configure Classic VPN with two tunnels and one Cloud Router.
- C. Use two Dedicated Interconnect connections in a single metropolitan area. Configure one Cloud Router and enable global routing in the VPC.
- D. Use HA VPN. Configure one tunnel from each interface of the VPN gateway to connect to the corresponding interfaces on the peer gateway on-premises. Configure one Cloud Router and enable global routing in the VPC.

Answer: D

Explanation:

The correct answer is **D**, which proposes using HA VPN with multiple tunnels for high availability. Here's why:

High availability, as defined by the 99.99% SLA requirement, necessitates redundancy and fault tolerance. Option A, using a single Dedicated Interconnect, presents a single point of failure. Should that connection fail,

connectivity is completely lost, violating the SLA. While Dedicated Interconnect provides high bandwidth, it doesn't inherently offer HA. Option B, Direct Peering coupled with Classic VPN, is not a robust solution for the SLA. Classic VPN relies on static routing and is not recommended for HA scenarios. Direct Peering itself doesn't address the need for highly available tunnels, and combining it with Classic VPN is an outdated approach. Option C, using two Dedicated Interconnect connections in the same metro area, does offer some redundancy. However, it's costly and doesn't provide the flexibility and dynamic routing that HA VPN offers.

HA VPN (option D) is specifically designed for high availability. By deploying a VPN gateway with two interfaces, and creating multiple tunnels to on-premises, you achieve active-active redundancy. If one tunnel or interface fails, traffic automatically switches to the other, maintaining connectivity. Cloud Router automatically manages routing using BGP, enabling dynamic route propagation and failover. Furthermore, enabling global routing in the VPC ensures that traffic can reach resources across different regions within the VPC network. This setup is the most appropriate and cost-effective way to achieve the required 99.99% SLA for hybrid connectivity compared to using Dedicated Interconnect.

Authoritative Links:

Google Cloud HA VPN Documentation: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/overview>

Google Cloud Cloud Router Documentation: <https://cloud.google.com/network-connectivity/docs/router/concepts/overview>

Google Cloud Dedicated Interconnect Documentation: <https://cloud.google.com/network-connectivity/docs/interconnect/concepts/dedicated-interconnect-overview>

Google Cloud Direct Peering Documentation: <https://cloud.google.com/network-connectivity/docs/direct-peering/concepts/overview>

Question: 165

CertyIQ

Your company is planning a migration to Google Kubernetes Engine. Your application team informed you that they require a minimum of 60 Pods per node and a maximum of 100 Pods per node. Which Pod per node CIDR range should you use?

- A./24
- B./25
- C./26
- D./28

Answer: A

Explanation:

Here's a detailed justification for selecting a /24 CIDR range for the Pod per node network in Google Kubernetes Engine (GKE), given the requirement of 60-100 Pods per node.

A /24 CIDR block provides 256 IP addresses (2^8). While not all 256 addresses are usable for pods due to reserved IPs for network services, a /24 comfortably accommodates the required 60-100 Pods per node. Using a smaller range would risk insufficient IP addresses, causing pod scheduling failures. A larger range than necessary could lead to inefficient IP address utilization but wouldn't directly hinder pod operation. Kubernetes assigns each pod an IP address from this subnet. With /24, GKE can accommodate the 60-100 pod requirements comfortably.

Options B, C, and D are unsuitable. A /25 has 128 usable IPs, not ideal for 100 pods. A /26 offers 64 IPs which is insufficient to satisfy the minimum 60 pods per node requirement. And a /28 provides just 16 IP addresses, which is far too few for the requirement. A /24 provides an optimal balance between IP address utilization and

accommodating the application team's needs. Therefore, the most appropriate choice is A, a /24 CIDR range. Choosing the correct range is crucial for a functional GKE setup and avoids potential bottlenecks during deployment.

For further reading, please see:

GKE Networking Concepts: <https://cloud.google.com/kubernetes-engine/docs/concepts/networking>

IP Ranges: <https://cloud.google.com/vpc/docs/vpc>

Question: 166

CertyIQ

You are designing a packet mirroring policy as part of your network security architecture for your gaming workload. Your infrastructure is located in the us-west2 region and deployed across several zones: us-west2-a, us-west2-b, and us-west2-c. The infrastructure is running a web-based application on TCP ports 80 and 443 with other game servers that utilize the UDP protocol. You need to deploy packet mirroring policies and collector instances to monitor web application traffic while minimizing inter-zonal network egress costs.

Following Google-recommended practices, how should you deploy the packet mirroring policies and collector instances?

- A. Create three packet mirroring policies: one for each zone. Create one group of collector instances for the us-west2 region. Configure each packet mirroring policy to match traffic for its zone based on instance-tags, and create a filter for TCP traffic.
- B. Create one packet mirroring policy for the us-west2 region. Create one group of collector instances for the us-west2 region. Configure the packet mirroring policy to match traffic for web server instances based on instance-tags, and create a filter for TCP traffic.
- C. Create three packet mirroring policies: one for each zone. Create three groups of collector instances: one group for each zone. Configure each policy to match traffic for its zone based on instance-tags, and create a filter for TCP traffic.
- D. Create three packet mirroring policies: one for each zone. Create three groups of collector instances: one group for each zone. Configure each policy to match traffic for its zone based on subnets, and create a filter for TCP traffic.

Answer: B

Explanation:

The correct answer is **B**. Here's why:

Minimizing Egress Costs: The primary goal is to minimize inter-zonal network egress costs. Packet mirroring involves copying network traffic from source instances to collector instances. Having collector instances in a different zone than the source instances results in cross-zone data transfer which incurs costs. Therefore, placing all collector instances within a single region, as suggested in option B, is crucial for cost optimization.

Regional Scope of Policies: Google Cloud packet mirroring policies are regional resources. This means that a single policy can be applied to any eligible instances within the same region. Option B takes advantage of this, deploying a single policy that covers all zones in us-west2. Creating a separate policy for each zone as suggested in options A, C, and D would be an unnecessary complication.

Targeting Traffic by Instance Tags: The question specifies that web application traffic on TCP ports 80 and 443 needs to be monitored. Instance tags provide a way to target mirroring to specific resources (web servers) regardless of which zone they reside in. Using instance tags simplifies configuration and ensures that only relevant traffic is monitored. Option B correctly uses instance tags for filtering.

Using a Single Filter: Since the same traffic type (TCP ports 80 and 443) needs to be monitored across all zones, a single filter can be defined in the regional packet mirroring policy (as suggested in option B). This eliminates the need for multiple filter configurations for each zone.

Why other options are incorrect:

Option A: Creating multiple policies and single group of collectors doesn't align to the principle of having a single region policy. Furthermore, the policy must match based on instance tags and traffic and not based on zones.

Option C and D: These options suggest creating multiple policies and collectors, each specific to a zone, which increases cost and complexity and is redundant because policies are regional by nature and instance tags can be used to filter the traffic. Option D also proposes using subnets to filter which isn't necessary since instance tags are already in use.

Authoritative Links for Research:

Google Cloud Packet Mirroring: <https://cloud.google.com/vpc/docs/packet-mirroring>

Packet Mirroring Concepts: <https://cloud.google.com/vpc/docs/packet-mirroring#concepts>

Google Cloud Networking Pricing: <https://cloud.google.com/vpc/network-pricing>

In summary, option B is the most efficient and cost-effective approach, adhering to Google's recommended practices by using a regional policy, instance tags for targeting specific resources, and a single group of collectors in the same region, minimizing inter-zonal traffic costs.

Question: 167

CertyIQ

Your company recently migrated to Google Cloud. You configured separate Virtual Private Cloud (VPC) networks for Department A and Department B. You need to configure both VPC networks to have access to the same on-premises location through separate links with full isolation between the VPC networks. Your design must also query on-premises DNS servers from workloads in Google Cloud using conditional forwarding. You want to minimize operational overhead. What should you do?

- A. Customize the operating system DNS configuration files to target the on-premises DNS servers.
- B. Keep the different VPC networks from both departments isolated with different on-premises links, and separate Cloud DNS private zones and Cloud DNS forwarding zones.
- C. Peer Department A's and Department B's VPC networks to have all on-premises connectivity via a single VPC network. Use separate Cloud DNS private zones and Cloud DNS forwarding zones.
- D. Configure a Cloud DNS Peering zone in Department A's VPC network pointing to Department B's VPC and a Cloud DNS outbound forwarding zone in Department B's VPC network. Use separate on-premises links in each VPC network.

Answer: B

Explanation:

Okay, here's a detailed justification for why option B is the correct solution, along with supporting explanations and links:

Justification for Option B:

Option B, "Keep the different VPC networks from both departments isolated with different on-premises links, and separate Cloud DNS private zones and Cloud DNS forwarding zones," is the most appropriate because it directly addresses all requirements of the scenario while minimizing operational overhead. Maintaining isolation between Department A's and Department B's VPC networks through distinct on-premises links aligns with the requirement for full isolation. By using separate Cloud DNS private zones and forwarding zones in each VPC, we avoid creating dependencies or shared infrastructure between the departments' DNS configurations. This approach prevents potential cross-departmental conflicts and reduces the risk of impacting one department's workloads due to issues in the other's environment. Each department has its own defined DNS resolution path, allowing them to use their specific on-premises DNS servers without routing

traffic through the other's network.

Why other options are incorrect:

Option A: Manually configuring OS DNS files is not a scalable or manageable solution, particularly in a dynamic cloud environment with potentially many VMs. This would require ongoing manual changes for every instance and greatly increase operational overhead, going against the requirement for minimal operational overhead. This is prone to configuration errors and requires specialized maintenance, which is not suitable for most enterprise settings.

Option C: Peering VPCs would create a single point of failure and violate the requirement for full isolation between the departments. While it could facilitate DNS resolution, it does not adhere to the isolation policy set out in the problem. This also complicates network management by combining traffic and potentially causing bottlenecks.

Option D: Cloud DNS peering only applies to resolving names within the same organization, and the question requires querying on-premises servers, and thus peering would be irrelevant. Moreover, this creates an unnecessary dependency and violates the isolation requirement between departments by routing one department's DNS requests through another's network. This would not provide full isolation of links and introduces unnecessary forwarding complexity.

Key Cloud Concepts and Links:

Virtual Private Cloud (VPC): VPCs provide isolated and configurable private networks in the cloud, enabling the separation of resources. (<https://cloud.google.com/vpc/docs/vpc>)

Cloud Interconnect: This allows you to connect your on-premises network to your Google Cloud VPC network. (<https://cloud.google.com/network-connectivity/docs/interconnect/concepts/overview>)

Cloud DNS: This is Google Cloud's scalable and reliable DNS service. (<https://cloud.google.com/dns/docs/overview>)

Private Zones: These provide name resolution within a VPC network. (<https://cloud.google.com/dns/docs/private-zones>)

Forwarding Zones: These direct DNS queries to specific servers outside the VPC network, like on-premises DNS servers. (<https://cloud.google.com/dns/docs/forwarding>)

In summary, option B best fulfills all the requirements through clear separation, DNS control, and minimised operational overhead by adhering to best practices of Google Cloud networking.

Question: 168

CertyIQ

You are planning to use Terraform to deploy the Google Cloud infrastructure for your company. The design must meet the following requirements:

- Each Google Cloud project must represent an internal project that your team will work on.
- After an internal project is finished, the infrastructure must be deleted.
- Each internal project must have its own Google Cloud project owner to manage the Google Cloud resources.
- You have 10-100 projects deployed at a time.

While you are writing the Terraform code, you need to ensure that the deployment is simple and the code is reusable with centralized management. What should you do?

- A.Create a single project and single VPC for each internal project.
- B.Create a single Shared VPC and attach each Google Cloud project as a service project.
- C.Create a single project and additional VPCs for each internal project.
- D.O Create a Shared VPC and service project for each internal project.

Answer: B

Explanation:

The best approach is **B. Create a single Shared VPC and attach each Google Cloud project as a service project**. Here's why:

Using a Shared VPC model allows for centralized management of the network infrastructure, addressing the need for simple and reusable code. A single Shared VPC hosts the network resources (subnets, firewalls, routes) used by all internal projects. Each internal project becomes a "service project" connected to the Shared VPC's network. This allows each project to have its own project owner responsible for their resources, satisfying that requirement. Service projects can be easily attached and detached, enabling clean deletion of finished project infrastructure by simply removing the service project connection, which aligns with the infrastructure lifespan requirement. Using a single VPC, rather than multiple as seen in options A and C, optimizes resource utilization and avoids unnecessary complexity, particularly for a fluctuating number of projects (10-100). This approach promotes consistent network configurations and simplifies network management tasks like firewall rules across all projects. This architecture allows teams to maintain autonomy over their project's cloud resources while adhering to a centrally managed and reusable network infrastructure via the Shared VPC. Option D creates redundant complexity and management overhead with multiple Shared VPCs. Option A and C fail in allowing centralized networking management.

Authoritative Links:

Shared VPC Overview: <https://cloud.google.com/vpc/docs/shared-vpc>

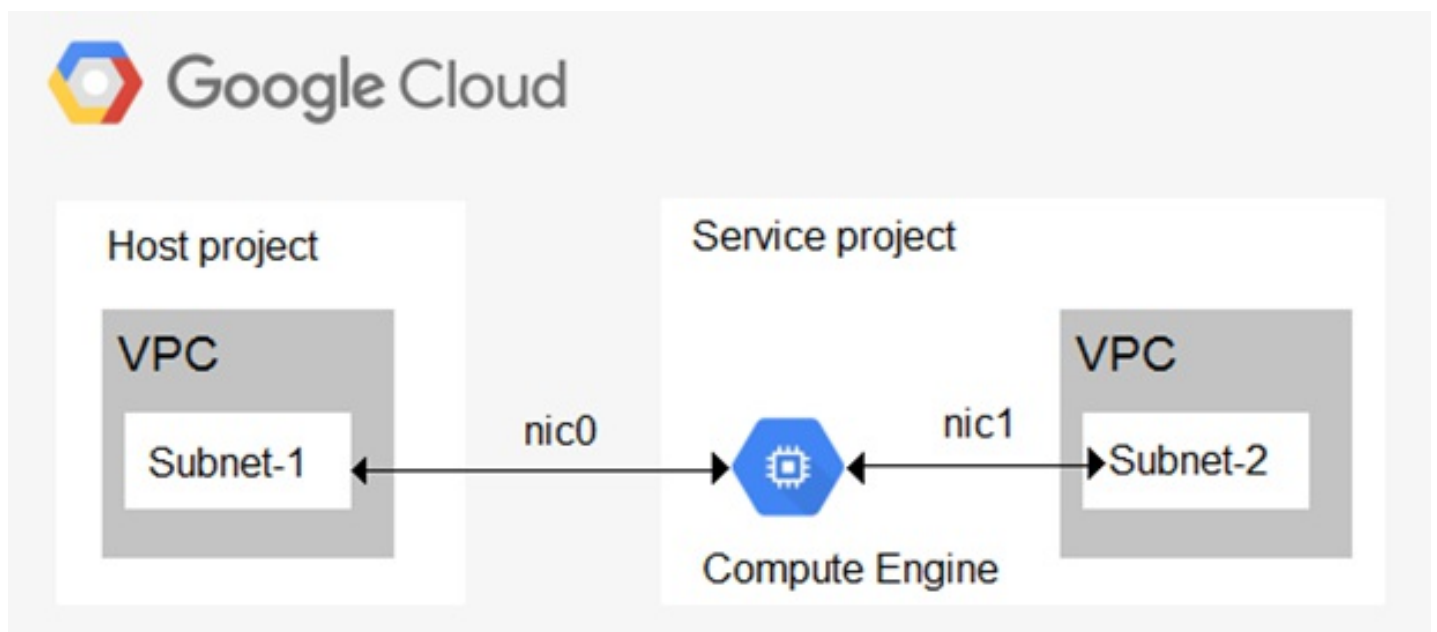
Terraform Google Provider Shared VPC Example:

https://registry.terraform.io/providers/hashicorp/google/latest/docs/resources/compute_shared_vpc_host_project

Question: 169

CertyIQ

You have the following Shared VPC design. VPC Flow Logs is configured for Subnet-1 in the host VPC. You also want to monitor flow logs for Subnet-2. What should you do?



- A. Configure a VPC Flow Logs filter for Subnet-2 in the host project VPC.
- B. Configure VPC Flow Logs in the service project VPC for Subnet-2.
- C. Configure Packet Mirroring in both the host and service project VPCs.
- D. Configure a firewall rule to permit Subnet-2 IP addresses outbound in the host project VPC.

Answer: A

Explanation:

Configure a VPC Flow Logs filter for Subnet-2 in the host project VPC.

Question: 170

CertyIQ

You are configuring your organization's Google Cloud environment to connect to your on-premises network, which does not support Border Gateway Protocol (BGP). Your on-premises network has 30 CIDR ranges that must be reachable from Google Cloud. Your VPN gateway creates a unique child security association (SA) per CIDR. You must ensure that the 30 CIDR ranges in your on-premises network are reachable from Google Cloud.

Following Google-recommended practices, which two methods can you use to accomplish this? (Choose two.)

- A. Create a single Cloud VPN tunnel that uses route-based VPN.
- B. Create a single Cloud VPN tunnel that uses policy-based routing with 30 CIDRs as the remote traffic selectors.
- C. Create multiple Cloud VPN tunnels that use policy-based routing so that each tunnel has one CIDR block for its local traffic selector and one CIDR block for its remote traffic selector. Connect each tunnel to unique peer IP addresses.
- D. Create multiple Cloud VPN tunnels that use policy-based routing with 10 CIDR per tunnel as the remote traffic selectors.
- E. Create multiple Cloud VPN tunnels that use policy-based routing so that each tunnel has one CIDR block for its local traffic selector and one CIDR block for its remote traffic selector. Connect each tunnel to the same peer IP address.

Answer: AC

Explanation:

Here's a detailed justification for why options A and C are the correct methods to connect a Google Cloud environment to an on-premises network lacking BGP support, with 30 CIDR ranges needing reachability, and unique SAs per CIDR:

Option A: Create a single Cloud VPN tunnel that uses route-based VPN.

This is a valid approach because route-based VPNs (also known as dynamic routing) rely on route advertisements rather than explicitly defined policy selectors for traffic routing. While the on-prem network doesn't support BGP, we can still configure the Google Cloud side to use a route-based Cloud VPN and statically define the necessary routes for the 30 CIDR ranges within Google Cloud. This way, Google Cloud learns about the on-premises routes via these static entries, and traffic is properly routed across the VPN tunnel. Though not dynamically learned from BGP, the effect is similar when static routes are configured on the Google Cloud side, thereby allowing connectivity.

Option C: Create multiple Cloud VPN tunnels that use policy-based routing so that each tunnel has one CIDR block for its local traffic selector and one CIDR block for its remote traffic selector. Connect each tunnel to unique peer IP addresses.

This approach works due to the nature of policy-based routing. With 30 unique CIDR ranges to reach, using policy-based VPN with single CIDR selectors per tunnel effectively builds 30 distinct tunnels. By creating one tunnel for each unique on-prem CIDR block, and associating the corresponding peer IP, Google Cloud can correctly route traffic to each on-prem destination through its respective tunnel. This ensures that each tunnel has dedicated traffic based on its selector rules and because each tunnel connects to a unique IP address, they are fully independent without the need for BGP route selection. The one-to-one mapping of the local and remote selectors per tunnel aligns with Google's recommendation of having unique SAs per CIDR

range, because each tunnel has its own specific security association.

Why other options are incorrect:

Option B: A single policy-based tunnel with 30 CIDRs as remote selectors does not create unique SAs, and this will cause issues. Also, the on-prem network does not support this.

Option D: Although it creates multiple tunnels, the grouping of 10 CIDR blocks per tunnel would not align with the unique SA per CIDR range requirement and is an inefficient usage of tunnels for this requirement.

Option E: Creating multiple tunnels to the same peer IP for distinct CIDR ranges can lead to routing conflicts, especially when using policy-based VPN, where the on-prem network has no knowledge of this setup and how to react, or cannot negotiate multiple tunnels with the same IP. Furthermore, it doesn't align with best practices and may cause issues with unique SAs.

Authoritative links:

Cloud VPN Overview: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/overview>

Choosing a VPN routing option: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/routing>

Cloud VPN IPsec Security: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/ipsec>

In summary, route-based VPN with static routes (option A) and policy-based VPN with one-to-one mapping between tunnels, CIDR ranges, and peer IPs (option C) are the two suitable approaches, that comply with the prompt's requirements and Google's best practices in a scenario where BGP isn't available.

Question: 171

CertyIQ

You have two VPCs: VPC A in Project A and VPC B in Project B. The VPCs are peered, and each VPC has VM instances in four zones. You are using the Network Intelligence Center Performance Dashboard to investigate the packet loss for traffic flows that start in VPC A and terminate in VPC B. You need the reported packet loss metric to have at least a 90% confidence level. What should you do?

- A. Ensure that each zone in each of the VPC networks has at least 10 compute instances. Look in Project A for the reported metric.
- B. Ensure that each zone in each of the VPC networks has at least 9 compute instances. Look in Project B for the reported metric.
- C. Ensure that each zone in each of the VPC networks has at least 9 compute instances. Look in Project A for the reported metric.
- D. Ensure that each zone in each of the VPC networks has at least 10 compute instances. Look in Project B for the reported metric.

Answer: B

Explanation:

Here's a detailed justification for why option B is the correct answer:

The Network Intelligence Center's Performance Dashboard calculates packet loss using data from Google Cloud's telemetry. To achieve a 90% confidence level in the reported packet loss metric, a sufficient number of compute instances (VMs) must participate in the observed traffic flows. Specifically, each zone in each of the involved VPC networks (VPC A and VPC B in this case) must have at least 9 instances actively generating and receiving traffic to provide a statistically robust sample. This requirement ensures that the measured packet loss is representative of the overall network behavior and not skewed by limited sampling. Option A and D specify 10 instances, this number is incorrect for achieving 90% confidence.

The Performance Dashboard reports metrics from the perspective of the destination project. Since the traffic flow starts in VPC A (Project A) and terminates in VPC B (Project B), you must examine the Performance

Dashboard within Project B to see the packet loss metric with the required confidence level. Looking in Project A, as suggested in options A and C, would not give you the correct metric for the traffic flow described in the question.

Therefore, the correct action is to ensure that each zone in each VPC network (A and B) has at least 9 compute instances, and then to view the reported packet loss metric within Project B.

Key concepts:

Statistical Confidence Level: A measure of the certainty that the calculated statistic accurately reflects the true underlying population. A higher confidence level requires a larger sample size.

Network Intelligence Center: A Google Cloud service for monitoring and troubleshooting network performance, which includes the Performance Dashboard for packet loss analysis.

Telemetry Data: Data collected from network infrastructure that provides insights into network traffic and performance.

Source vs Destination in network monitoring: When analyzing bidirectional traffic, a destination-centric view provides more relevant insights for troubleshooting based on packet flow path.

Authoritative links:

Network Intelligence Center Performance Dashboard Overview: <https://cloud.google.com/network-intelligence-center/docs/performance-dashboard/overview> - This document provides a general overview of the Performance Dashboard.

Network Intelligence Center Performance Dashboard Metrics: <https://cloud.google.com/network-intelligence-center/docs/performance-dashboard/metrics> - This document provides information on the metrics available in the Performance Dashboard, including the required minimum compute instance count for confidence levels.

Question: 172

CertyIQ

You are designing a new network infrastructure for your customer in Google Cloud. Your customer requires a connection between two Google Cloud VPCs that must include a VPN tunnel. You want to follow Google-recommended practices while ensuring maximum availability of the connection. Which VPN configuration should you choose?

- A. Policy-based VPN using Classic VPN between the two Google Cloud VPCs
- B. Border Gateway Protocol (BGP)-based VPN using Classic VPN between the two Google Cloud VPCs
- C. Route-based VPN using Classic VPN between the two Google Cloud VPCs
- D. Border Gateway Protocol (BGP)-based VPN using HA VPN between the two Google Cloud VPCs

Answer: D

Explanation:

The correct answer is **D. Border Gateway Protocol (BGP)-based VPN using HA VPN between the two Google Cloud VPCs**. Here's why:

High Availability (HA) VPN is Google Cloud's recommended solution for production VPN connections, prioritizing reliability and resilience. It employs two VPN gateways and automatically reroutes traffic if one fails, ensuring minimal downtime. This contrasts with Classic VPN, which does not offer this level of built-in redundancy. Using HA VPN aligns with the customer's requirement for maximum availability.

Moreover, Border Gateway Protocol (BGP) is the recommended routing protocol for HA VPN and dynamic routing. BGP enables dynamic route exchange between the VPN gateways and the on-premises or other network, allowing automatic adjustments to network topology changes. BGP is preferred over route-based or

policy-based VPNs because of its flexibility and efficiency in managing network traffic, offering granular control of routing decisions, and providing advanced metrics like route selection through AS path length. Route based VPNs offer less granular control and policy based VPNs are not recommended for a modern infrastructure.

Therefore, BGP-based HA VPN provides the best solution for a highly available connection between two Google Cloud VPCs.

Authoritative Links:

Google Cloud HA VPN Documentation: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/ha-vpn>

Google Cloud Classic VPN vs HA VPN: <https://cloud.google.com/network-connectivity/docs/vpn/concepts/classic-vs-ha-vpn>

Google Cloud BGP Overview: <https://cloud.google.com/network-connectivity/docs/router/concepts/bgp>

Question: 173

CertyIQ

Your company is moving to a hybrid cloud environment and needs to connect two on-premises data centers to Google Cloud. Your company has opted for no service level agreement (SLA) on the Dedicated Interconnect ports. You set up a single Dedicated Interconnect to connect each on-premises data center to Google Cloud: one Dedicated Interconnect in us-east1 and another Dedicated Interconnect in us-west1. You also configured a Cloud Router for each Dedicated Interconnect in each respective region. You now need to configure the Interconnect attachments to provide as much high availability diversity as possible based on this design. What should you do?

- A. •Build one VLAN attachment from each Dedicated Interconnect corresponding to the Cloud Router in that region.
•Enable global routing at the VPC layer.
- B. •Build one VLAN attachment from each Dedicated Interconnect corresponding to the Cloud Router in that region.
•Enable regional routing at the VPC layer.
- C. •Build two VLAN attachments from each Dedicated Interconnect: one connecting to the Cloud Router in us-east1, and one connecting to the Cloud Router in us-west1.
•Enable regional routing at the VPC layer.
- D. •Build two VLAN attachments from each Dedicated Interconnect: one connecting to the Cloud Router in us-east1, and one connecting to the Cloud Router in us-west1.
•Enable global routing at the VPC layer.

Answer: D

Question: 174

CertyIQ

Your company runs an enterprise platform on-premises using virtual machines (VMs). Your internet customers have created tens of thousands of DNS domains pointing to your public IP addresses allocated to the VMs. Typically, your customers hard-code your IP addresses in their DNS records. You are now planning to migrate the platform to Compute Engine and you want to use Bring Your Own IP. You want to minimize disruption to the platform. What should you do?

- A. Create a VPC and request static external IP addresses from Google Cloud. Assign the IP addresses to the Compute Engine instances. Notify your customers of the new IP addresses so they can update their DNS records.
- B. Verify ownership of your IP addresses. After the verification, Google Cloud advertises and provisions the IP prefix for you. Assign the IP addresses to the Compute Engine instances.
- C. Create a VPC with the same IP address range as your on-premises network. Assign the IP addresses to the Compute Engine instances.

D. Verify ownership of your IP addresses. Use live migration to import the prefix. Assign the IP addresses to the Compute Engine instances.

Answer: D

Explanation:

Comparing options B and D: Both involve verifying ownership and bringing the IPs to Google Cloud (BYOIP). Option B describes the general BYOIP process. Option D specifically mentions "live migration to import the prefix," which is the recommended method within BYOIP specifically designed for minimizing disruption when traffic is actively flowing to those IPs. Since the goal is to minimize disruption, using the live migration feature of BYOIP is the most appropriate approach.

Question: 175

CertyIQ

You need to create the technical architecture for hybrid connectivity from your data center to Google Cloud. This will be managed by a partner. You want to follow Google-recommended practices for production-level applications. What should you do?

- A. Ask the partner to install two security appliances in the data center. Configure one VPN connection from each of these devices to Google Cloud, and ensure that the VPN devices on-premises are in separate racks on separate power and cooling systems.
- B. Configure two Partner Interconnect connections in one metropolitan area (metro). Make sure the Interconnect connections are placed in different metro edge availability domains. Configure two VLAN attachments in a single region, and configure regional dynamic routing on the VPC.
- C. Configure two Partner Interconnect connections in one metro and two connections in another metro. Make sure the Interconnect connections are placed in different metro edge availability domains. Configure two VLAN attachments in one region and two VLAN attachments in another region, and configure global dynamic routing on the VPC.
- D. Configure two Partner Interconnect connections in one metro and two connections in another metro. Make sure the Interconnect connections are placed in different metro edge availability domains. Configure two VLAN attachments in one region and two VLAN attachments in another region, and configure regional dynamic routing on the VPC.

Answer: C

Explanation:

C. Configure two Partner Interconnect connections in one metro and two connections in another metro. Make sure the Interconnect connections are placed in different metro edge availability domains. Configure two VLAN attachments in one region and two VLAN attachments in another region, and configure global dynamic routing on the VPC.

This setup ensures high availability and redundancy by distributing connections across multiple metros and metro edge availability domains. Global dynamic routing allows traffic to be efficiently routed across regions, improving resilience and performance for production workloads.

Question: 176

CertyIQ

You are deploying your infrastructure in the us-central1 region. Your on-premises data center is located in New York City, and the Google Cloud region closest to New York City is us-east4. Your Cloud Interconnect is located in Ashburn, Virginia (VA), United States. You need to use Cloud Interconnect to connect your application infrastructure with backend systems in your data center location. You do not expect the application bandwidth to exceed 500 Mbps. You want to minimize latency and cost. What should you do?

- A. Create a Cloud Router and VLAN attachments in the us-east4 region attached to your physical Interconnect in Ashburn, VA. Enable global routing in your VPC. Set the bandwidth on the VLAN attachments to 500 Mbps.
- B. Create a Cloud Router and VLAN attachments in the us-east4 region attached to your physical Interconnect in Ashburn, VA. Enable global routing in your VPC.
- C. Create a Cloud Router in the us-central1 region and VLAN attachments in the us-east4 region attached to your physical Interconnect in Ashburn, VA. Enable global routing in your VPC.
- D. Create a Cloud Router and VLAN attachments in the us-central1 region attached to your physical Interconnect in Ashburn, VA.

Answer: A

Question: 177

CertyIQ

You have provisioned a Cloud Interconnect connection with a VLAN attachment. You configured Border Gateway Protocol (BGP) between your on-premises router and your Cloud Router. After deploying and testing the connection, you discover that the BGP session is not established between your on-premises router and the Cloud Router. Which two actions should you take to resolve this issue? (Choose two.)

- A. From the Google Cloud console, run `gcloud compute routers get-status` to verify the Address Resolution Protocol (ARP) learned.
- B. Verify that you have configured the on-premises router's subinterface with a subnet mask of /31.
- C. Verify that you have configured the on-premises router's eBGP multihop with a minimum hop length of 4.
- D. Verify that you have configured the on-premises router's BGP security parameters to use MD5 authentication.
- E. From the Google Cloud console, run `gcloud compute interconnects get-diagnostics` to verify the Address Resolution Protocol (ARP) learned.

Answer: AB

Question: 178

CertyIQ

Your company has a single on-premises data center that needs to be connected to a VPC in Google Cloud. The total bandwidth requirement is 10Gbps. The connection must be redundant and have a minimum SLA of 99.9%. Due to the sensitive nature of the workloads, you need to implement the solution with the lowest latency. What should you do?

- A. Order a 10Gbps Partner Interconnect VLAN attachment. Create a Cloud Router in your Google Cloud VPC.
- B. Order two 10Gbps Dedicated Interconnect connections in a single metropolitan area (metro). Distribute the connections across different edge availability domains. Create a Cloud Router and two 10Gbps VLAN attachments.
- C. Create one HA VPN gateway. Create two tunnels—one tunnel for each of the two interfaces of the HA VPN gateway. Terminate each of the two tunnels on the single public IP address that is configured on the VPN termination device that is located on-premises.
- D. Create one HA VPN gateway. Create two tunnels—one tunnel for each of the two interfaces of the HA VPN gateway. Terminate each of the two tunnels on different public IP addresses that are configured on the VPN termination device that is located on-premises.

Answer: B

Question: 179

CertyIQ

Your company deployed a hub and spoke architecture in Google Cloud to host their workloads. They use VPC

network peerings to connect the hub and the spokes. You need to replicate the design and use Network Connectivity Center. What should you do?

- A. Choose a Network Connectivity Center star topology. Deploy the hub VPC in the center group. Deploy the spoke VPCs in the edge group.
- B. Choose a Network Connectivity Center star topology. Deploy the spoke VPCs in the center group. Deploy the hub VPC in the edge group.
- C. Choose a Network Connectivity Center mesh topology. Configure the hub and the spokes as Network Connectivity Center spokes.
- D. Choose a Network Connectivity Center mesh topology. Configure the spokes as Network Connectivity Center spokes.

Answer: A

Question: 180

CertyIQ

You are deploying HA VPN within Google Cloud. You need to exchange routes dynamically between your on-premises gateway and Google Cloud. You have already created a HA VPN gateway and a peer VPN gateway resource. What should you do?

- A. Create a Cloud Router, add VPN tunnels, and configure BGP sessions.
- B. Create a Cloud Router, add VPN tunnels, and configure static routes to your subnet ranges.
- C. Create a second HA VPN gateway, add VPN tunnels, and create firewall rules to allow BGP traffic to the Cloud Router.
- D. Create a second HA VPN gateway, add VPN tunnels, and enable global dynamic routing.

Answer: A

Question: 181

CertyIQ

You are implementing a VPC architecture for your organization by using a Network Connectivity Center hub and spoke topology:

- There is one Network Connectivity Center hybrid spoke to receive on-premises routes.
- There is one VPC spoke that needs to be added as a Network Connectivity Center spoke.

Your organization has limited routable IP space for their cloud environment (192.168.0.0/20). The Network Connectivity Center spoke VPC is connected to on-premises with a Cloud Interconnect connection in the us-east4 region. The on-premises IP range is 172.16.0.0/16. You need to reach on-premises resources from multiple Google Cloud regions (us-west1, europe-central1, and asia-southeast1) and minimize the IP addresses being used. What should you do?

- A. 1. Configure a Private NAT gateway and NAT subnet in us-west1(192.168.1.0/24), europe-central1(192.168.2.0/24) and asia-southeast1(192.168.3.0/24).
2. Add the VPC as a spoke and configure an export include policy to advertise only 192.168.1.0/24, 192.168.2.0/24, and 192.168.3.0/24 to the hub.
3. Enable global dynamic routing to allow resources in us-west1, us-central1 and asia-southeast1 to reach the on-premises location through us-east4.
- B. 1. Configure a Private NAT gateway instance in us-west1(172.16.1.0/24), europe-central1(172.16.2.0/24), and asia-southeast1(172.16.3.0/24).
2. Add the VPC as a spoke and configure an export include policy on the VPC spoke to advertise only the NAT subnets 172.16.1.0/24, 172.16.2.0/24, and 172.16.3.0/24 to the hub.
3. Enable global dynamic to allow resources in us-west1, us-central1, and asia-southeast1 to reach the on-premises location through us-east4.
- C. 1. Configure a Private NAT gateway instance in us-east4(192.168.1.0/24).
2. Add the VPC as a spoke and configure an export include policy on the VPC spoke to advertise 192.168.1.0/24

to the hub.

3. Enable global dynamic routing to allow resources in us-west1, us-central1 and asia-southeast1 to reach the on-premises location through us-east4.

D.1. Configure a Private NAT gateway instance in us-west1(192.168.1.0/24), europe-central1(192.168.2.0/24), and asia-southeast1(192.168.3.0/24).

2. Add the VPC as a spoke and configure an export exclude policy on the VPC spoke to advertise only the NAT subnets 192.168.1.0/24, 192.168.2.0/24, and 192.168.3.0/24 to the hub.

3. Enable global dynamic routing to allow resources in us-west1, us-central1, and asia-southeast1 to reach the on-premises location through us-east4.

Answer: A

Question: 182

CertyIQ

You have several VMs across multiple VPCs in your cloud environment, which require access to internet endpoints. These VMs cannot have public IP addresses due to security policies, so you plan to use Cloud NAT to provide outbound internet access. Within your VPCs, you have several subnets in each region. You want to ensure that only specific subnets have access to the internet through Cloud NAT. You want to avoid any unintentional configuration issues caused by other administrators, and align to Google-recommended practices. What should you do?

A. Create a firewall rule in each VPC at priority 500 that targets all instances in the network and denies egress to the internet, 0.0.0.0/0. Create a firewall rule at priority 300 that targets all instances in the network, has a source filter that maps to the allowed subnets, and allows egress to the internet, 0.0.0.0/0. Deploy Cloud NAT, and configure all primary and secondary subnet source ranges.

B. Create a constraints/compute.restrictCloudNATUsage organizational policy constraint. Attach the constraint to a folder that contains the associated projects. Configure the allowedValues to only contain the subnets that should have internet access. Deploy Cloud NAT and select only the allowed subnets.

C. Create a firewall rule in each VPC at priority 500 that targets all instances in the network and denies egress to the internet, 0.0.0.0/0. Create a firewall rule at priority 300 that targets all instances in the network, has a source filter that maps to the allowed subnets, and allows egress to the internet, 0.0.0.0/0. Deploy Cloud NAT, and configure a custom source range that includes the allowed subnets.

D. Deploy Cloud NAT in each VPC, and configure a custom source range that includes the allowed subnets. Configure Cloud NAT rules to only permit the allowed subnets to egress through Cloud NAT.

Answer: B

Question: 183

CertyIQ

Your organization has five different VPCs across different projects in your Google Cloud organization that need high-throughput connectivity. You have performed an audit of the IP address utilization in each VPC, and there are two overlapping subnets that are used by two of the VPCs: 240.0.0.0/16 and 240.128.0.0/24. You have confirmed that no Class E subnets (240.0.0.0/4) will require inter-VPC connectivity, but all other subnets in the VPCs will need connectivity. You need to deploy a Google Cloud routing solution to meet the connectivity requirements. What should you do?

A. Create a full mesh of VPC Network Peering connections between all five VPCs. Make sure not to import or export subnet routes with public IP addresses. Add Cloud network firewall policy rules to allow traffic.

B. Create a Network Connectivity Center hub with a mesh topology. Add a VPC spoke for each of the five VPCs and configure an export exclude filter for 240.0.0.0/4. Add Cloud network firewall policy rules to allow traffic.

C. Create a series of multiple network interface VMs with an interface in each VPC. Place the VMs in an instance group. Create an internal passthrough Network Load Balancer in each VPC with the backend of the instance group. Configure custom static routes in each VPC with the next hop of the respective load balancer. Add Cloud network firewall policy rules to allow traffic.

D. Create a full mesh of VPC Network Peering connections between all five VPCs with an export exclude filter for 240.0.0.0/4 on every side. Add Cloud network firewall policy rules to allow traffic.

Answer: B

Question: 184

CertyIQ

You are attempting to establish a HA VPN to your on-premises network; however, the VPN connection is not establishing successfully. You have full administrative control over the Google Cloud networking environment and the on-premises firewalls that are acting as the VPN devices. The Google Cloud console shows "Negotiation failure" and "BGP is down". You check Cloud Logging by using a query for resource.type="vpn_gateway" and resource.labels.gateway_id="TUNNEL_ID_NUMBER". Logs Explorer shows frequent log entries:

```
log name: .../logs/cloud.googleapis.com%2Fipsec_events"  
type: "vpn_gateway"  
textPayload: "received NO_PROPOSAL_CHOSEN notify, no CHILD_SA built"
```

You need to troubleshoot the VPN failure and take corrective action based on the Cloud Logging entries. What should you do?

- A. Update the Google Cloud BGP session configuration to match the BGP peer ASN on the on-premises side.
- B. Compare and review the Phase 2 settings on the on-premises firewall. Make sure the settings match one of the supported cipher suites for HA VPN.
- C. Create a new Cloud VPN gateway in a region closer to the peer VPN gateway.
- D. Compare the Phase 1 settings and recreate the Cloud VPN tunnel by choosing a different IKE version and pre-shared key.

Answer: B

Question: 185

CertyIQ

Your team deployed two applications in GKE that are exposed through an external Application Load Balancer. When queries are sent to www.mountkirkgames.com/sales and www.mountkirkgames.com/get-an-analysis, the correct pages are displayed. However, you have received complaints that www.mountkirkgames.com yields a 404 error. You need to resolve this error. What should you do?

- A. Review the Service YAML file. Add a new path rule for the * character that directs to the base service. Reapply the YAML.
- B. Review the Ingress YAML file. Add a new path rule for the * character that directs to the base service. Reapply the YAML.
- C. Review the Ingress YAML file. Define the default backend. Reapply the YAML.
- D. Review the Service YAML file. Define a default backend. Reapply the YAML.

Answer: C

Question: 186

CertyIQ

Your multi-region VPC has had a long-standing HA VPN configured in "region 1" connected to your corporate network. You are planning to add two 10 Gbps Dedicated Interconnect connections and VLAN attachments in "region 2" to connect to the same corporate network. You need to plan for connectivity between your VPC and corporate network to ensure that traffic uses the Dedicated Interconnect connections as the primary path and the HA VPN as the secondary path. What should you do?

- A. Enable regional dynamic routing mode on the VPC. Configure BGP associated with the HA VPN in "region 1" to use a base priority value of 100. Configure BGP associated with the VAN attachments to use a base priority of 20000. Configure your on-premises routers to use similar multi exit discriminator (MED) values.

B.Enable regional dynamic routing mode on the VPC. Configure BGP associated with the HA VPN in "region 1" to use a base priority value of 20000. Configure BGP associated with the VLAN attachments to use a base priority of 100. Configure your on-premises routers to use similar multi exit discriminator (MED) values.

C.Enable global dynamic routing mode on the VPC. Configure BGP associated with the HA VPN in "region 1" to use a base priority value of 20000. Configure BGP associated with the VLAN attachments to use a base priority of 100. Configure your on-premises routers to use similar multi exit discriminator (MED) values.

D.Enable global dynamic routing mode on the VPC. Configure BGP associated with the HA VPN in "region 1" to use a base priority value of 100. Configure BGP associated with the VLAN attachments to use a base priority of 20000. Configure your on-premises routers to use similar multi exit discriminator (MED) values.

Answer: C

Question: 187

CertyIQ

Your organization is developing a landing zone architecture with the following requirements:

- There should be no communication possible between production and non-production environments.
- Communication between applications within an environment may be necessary.
- Network administrators should centrally manage all network resources, including subnets, routes, and firewall rules.
- Each application should be billed separately.
- Developers of an application within a project should have the autonomy to create their compute resources. They should not create or modify networking resources.
- Up to 1000 applications are expected per environment.

You need to create a design that accommodates these requirements. What should you do?

A.Create a design that has one Shared VPC host project for the production environment, and another Shared VPC host project for the nonproduction environment. Associate the various applications' service projects with the corresponding environment's host project.

B.Create a design that has a Shared VPC for each project. Implement hierarchical firewall policies to apply micro-segmentation between VPCs.

C.Create a design that implements a single Shared VPC. Use VPC firewall rules with secure tags to enforce micro-segmentation between environments.

D.Create a design where each project in each environment has its own VPC with its own subnets, routes, and firewall rules. Ensure all VPCs are added as spokes to a Network Connectivity Center hub.

Answer: A

Explanation:

A. Create a design that has one Shared VPC host project for the production environment, and another Shared VPC host project for the nonproduction environment. Associate the various applications' service projects with the corresponding environment's host project.

This setup follows Google Cloud best practices for Shared VPC architecture, ensuring centralized network management while maintaining clear separation between production and nonproduction environments. It allows service projects to securely use shared resources while enforcing consistent security policies across environments.

Question: 188

CertyIQ

Your company uses Compute Engine instances that are exposed to the public internet. Each compute instance has

a single network interface with a single public IP address. You need to block any connection attempt that originates from internet clients with IP addresses that belong to the BGP_ASN_TOBLOCK BGP ASN. What should you do?

- A. Create a new Cloud Armor backend security policy, and use the --network-src-asns parameter.
- B. Create a new Cloud Armor network edge security policy, and use the --network-src-asns parameter.
- C. Create a new Cloud Armor edge security policy, and use the --network-src-asns parameter.
- D. Create a new firewall policy ingress rule, and use the --network-src-asns parameter.

Answer: B

Question: 189

CertyIQ

Your frontend application VMs and your backend database VMs are all deployed in the same VPC but across different subnets. Global network firewall policy rules are configured to allow traffic from the frontend VMs to the backend VMs. Based on a recent compliance requirement, this traffic must now be inspected by network virtual appliances (NVAs) firewalls that are deployed in the same VPC. The NVAs are configured to be full network proxies and will source NAT-allowed traffic. You need to configure VPC routing to allow the NVAs to inspect the traffic between subnets. What should you do?

- A. Place your NVAs behind an internal passthrough Network Load Balancer named ILB1. Add the global network firewall policy rules to allow traffic through your NVAs. Create a policy-based route (PBR) with the source IP range of the backend VM subnet, destination IP range of the frontend VM subnet, and the next hop of ILB1. Scope the PBR to the VMs with the backend network tag. Add a backend network tag to your backend servers.
- B. Place your NVAs behind an internal passthrough Network Load Balancer named ILB1. Add global network firewall policy rules to allow traffic through your NVAs. Create a custom static route with the destination IP range of the backend VM subnet, frontend instance tag, and the next hop of ILB1. Add a frontend network tag to your frontend VMs.
- C. Create your NVA with multiple interfaces. Configure NIC0 for NVA in the backend subnet. Configure NIC1 for NVA in the frontend subnet. Place your NVAs behind an internal passthrough Network Load Balancer named ILB1. Add global network firewall policy rules to allow traffic through your NVAs. Create a custom static route with the destination IP range of the backend VM subnet, frontend instance tag, and the next hop of ILB1. Add a frontend network tag to your frontend VMs.
- D. Place your NVAs behind an internal passthrough Network Load Balancer named ILB1. Add global network firewall policy rules to allow traffic through your NVAs. Create a policy-based route (PBR) with the source IP range of the frontend VM subnet, destination IP range of the backend VM subnet, and the next hop of ILB1. Scope the PBR to the VMs with the frontend network tag. Add a frontend network tag to your frontend servers.

Answer: D

Question: 190

CertyIQ

Your organization wants to set up hybrid connectivity with VLAN attachments that terminate in a single Cloud Router with 99.9% uptime. You need to create a network design for your on-premises router that meets those requirements and has an active/passive configuration that uses only one VLAN attachment at a time. What should you do?

- A. Create a design that uses the LOCAL_PREF BGP attribute to influence the egress path from Google Cloud to the on-premises environment.
- B. Create a design that uses an equal-cost multipath (ECMP) with flow-based hashing on your on-premises devices.
- C. Create a design that uses a BGP multi-exit discriminator (MED) attribute to influence the egress path from Google Cloud to the on-premises environment.
- D. Create a design that uses the AS_PATH BGP attribute to influence the egress path from Google Cloud to the on-premises environment.

Answer: A

Question: 191

CertyIQ

You are implementing firewall controls to protect your computer resources in a newly created VPC. To make the protection process easier to manage and control, you've defined the hierarchical firewall policies, global network firewall policies, and VPC firewall rules. The configuration of rules defines the following characteristics:

- The hierarchical firewall policy, bound at the organization level, is allowing/denying specific external traffic.
- There is a global network firewall policy with rules that enforce intrusion prevention system (IPS) capabilities for specific external inbound/outbound traffic.
- The VPC firewall rules allow internal communication from RFC 1918 defined subnets communications.
- The VPC firewall contains an explicit deny rule with logs enabled.

This configuration was successful in multiple preexisting VPCs. However, you noticed that the logs were missing when you were reviewing a newly created VPC. All external communications are hanging, but internal traffic is working as expected. You want to fix the connectivity issue.

What should you do?

- A.Create a new VPC and migrate existing resources to the new VPC. Delete the old VPC, and reapply the firewall policies and rules in the new VPC.
- B.Raise the priority numbers of the firewall policy rules and lower the priority numbers of the VPC firewall rules.
- C.Review the order in which the VPC firewall rules and policies are evaluated. If the VPC firewall rules are being evaluated before firewall policies, switch the order.
- D.Lower the priority numbers of the firewall policy rules and raise the priority numbers of the VPC firewall rules.

Answer: C

Question: 192

CertyIQ

You are configuring the intrusion prevention service (IPS) feature on Cloud Next Generation Firewall Enterprise. You deployed your firewall endpoints and you need to inspect the traffic of the VMs. What should you do?

- A.Configure Packet Mirroring to match the source/destination IP addresses of the VMs.
- B.Configure a firewall rule to match the source/destination IP addresses of the VMs, and use the goto_next action.
- C.Configure a firewall rule to match the hostnames of the VMs, and use the apply_security_profile_group action.
- D.Configure a firewall rule to match the source/destination IP addresses of the VMs, and use the apply_security_profile_group action.

Answer: D

Question: 193

CertyIQ

Your organization recently exposed a set of services through a global external Application Load Balancer. After conducting some testing, you observed that responses would intermittently yield HTTP 4xx or 5xx error response codes. You already enabled and reviewed the health check logs. You need to identify the error. What should you do?

- A.Access a VM in the VPC through SSH to access the backend VM directly. If the request is successful from the VM, increase the quantity of backends.

B.Delete the load balancer and backend services. Create a new Passthrough Network Load Balancer. Configure a failover group of VMs for the backend.

C.Validate the health of the backend service. Enable logging for the backend service and identify the error response in Cloud Logging. Review the statusDetails log field.

D.Validate the health of the backend service. Disable any Cloud Armor policies on the backend service, and identify any error response in Cloud Logging. Review the statusDetails log field.

Answer: C

Question: 194

CertyIQ

Your company's current network architecture has two VPCs that are connected by a dual-NIC instance that acts as a bump-in-the-wire firewall between the two VPCs. Flows between pairs of subnets across the two VPCs are working correctly. Suddenly, you receive an alert that none of the flows between the two VPCs are working anymore. You need to troubleshoot the problem. What should you do? (Choose two.)

A.Verify that a VPC Service Controls perimeter has not been enabled for the project that contains the two VPCs and the dual-NIC instance.

B.Use Cloud Logging to verify that there were no modifications to the VPC firewall rules or policies that were applied to the two network interfaces of the dual-NIC instance.

C.Verify that a public IP address has not been assigned to any network interface of the dual-NIC instance.

D.Verify that the dual-NIC instance has the --can-Ip-Forward attribute enabled.

E.Verify that the dual-NIC instance has not been added to a backend service.

Answer: BD

Question: 195

CertyIQ

Your company deployed Cloud Next Generation Firewall Enterprise (Cloud NGFW Enterprise). You have already created a CA pool and a CA in Certificate Authority Service. You need to enable TLS inspection. What should you do?

A.Grant the network security service agent service account the privateca.certificateRequester role. Create a TLS inspection policy linking to the CA pool. Configure your VPC endpoint associations to use the TLS inspection policy. Flip the TLS inspection flag in your firewall policy rules to true.

B.Grant the network security service agent service account the privateca.poolReader role. Create a TLS inspection policy linking to the CA pool. Configure your VPC endpoint associations to use the TLS inspection policy. Flip the TLS inspection flag in your firewall policy rules to true.

C.Grant the network security service agent service account the privateca.certificateRequester role. Create a trust config in Certificate Manager Flip the TLS inspection flag in your firewall policy rules to true.

D.Grant the network security service agent service account the privateca.certificateRequester role. Create a trust config in Certificate Manager. Flip the TLS inspection flag in your firewall policy rules to true.

Answer: A

Question: 196

CertyIQ

You have recently taken over responsibility for your organization's Google Cloud network security configurations. You want to review your Cloud Next Generation Firewall (Cloud NGFW) configurations and ensure there are no rules that are allowing ingress traffic to your VMs and services from the internet. You want to avoid manual work. What should you do?

- A. Review the firewall policy rules associated with the VPC, and filter for rules that allow ingress from 0.0.0.0/0.
- B. Enable "Overly permissive rules insights" in Firewall Insights. Review results for rules that show allowed ingress traffic from internet sources.
- C. Run Connectivity Tests from multiple external sources to double-check ingress traffic settings.
- D. Enable the Network Analyzer API and review the "VPC Network" category insights.

Answer: B

Question: 197

CertyIQ

Your company's cloud network has hybrid connectivity to an on-premises environment through Cloud Interconnect in two regions (us-east4 and us-west1). You received complaints that some on-premises destinations are no longer reachable from us-east4, after changes were made to advertise additional routes to us-west1. You need to troubleshoot to see if any routes were dropped. What should you do?

- A. Query the dynamic_routes/learned_routes/dropped_unique_destinations metric and review the global routing_mode metric attribute.
- B. Query the dynamic_routes/learned_routes/unique_destinations_limit metric and review the global routing_mode metric attribute.
- C. Query the dynamic_routes/learned_routes/any_dropped_unique_destinations metric and review the regional routing_mode metric attribute.
- D. Query the dynamic_routes/learned_routes/dropped_unique_destinations metric and review the regional routing_mode metric attribute.

Answer: A

Question: 198

CertyIQ

Your organization has resources in two different VPCs, each in different Google Cloud projects, which require connectivity between them. You have already determined that there is no IP address overlap; however, one VPC uses privately used public IP (PUP) ranges. You would like to enable connectivity between these resources by using a lower cost and higher performance method. What should you do?

- A. Create a HA VPN between the two VPCs that includes the PUP ranges in the Custom Route Advertisements of the Cloud Router. Create the necessary ingress VPC firewall rules that target the specific resources by using network tags as the source filter.
- B. Create a HA VPN between the two VPCs that includes the PUP ranges in the Custom Route Advertisements of the Cloud Router. Create the necessary ingress VPC firewall rules that target the specific resources by using IP ranges as the source filter.
- C. Create a VPC Peering between the two VPCs that allows the export and import of custom routes. Create the necessary ingress VPC firewall rules that target the specific resources by using service accounts as the source filter.
- D. Create a VPC Peering between the two VPCs that allows the export and import of subnet routes with public IP addresses. Create the necessary ingress VPC firewall rules that target the specific resources by using IP ranges as the source filter.

Answer: D

Question: 199

CertyIQ

Your organization recently re-architected your cloud environment to use Network Connectivity Center. However, an error occurred when you tried to add a new VPC, named vpc-dev, as a spoke. The error indicated that there was

an issue with an existing spoke and the IP space of a VPC, named vpc-pre-prod. You must complete the migration quickly and efficiently. What should you do?

- A. Delete the VMs associated with the conflicting subnets, then delete the conflicting subnets in vpc-dev. Recreate the subnets with a new IP range and redeploy the previously-deleted VMs in the new subnets. Add the VPC spoke for vpc-dev.
- B. Exclude the conflicting IP range by using the --exclude-export-ranges flag when creating the VPC spoke for vpc-dev.
- C. Exclude the conflicting IP range by using the --exclude-export-ranges flag in the hub when attaching the VPC spoke for vpc-dev.
- D. Remove the conflicting VPC spoke for vpc-pre-prod from the set of VPC spokes in Network Connectivity Center. Add the VPC spoke for vpc-dev. Add the previously removed vpc-pre-prod as a VPC spoke.

Answer: B

Question: 200

CertyIQ

Recently, your networking team enabled Cloud CDN for one of the external-facing services that is exposed through an external Application Load Balancer. The application team has already defined which content should be cached within the responses. Upon testing the load balancer, you did not observe any change in performance after the Cloud CDN enablement. You need to resolve the issue. What should you do?

- A. Configure the CACHE_ALL_STATIC caching mode on Cloud CDN to ensure Cloud CDN caches all static content as well as content defined by the backends.
- B. Configure the FORCE_CACHE_ALL caching mode on Cloud CDN to ensure all appropriate content is cached.
- C. Configure the USE_ORIGIN_HEADERS caching mode on Cloud CDN to ensure Cloud CDN caches content depending on responses to requests from the backends.
- D. Configure the CACHE_ALL_STATIC caching mode on Cloud CDN to ensure Cloud CDN cache content depending on responses to requests from the backends.

Answer: C

Question: 201

CertyIQ

Your organization requires that all SMTP traffic to your cloud environment is blocked, except for traffic that originates from your corporate network. Your organization also requires that only specific VPCs across your Google Cloud projects will allow SMTP access from your corporate network. You need to configure a security policy that will enable this connectivity. What should you do?

- A.1. Configure an ingress hierarchical firewall rule with priority 10000 specifying the 0.0.0.0/0 source, TCP port 25, and the deny action.
- 2. Configure an egress hierarchical firewall rule with priority 10010 specifying the source of your corporate network as TCP port 25 and the goto_next action.
- 3. Associate the hierarchical firewall policy at the organization level.
- 4. Configure firewall policy rules allowing TCP port 25 in the firewall policies associated with the respective VPCs that require that access.
- B.1. Configure an ingress hierarchical firewall rule with priority 10000 specifying the 0.0.0.0/0 source, TCP port 25, and the allow action.
- 2. Associate the hierarchical firewall policy at the organization level.
- 3. Configure firewall policy rules to deny TCP port 25 in the firewall policies associated with the respective VPCs that do not require that access.
- C.1. Configure an ingress hierarchical firewall rule with priority 10000 specifying the source of your corporate network, TCP port 25, and the goto_next action.
- 2. Configure an ingress hierarchical firewall rule with priority 10010 specifying the 0.0.0.0/0 source, TCP port 25, and the deny action.
- 3. Associate the hierarchical firewall policy at the organization level.

4. Configure firewall policy rules allowing TCP port 25 in the firewall policies associated with the respective VPCs that require that access.

D.1. Configure an ingress hierarchical firewall rule with priority 10000 specifying the 0.0.0.0/0 source, TCP port 25, and the deny action.

2. Associate the hierarchical firewall policy at the organization level.

3. Configure firewall policy rules allowing TCP port 25 in the firewall policies associated with the respective VPCs that require that access.

Answer: C

Question: 202

CertyIQ

Your organization has a subset of applications in multiple regions that require internet access. You need to control internet access from applications to URLs, including hostnames and paths. The compute instances that run these applications have an associated secure tag. What should you do?

A. Deploy a Cloud NAT gateway. Use fully qualified domain name (FQDN) objects in the firewall policy rules to filter outgoing traffic to specific domains from machines that match a service account.

B. Deploy a Cloud NAT gateway. Use fully qualified domain name (FQDN) objects in the firewall policy rules to filter outgoing traffic to specific domains from machines that match the secure tag.

C. Deploy a single Secure Web Proxy instance with global access enabled. Apply a Secure Web Proxy policy to allow access from machines that match the secure tag to the URLs defined in a URL list.

D. Deploy a Secure Web Proxy instance in each region. Apply a Secure Web Proxy policy to allow access from machines that match the secure tag to the URLs defined in a URL list.

Answer: D

Question: 203

CertyIQ

You are implementing hybrid connectivity between your company's data center and Google Cloud. You've already deployed redundant Dedicated Interconnect connections, and are now deploying VLAN attachments in us-central1. You want to use an active/passive approach, where interconnect-1 is active and interconnect-2 is a passive backup. You need to deploy a Cloud Router to enable BGP connectivity. You want to follow Google-recommended practices. What should you do?

A.1. Configure the primary interconnect-1 BGP session on the Cloud Router with priority 0 and ASN 65101.

2. Configure the secondary interconnect-2 BGP session on the Cloud Router with priority 200 and ASN 65102.

3. Configure the on-premises ASN as 65000.

B.1. Configure the primary interconnect-1 BGP session on the Cloud Router with priority 0.

2. Configure the secondary interconnect-2 BGP session on the Cloud Router with priority 200.

3. Configure both Google-side BGP ASNs as 65100.

4. Configure the on-premises ASN as 65000.

C.1. Configure the primary and secondary interconnects of the BGP sessions on the Cloud Router with priority 100 and ASN 16550.

2. Configure the on-premises ASN as 65001 for primary interconnect-1.

3. Configure the on-premises ASN as 65002 for secondary interconnect-2.

D.1. Configure the primary and secondary interconnects of the BGP sessions on the Cloud Router with priority 100 and ASN 4200000001.

2. Configure the on-premises ASN as 4200000010.

3. Disable the BGP session on the on-premises router for the secondary interconnect-2.

Answer: B

Question: 204**CertyIQ**

Your organization has multiple VMs running on Google Cloud within a VPC. The VMs require connectivity to certain Google APIs. You need to enable Private Google Access for VM connectivity to Cloud Storage. What should you do?

- A. Enable Private Google Access on the project, remove the default route that points to the default internet gateway, and enable the Cloud Storage API.
- B. Enable Private Google Access on the VM, remove the default route that points to the default internet gateway, and enable the Cloud Storage API.
- C. Enable Private Google Access on the VPC, create a default route that points to the default internet gateway, and enable the Cloud Storage API.
- D. Enable Private Google Access on the subnet, create a default route that points to the default internet gateway, and enable the Cloud Storage API.

Answer: D**Explanation:**

D. Enable Private Google Access on the subnet, create a default route that points to the default internet gateway, and enable the Cloud Storage API.

Private Google Access is enabled at the subnet level, allowing VMs without external IP addresses to reach Google APIs and services, including Cloud Storage. This ensures secure connectivity while maintaining internal network isolation.

Question: 205**CertyIQ**

You are configuring the final elements of a migration effort where resources have been moved from on-premises to Google Cloud. While reviewing the deployed architecture, you noticed that DNS resolution is failing when queries are being sent to the on-premises environment. You login to a Compute Engine instance, try to resolve an on-premises hostname, and the query fails. DNS queries are not arriving at the on-premises DNS server. You need to use managed services to reconfigure Cloud DNS to resolve the DNS error. What should you do?

- A. Ensure that the operating systems of the Compute Engine instances are configured to send DNS queries to the on-premises DNS servers directly.
- B. Validate that there is network connectivity to the on-premises environment and that the Compute Engine instances can reach other on-premises resources. If errors persist, remove the VPC Network Peerings and recreate the peerings after validating the routes.
- C. Validate that the Compute Engine instances are using the Metadata Service IP address as their resolver. Configure an outbound forwarding zone for the on-premises domain pointing to the on-premises DNS server. Configure Cloud Router to advertise the Cloud DNS proxy range to the on-premises network.
- D. Review the existing Cloud DNS zones, and validate that there is a route in the VPC directing traffic destined to the IP address of the DNS servers. Recreate the existing DNS forwarding zones for . to forward all queries to the on-premises DNS servers.

Answer: C**Explanation:**

C. Validate that the Compute Engine instances are using the Metadata Service IP address as their resolver. Configure an outbound forwarding zone for the on-premises domain pointing to the on-premises DNS server. Configure Cloud Router to advertise the Cloud DNS proxy range to the on-premises network.

This ensures that Compute Engine instances correctly resolve DNS queries while maintaining connectivity with on-premises resources. The outbound forwarding zone directs queries to the appropriate DNS server, and

Cloud Router helps propagate the necessary routes.

Question: 206

CertyIQ

Your organization's security team recently discovered that there is a high risk of malicious activities originating from some of your VMs connected to the internet. These malicious activities are currently undetected when TLS communication is used. You must ensure that encrypted traffic to the internet is inspected. What should you do?

- A.Enable Cloud Armor TLS inspection policy, and associate the policy with the backend VMs.
- B.Use Cloud NGFW Essentials. Create a firewall rule for egress traffic, and enable VPC Flow Logs with the TLS inspect option. Analyze the output logs content and block the outputs that have malicious activities.
- C.Configure a TLS agent on every VM to intercept TLS traffic before it reaches the internet. Configure Sensitive Data Protection to analyze and allow/deny the content.
- D.Use Cloud NGFW Enterprise. Create a firewall rule for egress traffic with the --tls-inspect flag, and associate the firewall rules with the VMs.

Answer: D

Explanation:

D. Use Cloud NGFW Enterprise. Create a firewall rule for egress traffic with the --tls-inspect flag, and associate the firewall rules with the VMs.

Cloud NGFW Enterprise provides advanced TLS inspection capabilities, allowing you to analyze encrypted traffic for potential threats. By enabling TLS inspection on egress traffic, you can detect and mitigate malicious activities that would otherwise go unnoticed.

Question: 207

CertyIQ

Your organization has a hub and spoke architecture with VPC Network Peering, and hybrid connectivity is centralized at the hub. The Cloud Router in the hub VPC is advertising subnet routes, but the on-premises router does not appear to be receiving any subnet routes from the VPC spokes. You need to resolve this issue. What should you do?

- A.Create custom routes at the Cloud Router in the spokes to advertise the subnets of the VPC spokes.
- B.Create custom routes at the Cloud Router in the hub to advertise the subnets of the VPC spokes.
- C.Create a BGP route policy at the Cloud Router, and ensure the subnets of the VPC spokes are being announced towards the on-premises environment.
- D.Create custom learned routes at the Cloud Router in the hub to advertise the subnets of the VPC spokes.

Answer: B

Question: 208

CertyIQ

Your organization has a legacy VPN device that uses IKEv1 and does not support BGP. Connectivity from your on-premises environment to Google Cloud needs to be established. You are using 172.16.100.0/24, 172.16.101.0/24, and 172.16.102.0/24 in your on-premises environment, and 192.168.100.0/24, 192.168.101.0/24, and 192.168.102.0/24 in your Google Cloud environment. You have configured a VPN gateway and you need to configure a policy-based VPN tunnel. What should you do?

- A. Configure the tunnel with LOCAL_TS set to 172.16.100.0/22 and REMOTE_TS set to 192.168.100.0/22.
- B. Configure the tunnel with LOCAL_TS set to 192.168.100.0/22 and REMOTE_TS set to 172.16.100.0/22.
- C. Configure the tunnel with LOCAL_TS set to 172.16.100.0/24, 172.16.101.0/24, and 172.16.102.0/24, and REMOTE_TS set to 192.168.100.0/24, 192.168.101.0/24, and 192.168.102.0/24.
- D. Configure the tunnel with LOCAL_TS set to 172.16.100.0/24, 172.16.101.0/24, and 172.16.102.0/24, and REMOTE_TS set to 0.0.0.0/0.

Answer: B

Question: 209

CertyIQ

You plan to deploy Google Cloud Armor web application firewall (WAF) policies that use the preconfigured WAF rules. You want all Google Cloud Armor logs to be sent to Cloud Logging with the highest level of detail possible. You have enabled Cloud Load Balancing logs for all the backend services where Cloud Armor WAF policies are applied. What should you do?

- A. Set the sample rate of the Cloud Load Balancing logs to 0.5.
- B. Set the Google Cloud Armor logging option to VERBOSE.
- C. Enable Google Cloud Armor logging for all the backend services where Cloud Armor WAF policies are applied. Set the Google Cloud Armor logging option to VERBOSE.
- D. Set the sample rate of the Cloud Load Balancing logs to 1.0.

Answer: B

Question: 210

CertyIQ

Your organization has implemented Vertex AI online prediction in your Google Cloud environment, which is in the us-central1 region. Online prediction is available through private services access by using the IP CIDR range of 172.16.53.0/24. You need to configure access to Vertex AI without affecting the existing routes. You want to use the VLAN attachments that are located in the us-west1 region as primary. The interconnect VLAN attachments in the us-west2 region can only be used as a backup. What should you do?

- A. Create a custom route advertisement on VLAN attachments in the us-west1 region for prefix 172.16.53.0/24. Create a custom route advertisement on VLAN attachments in the us-west2 region for prefix 172.16.53.0/24.
- B. Create a custom learned route on VLAN attachments in the us-west1 region for prefix 172.16.53.0/24, and set the route priority on the BGP session as 100. Create a custom route advertisement on VLAN attachments in the us-west2 region for prefix 172.16.53.0/24, and set the route priority on the BGP session as 200.
- C. Create a custom route advertisement on VLAN attachments in the us-west1 region for prefix 172.16.53.0/24, and set the route priority on the BGP session as 100. Create a custom route advertisement on VLAN attachments in the us-west2 region for prefix 172.16.53.0/24, and set the route priority on the BGP session as 200.
- D. Create a custom route advertisement on VLAN attachments in the us-west1 region for prefix 172.16.53.0/24, and create a BGP route-policy to set the multi-exit discriminator (MED) to 100. Create a custom route advertisement on VLAN attachments in the us-west2 region for prefix 172.16.53.0/24, and create a BGP route-policy to set the multi-exit discriminator (MED) to 200.

Answer: C

Explanation:

Create a custom route advertisement on VLAN attachments in the us-west1 region for prefix 172.16.53.0/24, and set the route priority on the BGP session as 100. Create a custom route advertisement on VLAN attachments in the us-west2 region for prefix 172.16.53.0/24, and set the route priority on the BGP session as 200.

Question: 211

CertyIQ

As part of your organization's modernization efforts, the application teams are migrating services to GKE on Google Cloud (GKE). The GKE clusters will live in service projects. The teams have validated the applications and configurations in their sandbox projects. When moving to production, you noticed that GKE nodes were not being created. Users were able to create Compute Engine instances, but the operation failed when they tried to create a GKE cluster. You need to enable the application teams so they can create said GKE clusters. What should you do?

- A.Ensure that the service project's GKE service account has the `compute.securityAdmin`, `container.hostServiceAgentUser` and `compute.networkUser` IAM permissions in the host project.
- B.Ensure that the service project's GKE service account has the `compute.securityAdmin`, `container.hostServiceAgentUser` and `compute.networkUser` IAM permissions in the service project.
- C.Ensure that the service project's GKE service account has the `compute.networkUser` IAM permission in the service project.
- D.Review the firewall rules configuration in the VPC. Identify what rule is blocking node creation.

Answer: A

Question: 212

CertyIQ

You are implementing a Shared VPC network for your organization, which has distributed teams. One of the application developers works across several teams and notices that they can deploy applications in subnets that are reserved for another application's service projects. You want to ensure that developers can only deploy resources in the subnets that are reserved for their respective service project. What should you do?

- A.Specify which Shared VPC subnets each application's service projects can access by using the `constraints/compute.restrictSharedVpcSubnetworks` organizational constraint.
- B.Grant the `compute.NetworkViewer` role to the developer in the Shared VPC host project.
- C.Restrict another application's project from accessing specific subnets in the host project by using the `constraints/compute.restrictSharedVpcHostProject` organizational constraint.
- D.Grant the `compute.NetworkUser` role to the developer in the specific Shared VPC service project.

Answer: A

Question: 213

CertyIQ

You are configuring HA VPN for your organization to connect your on-premises environment to your Google Cloud network. Your on-premises environment is closest to the `us-west1` Google Cloud region. You have Google Cloud resources in `us-west2`, which requires a throughput of 300,000 packets per second (PPS) and an approximate bandwidth of 4 Gbps. You need to have predictable bandwidth management and maintain an SLA of 99.99% with minimal costs. What should you do?

- A.Create an HA VPN gateway with two tunnels. Configure BGP on both tunnels with tunnel 0 configured with a base routing priority metric of 100 and tunnel 1 with a base routing priority metric of 200. Configure the on-premises router with the corresponding multi-exit discriminator (MED) value.
- B.Create two HA VPN gateways, each with two tunnels. Configure BGP on each of the gateways' tunnels with tunnel 0 configured with a base routing priority metric of 100 and tunnel 1 with a base routing priority metric of 100. Configure the on-premises router with the same corresponding multi-exit discriminator (MED) value.
- C.Create an HA VPN gateway with two tunnels. Configure BGP on both tunnels with tunnel 0 configured with a base routing priority metric of 100 and tunnel 1 with a base routing priority metric of 100. Configure the on-premises router with the corresponding multi-exit discriminator (MED) value.
- D.Create an HA VPN gateway with four tunnels. Configure BGP on four tunnels with tunnel 0 configured with a

base routing priority metric of 100, tunnel 1 with a base routing priority metric of 200, tunnel 2 with a base routing priority of 300, and tunnel 3 with a base routing priority of 400. Configure the on-premises router with the corresponding multi-exit discriminator (MED) value.

Answer: C

Question: 214

CertyIQ

Your organization mandates that all internal IP addresses used by all database VMs must be statically allocated. While analyzing your VPC IP address allocations, you observed that the database VMs do not have static IP addresses. You need to configure the VPC to follow your organization's mandate without causing any disruption to current operations. What should you do?

- A. Promote the internal IP addresses to static assignments for all database VMs.
- B. Create a firewall rule to allow only traffic to the IP addresses allocated to your database VMs.
- C. Define a maintenance window to shut down the database VMs one at a time, promote the internal IP address to a static assignment, and restart the VM.
- D. Define an organization policy to allow only statically allocated IP addresses for VMs. Ensure the prefix matches your database VMs.

Answer: A

Question: 215

CertyIQ

Your organization deployed a mission critical application that is expected to be a new revenue source. As part of the planning and deployment process, you have recently implemented a security profile with the default set of threat signatures provided by Cloud Next Generation Firewall (Cloud NGFW). This application is the only application running on this project. You need to increase the security posture of the application to log the threat and drop the related packets. What should you do?

- A. Configure a new default threat signature with Deny All to all severity options. Review the logs to understand the impact.
- B. Set up a Linux VM as the frontend gateway for the application. Create iptables rules to drop all packets, excluding the application port.
- C. For all severity options (critical, high, medium, low and informational) in the security profile, change the default override action to Deny.
- D. Configure Cloud Scheduler to run a task that checks the Cloud NGFW logs to verify the threats. Configure the task to create a security profile with each signature ID set to override the default action.

Answer: B

Question: 216

CertyIQ

You are configuring a Cross-Cloud Interconnect connection for your Google Cloud organization with two public cloud service providers (CSPs)—CSP 1 and CSP 2. The CSP 1 and CSP 2 environments are closest to Frankfurt, Germany. You can choose between two common colocation locations, Frankfurt and Munich. Your organization's Google Cloud infrastructure is deployed in the North American region, us-east4, which is located in Virginia, USA. The VPC dynamic routing mode has been set to GLOBAL. Your organization requires 20 Gbps of protected bandwidth with a 99.9% Google Cloud SLA. You want to minimize costs where possible. What should you do?

- A.1. Create two Cross-Cloud Interconnect connections to CSP 1, with 40 Gbps of total bandwidth (20 Gbps in zone 1 and 20 Gbps in zone 2) in a common co-location facility located in Frankfurt, Germany.
- 2. Create two Cross-Cloud Interconnect connections to CSP 2, with 40 Gbps of total bandwidth (20 Gbps in

- zone 1 and 20 Gbps in zone 2) in a common co-location facility located in Frankfurt, Germany.
3. Create a Cloud Router in europe-west3 (Frankfurt), and configure two VLAN attachments for CSP 1 and two VLAN attachments for CSP 2.
- B.1. Create two Cross-Cloud Interconnect connections to CSP 1, with 20 Gbps of total bandwidth (10 Gbps in zone 1 and 10 Gbps in zone 2) in a common co-location facility located in Frankfurt, Germany.
2. Create two Cross-Cloud Interconnect connections to CSP 2, with 20 Gbps of total bandwidth (10 Gbps in zone 1 and 10 Gbps in zone 2) in a common co-location facility located in Frankfurt, Germany.
3. Create a Cloud Router in europe-west3 (Frankfurt), and configure two VLAN attachments for CSP 1 and two VLAN attachments for CSP 2.
- C.1. Create two Cross-Cloud Interconnect connections to CSP 1, with 40 Gbps of total bandwidth (20 Gbps in zone 1) in a common co-location facility located in Frankfurt, Germany and (20 Gbps in zone 2) in a common co-location facility located in Munich, Germany.
2. Create two Cross-Cloud Interconnect connections to CSP 2, with 40 Gbps of total bandwidth (20 Gbps in zone 1) in a common co-location facility located in Frankfurt, Germany and (20 Gbps in zone 2) in a common co-location facility located in Munich, Germany.
3. Create a Cloud Router in europe-west3 (Frankfurt), and configure two VLAN attachments for CSP 1 and two VLAN attachments for CSP 2.
- D.1. Create two Cross-Cloud Interconnect connections to CSP 1, with 40 Gbps of total bandwidth (20 Gbps in zone 1 and 20 Gbps in zone 2) in a common co-location facility located in Frankfurt, Germany.
2. Create two Cross-Cloud Interconnect connections to CSP 2, with 40 Gbps of total bandwidth (20 Gbps in zone 1 and 20 Gbps in zone 2) in a common co-location facility located in Frankfurt, Germany.
3. Create a Cloud Router in us-east4 (Ashburn, Virginia, USA), and configure two VLAN attachments for CSP 1 and two VLAN attachments for CSP 2.

Answer: A

Question: 217

CertyIQ

Your organization's application is running on a VPC-native GKE Standard cluster with public IP addresses. You need to configure access to the remote address range 35.100.0.0/16 through Cloud NAT, instead of using the GKE nodes' external IP addresses. SNAT is enabled on the cluster and needs to be configured. What should you do?

- A. Configure nonMasqueradeCIDRs in the ip-masq-agent ConfigMap. Include the 35.100.0.0/16 range in the list.
- B. Configure nonMasqueradeCIDRs in the ip-masq-agent ConfigMap. Remove the 35.100.0.0/16 range from the list.
- C. Configure Cloud NAT and create an exclusion rule for any SNAT address translation.
- D. Configure Cloud NAT with nonMasqueradeCIDRs, and enable SNAT with the same configuration to allow traffic to 35.100.0.0/16.

Answer: B

Question: 218

CertyIQ

Your organization has approximately 100 teams that need to manage their own environments. A central team must manage the network. You need to design a landing zone that provides separate projects for each team. You must also make sure the solution can scale. What should you do?

- A. Configure Policy-based Routing for each team.
- B. Configure a Shared VPC, and create a VPC network in the host project.
- C. Configure VPC Network Peering, and peer one of the VPC's to the service project.
- D. Configure a Shared VPC, and create a VPC network in the service project.

Answer: B

Question: 219**CertyIQ**

You are using Network Connectivity Center and you already have the hub configured. All VPCs in your environment need to have network connectivity to each other. All the subnet ranges are unique. You need to configure your topology accordingly. What should you do?

- A. Configure a star topology, add the VPC spokes to the hub, and specify all subnet ranges in the `excludeExportRanges` filter.
- B. Configure a mesh topology, add the VPC spokes to the hub, and specify all subnet ranges in the `excludeExportRanges` filter.
- C. Configure a mesh topology, and add the VPC spokes to the hub.
- D. Configure a star topology, and add the VPC spokes to the hub.

Answer: D**Question: 220****CertyIQ**

You are creating a design that will connect your single on-premises data center to a VPC in Google Cloud by using an IPsec VPN connection. The connection must have a minimum SLA of 99.99%. There is a single VPN termination device located in your on-premises data center. The VPN termination device can be configured only with a single public IP address. Your design must also have the least amount of setup effort. What should you do?

- A.1. Create two HA VPN gateways.
2. Create one tunnel on interface 0 of one gateway and create one tunnel on interface 1 of the other gateway.
3. Terminate each of the two tunnels on the single public IP address that is configured on the VPN termination device located in your on-premises data center.
- B.1. Create one Classic VPN gateway and one HA VPN gateway.
2. Create one tunnel on the interface of the Classic VPN gateway and one tunnel on interface 1 of the HA VPN gateway.
3. Terminate each of the two tunnels on the single public IP address that is configured on the VPN termination device located in your on-premises data center.
- C.1. Replace the existing on-premises VPN termination device with a new device that is configured with two different public IP addresses.
2. Create one HA VPN gateway.
3. Create one tunnel for each of the two HA VPN gateway interfaces.
4. Terminate each of the two tunnels on one of the two public IP addresses that is configured on the new VPN termination device located in your on-premises data center.
- D.1. Create one HA VPN gateway.
2. Create one tunnel for each of the two HA VPN gateway interfaces.
3. Terminate each of the two tunnels on the single public IP address that is configured on the VPN termination device located in your on-premises data center.

Answer: D**Question: 221****CertyIQ**

Your organization has a highly available application that is not HTTP-based. The application runs on multiple TCP ports and is hosted in multiple regions. You need to design a solution to load balance the application in the same Shared VPC where the service will be accessed. The IP address header must contain the client's true source IP address. No public internet access is required. What should you do?

- A. Configure multiple regional internal proxy Network Load Balancers and enable global access. Use DNS routing policies to balance traffic across regions.
- B. Configure multiple regional internal Application Load Balancers and enable global access. Use DNS routing policies to balance traffic across regions.

C.Configure a single cross region internal proxy Network Load Balancer.

D.Configure multiple regional internal passthrough Network Load Balancers and enable global access. Use DNS routing policies to balance traffic across regions.

Answer: C

Question: 222

CertyIQ

Your organization is using a Shared VPC model. Service project owners want to independently manage their DNS zones in service projects. All service project workloads must be able to resolve all private zones that are defined in other service projects. You need to create a solution that meets these goals. What should you do?

A.Create a Cloud DNS private zone in each service project. Use a Cloud DNS forwarding zone to forward queries to the Shared VPC in the host project.

B.Create a Cloud DNS private zone in each service project. Use Cloud DNS peering zones that target the Shared VPC in the host project.

C.Create a Cloud DNS response policy zone in each service project. Use Cloud DNS peering zones that target the Shared VPC in the host project.

D.Create a Cloud DNS private zone in each service project. Use cross-project binding to associate the zones to the Shared VPC in the host project.

Answer: D

Question: 223

CertyIQ

Your organization wants to deploy HA VPN over Cloud Interconnect to ensure encryption-in-transit over the Cloud Interconnect connections. You have created a Cloud Router and two encrypted VLAN attachments that have a 5 Gbps capacity and a BGP configuration. The BGP sessions are operational. You need to complete the deployment of the HA VPN over Cloud Interconnect. What should you do?

A.Create an HA VPN gateway and associate the gateway with your two encrypted VLAN attachments. Configure the HA VPN Cloud Router, peer VPN gateway resources, and HA VPN tunnels. Use the same encrypted Cloud Router used for the Cloud Interconnect tier.

B.Enable MACsec on Partner Interconnect.

C.Enable MACsec for Cloud Interconnect on the VLAN attachments.

D.Create an HA VPN gateway and associate the gateway with your two encrypted VLAN attachments. Create a new dedicated HA VPN Cloud Router, peer VPN gateway resources, and HA VPN tunnels.

Answer: D

Question: 224

CertyIQ

You have recently taken over responsibility for your organization's Google Cloud network security configurations. You want to review your Cloud Next Generation Firewall (Cloud NGFW) configurations to ensure that there are no rules allowing ingress traffic to your VMs and services from the internet. You want to avoid manual work. What should you do?

A.Export all your Cloud NGFW rules into a CSV file and search for 0.0.0.0/0.

B.Use Firewall Insights, and enable insights for Overly permissive rules.

C.Run Connectivity Tests from multiple external sources to confirm that traffic is not allowed to ingress to your most critical services in Google Cloud.

D.Review Network Analyzer insights on the VPC network category.

Answer: B

Question: 225

CertyIQ

Your organization is connecting their Shared VPC network to their on-premises data center by using Dedicated Interconnect to provide connectivity to all of its service projects. You need to create a design to configure your VLAN attachments and Cloud Routers. You also want to achieve a 99.9% Cloud Interconnect SLA based on Google Cloud's reference design. What should you do?

- A. Create two Cloud Interconnect connections in different edge availability domains of two different co-location facilities in a project that will contain your connections. Create one VLAN attachment and Cloud Router for each physical interconnect in the Shared VPC host project.
- B. Create two Interconnect connections in different edge availability domains of the co-location facility in a project that will contain your connections. Create one VLAN attachment for each physical Cloud Interconnect connection and a single Cloud Router in the Shared VPC host project.
- C. Create two Cloud Interconnect connections in different edge availability domains of the co-location facility in a project that will contain your connections. Create one VLAN attachment for each physical interconnect and a single Cloud Router in the service projects.
- D. Create two Cloud Interconnect connections in different edge availability domains of the co-location facility in a project that will contain your connections. Create a Cloud Router in the Shared VPC host project and the VLAN attachments in the Shared VPC service projects.

Answer: A

Question: 226

CertyIQ

Your organization's on-premises networking team is reporting frequent BGP session flaps toward your Google Cloud environment. You need to review the BGP configuration. What should you do?

- A. Switch to static routing.
- B. Increase the BGP hold timer to 36000 seconds max.
- C. Ensure that graceful restart is enabled on the on-premises router.
- D. Ask the on-premises team to enable Bidirectional Forwarding Detection (BFD).

Answer: C

Question: 227

CertyIQ

Your organization has over 250 autonomous business units that currently operate in a decentralized manner. Due to the organization's maturity, there is limited routable private IP address space, which is insufficient to accommodate all of the necessary workloads. You need to create a cloud-first network design that uses the same IP address space across business unit workloads where possible. These business units require communication between units, and access to their on-premises data center. What should you do?

- A. Create a hub and spoke model that incorporates VPC Network Peering with hybrid connectivity centralized within the hub.
- B. Create a Network Connectivity Center design that incorporates Private NAT to facilitate communication between VPC spokes, and a Routing VPC to exchange dynamic routes from the on-premises environment.
- C. Create a Network Connectivity Center design that incorporates Private Service Connect to provide bidirectional communication between VPC spokes, and a Routing VPC to exchange dynamic routes from the on-premises environment.

D.Create a hub and spoke design that incorporates a centralized network virtual appliance (NVA) in the hub to perform routing and NAT between spokes.

Answer: B

Question: 228

CertyIQ

You are configuring an Application Load Balancer. The backend resides in your on-premises data center and is connected by Dedicated Interconnect. You need to ensure the load balancer can reference these on-premises resources. You do not want the traffic to traverse the internet at all. What should you do?

- A.Configure an internet network endpoint group (NEG) as a backend service as part of the load balancer. Ensure firewalls are opened for the proxy-only subnet.
- B.Configure a zonal network endpoint group (NEG) as a backend service as part of the load balancer. Ensure firewalls are opened for the client source IPs.
- C.Configure a hybrid network endpoint group (NEG) as a backend service as part of the load balancer. Ensure firewalls are opened for the proxy-only subnet.
- D.Configure a Private Service Connect network endpoint group (NEG) as a backend service as part of the load balancer. Ensure firewalls are opened for the client source IPs.

Answer: C

Question: 229

CertyIQ

You are troubleshooting connectivity issues between Google Cloud and a public SaaS provider. Connectivity between the two environments is through the public internet. Your users are reporting intermittent connection errors when using TCP to connect; however, ICMP tests show no failures. According to users, errors occur around the same time every day. You want to troubleshoot and gather information by using Google Cloud tools that are most likely to provide insights to what is occurring within Google Cloud. What should you do?

- A.Enable and review Cloud Logging for Cloud Armor. Look for logs with errors matching the destination IP address of the public SaaS provider.
- B.Enable and review Cloud Logging on your Cloud NAT gateway. Look for logs with errors matching the destination IP address of the public SaaS provider.
- C.Enable the Firewall Insights API. Set the deny rule insights observation period to one day. Review the insights to assure there are no firewall rules denying traffic.
- D.Create a Connectivity Test by using TCP, the source IP address of your test VM, and the destination IP address of the public SaaS provider. Review the live data plane analysis and take the next steps based on the test results.

Answer: B

Question: 230

CertyIQ

You configured a single IPsec Cloud VPN tunnel for your organization to a third-party customer. You confirmed that the VPN tunnel is established. However, the BGP session status states that the BGP is not configured. The customer has provided you with their BGP settings:

- Local BGP address: 169.254.11.1/30
- Local ASN: 64515
- Peer BGP address: 169.254.11.2
- Peer ASN: 64517
- Base MED: 1000

•MD5 Authentication: Disabled

You need to configure the local BGP session for this tunnel based on the settings provided by the customer. You already associated the Cloud Router with the Cloud VPN Tunnel. What settings should you use for the BGP session?

A. Peer ASN: 64517 -
Advertised Route Priority (MED): 100

Local BGP IP: 169.254.11.2 -

Peer BGP IP: 169.254.11.1 -
MD5 Authentication: Disabled

B. Peer ASN: 64515 -
Advertised Route Priority (MED): 100

Local BGP IP: 169.254.11.1 -

Peer BGP IP: 169.254.11.2 -
MD5 Authentication: Disabled

C. Peer ASN: 64515 -
Advertised Route Priority (MED): 100

Local BGP IP: 169.254.11.2 -

Peer BGP IP: 169.254.11.1 -
MD5 Authentication: Disabled

D. Peer ASN: 64515 -
Advertised Route Priority (MED): 1000

Local BGP IP: 169.254.11.2 -

Peer BGP IP: 169.254.11.1 -
MD5 Authentication: Enabled

Answer: C

Question: 231

CertyIQ

Your organization's current architecture has one Shared VPC host project (SH_HOST_PRJ) that contains a single VPC (SH_VPC) and two Shared VPC service projects (SP_ONE_PRJ and SP_TWO_PRJ) that do not contain any VPCs. Each Shared VPC service project belongs to a different team: TEAM_ONE manages SP_ONE_PRJ and TEAM_TWO manages SP_TWO_PRJ.

You must design a solution that allows each team to create their own DNS private zones and DNS records only in their respective Shared VPC service projects. Workloads in SP_ONE_PRJ must be able to resolve all the DNS private zones defined in SP_TWO_PRJ and conversely. Your design must have the least amount of set up effort. What should you do?

A.1. TEAM_ONE uses cross-project binding and creates Cloud DNS private zones and DNS records in SP_ONE_PRJ, and binds the zones to the Shared VPC host project (SH_HOST_PRJ).

2. TEAM_TWO creates Cloud DNS private zones and DNS records in SP_TWO_PRJ, and uses cross-project binding to connect the zones to the Shared VPC host project (SH_HOST_PRJ).

B.1. TEAM_ONE uses cross-project binding and creates Cloud DNS private zones and DNS records in SP_ONE_PRJ, and binds the zones to the VPC (SH_VPC) in the Shared VPC host project (SH_HOST_PRJ).

2. TEAM_TWO creates DNS private zones and DNS records in SP_TWO_PRJ and uses cross-project binding to connect the zones to the VPC (SH_VPC) in the Shared VPC host project (SH_HOST_PRJ).

C.1. TEAM_ONE creates a new VPC (SP_ONE_VPC) in the Shared VPC service projects (SP_ONE_PRJ).

TEAM_ONE creates Cloud DNS private zones and DNS records in SP_ONE_PRJ, and binds the zones to the new VPC (SP_ONE_VPC). TEAM_ONE creates a Cloud DNS peering relationship between SP_ONE_VPC and the VPC (SH_VPC) in the Shared VPC host project (SH_HOST_PRJ).

2. TEAM_TWO completes the same actions for the SP_TWO_PRJ project.

D.1. TEAM_ONE creates a new VPC (SP_ONE_VPC) in the Shared VPC service projects (SP_ONE_PRJ).

TEAM_ONE creates Cloud DNS private zones and DNS records in SP_ONE_PRJ, and binds the zones to the new VPC (SP_ONE_VPC). TEAM_ONE creates a VPC Network Peering relationship between SP_ONE_VPC and the VPC (SH_VPC) in the Shared VPC host project (SH_HOST_PRJ).

2. TEAM_TWO completes the same actions for the SP_TWO_PRJ project.

Answer: B

Question: 232

CertyIQ

You are troubleshooting an application in your organization's Google Cloud network that is not functioning as expected. You suspect that packets are getting lost somewhere. The application sends packets intermittently at a low volume from a Compute Engine VM to a destination on your on-premises network through a pair of Cloud Interconnect VLAN attachments. You validated that the Cloud Next Generation Firewall (Cloud NGFW) rules do not have any deny statements blocking egress traffic, and you do not have any explicit allow rules. Following Google-recommended practices, you need to analyze the flow to see if packets are being sent correctly out of the VM to isolate the issue. What should you do?

A. Create a packet mirroring policy that is configured with your VM as the source and destined to a collector. Analyze the packet captures.

B. Enable VPC Flow Logs on the subnet that the VM is deployed in with SAMPLE_RATE = 1.0, and run a query in Logs Explorer to analyze the packet flow.

C. Verify the network/attachment/egress_dropped_packets_count Cloud Interconnect VLAN attachment metric.

D. Enable Firewall Rules Logging on your firewall rules and review the logs.

Answer: A

Question: 233

CertyIQ

Your organization is launching a new video game that will be available to all users globally through Cloud CDN. During the early release phase, you discovered that the wrong binary version was uploaded from Cloud Storage and cached in Cloud CDN. Thousands of users have downloaded the wrong version. Your marketing department has notified users that this was the wrong version of the game and asked all users to download the updated version using the same URL. You need to ensure users are downloading the updated version of the game. What should you do?

A. Create a security policy to block all Cloud CDN requests, review the logs, and filter which users are attempting to download the wrong game binary.

B. Create a new URL path for the updated game binary. Allow the cache to expire automatically through HTTP headers.

C. Upload the updated game binary to Cloud Storage. Invalidate the wrong game binary from the Cloud CDN cache.

D. Disable Cloud CDN. Reconfigure the load balancer with the updated game binary. Enable Cloud CDN.

Answer: C

Question: 234

CertyIQ

You recently reviewed the user behavior for your main application, which uses an external global Application Load Balancer, and found that the backend servers were overloaded due to erratic spikes in the rate of client requests. You need to limit the concurrent sessions and return an HTTP 429 Too Many Requests response back to the client

while following Google-recommended practices. What should you do?

- A. Create a Cloud Armor security policy, and associate the policy with the load balancer. Configure the security policy's settings as follows: action: throttle; conform action: allow; exceed action: deny-429.
- B. Configure the load balancer to accept only the defined amount of requests per client IP address, increase the backend servers to support more traffic, and redirect traffic to a different backend to burst traffic.
- C. Create a Cloud Armor security policy, and apply the predefined Open Worldwide Security Application Project (OWASP) rules to automatically implement the rate limit per client IP address.
- D. Configure a VM with Linux, implement the rate limit through iptables, and use a firewall rule to send an HTTP 429 response to the client application.

Answer: A

Question: 235

CertyIQ

Your company uses web application firewall (WAF) capabilities from a third-party cloud WAF provider. This WAF provider proxies all the HTTPS connections from internet clients, applies security policies, and then opens a new HTTPS connection to the public IP address of your global Application Load Balancer in Google Cloud. Your Google Cloud workloads are the backend of this global Application Load Balancer. Currently, Cloud Armor is not configured. You need to create a Cloud Armor security policy that blocks sessions that originate from internet clients with source IP addresses that belong to the IP_RANGE_BLOCK IP range. The block must be executed by the Cloud Armor security policy; it will not be done by the third-party cloud WAF provider. What should you do?

- A.1. Create a new Cloud Armor network edge security policy. In the policy, set the `userIpRequestHeaders[]` attribute.
- 2. Add a policy rule that denies traffic that matches `inIpRange(origin.user_ip, 'IP_RANGE_BLOCK')` statement.
- 3. Apply the policy to the backend service that includes all your Google Cloud workloads.
- B.1. Create a new Cloud Armor network edge security policy. In the policy, set the `userIpRequestHeaders[]` attribute.
- 2. Add a policy rule that denies traffic that matches the `inIpRange(origin.ip, 'IP_RANGE_BLOCK')` statement.
- 3. Apply the policy to the backend service that includes all your Google Cloud workloads.
- C.1. Create a new Cloud Armor backend security policy. In the policy, set the `userIpRequestHeaders[]` attribute.
- 2. Add a policy rule that denies traffic that matches the `inIpRange(origin.user_ip, 'IP_RANGE_BLOCK')` statement.
- 3. Apply the policy to the backend service that includes all your Google Cloud workloads.
- D.1. Create a new Cloud Armor backend security policy. In the policy, set the `userIpRequestHeaders[]` attribute.
- 2. Add a policy rule that denies traffic that matches the `inIpRange(origin.ip, 'IP_RANGE_BLOCK')` statement.
- 3. Apply the policy to the backend service that includes all your Google Cloud workloads.

Answer: A

Explanation:

The correct answer is A. Here's why:

1. **Network Edge vs. Backend Security Policy:** The key is where the policy is applied. Because the traffic is already proxied through a third-party WAF and arriving at Google Cloud's network edge, we need to apply the Cloud Armor policy at the network edge to inspect the true client IP. Network edge policies are designed for this scenario. Backend security policies are generally used when you want to protect your backend services directly, without a preceding proxy. Therefore, we need a network edge security policy.
2. **userIpRequestHeaders[]:** Since the third-party WAF is proxying the traffic, the original client IP address is likely not present in the standard `origin.ip` field that Cloud Armor uses. Instead, the WAF likely adds the original client IP to a header (e.g., `X-Forwarded-For`). The `userIpRequestHeaders[]` attribute is essential because it tells Cloud Armor which header contains the original client IP address. By setting it, Cloud Armor knows where to look for the client IP. If you don't set it, Cloud

Armor will only see the IP address of the WAF, making it impossible to block the originating client.

3. **origin.user_ip:** Given we are now instructing Cloud Armor to look in `userIpRequestHeaders[]`, the correct field to use in `inIpRange` is `origin.user_ip`. This is because `origin.user_ip` refers to the user IP address resolved from the headers specified in the `userIpRequestHeaders[]` attribute.
4. **Applying the Policy:** The policy needs to be applied to the backend service that serves your Google Cloud workloads behind the global Application Load Balancer. This ensures that the policy is enforced before the traffic reaches your backend.

Options B, C, and D are incorrect because of the following reasons:

B: `origin.ip` will not contain the actual client IP because of the proxy, making the rule ineffective. Also, applies to the backend service instead of network edge.

C: A backend security policy is the incorrect type. The proxy is sitting in front of Google Cloud, not within it. As in this case, we have a third-party WAF provider, hence we need to enforce at the EDGE network.

D: A backend security policy is incorrect, and `origin.ip` will not contain the actual client IP address due to the third-party WAF.

Here are links to Google Cloud documentation for further understanding:

Cloud Armor Overview: <https://cloud.google.com/armor/docs/overview>

Cloud Armor Security Policies: <https://cloud.google.com/armor/docs/configure/security-policies>

Cloud Armor Custom Rules: <https://cloud.google.com/armor/docs/rules-language>

Question: 236

CertyIQ

Your organization, TerramEarth, is launching a global application to manage credit card payments. There are some client VMs inside the same VPC as the application that need to access this application privately. Due to compliance requirements, the internal clients cannot use the global external IP address of the application. Currently, Cloud DNS only resolves `myglobalapp.terramearth.com` to the public IP address with a public zone. The clients will need to reach `myglobalapp.example.com`, without using its external IP address. You need to configure Cloud DNS to follow this requirement while following Google-recommended practices. What should you do?

- A. Create a sub-domain named `internal.terramearth.com`. Add the new DNS entry (`myglobalapp.internal.terramearth.com`) to the sub-domain pointing to the internal IP address from the application VM.
- B. Configure a query logic script inside Cloud DNS to check the source IP address from the VPC, and respond with a modified DNS record to include the internal IP address from the application VM.
- C. Configure a private zone for the application record (`myglobalapp.terramearth.com`) and point to the internal IP address of the application VM. Bind this zone to the VPC.
- D. Promote the ephemeral IP address from the application VM to static, add this static IP address to each internal client's host file, and change the `myglobalapp.terramearth.com` DNS record to this new static IP address.

Answer: C

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